Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY

Direct Testimony

Of

Charles F. Willard

Dated: April 28, 2023

Testimony of Charles F. Willard

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I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Charles F. Willard. My business address is 300 Erie Boulevard West, Syracuse, New York 13202.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc., a subsidiary of National Grid USA ("National Grid"), and currently hold the position of Director, NY Environmental. My responsibilities include overseeing the environmental and site investigation and remediation ("SIR") program along with other environmental programs for National Grid's New York operating companies, including The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY" or "Company") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI").

Q. Please describe your educational background and business experience.

A. I am a graduate of the State University of New York at Geneseo with a Bachelor of Arts degree in Geology. In addition, I hold a Master's in Engineering Geology with a concentration in Environmental Engineering from Drexel University and a Master's in Business Administration from LeMoyne University.

I have been employed by National Grid since 1996. Prior to my appointment as Director, I held the positions of Lead Environmental Engineer, Manager of New York SIR, and Director of SIR. Before joining National Grid, I held various management level positions in the field of environmental engineering and worked on projects such as environmental investigations, feasibility studies, remedial designs, and construction at large Environmental Protection Agency ("EPA") Superfund, Resource Conservation and Recovery Act, and New York State Superfund sites.

Q. Have you previously testified before the New York State Public Service Commission (the "Commission")?

A. Yes. I testified on behalf of KEDNY and KEDLI in Cases 16-G-0058 and 16-G-0059 (the 2016 KEDNY and KEDLI Rate Cases) and Cases 19-G-0309 and 19-G-0310 (the "2019 KEDNY and KEDLI Rate Cases"), and on behalf of Niagara Mohawk Power Corporation d/b/a National Grid in Cases 20-E-0380 and 20-G-0381 (the "2020 NMPC Rate Cases").

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to support the SIR costs included in KEDNY's revenue requirement. Specifically, I will:

(i) provide an overview of KEDNY's SIR program;

(ii) discuss the historic level of SIR costs incurred by KEDNY, including those incurred during the twelve months ended December 31, 2022 ("Historic Test Year"), as well as the forecast costs for the twelve months ending March 31, 2025 ("Rate Year") and the twelve months ending March 31, 2026 ("Data Year 1"), March 31, 2027 ("Data Year 2"), and March 31, 2028 ("Data Year 3") (collectively, the "Data Years");

(iii) present KEDNY's proposal for recovery of SIR costs in the Rate Year and Data Years, including its proposal to address the significant level of projected SIR costs at the Newtown Creek and Gowanus Canal Superfund sites; and

(iv) discuss KEDNY's cost control procedures.

I will also address KEDNY's compliance with the rate case filing requirements adopted by the Commission in its November 28, 2012 Order in Case 11-M-0034 ("SIR Generic Order").

Q. Do you sponsor any exhibits as part of your testimony?

A. Yes. Attached to my testimony are the following exhibits and appendix that were prepared under my direction and supervision:

(i) Exhibit __ (CFW-1) provides details on work progress at KEDNY's

manufactured gas plant ("MGP") and Superfund sites;

(ii) Exhibit __ (CFW-2) is an example of a New York State Department of Environmental Conservation ("DEC") Order on Consent;

(iii) Exhibit __ (CFW-3) is the Unilateral Administrative Order for the remedial design of the Gowanus Canal Superfund Site;

(iv) Exhibit __ (CFW-3A) is the Unilateral Administrative Order 2 on the Gowanus Canal Superfund Site;

(v) Exhibit _ (CFW-3B) is the Unilateral Administrative Order 2 Amendment on the Gowanus Canal Superfund Site

Exhibit __ (CFW-3C) is the Unilateral Administrative Order 3 on the Gowanus Canal Superfund Site;

(vi) Exhibit __ (CFW-4) is the Administrative Settlement Agreement and Order on Consent for the remedial investigation and feasibility study of the Newtown Creek Superfund Site;

(vii) Exhibit __ (CFW-5) provides examples of value engineering;

(viii) Exhibit __ (CFW-6) is a copy of the DEC work schedules for fiscal years ("FY") 2022, 2023 and 2024;

(ix) Exhibit __ (CFW-7) sets forth KEDNY's compliance with existing timetables and DEC requirements;

(x) Exhibit __ (CFW-8) provides KEDNY's past SIR program spend on an annual basis for 2019-2021;

(xi) Exhibit __ (CFW-9) sets forth KEDNY's SIR program spend in the Historic Test Year;

(xii) Exhibit __ (CFW-10) sets forth KEDNY's forecast SIR program spend in the Rate Year and Data Years; and

(xiii) Appendix 1 sets forth the Company's cost control efforts.

II. SIR Program Overview

Q. Please provide a brief overview of KEDNY's SIR program.

A. KEDNY's SIR program includes activities in connection with the management, investigation, and remediation of MGP and Superfund sites that have been contaminated by the past release of substances from property owned or formerly owned by KEDNY or its predecessors. KEDNY has responsibility for 27 MGP and two federal Superfund sites associated with waterways (Gowanus Canal and Newtown Creek) adjacent to former MGP sites. The Gowanus Canal is in active remedial construction and Newtown Creek is in the remedial planning process. Of the 27 MGP sites, four sites are in site characterization or remedial investigation, three sites are in remedial planning, three sites are in remedial action, one inactive site is being monitored and 12 sites have been either remediated but are subject to future monitoring and/or are subject to DEC site management plan obligations. Four sites have received a "No Further Action" determination

following site characterization and/or remediation. As I will discuss later in my testimony, remediation and investigation activities at the Gowanus Canal and Newtown Creek Superfund sites are projected to cause KEDNY to incur significant SIR costs in the Rate Year, Data Years, and beyond.

Exhibit __ (CFW-1) describes the sites currently being managed by KEDNY. The exhibit includes the background of each site and details the investigation and remediation activities that have taken place over the last three years from January 2019 through the end of the Historic Test Year.

Q. What types of costs does KEDNY incur under its SIR program?

A. Costs under KEDNY's SIR program include consultant and contractor costs, remediation activities aimed at reducing the volume, toxicity, or mobility of pre-existing contamination, and incremental external costs (including insurance and legal costs) incurred to seek recovery from third parties or to otherwise mitigate the Company's SIR costs or liabilities.

Q. What roles do the DEC and EPA play regarding the scope and timing of investigation and remediation work conducted at MGP and Superfund sites?

A. The DEC and EPA control the scope and timing of work at MGP and

Superfund sites. The scope of KEDNY's site investigations, work plans, clean-up, and field work decisions are reviewed, approved, and/or expanded by the DEC and/or the EPA pursuant to various orders. An example Order on Consent for 23 of KEDNY's MGP sites is provided in Exhibit __ (CFW-2). The EPA Unilateral Administrative Orders for the remedial design, remedial action of Remedial Target Area ("RTA") 1, and remedial design of RTAs-2 and 3 of the Gowanus Canal Superfund site and the Administrative Settlement Agreement and EPA Order on Consent for the remedial investigation and feasibility study of the Newtown Creek Superfund site are provided in Exhibit __ (CFW-3) and Exhibit __ (CFW-4), respectively. Although the Company has limited control over the scope and timing of SIR activities, KEDNY challenges the DEC and EPA when a more cost effective and equally protective remedy is available and, as described later in my testimony, manages costs in the areas it can control, such as contracting procedures. Exhibit __ (CFW-5) provides examples of changes to DEC and EPA remedy decisions and value engineering-related modifications following discussions with KEDNY that resulted in cost savings for customers.

Q. **How is the schedule for work at MGP sites determined?**

A. Prior to the beginning of each New York State fiscal year (April 1 through

March 31), KEDNY and the DEC meet to discuss the upcoming work schedule. The DEC approves the work schedule for the upcoming fiscal year only. As a result, the amount of spending in a given year is highly dependent upon the DEC and other third parties, including private property owners and permitting authorities. While the DEC only approves a one-year schedule, the Company and the DEC discuss the anticipated logical progression of work at each site for an additional year beyond the upcoming fiscal year, which gives the Company a sense of the projected work that will be required.

The DEC-approved schedule for the twelve months ended March 31, 2022, March 31, 2023 and March 31, 2024 is provided in Exhibit __ (CFW-6). The schedule includes KEDNY's 23 actively tracked MGP sites subject to DEC Orders on Consent. Three sites (Jamaica Holder Station, Keap Street Holder Station, and Rutledge Street Holder Station) have received "No Further Action" determinations and are no longer tracked with the DEC. Remediation at a fourth site, the Newtown Holder Station, was previously completed and the site closed in approximately 2002.

The schedule for work at the Gowanus Canal and Newtown Creek Superfund sites as required by the EPA is discussed later in my testimony.

Q. Has the Company prepared an exhibit setting forth the status of its remediation process in relation to existing timetables and DEC and EPA requirements?

A. Yes. Exhibit __ (CFW-7) shows the status as of the end of calendar year 2022 for each site, indicates whether there was a difference between the scheduled activities and the activities actually completed and, if there was a difference, explains the reason. Variances from the timetable have occurred at two of KEDNY's MGP sites and were due to delays resulting from site access constraints, and regulatory reviews or approvals, all of which were outside of KEDNY's control.

Q. Has KEDNY discussed the differences identified in Exhibit __ (CFW-7) with the DEC and EPA?

A. Yes. KEDNY updates the DEC quarterly and the EPA, monthly, on the status of each site. Further, the Company's project managers communicate with the DEC's and EPA's project managers on project schedule and progress of work as necessary. Based on these updates, it is my understanding that the DEC and EPA are satisfied with KEDNY's progress.

III. SIR Program Costs

Q. What level of SIR costs does KEDNY currently recover annually through base rates?

A. KEDNY's rate allowance currently includes: (i) historic, unrecovered SIR expenditures in an amount equal to one-tenth of the forecast SIR deferral balance as of December 31, 2016 ($18.521 million per year); and (ii) average forecast SIR costs of $70.92 million in FY 2021 – FY 2024) (the "Forecast Rate Allowance"). These amounts were approved in the Commission's August 12, 2021 Order in the 2019 KEDNY and KEDLI Rate Cases.

Because of the dynamic scope of the Gowanus Canal and Newtown Creek projects and the number of parties involved, it was difficult to predict the precise amount of these costs in the 2019 KEDNY and KEDLI Rate Cases. Consequently, the Forecast Rate Allowance reflected $20 million in each of the Rate Year and Data Years for costs associated with remediation of Gowanus Canal and Newtown Creek, based on the combined spending on these sites during the Historic Test Year.

Q. Does KEDNY fully reconcile SIR costs?

A. Yes. Each year, KEDNY fully reconciles its actual SIR expense (inclusive of Gowanus Canal and Newtown Creek costs) to the Forecast Rate Allowance. Any under or over expenditures are deferred for future refund to or recovery from customers (with the exception of the Citizens site). KEDNY agreed to absorb 10 percent of the remaining investigation costs for the Citizens site pursuant to the "Stipulation and Agreement Resolving Corporate Structure Issues and Establishing Multi-Year Rate Plan," dated June 25, 1996 in Case 95-G-0671.

Additionally, given the uncertain timing and potential magnitude of SIR costs, the 2016 and 2019 Joint Proposals in the 2016 KEDNY and KEDLI Rate Cases and the 2019 KEDNY and KEDLI Rate Cases authorized KEDNY to implement an SIR Recovery Surcharge.

Q. Please explain how the SIR Recovery Surcharge operates.

A. Beginning in calendar year 2018, if the difference between actual SIR expense (inclusive of Gowanus Canal and Newtown Creek costs) and the Forecast Rate Allowance exceeds $25 million on a cumulative basis, KEDNY is authorized to recover through the SIR Recovery Surcharge: (i) the difference between actual SIR expense and the Forecast Rate

Allowance, and (ii) any amount that was not recovered in the prior rate year's SIR Recovery Surcharge (if triggered) because the cumulative difference between actual SIR costs and the Forecast Rate Allowance did not exceed the $25 million threshold and/or the amount would have increased KEDNY's aggregate revenues by more than two percent.

To date, KEDNY has not implemented the SIR Recovery Surcharge. However, as explained below, given the size of the costs associated with Gowanus Canal and Newtown Creek, the same circumstances that supported Commission approval of KEDNY's SIR Recovery Surcharge still exist and warrant continuation of the mechanism.

Q. What have the Company's historic SIR costs been?

A. Prior to the Historic Test Year, during calendar years 2019 through 2021, KEDNY conducted investigation, remediation, and operations, maintenance, and monitoring ("OM&M") activities pursuant to Orders on Consent or EPA Administrative Agreements under the DEC's and EPA's remedial programs at 23 of its MGP and Superfund sites. KEDNY incurred approximately $42.173 million, $152.957 million, and $77.689 million of SIR costs in calendar years 2019, 2020 and 2021, respectively, for these activities. While investigation and OM&M activities contributed

to a steady and significant level of costs during this time, increased spending was primarily due to the remedial construction activities at the Citizens MGP site, Fulton MGP site, and Gowanus Canal Superfund site.

Q. What SIR costs did KEDNY incur in the Historic Test Year?

A. KEDNY incurred $76.080 million in SIR costs in the Historic Test Year, as shown in Exhibit __ (CFW-9). Of this amount, $37.829 million was incurred for design work and RTA-1 remedial construction at Gowanus Canal, and $6.573 million for remedial investigation activities at Newtown Creek (totaling $44.403 million). The remaining $31.678 million was for investigation and remediation activities at 23 of KEDNY's MGP sites. These activities included: remedial planning, remedial action, or OM&M activities at 19 sites; site characterization or remedial investigation activities at four sites; and SIR Program activities.

Q. What is KEDNY's forecast of SIR costs for work at the Company's MGP sites?

A. As shown in Exhibit __ (CFW-10), KEDNY forecasts SIR costs for work at MGP sites to be $58.100 million in the Rate Year and $49.618 million, $61.858 million, and $70.441 million in Data Years 1, 2 and 3, respectively. These forecasts do not include projected costs for Gowanus Canal or

Newtown Creek, which are addressed separately below.

Q. Please explain how KEDNY developed its forecast of SIR costs for work at its MGP sites.

A. The DEC schedule is not yet available for the Rate Year or Data Years. Therefore, to develop the forecast, KEDNY utilized the current approved DEC schedule as well as the anticipated work schedule for FY 2024. KEDNY considered the status of each site and knowledge derived from project managers regarding site progress and potential delays to determine the projected work on a site-by-site basis. KEDNY projected costs for each site based on the nature of the work at the site and the expected remedy for those sites where a remedy has been selected (*e.g.,* excavation of site soils, containment recovery, installation of barrier walls, treatment of groundwater, *etc.*), site specific conditions, and past spending and estimates provided by consultants for the anticipated work. In instances where the project manager anticipated delays (*e.g.,* potential site access issues, permitting delays, or onsite condition constraints), the timing of the work was adjusted in the forecast.

As shown in Exhibit __ (CFW-10), following the development of the MGP site forecasts, the total MGP site forecast was reduced by 20 percent to

account for unforeseen delays. This reduction was based on the average underspend over the past three calendar years. The forecast for the Gowanus and Newtown sites is the average of the past spend over the past three calendar years (with a three percent inflation factor applied).

Q. What are the major drivers of SIR costs in the Rate Year for KEDNY's MGP sites?

A. While KEDNY has made progress at most of its MGP sites, SIR spending above the Historic Test Year level is expected to continue through the Rate and Data Years. The increase in SIR costs at KEDNY's MGP sites (*i.e.,* excluding Gowanus Canal and Newtown Creek) from $31.678 million in the Historic Test Year to $58.100 million in the Rate Year is due to increased remedial action work that KEDNY anticipates will take place in the Rate Year. While the Company will continue its investigation and remediation activities at all 23 MGP sites worked on in the Historic Test Year, approximately 90 percent of the spending forecast for the Rate Year is due to activity at four sites. Specifically, KEDNY will be conducting significant remedial activities at the Citizens Gas Works, Fulton Municipal Gas Works, Metropolitan Gas Works, and Equity Works. Portions of the remedial plans at three of these sites (*i.e.,* Citizens, Fulton Municipal Gas Works, and Metropolitan Gas Works) must be completed on a schedule

coordinated with the EPA's schedule for the Gowanus Canal remediation.

The specific work at the Citizens Gas Works, Fulton Municipal Gas Works, Metropolitan Gas Works, and Equity Works, along with general SIR program activity, that will take place in the Rate Year includes the following:

Citizens Gas Works Site – The Citizens Site remediation of Parcels I, II and III was completed in 2021 in accordance with the DEC-approved 100 percent Remedial Design Plan and Decision Document. Following completion of the remediation, however, the DEC, in consultation with the EPA, determined that the contamination identified during the remedial construction phase was more significant than was anticipated during the design phase and, therefore, additional remediation is required. The extent of the additional remediation is still undetermined but will likely include some soil excavation and groundwater treatment during the construction of the new development. The Company anticipates that costs for these activities will be incurred through the Rate Year.

Fulton Municipal Gas Works – The EPA required that a cutoff barrier wall be constructed to separate the site from the Gowanus Canal. This barrier

wall was completed in 2021. Current work involves remedial support of private owners constructing buildings on the former MGP footprint during the Rate Year. The remediation that New York City is expected to perform related to the combined sewer overflow tank construction is anticipated to run through the Rate Year and the Data Years. The New York City park and pool-related remedial activities are anticipated to start in the Data Years.

Metropolitan Gas Works - The installation of the cutoff barrier wall to separate the site from the Gowanus Canal is anticipated to be performed under an expedited Interim Remedial Measure and is planned during the Rate Year. The work is expected to include remedial measures that are protective of the Gowanus Canal

Equity Works Site – Remedial construction is planned during the Rate Year. Following this remediation, the site will be returned to the property owner and future site management will fall under the site management and OM&M plans.

SIR Program Activities – Activities including pursuit of cost recovery from other parties and legacy insurance policies are expected to continue along

with other SIR program activities.

Q. What are the major drivers of SIR costs in the Data Years for KEDNY's MGP sites?

A. For Data Year 1, spending is forecast to remain significant, but lower than the Rate Year, at approximately $49.618 million. In Data Year 1, the Citizens and Fulton Site will continue to incur significant remediation costs. The Greenpoint Site and Williamsburg Site are forecast to have moderate costs for investigation and site management support activities, remedial design and remedial construction. The Metropolitan and Equity Site costs will decrease as construction activities are completed.

In Data Year 2, KEDNY anticipates spending approximately $61.858 million. This encompasses continued work at the Fulton site and increased spending at the Williamsburg Site due to construction activities, as well as moderate spending at the Greenpoint Site due to design, investigation and site management activities. Citizens Site spending decreases as construction activities are completed.

For Data Year 3, KEDNY forecasts spending to remain higher than the Rate Year, Data Year 1, and Data Year 2 levels, at approximately $70.441

million. The Company expects to continue to incur costs at the Fulton, and Williamsburg sites, and increased spending is expected at the Jamaica Gas Light site to support remedial action. The Company will also continue moderate spending at the Greenpoint Site in support of site management activities.

Q. Could actual SIR costs differ from the Company's forecast for the MGP sites?

A. Yes. The forecast reflects the Company's best estimate of SIR costs. While KEDNY believes that its forecast is reasonable, SIR costs are subject to a high degree of variability, through no fault of the Company, as projects are subject to schedule and scope modifications by the DEC and the EPA, as well as site access issues with property owners or delays resulting from onsite condition constraints. In addition, estimates of future costs for SIR activities can be significantly influenced by the pace of redevelopment in areas where former MGP facilities are located. Areas of Brooklyn that historically were used for industrial and commercial purposes, particularly along waterfronts, are being redeveloped rapidly for residential and mixed use. Redevelopment at these properties or adjacent properties could significantly influence the schedule and scope of remedial activities at a number of sites. KEDNY's forecast reflects costs that it expects will be

incurred in the Rate Year and Data Years; however, there is a risk that actual costs could be more or less. Accordingly, as discussed below, KEDNY is proposing to continue fully reconciling these costs.

Q. Turning to the Gowanus Canal and Newtown Creek Superfund sites, please provide some background on these sites.

A. Gowanus Canal - KEDNY had three former MGP facilities located along the Gowanus Canal – Fulton, Citizens, and Metropolitan. These former MGP facilities are individual remediation projects (as discussed above) and, along with other industries and processes located on or near the canal, have contributed to the need for environmental remediation of the canal itself, which is listed on the EPA's Superfund National Priorities List.

In 2013, the EPA identified a remediation plan for the Gowanus Canal that involves removing contaminated sediment via dredging, installing a cap, and other activities. In 2014, the EPA issued a Unilateral Administrative Order ("UAO"), provided in Exhibit __ (CFW-3), to KEDNY and 26 other parties to perform the remedial design of the EPA's selected remedy in accordance with the schedule set by the EPA. The EPA led and expedited the completion of the remedial investigation and feasibility study in a little more than two years. The cleanup is being designed in multiple phases

(RTAs 1,2, and 3). Each RTA is approximately one-third of the canal length with RTA-1 being at the head of the canal, or furthest north, and RTA-2 in the middle, and RTA-3 at the southern extent where the canal connects to Gowanus Bay. The design sequence requires RTA-1 construction be completed first, followed by RTA-2, and finally, RTA-3.

In 2019, the EPA issued UAO 2, provided in Exhibit __ (CFW-3A), to KEDNY and the same 26 parties, ordering the parties to perform the tasks necessary to provide structural support for bulkheads and infrastructure in preparation for implementation of the dredging and capping remedy selected in the EPA's September 27, 2013 Record of Decision for the Gowanus Canal. An amendment to the UAO, provided in Exhibit _ (CFW-3B), was issued in April 2019 to KEDNY and the same 26 parties, assigning additional bulkheads and infrastructure requiring support during the remedial construction of RTA-1. In 2020, the EPA issued UAO 3, provided in Exhibit __ (CFW-3C), ordering six of the original parties, including KEDNY, to implement the remedial action in RTA-1 and procure a site staging area for future RTA-2 dredging and capping work. The final design inclusive of the remedial requirements in the Record of Decision and UAOs was completed in September 2020 and construction began in RTA-1 in November 2020 with the site mobilization. As of December 2022,

RTA-1 remedial action is underway and more than 50 percent complete. It is important to note that at least 26 other parties are responsible for the costs of the remedial design of the Gowanus Canal under the UAO and at least a subset of this group has been named responsible for construction and cleanup under the existing remedial action UAOs, and presumably, any future EPA orders. KEDNY has been proactive in its efforts to research and work with the EPA to identify and engage these and other potentially responsible parties to equitably share the cost of investigation and remediation.

Newtown Creek - At the Newtown Creek Superfund site, the remedial investigation and feasibility study commenced in 2011 and is in progress. This project is proceeding more slowly than Gowanus Canal because the EPA is pursuing a traditional remedial investigation conducted by a group of responsible parties. KEDNY and five other responsible parties have entered an Administrative Settlement Agreement and Order on Consent, provided in Exhibit __ (CFW-4), with the EPA to conduct the Newtown Creek remedial investigation and feasibility study. The remedial investigation of this site is still in review with the EPA as of January 2023. The feasibility study to develop and evaluate remedial alternatives has commenced. However, it is likely to be some time before a remedy is

evaluated, selected, and designed, and construction commences. During the remedy evaluation process, it is expected that some supplemental investigation will be required. Given the size and complexity of this site, the remainder of the investigation, design, and construction is likely to be lengthy, extensive, and costly. The responsible parties have proposed an Early Action to the EPA and are working to develop the scope and objectives for this expedited remedial work that will target a portion of Newtown Creek.

KEDNY continues to be very active in discussions with the other responsible parties and with the EPA to manage the process and ensure a fair and equitable characterization of impacts and allocation of responsibility. KEDNY also is working with the other responsible parties who executed the settlement agreement to develop information regarding the operations and resulting environmental impacts of still other responsible parties to enable KEDNY to seek cost recovery proportionate to its equitable share of liability.

Q. Does the Company expect to incur SIR costs in the Rate Year and Data Years for the Gowanus Canal and Newtown Creek sites?

A. Yes. KEDNY expects to incur significant SIR costs for work performed

in the Rate Year and Data Years for both Gowanus Canal and Newtown Creek. The combined spending for the Gowanus Canal and Newtown Creek during the three years ended December 31, 2022 averaged approximately $62.400 million. While it is difficult to estimate the precise costs for the Gowanus Canal and Newtown Creek Superfund sites, the Company anticipates that the three-year average is representative of the spending that will be incurred for these two sites over the next several years. Therefore, as shown in Exhibit ___ (CFW-10), the Company's forecast includes an average of $67.225 million/year (with a three percent annual inflation applied) as the reasonable estimate of costs for the Gowanus Canal and Newtown Creek Superfund sites.

With respect to the Gowanus Canal, the RTA-1 construction should be substantially completed, with the exception of turning basin one construction, which will likely start and continue during the Rate Year and Data Years. RTA-2 construction is anticipated to start during the Rate Year and continue throughout the Data Years. Remedial Design for RTA-3 and associated pre-construction field activities will also be underway during the Rate Year and Data Years. The work will be conducted in accordance with the EPA Record of Decision, which estimated the total project cost at $506 million. The pace of progress and the associated costs

are highly dependent on regulatory drivers and third-party access and cooperation along the entire length of the canal.

As for Newtown Creek, work is expected to continue through the Rate Year and Data Years and involve the completion of both the Remedial Investigation, Feasibility Study process, and planning for implementation of Early Action. The Newtown Creek Record of Decision is scheduled to be issued by EPA in 2028. The longer-term costs associated with the feasibility study and remedy selection process, the Early Action in a portion of Newtown Creek, and other project costs are highly uncertain given the changing scope and schedule of the project, which is being driven by the EPA. While near-term costs at the Gowanus Canal are expected to be higher, the total costs for Newtown Creek are likely to exceed those of Gowanus Canal, as Newtown Creek is larger, more polluted, and will require extensive design and remediation work.

Q. What is KEDNY's proposal for recovery of its SIR costs?

A. KEDNY proposes to include the following cost components in base rates:

(i) a forecast of $100 million per year in each of the Rate Year and Data Years for the MGP sites and Gowanus Canal and Newtown Creek SIR work; and

(ii) the current amortization of $18.521 million annually, representing one-tenth of the forecast deferral balance at December 31, 2016 through December 31, 2026 ("Historic Amortization").

While the average forecast for the MGP sites and Gowanus Canal and Newtown Creek SIR work is $127 million per year, the Company believes that its $100 million per year forecast is a more reasonable estimate for inclusion in rates. KEDNY will continue to pursue offsets to remediation costs, including insurance recoveries, the identification of potentially responsible parties, and property sales. Some of these potential offsets are expected to occur in the Rate Year and Data Years but are not known or measurable at this time. The forecast rate allowance accounts for these potential future offsets in a manner intended to mitigate bill impacts.

Because of the uncertainty and magnitude of SIR costs, KEDNY proposes to continue the current SIR reconciliation mechanism, whereby actual SIR expenses are reconciled to the base rate allowance (exclusive of the amortization), with any difference deferred for future recovery from, or credit to, customers. The reconciliation mechanism is critical as a number of the Company's MGP sites are anticipated to continue to have significant expense variability. The provision in the current reconciliation related to

the Citizen's Site, requiring the Company to absorb ten percent of the costs, would also continue. KEDNY also proposes to continue the current SIR Recovery Surcharge.

Q. Why does KEDNY believe that its proposal is consistent with the Commission's guidance and in the best interests of customers?

A. The rate recovery proposal is consistent with the SIR Generic Order as it is tailored to address the specific conditions at KEDNY – namely, the rate allowance reflects significant forecast SIR costs combined with a means to recover a portion of KEDNY's historic deferral balance. In the SIR Generic Order, the Commission noted its intent to retain the flexibility to "tailor the rate treatment of SIR costs to the concrete conditions of each utility" and moderate annual bill impacts by spreading the amortization of costs over a longer period "as needed and as dictated by the costs in specific rate cases." Increasing the rate allowance for forecast MGP-related SIR costs and including a projection of costs for Gowanus Canal and Newtown Creek allows the Company to recover costs going forward and hopefully avoids adding to an already substantial deferral balance. The projected rate allowance also factors in potential offsets to remediation costs that may occur but are unknown at this time. Maintaining the SIR Recovery Surcharge enables the Company to mitigate bill impacts by not reflecting

an overly aggressive forecast in base rates, while addressing the uncertainty in forecasting. These same circumstances justified the Commission's adoption of the SIR Recovery Surcharge in the 2019 KEDNY and KEDLI Rate Cases. The recovery proposal reflects the specific SIR cost issues affecting the Company and is intended to avoid future rate shock to customers.

Q. What is the Company's proposal for the Historic Amortization that ends December 31, 2026?

A. The Historic Amortization was calculated based on an estimate of the SIR deferral balance at December 31, 2016. At December 31, 2026, KEDNY forecasts a remaining regulatory asset of $19.188 million associated with the Historic Amortization. As it was a forecast, the Company will reconcile the amount at December 31, 2016 for future disposition.

IV. SIR Cost Control Efforts

Q. What steps has the Company undertaken to control its SIR costs and liabilities?

A. KEDNY follows the cost management practices set forth in the Inventory of Best Practices for Utility SIR Programs established in the SIR Generic Order and as clarified by the DEC. Appendix 1 describes the Company's

cost control efforts.

Q. Has the Commission previously reviewed KEDNY's cost control measures?

A. Yes. The Commission has reviewed the Company's cost control measures many times. In its November 28, 2012 Order in Case 06-G-1185, which authorized KEDNY to recover a portion of its SIR deferral balance, the Commission found that KEDNY had employed cost effective measures and that the costs through December 31, 2009 had been prudently incurred. In its October 19, 2015 Order in Case 15-G-0323, the Commission reviewed the Company's oversight and cost containment procedures and found that they appeared reasonable. In its December 16, 2016 Order in the 2016 KEDNY and KEDLI Rate Cases, the Commission noted that Staff had reviewed and concurred that the Company was pursuing all appropriate cost control efforts.

V. Compliance with Rate Case Filing Requirements

Q. Are you familiar with the rate case filing requirements adopted by the Commission in the SIR Generic Order?

A. Yes, I am. At page 30 of the SIR Generic Order, the Commission required that "[in] any future rate filing in which a utility seeks to recover SIR expenses, the utility must provide sworn testimony (1) establishing that the

remediation process is in compliance with existing timetables and DEC requirements, or providing explanations for any divergence; (2) discussing the utility's SIR cost control efforts, including an attestation to utility compliance with the best practices inventory; and (3) indicating the results of any internal process the utility may have conducted with respect to review of SIR procedures, and in particular explaining how internal controls are brought to bear on site investigation and remediation projects."

In addition, in the 2019 KEDNY and KEDLI Rate Cases, the Commission required that "the Companies submit Annual Reports in Case 11-M-0034 that: detail the specific remedial and programmatic activities scheduled and performed at all contaminated sites during each of the rate years and the Stay Out period; identify and include the DEC or USEPA administrative orders and approved work plans and timetables for implementation of remedial activities; and break down the remedial costs incurred for such activities, including but not limited to a detailed identification of the costs associated with the General SIR Program and cost expended on various SIR participants (*i.e.,* consultants, engineers, contractors, attorneys, and others)."

Q. Has the Company complied with each of these requirements in this filing?

A. Yes, it has.

Q. Please discuss the Companies' compliance with each of these requirements.

A. Earlier in my testimony, I discussed the Company's compliance with existing timetables and DEC requirements, with reference to Exhibit __ (CFW-7). The Company's cost control efforts are also discussed earlier in my testimony and in Appendix 1, which details SIR cost control efforts. With respect to the Company's internal controls and how they apply to SIR projects, National Grid has conducted internal reviews of the administration and management of the SIR group, as well as the adequacy of controls to meet significant environmental obligations. Further, as discussed above and in Appendix 1, the Company utilizes established procedures for selecting environmental consultants and contractors as well as a rigorous process for reviewing and approving consultant and contractor invoices as additional internal controls on SIR projects.

Q. Does this conclude your testimony?

A. Yes.

APPENDIX 1 – KEDNY SIR COST CONTROL EFFORTS

The Company follows the cost management practices set forth in the Inventory of Best Practices for Utility SIR Programs established in Case 11-M-0034 and as clarified by the New York State Department of Environmental Conservation ("DEC"). Examples of the Company's SIR cost control efforts are discussed below.

Experienced Staff and Contractors

The Company staffs its SIR program with experienced project managers. The project managers are responsible for, among other things, monitoring project activity, ensuring that work is performed in accordance with work plans and project specifications, and managing project costs, including reviewing contractor invoices. The Company also utilizes specialized environmental consultants and contractors to perform investigation, engineering design, remedial evaluation and construction, construction oversight, and operations and maintenance activities in accordance with the Company's policies and accepted industry practices.

Competitive Bidding Process

Procurement of consultants and contractors differ due to nature and scope of the work. The scope of work for contractor work is typically well defined in bid

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specifications based on prior field studies and approved by the DEC. The scope of work for consultants is defined based on the immediate project scope of work which may expand due to the progressive nature of the work. For instance, the scope could change following the collection of new data and direction set by the DEC. As more data is collected, it is often advantageous to keep the same consultant on the project due to reduced transaction costs and the benefit of consultant's site knowledge.

Contractors

Contractor work relating to environmental response activities are subject to a project-specific competitive bid selection process, with the exception of a select number of small projects performed under Blanket Construction Contracts. Blanket Construction Contracts are established through competitive bids for rates and equipment and are utilized for smaller, time and material construction projects and routine site support work at select sites. During the project-specific competitive bid process on larger projects, a request for proposal ("RFP") is issued to usually three to six pre-qualified vendors. Final vendor selection is based on specific criteria such as cost, technical merit, and personnel qualifications.

Exceptions to these standard procedures are made in situations where it is necessary to maintain vendor consistency through multi-phased assessment/remediation

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projects or to implement immediate or short-term response to imminent hazards when there is not time for a competitive bid process. For work not competitively bid or performed under a Blanket Construction Contract, the reasons for not competitively bidding the work are clearly documented and must be approved by Procurement as an exception to the usual bid process.

Consultants

In 2018, the Company concluded a procurement to secure competitive pricing and terms to establish new Master Services Agreements ("MSAs") for routine SIR related environmental consulting work, as well as, construction management work. Consultants were initially evaluated based on their qualifications and competitive rates. The Company negotiated MSAs with the consulting firms that offered the best value for the services needed while taking advantage of the work volume discounts afforded to National Grid. As a result, the consulting pool was expanded from five to seven firms providing a wider range of expertise to support program activities.

Where specialty consulting is needed, particularly in support of legal matters, interviews are conducted and rates established with a single vendor in consideration of the specialty services required.

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Consultants provide site-specific proposals, along with Work Authorization Forms for new assignments, followed by Change Notifications to address changes to the scope of work. The proposed scope is reviewed by the SIR Company Project Manager and passed onto management for review and approval. Because the Company Project Manager is in frequent communication with the consultant, receives work products and periodically observes the field work, they verify the work completed in the monthly invoice. A project analyst is tasked with verifying that the invoiced items and rates are consistent with the MSA contract.

Invoice Review

Vendors must prepare detailed monthly invoices that include the following information:

- a project description that meets applicable regulatory requirements and identifies the purchase order number and work authorization number;

- the budget for each task, the amount invoiced for each task on the current invoice, the total amount invoiced to date, and the amount remaining in the purchase order or work authorization for completion of the task;

- backup to the invoice that contains a detailed breakdown of costs for each task must be attached to each monthly invoice. Required details include

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where appropriate labor charges broken down by employee name, labor category, billable rate, current and cumulative hours worked on the task, current and cumulative dollars billed, and a breakdown of other direct charges; and

- for lengthy projects, the vendor must submit a monthly progress report that identifies, by task, the activities completed in the period, the task budget, the amount expended during the current billing period, and the total amount expended to date. The vendor must also estimate the percent completion of each task.

In addition, contractor invoices are reviewed by comparing the summary and detailed invoices to the original bid spreadsheet to determine if the vendor's level of effort corresponds to the bid estimate. Special attention is directed to those tasks where the level of effort and/or the cost exceeds the estimate. In such cases, contractors are required to submit change order requests with revised cost detail. Following SIR Project Manager review and, if appropriate, acceptance of all or a portion of the proposed increase, the contractor updates the spreadsheet with the approved change.

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Cost Effective Alternatives

Although the Company has limited control over the scope and timing of its SIR activities, it does manage costs in the areas it can control, such as contracting procedures, and challenges the DEC and the United States Environmental Protection Agency ("EPA") when a more cost effective and equally protective remedy is available. The Company manages its SIR programs to eliminate risk to human health and the environment in a cost effective manner, and works closely with the environmental regulatory agencies to accomplish this goal. Examples of instances where the Company was successful in developing and receiving approval for more cost-effective alternatives than the DEC's initial remedy decisions are provided in Exhibit __ (CFW-5).

In addition, to ensure appropriate disposal of materials from all MGP sites, the Company uses only facilities approved by the National Grid Vendor Advisory Group, a cross-functional group comprised of National Grid employees from the SIR, Environmental Compliance, Legal and Procurement Departments. The group employs a pre-qualification and auditing process that reduces the likelihood that disposal facilities themselves ultimately become the subject of future environmental proceedings.

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Outreach Efforts

The Company provides MGP site progress reports to the DEC and EPA on a quarterly, and monthly basis, respectively. The Company also participates in outreach to local communities and officials, citizen groups, and property owners, and established public websites for major MGP sites that contain information about the sites and the respective investigation and remediation activities. The Company strongly believes that purposeful community outreach efforts are critical to a successful remediation effort.

Insurance Cost Recovery

The Company attempts to reduce its SIR cost responsibility by seeking recoveries from insurers and other responsible third parties. Efforts to recover outstanding insurance claims have involved a combination of litigation and settlement activities associated with the Company's SIR sites. To date, the Company has settled its MGP claims with a number of insurance companies for approximately $69.8 million. The Company is also engaged in litigation with other insurers.

Identification of Other PRPs

The Company seeks aggressively to contribute only its equitable share of remedial costs. For example, the Company filed a federal Superfund lawsuit in the Eastern District of New York against 15 parties seeking cost recovery and contribution for

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environmental remediation costs in the area of the former Williamsburg MGP site. The Company uses a similar strategy to seek other responsible parties at a number of its former MGP sites. The Company also works collaboratively, where possible, with property owners to conduct the construction phases of SIR projects during property redevelopment or construction activities to minimize the overall contribution of costs. This approach provides opportunities for synergies that could yield cost savings for all parties.

In addition, the Company has taken a proactive approach to ensure that it pays only an equitable share of the remediation costs at Gowanus Canal and Newtown Creek, two federal Superfund sites with multiple responsible parties. The Company has performed research to identify potentially responsible parties, worked with both EPA and the cooperative parties to help ensure equitable participation in the projects, and worked through a mediation process to establish liability shares for site-related costs.

INDEX OF EXHIBITS

Exhibit __ (CFW-1)	Details on the progress of work at KEDNY's manufactured gas plant ("MGP") sites
Exhibit __ (CFW-2)	New York State Department of Environmental Conservation ("DEC") Order on Consent example
Exhibit __ (CFW-3)	Unilateral Administrative Order for the remedial design of the Gowanus Canal Superfund Site
Exhibit __ (CFW-3A)	Unilateral Administrative Order 2 on the Gowanus Canal Superfund Site
Exhibit __ (CFW-3B)	Unilateral Administrative Order 2 Amendment on the Gowanus Canal Superfund Site
Exhibit __ (CFW-3C)	Unilateral Administrative Order 3 on the Gowanus Canal Superfund Site
Exhibit __ (CFW-4)	Administrative Settlement Agreement and Order on Consent for the remedial investigation and feasibility study of the Newtown Creek Superfund Site
Exhibit __ (CFW-5)	Value Engineering examples
Exhibit __ (CFW-6)	DEC work schedules for fiscal years ("FY") 2022, 2023 and 2024
Exhibit __ (CFW-7)	KEDNY's compliance with existing timetables and DEC requirements
Exhibit __ (CFW-8)	KEDNY's past SIR program spend on an annual basis for 2019-2021
Exhibit __ (CFW-9)	KEDNY's SIR program spend in the Historic Test Year
Exhibit __ (CFW-10)	KEDNY's forecast SIR program spend in the Rate Year and Data Years

Exhibit ___ (CFW-1)

Details on the progress of work at KEDNY's manufactured gas plant ("MGP") sites

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Bay Ridge Holder Stations A & B	NYSDEC Site ID:	224058
Site location:	A: 837 64th Street, Brooklyn, NY; B: 829-884 65th Street, Brooklyn, NY	PRP:	NA

Site Background

The Site is a former holder station located on two non-contiguous parcels in a primarily commercial area of Brooklyn, NY. Station A, or OU-1, is non-owned and is currently a parking lot for an adjacent supermarket. Station B, or OU-2, is owned by National Grid. Station B contains little league ball fields, the National Grid Bay Ridge gate station, and a currently vacant area.

OU-1 requires no further action as stated in the February 2012 SC Report approval letter. NYSDEC requested an Interim SMP be prepared and an environmental easement be used to restrict the future uses of the gate station area and an adjacent vacant area of Station B (OU-2).

Site Investigation and Remediation Progress January 2019 through December 2022

Per direction of NYSDEC, annual groundwater monitoring is on-going. The emerging contaminants groundwater sampling summary report was submitted to NYSDEC in February 2019. Sampling and monitoring support for National Grid's capital improvement project commenced in 2019. A NOIA regarding a capital improvement project was submitted to NYSDEC in March 2019. An intrusive activity summary report was submitted to NYSDEC in June 2019.

The Interim SMP was revised in April 2020 and approved in October 2021. The property was re-surveyed for the Interim SMP Environmental Easement in November 2020. A recommendation to reduce groundwater sampling was submitted to NYSDEC in the 2020 Annual Groundwater Summary Report in February 2021 and approved in March 2021 from annual to biennial. A soil import form for the vacant portion of the property was submitted to NYSDEC in May 2021. Import request forms were submitted to NYSDEC in March 2022.

SITE INFORMATION AND PROGRESS SUMMARY				
General Site Information				
Site Name:	Belmont Holder Station	NYSDEC Site ID:	224060	
Site location:	290 Belmont Avenue, Brooklyn, NY 11207		PRP:	NA
Site Background				
The approximately 1-acre Site was a former holder station and is located in a low density residential area of Brownsville, Brooklyn, NY. Owned by Con Edison, the site is zoned for light manufacturing and commercial use and is part of Con Edison's Brownsville Substation. A SC is required.				
Site Investigation and Remediation Progress January 2019 through December 2022				
National Grid has prepared a SC plan, and it has been approved by the NYSDEC. The SC will be conducted once the site is no longer "on hold" with the NYSDEC and access to the site is obtained.				

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Brooklyn Gas Light Works MGP Site	NYSDEC Site ID:	224048	
Site location:	1 Hudson Avenue, Brooklyn, NY 11201		PRP:	NA

Site Background

The approximately 5-acre Site was a former MGP and is located along the East River water front in a low density residential area in the Vinegar Hill neighborhood of Brooklyn, NY. The site is situated on portions of 2 properties, one being owned by the Department of the Navy and the other being owned by Con Edison. The site parcels are zoned for industrial use. Most of the site is on the Con Edison property, which is the site of a former power generating station. Demolition of the power station had begun as of January 2012. The smaller portion of the former site footprint falls within the Brooklyn Navy Yard.

A SC is required.

Site Investigation and Remediation Progress January 2019 through December 2022

National Grid has prepared a SC plan, and it has been approved by the NYSDEC. The SC will be conducted once the site is no longer "on hold" with the NYSDEC and access to the site is obtained.

The property owner has begun demolition of the former generating station.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Citizens Gas Works MGP Site	NYSDEC Site ID:	C224012
Site location:	Smith Street and Luquer Street, Brooklyn, NY	PRP:	NA

Site Background

Citizens MGP Site, also referred to as the Carroll Gardens Site, is an 11-acre former MGP site located in Brooklyn, NY on the Gowanus Canal. The site OU-1 parcels are owned by a 3rd party and NYC. The NYC owned property included a concrete production facility operated by a 3rd party which stopped operating in 2018. The off-site areas currently subject to remedial investigation are referred to as OU-2. The site was under the VCA until 2007 when it became part of the NYS BCP with the other property owners to accelerate redevelopment of the site after the MGP remediation. The property is mostly vacant and surrounded by commercial and residential properties and the Gowanus Canal which is under the federal superfund program. National Grid previously received information regarding properties within OU-2 that may be subject to redevelopment in the future. The redevelopment of Parcel 3 is expected to start in 2023, and potentially in 2024 for Parcel 1.

The RDD for OU-1 was issued by NYSDEC in April 2007. In March 2012, the NYSDEC requested and approved the commencement of a DNAPL O&M program to collect DNAPL at the site.

Site Investigation and Remediation Progress January 2019 through December 2022

Addenda No 1 and 2 to the NYSDEC-approved 100% Design regarding changes to some of the bulkhead tie-backs was submitted to NYSDEC in February 2019. Addendums numbered 5 and 6 to the RD were submitted to NYSDEC in July and August 2021, respectively. A Data Summary Report for Parcel IV was submitted to NYSDEC in August 2021. The Supplemental Soil Investigation Work Plan for Parcel IV was submitted to NYSDEC in October 2021. A Baseline Groundwater Monitoring Work Plan was submitted to NYSDEC in December 2021. A Data Summary report for OU-2 offsite areas was submitted to NYSDEC in November 2021.

National Grid worked with NYC to coordinate the Site remediation schedule and the timing of the City's former tenant completing the full demobilization of their former concrete production facility in the Summer of 2019. NYSDEC issued a fact sheet in May 2019 and held an availability session in June 2019 to discuss the start of the remediation project at the site. National Grid contactors mobilized to the site and initiated the remedial construction activities for Parcels I, II, and III in July 2019. Remedial activities were completed in Q1 2022. Site restoration activities were completed in Q2 2022.

The DNAPL recovery program continued until June 2019 when it was suspended in anticipation of the remedial construction efforts and removal of the concrete pile/debris. The installation of new piezometers and recovery wells at OU-1 on Parcels II and III of the Site was completed in Q1 2022. The DNAPL recovery program resumed for OU-1 in Q1 2022.

Coordination with the Parcel IV property owner to obtain access for the implementation of the NYSDEC-approved supplemental soil investigation occurred throughout 2022. In Q1 and 2 2022, monitoring wells were installed, existing wells were developed, and wells were sampled in accordance with the Baseline Groundwater Monitoring Work Plan. The draft Interim SMP for Parcels I and II of the Site was submitted to NYSDEC in April 2022. The NYSDEC-requested survey and available gauging data for Site-related wells were submitted to NYSDEC in May 2022. The Baseline Groundwater Monitoring Report was submitted to NYSDEC in May 2022. The Supplemental Off-Site Investigation Work Plan for OU-2 was submitted to NYSDEC in August 2022. Semi-annual post-remediation groundwater sampling commenced in October 2022. The NYSDEC-approved supplemental off-Site investigation field activities for OU-2 were initiated in October 2022. Groundwater samples were collected from newly-installed off-Site groundwater monitoring wells in December 2022.

SITE INFORMATION AND PROGRESS SUMMARY				
General Site Information				
Site Name:	Clifton Former MGP Site	NYSDEC Site ID:	243023	
Site location:	25 and 40 Willow Avenues, Staten Island, NY 10305		PRP:	NA

Site Background

The MGP site is located on Staten Island. The site is divided into two OUs. OU-1 is located at 40 Willow Avenue and is the location of a former relief holder and has a gas regulator station operating on part of it. The OU-2 is located on 25 Willow Avenue and is the location of the former MGP plant. The property was leased to Saturn as a new vehicle preparation facility until 2005. Both OUs are owned by National Grid. The surrounding properties are mixed commercial and residential. Remedial activities were completed at the site from 2009 through 2015 in compliance with the ROD and RD.

Site Investigation and Remediation Progress January 2019 through December 2022

Oversight and CAMP monitoring of geotechnical borings for NYCDDC for upcoming sewer and water utility installation in Willow Avenue was started in June 2019. A summary report regarding the NYCDDC geotechnical project was submitted to NYSDEC in August 2019. Discussions and outreach with NYCDDC for the upcoming utility installation project are ongoing. A NOIA regarding the National Grid Gas Operations project was submitted to NYSDEC in April 2020 and CAMP, oversight, and soil and groundwater disposal were provided during this work and was completed March 2021

SMP reporting, NAPL monitoring and recovery, and groundwater sampling and management is ongoing. Semi-annual reports are submitted to NYSDEC in January and July. Operations, monitoring and sampling of the Groundwater Treatment System for the Containment Cell is ongoing. The emerging contaminants groundwater sampling summary report was submitted to the NYSDEC in February 2019. During Q4 2022, three piezometers and two vapor intrusion sampling points were abandoned at an offsite property.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Coney Island MGP Site (Brooklyn Borough Gas Works)	NYSDEC Site ID:	224026	
Site location:	Shell Road and Shore Parkway, Brooklyn, NY		PRP:	NA

Site Background

The site is a former MGP site located on Neptune Avenue and Shore Parkway, Brooklyn, Coney Island New York. The property is currently owned by National Grid and two small parcels are owned by the NYCT. OU-1 ROD to install barrier wall around the perimeter of the site was issued in March 2001. The OU-2 ROD to remediate the creek and uplands impacted areas was issued in March 2002 and was subsequently split into OU-2 and OU-3, - respectively. The adjacent properties are primarily commercial/residential. There are no active utility operations on the property.

The OU-1 barrier wall installation was completed in 2004. The OU-2 creek dredge and OU-3 uplands excavation remediation were completed in November 2008. In 2010, the NYSDEC approved the SMP governing O&M activities at the site which includes gauging and sampling select wells, measuring the engineered cap thickness in the creek and confirming the uplands engineered cap system and ecological buffer are intact. In February 2012, the NYSDEC issued an Explanation of Significant Difference removing the groundwater and NAPL collection and treatment system components from the O&M activities after National Grid proved that NAPL recovery via automated system was not feasible. In June 2012, the Environmental Notice for the site was recorded with the NYC Registrar's Office.

Site Investigation and Remediation Progress January 2019 through December 2022

The emerging contaminants groundwater sampling report was submitted to the NYSDEC in February 2019. A letter from NYSDEC was received June 2020 that the sampling for the site has been satisfied and no further sampling is required at this time. A recovery well installation and monitoring well abandonment request letter was submitted to NYSDEC in March 2019. New recovery wells were installed in locations with accumulating DNAPL and monitoring wells were abandoned in September 2019. A work plan for the abandonment and re-installation of two damaged monitoring wells was submitted to NYSDEC in September 2019 and performed in December 2019.Currently, O&M in accordance with the approved SMP (Revision 5, June 2019) is being performed on a periodic basis. Periodic review reports for O&M activities are submitted to NYSDEC annually. Semi-annually sampling was reduced to annual sampling in 2020 in accordance with the SMP.A letter requesting the postponement of additional evaluation of scouring in Coney Island Creek was submitted to NYSDEC in March 2019. NOIAs regarding intrusive NYCDDC projects were submitted to NYSDEC in April 2019 and March 2021. A NOIA regarding Coney Island Creek sampling conducted by the USEPA was submitted to NYSDEC in April 2021.To support upcoming construction activities to be performed by NYCDDC and the MTA, a work plan for a boring program within the Coney Island Creek was submitted and approved by NYSDEC in April 2019. The borings were conducted in June 2019. A summary report for the borings was submitted in July 2019. Several monitoring wells were modified from stick-up to flush mount prior to the start of the MTA project. Environmental oversight and CAMP monitoring of intrusive activities for the MTA project began in September 2019 and was completed in 2021. A summary report for the CAMP monitoring of NYCDDC projects was submitted in December 2021.

A work plan for the abandonment of recovery wells and an evaluation of the passive vent system were submitted to NYSDEC in May 2022 and a revised work plan was submitted to NYSDEC in August 2022. A NOIA was submitted to NYSDEC regarding investigation of the property related to due diligence in September 2022. In Q4 2022, the passive vent system evaluation was initiated, and recovery wells were abandoned, as approved by NYSDEC; the passive vent system evaluation is ongoing.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site name:	Dangman Park MGP	NYSDEC Site ID:	224047	
Site location:	Neptune Avenue, Brooklyn NY		PRP:	NA

Site Background

The site is a former MGP site located on Neptune Avenue in Brighton Beach, Brooklyn, New York. The property is privately owned. The adjacent properties are primarily commercial/residential. There are no utility operations on the property.

The RI Report was approved by the NYSDEC in 2014. Based on the RI data, the Company will manage MGP-related environmental aspects of the property owner's redevelopment. The property owner of one of the parcels initiated redevelopment activities in 2014.

Site Investigation and Remediation Progress January 2019 through December 2022

National Grid met with the NYSDEC Project Manager on-site in December 2019. NYSDEC approved the CCR in January 2020 with minor modification. A final CCR was submitted to the NYSDEC in June 2020. NYSDEC issued a Fact Sheet and PRAP in February 2020, however the comment period and public meeting were postponed as a result of the COVID-19 response. In September 2021, NYSDEC issued a revised Fact Sheet announcing the OU-1 PRAP public meeting and that comments would be accepted through October 30, 2021. National Grid provided comments on the PRAP to NYSDEC in October 2021. A final Supplemental RI Work Plan for OU-2 was submitted to NYSDEC in July 2021.

The Monitoring Wells MW-14 and MW-20 Groundwater Sampling and Decommissioning Work Plan was submitted to and approved by NYSDEC in February 2022. A letter requesting that the OU-1 property boundary be defined as the entirety of Block 7273, Lot 1001 was submitted, and MW-14 was decommissioned, in February 2022. A groundwater sample was collected from monitoring well MW-20, and the well was decommissioned in May 2022 in accordance with the Work Plan. The MW-14 Decommissioning Activities letter was submitted to the NYSDEC in May 2022. The MW-20 Sampling and Decommissioning Activities letter was submitted to the NYSDEC in September 2022. National Grid attempted to negotiate access to OU-2 for implementation of the July 2021 Supplemental RI Work Plan in 2022.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Equity Works MGP Site	NYSDEC Site ID:	224050	
Site location:	222, 252, and 254 Maspeth Avenue, Brooklyn, NY 11211		PRP:	NA

Site Background

The approximately 2-acre site, situated on three non-owned properties, was a former MGP and is located in an industrial area of Greenpoint, Brooklyn, NY.

The 222 Maspeth Ave property has been operated as a waste transfer station by Cooper Tank Recycling, which has been permitted to expand their waste transfer operations onto the 252 and 254 Maspeth Ave properties. In late 2017 Cooper Tank moved their transfer station to another location. They still own parcels 222 and 254 and maintain the NYSDEC permit to operate a transfer station. However, they no longer control parcel 252, which is currently being used for bus parking.

The site is in the FS phase. However, NYSDEC has required supplemental investigation in advance of the FS being approved. An interim SMP has been prepared and approved by NYSDEC in December 2012. A soil removal IRM was completed in 2012 and DNAPL recovery system installed in 2015.

Site Investigation and Remediation Progress January 2019 through December 2022

The final Supplemental Investigation Report for the 222 Maspeth Avenue parcel was submitted in March 2019. Quarterly DNAPL gauging and recovery activities are ongoing. Summary Reports for the NAPL Recovery Collection IRM are submitted to NYSDEC annually.

A PDI Work Plan and RD Work Plan for an IRM was submitted to the NYSDEC in April 2019. The PDI work was completed in May and June 2019. The IRM work plan was re-submitted to the NYSDEC in December 2019. In December 2019, NYSDEC, National Grid, and the property owner conducted a conference call to resolve how the property owner would participate in the IRM design process. Revised IRM Work Plans were submitted in January and April 2020. NYSDEC submitted a property owner's comments to the revised IRM Work Plan in February 2020. In March 2020, NYSDEC, National Grid, and a property owner conducted conference calls to resolve how to address NYSDEC's February 2020 request to install two new monitoring wells and to complete groundwater monitoring following completion of the IRM work. The Final IRM Work Plan was submitted in June 2020. An IRM WP Addendum was submitted to NYSDEC in July 2021 and implementation commenced in August 2021. The 60% design for the IRM was submitted to NYSDEC in October 2020 and the 90% design was submitted in July 2021. The IRM 95% Design in March 2022. In Q2 2022, a bid walk with remedial contractors was completed to support contractor bids for the IRM. The 100% 222 Maspeth Avenue IRM design was submitted to and approved by the NYSDEC in August 2022.

Throughout 2022, weekly gauging and NAPL recovery from pilot recovery wells were performed, as outlined in the 222 Maspeth Avenue IRM Work Plan Addendum. The first quarterly DNAPL recovery O&M event was completed in January 2022, and quarterly O&M events were conducted throughout the year. Upgrades to the automated DNAPL recovery system were completed in November 2022. NYSDEC approved modifications to the frequency of DNAPL gauging and recovery at select pilot recovery wells originally outlined in the IRM Work Plan Addendum in November 2022.

SITE INFORMATION AND PROGRESS SUMMARY				
General Site Information				
Site name:	Flatbush Holders Site	NYSDEC Site ID:	224061	
Site location:	329 Clarkson, 741 Parkside and 760 Parkside Avenues, Brooklyn, NY		PRP:	NA

Site Background

The site is a former holder site located on the north and south side of Parkside Avenue, south of Winthrop Street, east of Nostrand Ave, west of New York Ave, and north of Clarkson Ave, in Flatbush, Brooklyn, New York. The property is currently owned by multiple private owners. The adjacent properties are primarily commercial/ residential. There is an active gas line and underground regulator on a portion of the property.

Site Investigation and Remediation Progress January 2019 through December 2022

Annual groundwater sampling is ongoing. The emerging contaminants groundwater sampling summary report was submitted to the NYSDEC in February 2019. A letter was received from NYSDEC in July 2020 that the sampling for emerging contaminants has been satisfied and no further sampling is required at this time. National Grid participated in discussions with SUNY Downstate and NYSDEC about future Environmental Easement and redevelopment. A supplemental SC Work Plan for the northern parcel was submitted to the NYSDEC in May 2020 and finalized in July 2020. The supplemental SC field work was completed between August and December 2020. A RSO Investigation WP for the 329 Clarkson Avenue property on the southern parcel was submitted to NYSDEC in May 2021 and implemented in Q3 2021. The draft RSO Investigation (formerly the supplemental SC) report for the northern parcel was submitted to NYSDEC in November 2021. CAMP monitoring of geotechnical borings for a third-party was completed during Q4 2021.

The draft RSOI report for the Southern Parcel of the Site was submitted in February 2022 and the final report was submitted in May 2022; the report was approved in June 2022. The final RSOI Report for the Northern Parcel of the Site was submitted in June 2022; the report was approved in August 2022. During Q4 2022, CAMP monitoring was conducted during geotechnical borings advanced by a third party on the eastern portion of the northern property on behalf of the Property Owner in anticipation of redevelopment. An Excavation Work Plan was submitted by a third party and approved by the NYSDEC in September 2022 in anticipation of redevelopment of the southern property.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Front Street Holder Station	NYSDEC Site ID:	224063	
Site location:	218 Front Street, Brooklyn, NY 11202		PRP:	NA

Site Background

The approximately 1.1-acre site was a former holder station and is now located in a mixed commercial/residential area of the heavily populated Vinegar Hill neighborhood in Brooklyn, NY. The site is situated on two non-owned properties. The property operated as a warehouse and building material storage yard and is currently being redeveloped.

The final SC Report was submitted in March 2013, finalized and approved in July 2013, and revised in 2015. In May 2015, the NYSDEC approved a request to modify the site boundary, making it coterminous with the boundary of the 218 Front Street property.

Site Investigation and Remediation Progress January 2019 through December 2022

In March 2019, National Grid received comments from the property owner on the draft ISMP. National Grid responded to the property owner's comments in August 2019. In May 2020, the property owner announced that they had entered into a sales contract. In July 2020, contact was made with the prospective property purchaser. The prospective developer provided comments on the draft ISMP in August 2020. The revised draft ISMP was submitted to NYSDEC in November 2020 and approved in December 2020. A draft Further Investigation work plan was submitted to NYSDEC in October 2020, comments were received in December 2020, and the work plan was revised in February 2021 and approved in March 2021. The fieldwork was initiated in Q1 2021 and completed in April 2021. The RSOI Report, which reported on the Further Investigation work plan implementation, was submitted to NYSDEC in June 2021. In August 2021, NYSDEC approved the RSOI report and requested a RA Work Plan. A summary of the proposed RA Work Plan was submitted to NYSDEC in August 2021. In September 2021, NYSDEC concurred with the proposed work and requested a full RA Work Plan. Preparation of the Environmental Easement is on hold until redevelopment of the property is complete. The property transaction was completed in October 2021.

The draft RA Work Plan and the Waste Characterization Work Plan were submitted in January 2022. During Q1 2022, the Waste Characterization Work Plan was implemented for transport and disposal of material in preparation of the RA Work Plan. The remediation of Holder No. 5, CAMP and ISMP oversight of intrusive activity at 218 Front St/Holder No. 4 commenced in Q4 2022. The Daily Field Report and CAMP reports related to Holder 5 remediation and intrusive activity at the rest of 218 Front St, including Holder No. 4, were submitted in Q4 2022.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site name:	Fulton Municipal Works MGP & 3rd Ave Holder Station	NYSDEC Site ID:	224051	
Site location:	• Fulton Municipal Works MGP: 270 Nevins St; 560 Degraw Street; 537 Sackett St; Nevins St., 3rd Ave., bet. Degraw and Douglas St. •3rd Avenue Holder Station: 191 3rd Avenue	PRP:	NA	

Site Background

The approximately 5.5-acre site is a former MGP and associated holder station located along the Gowanus Canal (Federal Superfund site) in a commercial/industrial area in Brooklyn, NY. Surrounding land uses include residential properties. None of the six site parcels is owned by the company. Current site land uses include a NYC park and swimming pool on Parcel 2, a rock climbing gym on Parcel 5, a film studio operated on Parcel 1 until it was demolished in 2022, equipment storage/maintenance, and offices previously existed on Parcel's 3, 4, 6 and 7 until they were demolished in 2022. Parcels 6-7 is the location for the USEPA required NYCDEP CSO Sewage Retention Tank. Remediation on Parcels 3 and 4 started in 2022 in advance of planned redevelopment.

Site Investigation and Remediation Progress January 2019 through December 2022

USEPA approved the 100% Design Report and it was submitted to the NYSDEC in March 2019. The PDI Data Report for Parcel 3 was submitted to the NYSDEC in February 2019. Supplemental Parcel 3 PDI work was completed during Q1 and 2 and a PDI Data Report Addendum was submitted to NYSDEC in Q2 2021. The RA Work Plan and ISS Characterization Work Plan for Parcel 3 were submitted to NYSDEC by the property owner in July 2021. The ISS Characterization Report and Contained-In determination letters for Parcel 3 were submitted to NYSDEC by the property owner in October and November 2021, respectively. Performance of the additional Soil Characterization work at Parcel 3 was initiated in Q3, 2021 and is ongoing. Access negotiations for Fulton Barrier Wall construction were completed in December 2019. Mobilization, site setup, staging area construction, temporary control installations and air quality monitoring for Fulton Barrier Wall construction commenced in September 2019. The construction phase of Fulton Barrier Wall commenced in December 2019 and was completed in Q2 2021.

The emerging contaminant groundwater analytical results summary report was submitted in February 2019. NAPL monitoring and recovery and groundwater elevation monitoring is ongoing. Groundwater elevation monitoring, NAPL monitoring, and NAPL recovery continued throughout 2022. During Q2 2022, a monitoring well was successfully replaced. Remedial activities at Parcel 3 and 4 were initiated in Q3-Q4 and are ongoing.

SITE INFORMATION AND PROGRESS SUMMARY			
General Site Information			
Site Name:	Gowanus Canal	NYSDEC Site ID: USEPA ID:	CERCLA 02-2009-2010
Site location:	1.8 Mile NYC man-made canal between Butler Street and Hamilton Avenue Bridge in Brooklyn, NY	PRP:	Yes

Site Background

The Gowanus Canal is located in Brooklyn. Three MGP sites are located on the Gowanus Canal. The MGP sites are subject to an Order on Consent with the NYSDEC. The USEPA has included the Gowanus Canal on the National Priorities List and has completed RI and FS for the site. USEPA issued a ROD for the Site in September 2013. National Grid is participating in the performance of the remedial design with other responsible parties under a UAO with the USEPA.

Site Investigation and Remediation Progress January 2019 through December 2022

The site is under USEPA Superfund Program and not under NYSDEC Division of Environmental Remediation's Program. Due to technical complexities of managing a project of this scale and managing expectations of third-party stakeholders and the USEPA, multiple experts were contracted to review the approach and ensure compliance with CERCLA regulations. The Gowanus Remediation Team continues to develop the RD plans for the project. In September 2019, the 90% RD for RTA-1 was submitted to USEPA for review. Major elements were accepted by USEPA December 2019. In February and March 2020, the final RTA-1 and Turning Basin 1 remedial designs were submitted to USEPA. RTA-1 mobilization began in September 2020 and on-water construction began in October 2020 with Phase 1 dredging commencing on November 16, 2020. Through December 2022, dredging (Phase 1 through 3), bulkhead installation, permanent submarine cable installation at Union Street Bridge, and ISS activities have been completed. USEPA approved the 100% RD for the RTA-1 work in October 2022. Equipment and material mobilization to support capping activities started in November 2022.

SITE INFORMATION AND PROGRESS SUMMARY				
General Site Information				
Site name:	Greenpoint MGP	NYSDEC Site ID:	224052	
Site location:	287 Maspeth Avenue, Brooklyn, NY		PRP:	N/A

Site Background

The Greenpoint MGP site is located beneath National Grid infrastructure (Liquefied Natural Gas tanks, transportation garage, offices, etc.) on Maspeth Ave in Brooklyn, NY. The site encompasses approximately 117 acres. The site is set in a in a predominately industrial and manufacturing area with some commercial and residential buildings to the northwest. The RI was performed in several phases due to the large size of the site and was completed in June 2019.

Site Investigation and Remediation Progress January 2019 through December 2022

Phase 3 of the RI in Area 8 of the site resumed in March 2019 and was completed in June 2019. The Draft RI Report was submitted in January 2020 and NYSDEC comments on the RI Report were received in November 2020. A letter in response to comments on the draft RI Report was submitted to NYSDEC in December 2020 and the revised RI Report was submitted was submitted to NYSDEC in March 2021. A second version of the RI Report was submitted to NYSDEC in May 2022 based on additional comments.

The emerging contaminant groundwater analytical results summary report was submitted to the NYSDEC in February 2019.

Revised Interim SMPs were submitted to NYSDEC in 2020 and 2021 based on comments from NYSDEC and NYSDOH.

Based on previous results, follow-up SVI sampling was performed in March through June 2019 and the SVI investigation resampling summary letter was submitted to NYSDEC and NYSDOH in October 2019. Soil vapor intrusion samples were collected during Q1 2021, and the report was submitted in April 2021 and approved in June 2021.

A work plan for the investigation of USTs was submitted to NYSDEC in February 2020, field work has been completed, and a summary report was submitted in November 2020. A work plan for the removal of the USTs was submitted to NYSDEC in January 2021. Seven USTs were abandoned in Q2, 2021. The draft UST closure report was submitted to NYSDEC in September 2021 with revisions submitted in November 2021.

An IRM Work Plan was submitted to NYSDEC in June 2020 regarding the removal of structures and the installation of a clean cover prior to a capital improvement project. The capital improvement project was deferred prior to approval of the Work Plan.

NAPL gauging and recovery is ongoing and is conducted monthly.

A work plan for surface soil sampling on the public ballfield area of the site was submitted to NYSDEC in December 2020. A revised draft work plan based on NYSDEC comments for additional surface soil sampling was submitted to and approved by NYSDEC in May 2021. The surface soil sampling was conducted in Q4 2021. A summary report was submitted to NYSDEC in March 2022 regarding the soil sampling conducted on the ballfield. NYSDEC provided comments to the report in March 2022 requesting an IRM to address the impacts identified in the report. National Grid decided to discontinue the use of the area as a ballfield; therefore, eliminating the need for an IRM.

A PDI work plan for the installation of SSDS systems was submitted to NYSDEC in July 2021, which was approved in August 2021. Soil vapor samples included in the work plan were collected in Q3 2021 and, pilot testing was performed in several on-site buildings to facilitate the design of the planned sub-slab depressurization systems in Q1 2022. The draft Soil Vapor Mitigation PDI Summary and Sub-Slab Depressurization System Design Report

were submitted to the agencies in August 2022 and an updated version was submitted and approved in December 2022.

National Grid routinely performs oversight activities in support of gas operations related work and capital improvement projects which are performed in areas of the site where MGP related contamination is present. These activities are managed under a NYSDEC required site management process and generally include soil disposal, monitoring and regulatory reporting. Soil Import Request forms are submitted to NYSDEC as necessary.

The installation and development of a new well and development of two existing wells occurred in Q3 2022 at the request of the USEPA.

SITE INFORMATION AND PROGRESS SUMMARY				
General Site Information				
Site name:	Jamaica Gas Light Co. MGP	NYSDEC Site ID:	241063	
Site location:	58-18 Beaver Road, Jamaica, NY 11433		PRP:	NA

Site Background

The approximately 1-acre site was originally a MGP and subsequently a holder station. The non-owned site parcel is located in a moderate density residential area of Jamaica, Queens, New York. The current site usage is roll off storage. Adjacent property uses include a cemetery, York College, the LIRR and a glass recycling facility. The Final RI Report was issued in December 2015.

Site Investigation and Remediation Progress January 2019 through December 2022

Efforts to secure site access for emerging contaminants sampling requested by NYSDEC are continuing. The property owner has not been responsive to National Grid's request for access. National Grid contacted NYSDEC for assistance with access in October 2019. Site access discussions are on going between NYSDEC and the property owner.

Annual Site Inspections are ongoing per the Interim SMP. The Revised Interim SMP was submitted to the NYSDEC in September 2022.

SITE INFORMATION AND PROGRESS SUMMARY				
General Site Information				
Site name:	Kings County MGP	NYSDEC Site ID:	224056	
Site location:	5400-5600 1st Avenue, Brooklyn NY		PRP:	NA
Site Background				
The site is a former MGP of 16 acres on the New York Bay waterfront of Brooklyn, NY. It consists of two non-owned parcels, zoned for industrial use. The major portion of the site is operated by the New York City Economic Development Commission and is occupied by the Brooklyn Wholesale Meat Market and another tenant. The smaller portion of the site is owned by US Power Generating Co. and operated as a power station.				
Site Investigation and Remediation Progress January 2019 through December 2022				
The emerging contaminant groundwater analytical results summary report was submitted in February 2019. A letter from the NYSDEC was received June 2020 that the sampling request has been satisfied and no further sampling is required at this time. The SC Addendum report was submitted to NYSDEC, and NYSDEC approved the report in August 2020. Efforts to secure site access are continuing.				

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Metropolitan Works MGP Site	NYSDEC Site ID:	224046
Site location:	124-136 Second Avenue, Brooklyn, NY 11215	PRP:	NA

Site Background

This former MGP site is located in a commercial and residential area along the Gowanus Canal (GC Superfund site) in the Gowanus neighborhood of Brooklyn, NY. The 10 site parcels are non-owned. Two of the 10 parcels were previously remediated and are now occupied by portions of a Lowes store and its associated parking lot. The remaining 8 site parcels are primarily occupied by other retail stores and a vacant Pathmark shopping center. The Pathmark supermarket, which occupied a parcel in the area of the former MGP, stopped operating and several other businesses have taken over the Pathmark space. The property was purchased by Two Trees in 2018 and they have not yet identified any plans about potential site redevelopment in the future.

The NYSDEC approved the initial RI report for the site in October 2014. A draft FS was submitted to NYSDEC in June 2015. A Supplemental RI Work Plan was prepared to investigate previously inaccessible areas of the site (around the former Pathmark supermarket) and was approved by the NYSDEC in December 2015.

Site Investigation and Remediation Progress January 2019 through December 2022

The SVI sampling was implemented in March 2019 and the sampling results were submitted to the agencies in May 2019. The revised Supplemental RI Report was submitted to NYSDEC in January and April 2020. NYSDEC and National Grid had a conference call in March 2020 to clarify NYSDEC comments on the revised Supplemental RI; the final Supplemental RI was submitted and approved in May 2020. A PDI work plan for the Bulkhead/Barrier Wall IRM was submitted to NYSDEC in May 2021. In-canal borings and existing bulkhead pre-condition inspection were completed in Q2 2021. An in-canal obstruction evaluation commenced in Q4 2021 and completed in 2022. The Uplands PDI work scope was completed during 2022. The preliminary draft design figure for the bulkhead was submitted in September 2022, and the 60% design drawings and specifications for the bulkhead barrier wall were submitted to NYSDEC and USEPA in November 2022.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site name:	Nassau MGP Site	NYSDEC Site ID:	224019B
Site location:	Brooklyn Navy Yard, Kent Ave, Brooklyn, NY	PRP:	NA

Site Background

The site is a former MGP site located on Kent Avenue, Navy Yard, Brooklyn, New York and borders Wallabout Channel. The property is currently owned by the Brooklyn Navy Yard and is vacant. The adjacent properties are primarily industrial. There are no active utility operations on the property.

The RI was completed, and a draft FS was submitted, however NYSDEC required additional investigation prior to approving the FS. National Grid has received information that this site may be subject to redevelopment in the future.

Site Investigation and Remediation Progress January 2019 through December 2022

The revised FS report was submitted to NYSDEC in November 2019. The PRAP was issued in February 2020 and the ROD was issued in September 2020. The FS was approved, and the final FS was submitted to the NYSDEC, in June 2020. The site survey was completed in August 2020. A boundary modification request, draft Interim SMP, draft PDI work plan, and revised CPP were submitted to NYSDEC in December 2020. A revised PDI Work Plan was submitted to NYSDEC in March 2021. Responses to NYSDEC comments on the Draft Interim SMP were submitted in December 2021. The revised Interim SMP was submitted to NYSDEC in February 2022 and NYSDEC provided comments in November 2022.

The Brooklyn Navy Yard completed geotechnical borings in the vicinity of the future PDI offsite boring locations in March 2022. The PDI Work Plan was revised based on observations of impacts during these geotechnical borings and was submitted to NYSDEC in March 2022. Revisions to the draft Interim SMP were submitted to NYSDEC in February 2022. NYSDEC approved the PDI Work Plan in March 2022. The PDI was implemented in Q2 2022. NAPL gauging and recovery commenced in Q2 2022 and continue monthly.

SITE INFORMATION AND PROGRESS SUMMARY			
General Site Information			
Site name:	Newtown Creek Superfund Site	NYSDEC Site ID:	CERCLA-02-2011-2011
Site location:	Newtown Creek borders Brooklyn and Queens in New York City	PRP:	Y

Site Background

Scope of SIR Work for the Site:

Newtown Creek is a 3.8-mile waterbody located on the border of Brooklyn and Queens in New York City. Two former MGP sites and a former gas holder are located in the vicinity of Newtown Creek. National Grid is a Potentially Responsible Party for addressing impacted sediments along with various oil companies, Phelps Dodge and NYC.

Site Investigation and Remediation Progress January 2019 through December 2022

The site is under USEPA Superfund Program and not under NYSDEC Division of Environmental Remediation's Program. The RI/FS for the entire Newtown Creek site commenced in 2011 and is scheduled for completion in 2027. The RI/FS will: a) evaluate the nature and extent of any identified contamination, b) identify potential pathways of contaminant migration and possible receptors, c) develop quantitative ecological and human health risk assessments and, d) develop and evaluate remedial alternatives and assess whether they will meet proposed remedial goals and remedial action objectives.

Drafts of the RI report were submitted by the NCG to EPA in April 2019, July 2020, October 2021, and December 2022 with comments received from USEPA after each draft. FS field study evaluation reports were prepared and revised. Field work including NAPL delineation refinement and mobility sampling, groundwater seepage rate measurements, geotechnical coring, biota tissue sampling, nearshore sediment sampling, and definitive tests for ebullition-mediated NAPL transport were completed. Initial risk-based preliminary remediation goals (PRGs) were presented to USEPA in December 2019 and were finalized and approved in December 2021. In addition to the OU-1 RI/FS, an AOC was entered into voluntarily by USEPA and the NCG in 2019 to prepare a focused FS to evaluate the potential for an interim RA for the lower two miles of Newtown Creek which is now formally designated as OU-3 of the Site. A focused FS for OU-3 was prepared in 2020, but USEPA ultimately decided to not move forward with this early RA and terminated the AOC for OU-3 in July 2021. Also, a treatability study was initiated in 2019 to support the screening and evaluation of various remedial technologies for the FS, but this work was put on hold in 2022 in lieu of another interim RA project in East Branch. Planning for the East Branch Early Action was initiated in 2022.

SITE INFORMATION AND PROGRESS SUMMARY			
General Site Information			
Site Name:	People's Works MGP Site	NYSDEC Site ID:	224053
Site location:	470-490 Kent Avenue Brooklyn, NY 11211	PRP:	NA
Site Background			
The 2-acre site was a MGP located along the Wallabout Channel in the Williamsburg neighborhood of Brooklyn, NY. The site footprint is situated on portions of two non-owned properties that were operated as a building supply store with lumber yard and are currently being redeveloped. Surrounding property uses are residential and commercial.			
Site Investigation and Remediation Progress January 2019 through December 2022			
The emerging contaminant groundwater sampling results summary report was submitted to the NYSDEC in February 2019. In November 2019 the NYSDEC requested an investigation of the adjacent parcels to the north of the former MGP site. A No Further Action Response to the SC report was received in 2020. A meeting with NYSDEC was held in January 2020 to discuss off-site investigation scope of work. A termination letter for People's Works MGP Site was submitted to the NYSDEC in September 2020. The site was terminated from its order so that it could be entered into the BCP by a developer. A draft Schaefer Landing (off-site area) SC work plan was submitted in September 2020, after the site had been terminated from its order.			

SITE INFORMATION AND PROGRESS SUMMARY					
General Site Information					
Site Name:	Plymouth Street Holder Station		NYSDEC Site ID:	224065	
Site location:	49 Gold Street, Brooklyn, NY 11201			PRP:	NA
Site Background					
The 1+ acre site was a holder station and is currently located in a low density residential area in the Vinegar Hill neighborhood of Brooklyn, NY. The former site footprint occupies a portion of a parcel zoned for light manufacturing and commercial use and which is owned by Con Edison. Con Edison operates the site as a substation. Several adjacent properties are residential. Other adjacent uses are listed as industrial or transportation/utilities.					
Site Investigation and Remediation Progress January 2019 through December 2022					
National Grid has prepared a SC plan, and it has been approved by the NYSDEC. The SC will be conducted once the site is no longer "on hold" with the NYSDEC and access to the site is obtained.					

SITE INFORMATION AND PROGRESS SUMMARY				
General Site Information				
Site Name:	Scholes Street Holder	NYSDEC Site ID:	224067	
Site location:	338 Scholes Street, Brooklyn, NY		PRP:	NA
Site Background				
The approximately 1-acre site was a holder station and is zoned for light manufacturing and commercial uses. The site covers three non-owned properties that have been developed with concrete and brick buildings. The site is located approximately 0.17 miles west of the English Kills, part of the Newtown Creek (Superfund Site). The final SC report was approved in 2014. National Grid has received information that this site may be subject to redevelopment in the future. The timing and scope are unknown.				
Site Investigation and Remediation Progress January 2019 through December 2022				
The Site Boundary Modification Request was not approved by NYSDEC in September 2020. A survey was completed in Q1 2021 to support the Environmental Easement. Three Draft Environmental Easement packages were submitted to NYSDEC in October 2021. The revised Draft Interim SMP was submitted to NYSDEC in March 2022 and an additional revision was submitted in September 2022.				

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Skillman Street Holder	NYSDEC Site ID:	224068	
Site location:	OU 1: 7 Skillman Street and 744 Bedford Avenue, Brooklyn, NY OU 2: 39 Skillman Street, Brooklyn, NY		PRP:	NA

Site Background

The 1-acre site was a holder station and is located in a commercial and residential area in Brooklyn, NY. The site is comprised of two non-contiguous and non-owned parcels. Post-holder station, OU 2 was an industrial clothes cleaning business. OU 1 has been occupied by two catering halls. A SC was completed in 2012. A no further action letter and a Release and Covenant Not to Sue were obtained for OU 2 and subsequently the OU was acquired by a developer who remediated and redeveloped the property under the BCP. OU 2 has been released from the February 2, 2007 Order on Consent and Administrative Settlement, Index # A2-0552-0606. A SC was completed for OU 1 in 2014. A PRAP was issued in February 2015 and a ROD was issued in March 2015. Prior to issuance of the PRAP, the OU 1 designation was dropped by NYSDEC.

In 2015, the catering hall at 744 Bedford Avenue was demolished, in preparation for redevelopment. Surrounding area is mixed residential/commercial. In March 2015, NYSDEC required additional investigation for the 744 Bedford Avenue property. The PDI Work Plan was submitted to, and approved by, NYSDEC in April 2015. Redevelopment at 7 Skillman Street is anticipated, but the timing is unknown.

Site Investigation and Remediation Progress January 2019 through December 2022

In January 2019, the easement packages were resubmitted to NYSDEC with revised language of "Interim Site Management Plan or any future Site Management Plan." The easements and NYC-RPTs need to be signed for each property; the owners' attorney has been contacted for assistance in having these documents signed. In spring/summer of 2019, Owner stated that there would be no easements until redevelopment of the site was completed. NYSDEC support may be required for execution of easements.

The emerging contaminant groundwater analytical results summary report was submitted to the NYSDEC in February 2019. A No Further Action Response to the emerging contaminant groundwater analytical results summary was received in 2020.

In July 2019, per the ROD and in anticipation of upcoming redevelopment, a draft Further Investigation Work Plan for 7 Skillman Street was submitted to NYSDEC. On July 26, 2019, National Grid received comments from Owner's consultant, via Owner, on the Draft Further Investigation Work Plan; the comments recommended not proceeding with the Further Investigation at the time, until discussions can be held with National Grid. The Owner has since been unresponsive to National Grid. In 2020 the property owner, through their council, approached National Grid about moving the site into the BCP. In June 2022, the National Grid outside counsel was contacted by the counsel for the owner about the status of the work and a site visit.

Annual Interim SMP activities, including site inspections and annual groundwater monitoring are ongoing.

SITE INFORMATION AND PROGRESS SUMMARY				
General Site Information				
Site Name:	Union Station (Citizens Branch) Holder Station	NYSDEC Site ID:	224054	
Site location:	2940 Atlantic Avenue, Brooklyn, NY 11207		PRP:	NA

Site Background

The approximately 1.2-acre site was briefly (for approximately 1 year) a MGP before being converted to a holder station. The non-owned site parcel is located in a commercial area of Brooklyn, NY and operated by Con Edison as a service center. Adjacent property uses include residential, commercial, and manufacturing. National Grid prepared a Revised SC Work Plan that was submitted to and approved by NYSDEC in October 2015.

Site Investigation and Remediation Progress January 2019 through December 2022

The SC Report was submitted to NYSDEC in February 2019 and approved by NYSDEC in March 2019. In December 2019 the NYSDEC requested that an Interim SMP be prepared for to govern future on-site activities. The draft Interim SMP was submitted to NYSDEC in July 2020. Comments were provided in response to NYSDEC's November 2020 comment letter in December 2020. The revised Interim SMP was submitted in October 2021. A Groundwater Sampling Work Plan was submitted to NYSDEC in March 2021. The Baseline groundwater sampling event was conducted in March 2021 and the Baseline Groundwater Sampling Report was submitted to NYSDEC in May 2021.

Groundwater sampling for emerging contaminants was requested by NYSDEC in May 2018. A work plan was submitted to NYSDEC and approved in September 2018. Groundwater sampling for emerging contaminants was completed in May and June 2019 and the report was submitted to NYSDEC in June 2020. In July 2020, a letter was received from NYSDEC that the sampling request has been satisfied and no further sampling is required at this time.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Williamsburg Works MGP Site	NYSDEC Site ID:	224055	
Site location:	2 North 11th Street, 20 North 12th Street, 35 Kent Avenue, 50 Kent Avenue, Brooklyn, NY		PRP:	Yes

Site Background

The approximately 5.7-acre site was a MGP and is located along the East River in a commercial area of the Williamsburg neighborhood, Brooklyn, NY. The site is located on 4 non-owned properties with uses as follows: Parcel 1: Warehouse/studio, Parcel 2: open space/park, Parcels 3 and 4 used for warehouses until January 31, 2015, when the warehouses burned down. Surrounding property uses are commercial and residential. Three of the site properties are within the footprint of the planned NYC Bushwick Inlet Park. In 2013, Counsel for the owner of the fourth site property (35 Kent Avenue) indicated that it likely will be re-developed, in 5 to 10 years, as a hotel.

The RI Report was submitted to NYSDEC in January 2015. A recovery well component of the IRM for the 50 Kent Avenue property was completed in 2014 and NAPL recovery is on-going. The 100% Design for the soil component of the 50 Kent Avenue Property IRM was approved in August 2015, and the IRM was commenced in December 2015. The soil component of the IRM was completed in June 2017 and the IRM CCR was issued to and approved by NYSDEC in 2018.

On November 7, 2016, National Grid sent a letter to NYSDEC expressing the intent to withdraw the Williamsburg Works MGP site from the Multi-site Order on Consent in an effort to secure a separate order for the site, which would provide greater flexibility to pursue additional responsible parties due to the complex industrial history of the site and surrounding properties.

Site Investigation and Remediation Progress January 2019 through December 2022

NAPL gauging and recovery on the 50 Kent Avenue property is ongoing. Well rehabilitation work to restore sump capacity was conducted in 2019; two damaged recovery wells were abandoned and replaced, as approved by the NYSDEC. The 2019 NAPL Recovery Well Rehabilitation and Reinstallation report was submitted to NYSDEC in September 2019. A draft Interim SMP was submitted to NYSDEC in April 2020 and revised in October 2020. A final Interim SMP was submitted to NYSDEC in July 2021. A draft RI Work Plan Addendum for Parcels 3 and 4 was provided to NYSDEC in June 2020, revised in December 2020, and finalized in July 2021. A draft revised CPP was submitted to NYSDEC in July 2021.

Groundwater sampling for emerging contaminants was completed in March 2019 and the report was submitted to NYSDEC in June 2020.

A new order was executed on September 21, 2021. The effective date of the new consent order is September 31, 2021.

The Addendum to the RI Work Plan Addendum for Parcels 3 & 4 was submitted to the NYSDEC in April 2022. The Parcels 3 and 4 RIWPA access agreement was executed in June 2022. The RI for Parcels 3&4 was conducted in Q3 and Q4 2022. The Addendum to the RI Work Plan Addendum for NAPL recovery was submitted to NYSDEC in December 2022.

SITE INFORMATION AND PROGRESS SUMMARY

General Site Information

Site Name:	Wythe Ave (Berry Street) Holder Station	NYSDEC Site ID:	224069	
Site location:	OU 1: 94 N. 13th Street, and 121 N. 12th Street (now consolidated through redevelopment as 55 Wythe Avenue), Brooklyn, NY OU 2: 120 and 132 N. 13th Street; 20, 24, 28, and 32 Berry Street; and 125 and 129 N. 12th Street, Brooklyn, NY	PRP:	Yes	

Site Background

The approximately 2-acre site was a holder station occupying an entire block within a commercial area in the Williamsburg neighborhood of Brooklyn, NY. In 2007, the ten site properties were divided between three non-National Grid owners and were used for manufacturing, storage, and warehousing. A SC has been completed and the report was approved in 2013. In 2014, two of the three owners sold their properties to a developer, who commenced redevelopment of the properties as a hotel and commercial space. An Interim SMP was approved by NYSDEC in June 2014. An IRM for the 55 Wythe Avenue properties was performed as part of its redevelopment in 2014. These properties have since become identified as OU 1, with the remainder of the site being identified as OU 2. Circa 2015, the owner of the OU 2 properties indicated that over the next 1.5 to two years he would rezone his eight parcels and then either redevelop or sell them. However, since then, the property owner has indicated that rezoning and redevelopment are in the planning stages. Surrounding property uses are commercial, parkland, and residential.

Site Investigation and Remediation Progress January 2019 through December 2022

The emerging contaminant groundwater analytical results summary report was submitted to NYSDEC in February 2019. A No Further Action Response to the report was received in 2020. The ROD for OU 2 was released in February 2019. The RSOI Work Plan was submitted to NYSDEC in September 2021 and approved in December 2021. Annual Interim SMP activities, including site inspections and groundwater monitoring, are on-going.

Environmental Easements are required by the RODs. The OU 1 and OU 2 easements, including the NYSDEC's suggested updated language, were sent to the owners in January 2019. The owner's signature is still needed to complete and file the OU 1 environmental easement. NYSDEC support will be required for execution of the OU 1 easement. A copy of the OU 2 easement, signed by the owner, was sent to NYSDEC in March 2019 for final approval. The OU 2 Environmental Easement was recorded in March 2020.

Exhibit ___ (CFW-2)

New York State Department of Environmental Conservation ("DEC")
Order on Consent example

NEW YORK STATE DEPARTMENT OF ENVIRONMENTAL CONSERVATION

In the Matter of the Development and Implementation of Remedial Programs for Former Manufactured Gas Plants and Gas Holder Locations under Article 27, Title 13 of the Environmental Conservation Law by Brooklyn Union Gas Company d/b/a KeySpan Energy Delivery New York & KeySpan Gas East Corporation d/b/a KeySpan Energy Delivery Long Island, Respondents.	**ORDER ON CONSENT and ADMINISTRATIVE SETTLEMENT** Index # A2-0552-0606 Multiple Sites

WHEREAS,

1. A. The New York State Department of Environmental Conservation ("Department") is responsible for inactive hazardous waste disposal site remedial programs pursuant to Article 27, Title 13 of the Environmental Conservation Law ("ECL") and may issue orders consistent with the authority granted to the Commissioner by such statute.

B. The Department is responsible for carrying out the policy of the State of New York to conserve, improve and protect its natural resources and environment and control water, land, and air pollution consistent with the authority granted to the Department and the Commissioner by Article 3, Title 1 of the ECL.

C. This Order on Consent and Administrative Settlement ("Order & Settlement Agreement") is issued pursuant to the Department's authority under, *inter alia*, ECL Article 27, Title 13, ECL Article 71, Title 27, and ECL § 3-0301, and resolves Respondent's liability to the State under the "Comprehensive Environmental Response, Compensation, and Liability Act of 1980", as amended ("CERCLA"), 42 U.S.C. § 9601 *et seq.*, to the extent set forth herein. It is the intent of the Department and KeySpan Energy Delivery New York and KeySpan Energy Delivery Long Island (together and individually "KeySpan" or the "Respondent") that this Order & Settlement Agreement constitutes an administrative settlement within the meaning of CERCLA § 113(f)(2),§ 113(f)(3)(B), 42 U.S.C. § 9613(f)(2), § 9613(f)(3)(B) and is a settlement under, *inter alia*, ECL Article 27, Title 13, ECL Article 71, Title 27, and ECL § 3-0301. Accordingly, pursuant to CERCLA § 113 (f)(3)(B), 42 U.S.C. § 9613(f)(3)(B), Respondent may seek

contribution from persons not parties to this Order & Settlement Agreement to the extent set forth in Subparagraph XIV.I. Such efforts by Respondent will be supported by the Department in such manner as the Department reasonably deems appropriate.

2. KeySpan is a New York corporation with offices at One MetroTech Center, Brooklyn, New York 11201.

3. Respondent is currently subject to the following Orders and/or Agreements with respect to the following sites which are more particularly described in Exhibit "A" Table 2:

A. Order on Consent D2-001-94-12 for Investigation and Interim Remedial Measures with respect to the Coney Island Manufactured Gas Plant ("MGP") Site (listed in the *Registry of Inactive Hazardous Waste Sites* In New York State ("*Registry*") as a Class 2 Site) in Brooklyn, NY;

B. Order on Consent Index # D2-0001-98-04 for Investigation and Remediation with respect to the Clifton MGP Site in Staten Island NY;

C. Order on Consent Index # D1-0001-98-11 for Investigation and Remedial Response Program with respect to the Bay Shore MGP Site in the Town of Islip, NY; Hempstead MGP Site in the Town of Hempstead, NY; and Glen Cove MGP Site in the City of Glen Cove, NY; and Halesite MGP Site in the Town of Huntington, NY;

D. Order on Consent Index # D1-0002-98-11 for Remedial Investigation/Feasibility Studies for Sag Harbor MGP Site (listed in the *Registry* as a Class 2 Site) in the Town of Southampton, NY; and Rockaway Park MGP Site (listed in the *Registry* as a Class 2 Site) in Queens, NY;

E. Order on Consent Index # D1-0001-99-05 for Preliminary Site Assessment of Patchogue MGP Site in the Town of Brookhaven, NY; Far Rockaway MGP Site in Queens, NY; Clinton (Hempstead) Road MGP Site in the Town of Hempstead, NY; Babylon MGP Site in the Town of Babylon, NY; and Garden City MGP Site in the Town of Hempstead, NY;

F. Agreement Index # D1-0002-99-05 for Investigation of Watchogue Creek MGP Site in the Town of Islip, NY;

G. Voluntary Cleanup Agreement Index # R1-0001-01-01 for a Remedial Response Program for Glenwood Landing Propane Plant and Compressor Station in the town of Oyster Bay, NY;

H. Voluntary Cleanup Agreement Index # A2-0460-0502 for Remedial Investigation/Feasibility Study with respect to the Citizens Gas Works MGP Site in Brooklyn, NY;

I. Voluntary Cleanup Agreement Index # W2-1062-05-03 for the remediation of four (4) non-contiguous parcels related to the Liquified Natural Gas Plant within the Greenpoint Energy Center in Brooklyn, NY;

J. Order on Consent and Administrative Settlement Index # A2-0523-0705 for a Remedial Investigation Program with respect to the Gowanus Canal in Brooklyn, NY;

K. Voluntary Cleanup Agreement Index # D2-0002-99-10 for Investigation and, if needed, Remediation of the Newtown Holder Station in Elmhurst, NY; and

L. Order on Consent Index # W2-1090-06-06, for Remediation of the Nassau Works MGP Site, Brooklyn, NY (part of Brooklyn Navy Yard OU-2; listed on the *Registry* as a Class 2 Site).

M. Voluntary Cleanup Agreement Index # R2-0330-98-01, for Implementation of a Response Program for LILCO - Edgemere Substation, in Rockaway, Queens, NY.

4. A. Respondent is the owner or formerly owned and/or operated MGP Sites (individually "Site"; collectively "Sites") at the following locations at which, *inter alia,* coal tar and associated hazardous substances ("MGP Wastes") were or may have been disposed at various times in the past and which are the subject of this Order & Settlement Agreement: (1) Brooklyn, NY: Peoples Works MGP Site; Dangman Park MGP Site; (2) Queens, NY: Far Rockaway MGP Site; Jamaica Gas Light Co. MGP Site; (3) Town of Brookhaven, NY: Patchogue MGP Site; (4) Babylon, NY: Babylon MGP Site. These MGP Sites, are more particularly described in Exhibit "A" - Table 1 attached hereto and are not currently listed in the *Registry*.

B. Respondent is the owner or formerly owned and/or operated gas holder ("Holder") Stations (individually "Site"; collectively "Sites") at the following locations at which, *inter alia,* MGP Wastes were or may have been disposed at various times in the past and which are the subject of this Order & Settlement Agreement: (1) Brooklyn, NY: Front Street Holder Station; Rutledge Street Holder Station; Skillman Street Holder Station; Keap Street Holder Station; Bay Ridge Holder Stations A & B; (2) Queens, NY: Jamaica Holder Station; (3) E. Garden City, NY: E. Garden City Holder Station; (4) Long Beach, NY: Long Beach Holder Station. These Holder Sites are more particularly described in Exhibit "A" - Table 1 attached hereto and are not currently listed in the *Registry.*

C. The Department and Respondent recognize that these MGP and Holder Sites have the potential to release MGP Wastes. The Department and KeySpan recognize that implementation of this Order & Settlement Agreement will expedite the cleanup of the Sites. The Department and Respondent further agree that the settlement set forth herein will avoid potentially prolonged and complicated litigation, and that this Order & Settlement Agreement is mutually acceptable, fair, reasonable, and in the public interest.

D. The Department and Respondent acknowledge that Respondent is regulated by the Public Service Commission ("PSC") of the State of New York. Costs incurred for site

investigation and remediation activities must be reviewed and approved by the PSC in order to be recoverable through rates. The Department will support Respondent's position that any necessary and appropriate response actions by Respondent were required to address Respondent's liability for such activities.

5. In addition to the MGP and Holder sites listed in Paragraphs 3 and 4 above, Respondent is the owner or formerly owned and/or operated a number of other sites, including MGP and Holder Sites, not currently the subject of existing orders or agreements and not included in this Order & Settlement Agreement at this time, but which may be included in this Order & Settlement Agreement in the future. These other known sites are more particularly described in Exhibit "A" - Table 3.

6. A. The Department alleges that Respondent is a "covered person" as defined in CERCLA § 107(a), 42 U.S.C. § 9607(a), that each Site is a "facility" as defined in CERCLA § 101(9), 42 U.S.C. § 9601(9) and that Respondent is liable to the State of New York under CERCLA § 107(a) for the response actions related to investigation work activities required by this Order & Settlement Agreement.

B. Respondent consents to the Department's issuance of this Order & Settlement Agreement with respect to the MGP Sites listed in Paragraph 4.A above and Holder Sites listed in Paragraph 4.B above, and on Exhibit "A" - Table 1, without (i) an admission or finding of liability, fault, wrongdoing, or violation of any law, regulation, permit, order, requirement, or standard of care of any kind whatsoever; (ii) an acknowledgment that there has been a release or threatened release of hazardous waste at or from any of the Sites; or (iii) an acknowledgment that a release or threatened release of hazardous waste at or from any of the Sites constitutes a significant threat to the public health or the environment.

7. Solely with regard to the matters set forth below, Respondent hereby waives any right to a hearing as may be provided by law, consents to the issuance and entry of this Order & Settlement Agreement, and agrees to be bound by its terms. Respondent consents to and agrees not to contest the authority or jurisdiction of the Department to issue or enforce this Order & Settlement Agreement, and agrees not to contest the validity of this Order & Settlement Agreement or its terms or the validity of data submitted to the Department by Respondent pursuant to this Order & Settlement Agreement.

NOW, having considered this matter and being duly advised, **IT IS ORDERED THAT**:

I. Initial Submittal- Records Search Reports

Respondent shall submit to the Department a Records Search Report for each of the Sites, in accordance with the requirements of Exhibit "F" attached hereto, as follows:

1. Within thirty (30) Days after the effective date of this Order & Settlement Agreement, for Sites 1 through 7 listed on Exhibit "A" – Table 1.

2. Within sixty (60) Days after the effective date of this Order & Settlement Agreement, for Sites 8 through 14 listed on Exhibit "A" – Table 1.

The Records Search Report can be limited if the Department notifies Respondent that prior submissions satisfy specific items required for the Records Search Report. Such Records Search Report shall be submitted in a format acceptable to the Department.

II. Development, Performance, and Reporting of Work Plans

A. Work Plans

All activities at any Site that comprise any element of an Inactive Hazardous Waste Disposal Site Remedial Program shall be conducted pursuant to one or more Department-approved work plans ("Work Plan" or "Work Plans") for each Site and this Order & Settlement Agreement, and all activities shall be consistent with the National Oil and Hazardous Substances Pollution Contingency Plan ("NCP"), 40 C.F.R. Part 300, as required under CERCLA, 42 U.S.C. § 9600 *et seq*. The Work Plan(s) under this Order & Settlement Agreement shall address both on-Site and off-Site conditions and shall be developed and implemented in accordance with 6 NYCRR Part 375, "DER-10, Technical Guidance for Site Investigation and Remediation," and/or Exhibits G, H, I, J, and K of this Order & Settlement Agreement. All Department-approved Work Plans shall be incorporated into and become enforceable parts of this Order & Settlement Agreement and shall be attached as Exhibit "B." Upon approval of a Work Plan by the Department, Respondent shall implement such Work Plan in accordance with the schedule contained in such Work Plan. Nothing in this Subparagraph shall mandate that any particular Work Plan be submitted.

Each Work Plan submitted for each Site shall use one of the following captions on the cover page:

1. "Site Characterization Work Plan" ("SC Work Plan"): a Work Plan whose objective is to identify the presence of any hazardous waste disposed of at the Site. Such Work Plan shall be developed in accordance with Exhibit "G";

2. "Remedial Investigation/Feasibility Study Work Plan" ("RI/FS Work Plan"): a Work Plan whose objective is to perform a Remedial Investigation and a Feasibility Study. Such Work Plan shall be developed and implemented in accordance with Exhibit "H";

3. "Interim Remedial Measure Work Plan" ("IRM Work Plan"): a Work Plan whose objective is to provide for an Interim Remedial Measure. Such Work Plan shall be developed in accordance with Exhibit "I";

4. "Remedial Design/Remedial Action Work Plan" ("RD/RA Work Plan"): a Work Plan whose objective is to provide for the development and implementation of final plans and specifications for implementing the remedial alternative set forth in the ROD. Such Work Plan shall be developed in accordance with Exhibit "J"; or

5. "Site Management Work Plan" ("SM Work Plan"): a Work Plan whose objective is to provide for all activities required to maintain and monitor the effectiveness of the Remedial Action or an IRM. Such Work Plan shall be developed in accordance with Exhibit "K."

B. Submission/Implementation of Work Plans

1. (a) The first Work Plan for each of the Sites referred to in Paragraphs 4.A and 4.B and listed on Exhibit "A" shall be submitted to the Department in accordance with the schedule on Exhibit "A" Table 1.

(b) The Department may request that Respondent submit additional or supplemental Work Plans for each of the Sites. Within thirty (30) Days after the Department's written request, Respondent shall advise the Department in writing whether it will submit and implement the requested additional or supplemental Work Plan for the Site or whether it elects to terminate this Order & Settlement Agreement with respect to the Site pursuant to Paragraph XIII. If Respondent elects to submit and implement such Work Plan, Respondent shall submit the requested Work Plan within sixty (60) Days after such election. If Respondent elects to terminate this Order & Settlement Agreement with respect to the Site or fails to make a timely election, this Order & Settlement Agreement shall terminate with respect to the Site pursuant to Paragraph XIII.

(c) Respondent may, at Respondent's option, propose one or more additional or supplemental Work Plans (including one or more IRM Work Plans) at any time, which the Department shall review for appropriateness and technical sufficiency.

(d) Any request made by the Department under Subparagraph II.B.1.(b) shall be subject to dispute resolution pursuant to Paragraph XII.

2. A Professional Engineer must stamp and sign all Work Plans other than a Work Plan for an RI/FS or an SC.

3. During all field activities, Respondent shall have on-Site a representative who is qualified to supervise the activities undertaken. Such representative may be an employee or a consultant retained by Respondent to perform such supervision.

C. Modifications to Work Plans

The Department shall notify Respondent in writing if the Department determines that any element of a Department-approved Work Plan needs to be modified in order to achieve the objectives of the Work Plan as set forth in Subparagraph II.A or to ensure that the Remedial Program otherwise protects human health and the environment. Upon receipt of such notification, Respondent shall, subject to Respondent's right to invoke dispute resolution pursuant to Paragraph XII or to terminate with respect to the Site pursuant to Paragraph XIII,

submit a Work Plan for such requested work to the Department within sixty (60) Days after the date of the Department's written notice pursuant to this Subparagraph.

D. Submission of Final Reports and Annual Reports

1. In accordance with the schedule contained in a Work Plan, Respondent shall submit a final report that includes the caption of that Work Plan on the cover page and a certification that all requirements of the Work Plan have been complied with and all activities have been performed in full accordance with such Work Plan. Such certification shall be by the person with primary responsibility for the day to day performance of the activities under this Order & Settlement Agreement and, except for RI and SC final reports, shall be by a Professional Engineer.

2. Any final report that includes construction activities shall include "as built" drawings showing any changes made to the remedial design or the IRM.

3. In the event that any ROD or Work Plan for any of the Sites requires site management, including reliance upon institutional or engineering controls, Respondent shall submit an annual report by the 1st Day of the month following the anniversary of the start of the site management. Such annual report shall be signed by a Professional Engineer or by such other expert as the Department may find acceptable and shall contain a certification under penalty of perjury that any institutional and/or engineering controls required by this Order & Settlement Agreement are unchanged from the previous certification and that nothing has occurred that would impair the effectiveness of such control or constitute a violation of or failure to comply with the approved SM Work Plan. Respondent or owner of the Site may petition the Department for a determination that the institutional and/or engineering controls may be terminated. Such petition must be supported by a statement by a Professional Engineer that such controls are no longer necessary for the protection of public health and the environment. The Department shall not unreasonably withhold its approval of such petition.

E. Review of Submittals other than Progress Reports and Health and Safety Plans

1. The Department shall make a good faith effort to review and respond in writing to each submittal Respondent makes pursuant to this Order & Settlement Agreement within sixty (60) Days. The Department's response shall include an approval, modification or disapproval of the submittal. For the purposes of this Order & Settlement Agreement, "approval" shall mean acceptance of the document by the Department without conditions. All Department-approved submittals shall be incorporated into and become an enforceable part of this Order & Settlement Agreement.

2. If the Department requests modification or disapproves a submittal, it shall specify the reasons for its requested modification or disapproval. Within fifteen (15) Days after the date of the Department's written notice that Respondent's submittal has been disapproved or

a modification was requested, Respondent shall elect, in writing, to either (i) modify the submittal to address the Department's comments, (ii) invoke dispute resolution pursuant to Paragraph XII, or (iii) in the event the rejected submittal is a Work Plan submitted prior to the Department's approval of the RD/RA Work Plan, terminate this Order & Settlement Agreement with respect to the Site pursuant to Paragraph XIII. If Respondent elects to modify the submittal, Respondent shall, within thirty (30) Days after such election, make a revised submittal that addresses all of the Department's stated reasons for disapproving the first submittal. In the event that Respondent's revised submittal is disapproved, the Department shall set forth its reasons for such disapproval in writing and Respondent shall be in violation of this Order & Settlement Agreement unless it invokes dispute resolution pursuant to Paragraph XII and its position prevails. Failure to make an election or failure to comply with the election is a violation of this Order & Settlement Agreement.

3. Within thirty (30) Days after the Department's approval of a final report, Respondent shall submit such final report, as well as all data gathered and drawings and submittals made pursuant to such Work Plan, in an electronic format acceptable to the Department. If any document cannot be converted into electronic format, Respondent shall submit such document in an alternative format acceptable to the Department.

F. Department's Issuance of a ROD

Respondent shall cooperate with the Department and provide reasonable assistance, consistent with the Citizen Participation Plan for each of the Sites, in soliciting public comment on the proposed remedial action plan ("PRAP"), if any, for each of the Sites. After the close of the public comment period, the Department shall select a final remedial alternative for each of the Sites in a ROD. Nothing in this Order & Settlement Agreement shall be construed to abridge the rights of Respondent, as provided by law, to judicially challenge the Department's ROD.

G. Release and Covenant Not to Sue

Upon (i) the Department's approval of either the RD/RA Work Plan final report or an IRM Work Plan final report evidencing that no further remedial action (other than site management activities) is required to meet the goals of the Remedial Program for a Site, and (ii) the Department's acceptance of any environmental easement required pursuant to Paragraph X, then, except for the provisions of Paragraphs VI and VIII, and except for the future site management of the Site and any Natural Resource Damage claims, such acceptance shall constitute a release and covenant not to sue with respect to the Site for each and every claim, demand, remedy, or action whatsoever against Respondent, its directors, officers, employees, agents, servants, successors, and assigns (except successors and assigns who were responsible under law for the development and implementation of a Remedial Program at the Site prior to the effective date of this Order & Settlement Agreement), and their respective secured creditors, which the Department has or may have pursuant to Article 27, Title 13 of the ECL or pursuant to any other provision of State or Federal statutory or common law, including but not limited to §

107(a) of CERCLA, 42 U.S.C. § 9607(a), involving or relating to investigative or remedial activities relative to or arising from the disposal of hazardous wastes (or other contaminants remediated by Respondent to the Department's satisfaction pursuant to the ROD or Work Plans) at the Site; provided, however, that the Department specifically reserves all of its rights concerning, and any such release and covenant not to sue shall not extend to any further investigation or remediation the Department deems necessary due to newly discovered environmental conditions on-Site or off-Site which are related to the disposal of hazardous wastes at the Site and which indicate that the Remedial Program is not protective of public health and/or the environment. The Department shall notify Respondent in writing of such environmental conditions or information and its basis for determining that the Remedial Program is not protective of public health and/or the environment.

This release and covenant not to sue shall be null and void, *ab initio*, in the event of fraud relating to the execution or implementation of this Order & Settlement Agreement or in the event of Respondent's failure to materially comply with any provision of this Order & Settlement Agreement subsequent to issuance of a release and covenant not to sue, in each case with respect to that particular Site. The Department's determination that Respondent has committed fraud or has materially failed to comply with this Order & Settlement Agreement shall be subject to dispute resolution pursuant to Paragraph XII.

Nothing herein shall be construed as barring, diminishing, adjudicating, or in any way affecting any legal or equitable rights or claims, actions, suits, causes of action, or demands whatsoever that (i) Respondent may have against anyone other than the Department, including but not limited to rights of contribution under § 113(f)(3)(B) of CERCLA, 42 U.S.C. § 9613(f)(3)(B), and (ii) the Department may have against anyone other than Respondent, its directors, officers, employees, agents, and servants, and those successors and assigns of Respondent who were not responsible under law for the development and implementation of a Remedial Program at the Site prior to the effective date of this Order & Settlement Agreement, and their respective secured creditors.

III. Progress Reports

Respondent shall submit written progress reports for each Site to the parties identified in Subparagraph XI.A.1 by the 10th Day of each month commencing with the month subsequent to the approval of the first Work Plan and ending with the Termination Date, unless a different frequency is set forth in a Work Plan. Such reports shall, at a minimum, include: all actions taken pursuant to this Order & Settlement Agreement during the reporting period and those anticipated for the upcoming reporting period; all approved modifications to work plans and/or schedules; all results of sampling and tests and all other data received or generated by or on behalf of Respondent in connection with the Site, during the reporting period, including quality assurance/quality control information; and information regarding percentage of completion, unresolved delays encountered or anticipated that may affect the future schedule, efforts made to mitigate such delays, information regarding activities undertaken in support of the Citizen

Participation Plan during the reporting period and those anticipated for the upcoming reporting period and the name of the Department Project Manager for the Site.

IV. Penalties

A. 1. Respondent's failure to comply with any term of this Order & Settlement Agreement constitutes a violation of this Order & Settlement Agreement, the ECL, and 6 NYCRR Part 375. Nothing herein abridges Respondent's right to contest any allegation that it has failed to comply with this Order & Settlement Agreement.

2. Payment of any penalties shall not in any way alter Respondent's obligations under this Order & Settlement Agreement.

B. 1. Respondent shall not suffer any penalty or be subject to any proceeding or action in the event it cannot comply with any requirement of this Order & Settlement Agreement as a result of any event arising from causes beyond the reasonable control of Respondent, of any entity controlled by Respondent, and of Respondent's contractors, that delays or prevents the performance of any obligation under this Order & Settlement Agreement despite Respondent's best efforts to fulfill the obligation ("Force Majeure Event"). The requirement that Respondent exercise best efforts to fulfill the obligation includes using best efforts to anticipate the potential Force Majeure Event, best efforts to address any such event as it is occurring, and best efforts following the Force Majeure Event to minimize delay to the greatest extent possible. "Force Majeure" does not include Respondent's economic inability to comply with any obligation, the failure of Respondent to make complete and timely application for any required approval or permit, and non-attainment of the goals, standards, and requirements of this Order & Settlement Agreement.

2. Respondent shall notify the Department in writing within seven (7) Days after it obtains knowledge of any Force Majeure Event. Respondent shall include in such notice the measures taken and to be taken to prevent or minimize any delays and shall request an appropriate extension or modification of this Order & Settlement Agreement. Failure to give such notice within such seven (7) Day period constitutes a waiver of any claim that a delay is not subject to penalties. Respondent shall be deemed to know of any circumstance which it, any entity controlled by it, or its contractors knew or should have known.

3. Respondent shall have the burden of proving by a preponderance of the evidence that (i) the delay or anticipated delay has been or will be caused by a Force Majeure Event; (ii) the duration of the delay or the extension sought warranted under the circumstances; (iii) best efforts were exercised to avoid and mitigate the effects of the delay; and (iv) Respondent complied with the requirements of Subparagraph IV.B.2 regarding timely notification.

4. If the Department agrees that the delay or anticipated delay is attributable to a Force Majeure Event, the time for performance of the obligations that are affected by the Force Majeure Event shall be extended for such time as is reasonably necessary to complete those obligations.

5. If the Department rejects Respondent 's assertion that an event provides a defense to non-compliance with this Order & Settlement Agreement pursuant to Subparagraph IV.B, Respondent shall be in violation of this Order & Settlement Agreement unless it invokes dispute resolution pursuant to Paragraph XII and Respondent's position prevails.

V. Entry upon Site

A. Respondent hereby consents, upon reasonable notice under the circumstances presented, to entry upon the Site (or areas in the vicinity of the Site which may be under the control of Respondent) by any duly designated officer or employee of the Department or any State agency having jurisdiction with respect to matters addressed pursuant to this Order & Settlement Agreement, and by any agent, consultant, contractor, or other person so authorized by the Commissioner, all of whom shall abide by the health and safety rules in effect for the Site, for inspecting, sampling, copying records related to the contamination at the Site, testing, and any other activities necessary to ensure Respondent's compliance with this Order & Settlement Agreement. Upon request, Respondent shall (i) provide the Department with suitable office space at each of the Sites, including access to a telephone, to the extent available, and (ii) permit the Department full access to all non-privileged records relating to matters addressed by this Order & Settlement Agreement. Raw data is not considered privileged and that portion of any privileged document containing raw data must be provided to the Department. In the event Respondent is unable to obtain any authorization from third-party property owners necessary to perform its obligations under this Order & Settlement Agreement, the Department may, consistent with its legal authority, assist in obtaining such authorizations.

B. The Department shall have the right to take its own samples and scientific measurements and the Department and Respondent shall each have the right to obtain split samples, duplicate samples, or both, of all substances and materials sampled. The Department shall make the results of any such sampling and scientific measurements available to Respondent.

VI. Payment of State Costs

A. Within forty-five (45) Days after the effective date of this Order & Settlement Agreement, Respondent shall pay to the Department the sum of $0.00, which shall represent reimbursement for State Costs as set forth on the cost summary attached as Exhibit "C." Respondent acknowledges that all past State Costs are not itemized on the cost summary and that additional charges may be billed at a later date for State Costs incurred prior to the effective date of this Order & Settlement Agreement.

B. Within forty-five (45) Days after receipt of an itemized invoice from the Department, Respondent shall pay to the Department a sum of money which shall represent reimbursement for State Costs, other than those identified in Subparagraph VI.A, for work performed at or in connection with each of the Sites through and including the Termination Date.

C. Personal service costs shall be documented by reports of Direct Personal Service, which shall identify the employee name, title, biweekly salary or billable hourly rate, and time spent (in hours) on the project during the billing period, as identified by an assigned time and activity code for each of the Sites. Approved agency fringe benefit and indirect cost rates shall be applied. Non-personal service costs shall be summarized by category of expense (*e.g.*, supplies, materials, travel, contractual) and shall be documented by expenditure reports. The Department shall not be required to provide any other documentation of costs, provided however, that the Department's records shall be available consistent with, and in accordance with, Article 6 of the Public Officers Law.

D. Such invoice shall be sent to Respondent at the following address:

Lawrence Liebs
KeySpan Corporation
One MetroTech Center
Brooklyn, New York 11201

E. Each such payment shall be made payable to the Department of Environmental Conservation and shall be sent to:

Bureau of Program Management
Division of Environmental Remediation
New York State Department of Environmental Conservation
625 Broadway
Albany, New York 12233-7012.

F. Each party shall provide written notification to the other within ninety (90) Days of any change in the foregoing addresses.

G. Respondent may contest, in writing, invoiced costs under Subparagraph VI.B if it believes that (i) the cost documentation contains clerical, mathematical, or accounting errors; (ii) the costs are not related to the State's activities with respect to the Remedial Program for the Site; or (iii) the Department is not otherwise legally entitled to such costs. If Respondent objects to an invoiced cost, Respondent shall pay all costs not objected to within the time frame set forth in Subparagraph VI.B and shall, within thirty (30) Days after its receipt of an invoice, identify, in writing, all costs objected to and the basis of the objection. This objection shall be filed with the Bureau of Program Management ("BPM") Director. The BPM Director or the BPM Director's designee shall have the authority to relieve Respondent of the obligation to pay invalid costs. Within forty-five (45) Days after the date of the Department's determination of the objection,

Respondent shall either pay to the Department the amount which the BPM Director or the BPM Director's designee determines Respondent is obligated to pay or commence an action or proceeding seeking appropriate judicial relief.

H. If any negotiable instrument submitted to the Department pursuant to this Order & Settlement Agreement is not honored when presented for payment, Respondent shall be in violation of this Order & Settlement Agreement, provided that (i) the Department gives Respondent written notice of same, and (ii) the Department does not receive a certified check or bank check in the amount of the uncollected funds within fourteen (14) Days after the date of the Department's written notification.

VII. Reservation of Rights

A. Except as provided in Subparagraph II.G, nothing contained in this Order & Settlement Agreement shall be construed as barring, diminishing, adjudicating, or in any way affecting any of the Department's rights or authorities, including, but not limited to, the right to require performance of further investigations and/or response action(s), to recover natural resource damages, and/or to exercise any summary abatement powers with respect to any person, including Respondent.

B. Except as otherwise provided in this Order & Settlement Agreement, Respondent specifically reserves all rights and defenses under applicable law respecting any Departmental assertion of remedial liability and/or natural resource damages against Respondent, and further reserves all rights respecting the enforcement of this Order & Settlement Agreement, including the rights to notice, to be heard, to appeal, and to any other due process. The existence of this Order & Settlement Agreement or Respondent's compliance with it shall not be construed as an admission of liability, fault, wrongdoing, or breach of standard of care by Respondent, and shall not give rise to any presumption of law or finding of fact, or create any rights, or grant any cause of action, which shall inure to the benefit of any third party. Further, Respondent reserves such rights as it may have to seek and obtain contribution, indemnification, and/or any other form of recovery from its insurers and from other potentially responsible parties or their insurers for past or future response and/or cleanup costs or such other costs or damages arising from the contamination at the Site as may be provided by law, including but not limited to rights of contribution under §113(f)(3)(B) of CERCLA, 42 U.S.C. § 9613(f)(3)(B).

VIII. Indemnification

Respondent shall indemnify and hold the Department, the State of New York, and their representatives and employees harmless for all third-party claims, suits, actions, damages, and costs of every name and description arising out of or resulting from the fulfillment or attempted fulfillment of this Order & Settlement Agreement by Respondent and/or any of Respondent's directors, officers, employees, servants, agents, successors, and assigns except for liability arising from (i) vehicular accidents occurring during travel to or from the Site; or (ii) willful, wanton, or malicious acts or omissions, and acts or omissions constituting gross negligence or criminal behavior by the Department, the State of New York, and/or their representatives and employees

during the course of any activities conducted pursuant to this Order & Settlement Agreement. The Department shall provide Respondent with written notice no less than thirty (30) Days prior to commencing a lawsuit seeking indemnification pursuant to this Paragraph.

IX. Public Notice

A. Respondent shall cause to be filed a Department-approved Notice of Order & Settlement Agreement for each of the Sites, in accordance with the schedule below, which Notices shall be substantially similar to the Notice of Order & Settlement Agreement attached to this Order & Settlement Agreement as Exhibit "D," with the recording officer of the county wherein the Site is to give all parties who may acquire any interest in the Site notice of this Order & Settlement Agreement. Within sixty (60) Days of each such filing, Respondent shall also provide the Department with a copy of such instrument certified by the recording officer to be a true and faithful copy.

1. Within thirty (30) Days after the effective date of this Order & Settlement Agreement, for Sites 1 through 6, 11 and 13 listed on Exhibit "A" – Table 1.

2. Within sixty (60) Days after the effective date of this Order & Settlement Agreement, for Sites 7 through 10, 12 and 14 listed on Exhibit "A" – Table 1.

B. If Respondent proposes to convey, or becomes aware of a proposal to convey, the whole or any part of Respondent's ownership interest in the Site, or becomes aware of such conveyance, Respondent shall, not fewer than forty-five (45) Days before the date of conveyance, or within forty-five (45) Days after becoming aware of such conveyance, whichever shall be applicable, notify the Department in writing of the identity of the transferee and of the nature and proposed or actual date of the conveyance, and shall notify the transferee in writing, with a copy to the Department, of the applicability of this Order & Settlement Agreement. However, such obligation shall not extend to a conveyance by means of a corporate reorganization or merger or the granting of any rights under any mortgage, deed, trust, assignment, judgment, lien, pledge, security agreement, lease, or any other right accruing to a person not affiliated with Respondent to secure the repayment of money or the performance of a duty or obligation.

X. Environmental Easement

A. 1. If a Department-approved Work Plan or the ROD for the Site, if any, relies upon one or more institutional and/or engineering controls, Respondent (or the owner of the Site) shall execute an environmental easement pursuant to ECL Article 71, Title 13 which shall be substantially similar to Exhibit "E." Respondent shall cause such instrument to be recorded with the recording officer of the county wherein the Site is located within thirty (30) Days of the Department's approval of such instrument. Respondent shall provide the Department with a copy of such instrument certified by the recording officer to be a true and faithful copy within sixty (60) Days after such recording.

B. If the ROD provides for "no action" other than implementation of one or more institutional controls, Respondent shall cause an environmental easement to be recorded under the provisions of Subparagraph X.A.1. If Respondent does not cause such environmental easement to be recorded, Respondent cannot obtain a release and covenant not to sue pursuant to Subparagraph II.G.

XI. Communications

A. All written communications required by this Order & Settlement Agreement shall be transmitted by United States Postal Service, by private courier service, or hand delivered as follows:

1. Communication from Respondent shall be sent to:

Robert Schick
Division of Environmental Remediation
New York State Department of Environmental Conservation
625 Broadway
Albany, New York 12233
(2 bound copies, 1 unbound copy and 1 disk)

with copies to:

Gary Litwin
Bureau of Environmental Exposure Investigation
New York State Department of Health
Flanigan Square
547 River Street
Troy, New York 12180-2216
(2 bound copies)

Regional Director
Region 2
NYS Dept of Environmental Conservation
One Hunter's Point Plaza
47-40 21st Street
Long Island City, NY 11101-5407
(Correspondence only)

Larry S. Eckhaus, Senior Attorney
Division of Environmental Enforcement
Superfund & Brownfields Restoration Bureau
NYS Dept of Environmental Conservation
625 Broadway, 14th Floor
Albany, NY 12233-5500
(Correspondence only)

2. Communication from the Department to Respondent shall be sent to:

Lawrence Liebs
KeySpan Corporation
One MetroTech Center
Brooklyn, New York 11201

with copy to:

Donna Riccobono, Esq.
KeySpan Corporation
One MetroTech Center
Brooklyn, New York 11201

B. The Department and Respondent reserve the right to designate additional or different addressees for communication upon written notice to the other.

C. Each party shall notify the other within ninety (90) Days after any change in the addresses in this Paragraph XI or in Paragraph VI.

XII. Dispute Resolution

A. If Respondent disagrees with the Department's notice under (i) Subparagraph II.B requesting additional or supplemental Work Plans; (ii) Subparagraph II.C requesting modification of a Department-approved Work Plan; (iii) Subparagraph II.E disapproving a submittal, a proposed Work Plan, or a final report; (iv) Subparagraph II. G. finding that Respondent materially failed to comply with the Order & Settlement Agreement; (v) Subparagraph IV.B rejecting Respondent's assertion of a Force Majeure Event; or (vi) Subparagraph XIV.G.2.iii requesting modification of a time frame, Respondent may, within thirty (30) Days of its receipt of such notice, make a written request for informal negotiations with the Department in an effort to resolve the dispute. A copy of such request shall be sent by Respondent to the appropriate Remedial Bureau Chief in the Department's Central Office. The Department and Respondent shall consult together in good faith and exercise best efforts to resolve any differences or disputes without resort to the procedures described in Subparagraph XII.B. The period for informal negotiations shall not exceed thirty (30) Days from the date of the Department's initial response to the Respondent's request for informal negotiations. If the parties cannot resolve a dispute by informal negotiations during this period, the Department's position shall be considered binding unless Respondent notifies the Department in writing within thirty (30) Days after the conclusion of the thirty (30) Day period for informal negotiations that it invokes the dispute resolution provisions provided under Subparagraph XII.B.

B. 1. Respondent shall file with the Office of Hearings and Mediation Services ("OH&M") a request for formal dispute resolution and a written statement of the issues in dispute, the relevant facts upon which the dispute is based, factual data, analysis, or opinion supporting its position, and all supporting documentation upon which Respondent relies (hereinafter called the "Statement of Position"). A copy of such request and written statement

shall be provided contemporaneously to the Director and to the parties listed under Subparagraph XI.A.1.

2. The Department shall serve its Statement of Position no later than twenty (20) Days after receipt of Respondent's Statement of Position.

3. Respondent shall have the burden of proving by substantial evidence that the Department's position does not have a rational basis and should not prevail. The OH&M can conduct meetings, in person or via telephone conferences, and request additional information from either party if such activities will facilitate a resolution of the issues.

4. The OH&M shall prepare and submit a report and recommendation to the Director. The Director shall issue a final decision in a timely manner. The final decision shall constitute a final agency action and Respondent shall have the right to seek judicial review of the decision pursuant to Article 78 of the Civil Practice Law and Rules ("CPLR") provided that Respondent notifies the Department within thirty (30) Days after receipt of a copy of the final decision of its intent to commence an Article 78 proceeding and commences such proceeding within sixty (60) Days after receipt of a copy of the Director's final decision. Respondent shall be in violation of this Order & Settlement Agreement if it fails to comply with the final decision resolving this dispute within forty-five (45) Days after the date of such final decision, or such other time period as may be provided in the final decision, unless it seeks judicial review of such decision within the sixty (60) Day period provided. In the event that Respondent seeks judicial review, Respondent shall be in violation of this Order & Settlement Agreement if it fails to comply with the final Court Order or any settlement within thirty (30) Days after the effective date of such Court Order or settlement, unless otherwise directed by the Court. For purposes of this Subparagraph, a Court Order or settlement shall not be final until the time to perfect an appeal of same has expired.

5. The invocation of dispute resolution shall not extend, postpone, or modify Respondent's obligations under this Order & Settlement Agreement with respect to any item not in dispute unless or until the Department agrees or a Court orders otherwise. Except as otherwise provided in this Order & Settlement Agreement, the invocation of the procedures set forth in this Paragraph XII shall constitute an election of remedies and such election shall constitute a waiver of any and all other administrative remedies which may otherwise be available to Respondent regarding the issue in dispute.

6. The Department shall keep an administrative record of any proceedings under this Paragraph XII that shall be available consistent with Article 6 of the Public Officers Law.

7. Nothing in this Paragraph XII shall be construed as an agreement by the parties to resolve disputes through administrative proceedings pursuant to the State Administrative Procedure Act, the ECL, or 6 NYCRR Part 622 or 6 NYCRR Part 375.

8. Nothing contained in this Order & Settlement Agreement shall be construed to authorize Respondent to invoke dispute resolution with respect to the remedy

selected by the Department in the ROD or any element of such remedy, nor to impair any right of Respondent to seek judicial review of the Department's selection of any remedy with respect to each Site.

XIII. Termination of Order & Settlement Agreement

A. This Order & Settlement Agreement will terminate with respect to a Site upon the earlier of the following events:

1. Respondent's election to terminate with respect to a Site pursuant to Subparagraphs II.B.1.b, II.C or II.E.2 so long as such election is made prior to the Department's approval of the RD/RA Work Plan for that Site. In the event of termination in accordance with this Subparagraph XIII.A.1, this Order & Settlement Agreement with respect to such Site shall terminate effective the 5th Day after the Department's receipt of the written notification terminating this Order & Settlement Agreement with respect to such Site or the 5 th Day after the time for Respondent to make its election has expired, whichever is earlier, provided, however, that if there are one or more Work Plan(s) with respect to such Site for which a final report has not been approved at the time of Respondent's notification of its election to terminate this Order & Settlement Agreement with respect to such Site pursuant to Subparagraphs II.B.1.b or II.E.32 or its failure to timely make such an election pursuant to Subparagraphs II.B.1.b or II.E.32, Respondent shall promptly complete the activities required by such previously approved Work Plan(s)consistent with the schedules contained therein. Thereafter, this Order & Settlement Agreement shall terminate with respect to such Site effective the 5 th Day after the Department's approval of the final report for all previously approved Work Plans with respect to such Site ; or

2. the Department's written determination that Respondent has completed all phases of the Remedial Program (including site management) for all the Sites, in which event the termination with respect to all Sites shall be effective on the 5 th Day after the date of the Department's approval of the final report relating to the final phase of the Remedial Program for all the Sites.

B. Notwithstanding the foregoing, the provisions contained in Paragraphs VI and VIII shall survive the termination of this Order & Settlement Agreement and any violation of such surviving Paragraphs shall be a violation of this Order & Settlement Agreement, the ECL, and 6 NYCRR Part 375, subjecting Respondent to penalties as provided under Paragraph IV so long as such obligations accrued on or prior to the Termination Date.

C. If the Order & Settlement Agreement is terminated with respect to a Site or all Sites pursuant to Subparagraph XIII.A.1, neither this Order & Settlement Agreement nor its termination shall affect any liability of Respondent may have for remediation of the Site and/or for payment of State Costs, including implementation of removal and remedial actions, interest, enforcement, and any and all other response costs as defined under CERCLA, nor shall it affect any defenses to such liability that may be asserted by Respondent. Respondent shall also ensure that it does not leave the Site in a condition, from the perspective of human

health and environmental protection, worse than that which existed before any activities under this Order & Settlement Agreement were commenced. Further, the Department's efforts in obtaining and overseeing compliance with this Order & Settlement Agreement shall constitute "reasonable efforts" under law to obtain a voluntary commitment from Respondent for any further activities to be undertaken as part of a Remedial Program for the Site.

XIV. Miscellaneous

A. Respondent shall retain professional consultants, contractors, laboratories, quality assurance/quality control personnel, and third party data validators ("Respondent's Contractors") acceptable to the Department to perform its obligations under this Order & Settlement Agreement. If the Department has not previously approved Respondent's Contractors for the work required by this Order & Settlement Agreement, Respondent shall submit the Contractors' qualifications to the Department a minimum of thirty (30) Days before the start of any activities for which each such Contractor will be responsible. The Department's approval of each such Contractor shall be obtained prior to the start of work by that Contractor. The responsibility for the performance of all Contractors retained by Respondent shall rest solely with Respondent. Subject to the requirements of this Subparagraph, Respondent retains the right to select or change firms or individuals in its sole discretion.

B. Respondent shall allow the Department to attend and shall notify the Department at least seven (7) Days in advance of any field activities as well as any pre-bid meetings, job progress meetings, the substantial completion meeting and inspection, and the final inspection and meeting; nothing in this Order & Settlement Agreement shall be construed to require Respondent to allow the Department to attend portions of meetings where privileged matters are discussed.

C. Respondent shall use "best efforts" to obtain all Site access, permits, easements, rights-of-way, rights-of-entry, approvals, institutional controls, or authorizations necessary to perform Respondent's obligations under this Order & Settlement Agreement.

1. The Department may exempt Respondent from the requirement to obtain any state or local permit or other authorization for any activity on the Site needed to implement this Order & Settlement Agreement that the Department determines is conducted in a manner which satisfies all substantive technical requirements applicable to like activity conducted pursuant to a permit.

2. If, despite Respondent's best efforts, any necessary Site access, easements, rights-of-way, rights-of-entry, approvals, institutional controls, or authorizations required to perform this Order & Settlement Agreement are not obtained within forty-five (45) Days after the effective date of this Order & Settlement Agreement, or within forty-five (45) Days after the date the Department notifies Respondent in writing that additional access beyond that previously secured is necessary, Respondent shall promptly notify the Department, and shall include in that notification a summary of the steps Respondent has

taken to obtain access. The Department may, as it deems appropriate and within its authority, assist Respondent in obtaining access. If any interest in property is needed to implement an institutional control required by a Work Plan and such interest cannot be obtained, the Department may require Respondent to modify the Work Plan pursuant to Subparagraph II.C of this Order & Settlement Agreement to reflect changes necessitated by the lack of access and/or approvals.

D. Respondent and Respondent's successors and assigns shall be bound by this Order & Settlement Agreement. Any change in ownership or corporate status of Respondent shall in no way alter Respondent's responsibilities under this Order & Settlement Agreement.

E. Respondent shall provide a copy of this Order & Settlement Agreement to each contractor hired to perform work required by this Order & Settlement Agreement and shall condition all contracts entered into pursuant to this Order & Settlement Agreement upon performance in conformity with the terms of this Order & Settlement Agreement. Respondent or its contractor(s) shall provide written notice of this Order & Settlement Agreement to all subcontractors hired to perform any portion of the work required by this Order & Settlement Agreement. Respondent shall nonetheless be responsible for ensuring that Respondent's contractors and subcontractors perform the work in satisfaction of the requirements of this Order & Settlement Agreement.

F. The paragraph headings set forth in this Order & Settlement Agreement are included for convenience of reference only and shall be disregarded in the construction and interpretation of any provisions of this Order & Settlement Agreement.

G. 1. The terms of this Order & Settlement Agreement constitute the entire agreement between the Department and Respondent concerning implementation of the activities required by this Order & Settlement Agreement. No term, condition, understanding, or agreement purporting to modify or vary any term of this Order & Settlement Agreement shall be binding unless made in writing and subscribed by the party to be bound. No informal advice, guidance, suggestion, or comment by the Department shall be construed as relieving Respondent of Respondent's obligation to obtain such formal approvals as may be required by this Order & Settlement Agreement. In the event of a conflict between the terms of this Order & Settlement Agreement and any Work Plan submitted pursuant to this Order & Settlement Agreement, the terms of this Order & Settlement Agreement shall control over the terms of the Work Plan(s) attached as Exhibit "B."

2. i. Except as set forth herein, if Respondent desires that any provision of this Order & Settlement Agreement be changed, other than a provision of a Work Plan or a time frame, Respondent shall make timely written application to the Commissioner with copies to the parties listed in Subparagraph XI.A.1. The Commissioner or the Commissioner's designee shall timely respond.

ii. Changes to a Work Plan shall be accomplished as set forth in Subparagraph II.C of this Order & Settlement Agreement.

iii. Changes to a time frame set forth in this Order & Settlement Agreement shall be sought by a written request to the Department's project attorney and project manager, which request shall be timely responded to in writing. The Department's decision relative to the request for a time frame change shall be subject to dispute resolution pursuant to Paragraph XII.

H. 1. If multiple parties sign this Order & Settlement Agreement, the term "Respondent" shall be read in the plural where required to give meaning to this Order & Settlement Agreement. Further, the obligations of such Respondents under this Order & Settlement Agreement are joint and several and the insolvency of or failure by any Respondent to implement any obligations under this Order & Settlement Agreement shall not affect the obligations of the remaining Respondent(s).

2. If Respondent is a partnership, the obligations of all general partners, including limited partners who act as general partners, to finance and perform obligations under this Order & Settlement Agreement and to pay amounts owed to the Department under this Order & Settlement Agreement are joint and several. In the event of the insolvency of or the failure of any of the general partners to implement the requirements of this Order & Settlement Agreement, the remaining general partners shall implement all such requirements.

3. Notwithstanding the foregoing Subparagraphs XIV.H. 1 and 2, if multiple parties sign this Order & Settlement Agreement as Respondents but not all of the signing parties elect, pursuant to Subparagraph II.B, to implement a Work Plan, then all Respondents are jointly and severally liable for each and every obligation under this Order & Settlement Agreement through the completion of the activities in such Work Plan that all such parties consented to; thereafter, only those Respondents electing to perform additional work shall be jointly and severally liable under this Order & Settlement Agreement for the obligations and activities under such additional Work Plan(s). The parties electing not to implement the additional Work Plan(s) shall have no obligations under this Order & Settlement Agreement relative to the activities set forth in such Work Plan(s). Only those Respondents electing to implement such additional Work Plan(s) shall be eligible to receive the release and covenant not to sue provided under Subparagraph II.G.

I. To the extent authorized under CERCLA § 113, 42 U.S.C. § 9613, New York General Obligations Law § 15-108, and any other applicable law, Respondent shall be deemed to have resolved its liability to the State for purposes of contribution protection provided by CERCLA § 113(f)(2), 42 U.S.C. § 9613(f)(2) for "matters addressed" pursuant to and in accordance with this Order & Settlement Agreement. "Matters addressed" in this Order & Settlement Agreement shall mean all response actions taken by Respondent to implement this Order & Settlement Agreement for the Sites and all response costs incurred and to be incurred by any person or party in connection with the work performed under this Order & Settlement Agreement, which costs have been paid by Respondent, including reimbursement of State Costs pursuant to this Order & Settlement Agreement. Furthermore, to the extent authorized under CERCLA § 113(f)(3)(B), 42 U.S.C. Section 9613(f)(3)(B), by entering into this administrative settlement of liability, if any, for some or all of the response action and/or for

some or all of the costs of such action, Respondent is entitled to seek contribution under CERCLA from any person except those who are entitled to contribution protection under CERCLA § 113(f)(2), 42 U.S.C. § 9613(f)(2).

J. Unless otherwise expressly provided herein, terms used in this Order & Settlement Agreement which are defined in ECL Article 27, Title 13, ECL Article 71, Title 36, or in regulations promulgated under such statute shall have the meaning assigned to them under said statute or regulations. Whenever terms listed in the Glossary attached hereto are used in this Order & Settlement Agreement or in the attached Exhibits, the definitions set forth in the Glossary shall apply. In the event of a conflict, the definition set forth in the Glossary shall control.

K. Respondent's obligations under this Order & Settlement Agreement represent payment for or reimbursement of response costs, and shall not be deemed to constitute any type of fine or penalty.

L. This Order & Settlement Agreement may be executed for the convenience of the parties hereto, individually or in combination, in one or more counterparts, each of which for all purposes shall be deemed to have the status of an executed original and all of which shall together constitute one and the same.

M. The effective date of this Order & Settlement Agreement is the 10th Day after the date the Commissioner or the Commissioner's designee signs this Order & Settlement Agreement.

DATED: FEB 22 2007

CARL JOHNSON,
ACTING EXECUTIVE DEPUTY COMMISSIONER
New York State Department
of Environmental Conservation

By: _____
Dale A. Desnoyers, Director
Division of Environmental Remediation

CONSENT BY RESPONDENT

Respondent KeySpan Gas East Corporation d/b/a KeySpan Energy Delivery Long Island hereby consents to the issuing and entering of this Order on Consent and Administrative Settlement, waives Respondent's right to a hearing herein as provided by law, and agrees to be bound by this Order on Consent and Administrative Settlement.

By: _____
William J. Akley

Title: Vice President, Gas Operations, LI

Date: 2-2-07

STATE OF NEW YORK)
) s.s.:
COUNTY OF Nassau)

On the 2nd day of February in the year 2007, before me, the undersigned, personally appeared William J. Akley, personally known to me or proved to me on the basis of satisfactory evidence to be the individual(s) whose name is (are) subscribed to the within instrument and acknowledged to me that he/she/they executed the same in his/her/their capacity(ies), and that by his/her/their signature(s) on the instrument, the individual(s), or the person upon behalf of which the individual(s) acted, executed the instrument.

_____
Signature and Office of individual
taking acknowledgment

JOAN M. JERZ
Notary Public, State of New York
No. 01JE4737272
Qualified in Nassau County
Commission Expires May 31, 2007

CONSENT BY RESPONDENT

Respondent The Brooklyn Union Gas Company d/b/a KeySpan Energy Delivery New York hereby consents to the issuing and entering of this Order on Consent and Administrative Settlement, waives Respondent's right to a hearing herein as provided by law, and agrees to be bound by this Order on Consent and Administrative Settlement.

By: _____
John J. Bishar Jr.

Title: Executive Vice President and Secretary

Date: 2/2/07

STATE OF NEW YORK)
) s.s.:
COUNTY OF Nassau)

On the 2nd day of February in the year 2007, before me, the undersigned, personally appeared John J. Bishar, Jr, personally known to me or proved to me on the basis of satisfactory evidence to be the individual(s) whose name is (are) subscribed to the within instrument and acknowledged to me that he/she/they executed the same in his/her/their capacity(ies), and that by his/her/their signature(s) on the instrument, the individual(s), or the person upon behalf of which the individual(s) acted, executed the instrument.

_____
Signature and Office of individual taking acknowledgment

JOAN M. JERZ
Notary Public, State of New York
No. 01JE4737272
Qualified in Nassau County
Commission Expires May 31, 2007

EXHIBIT "A"

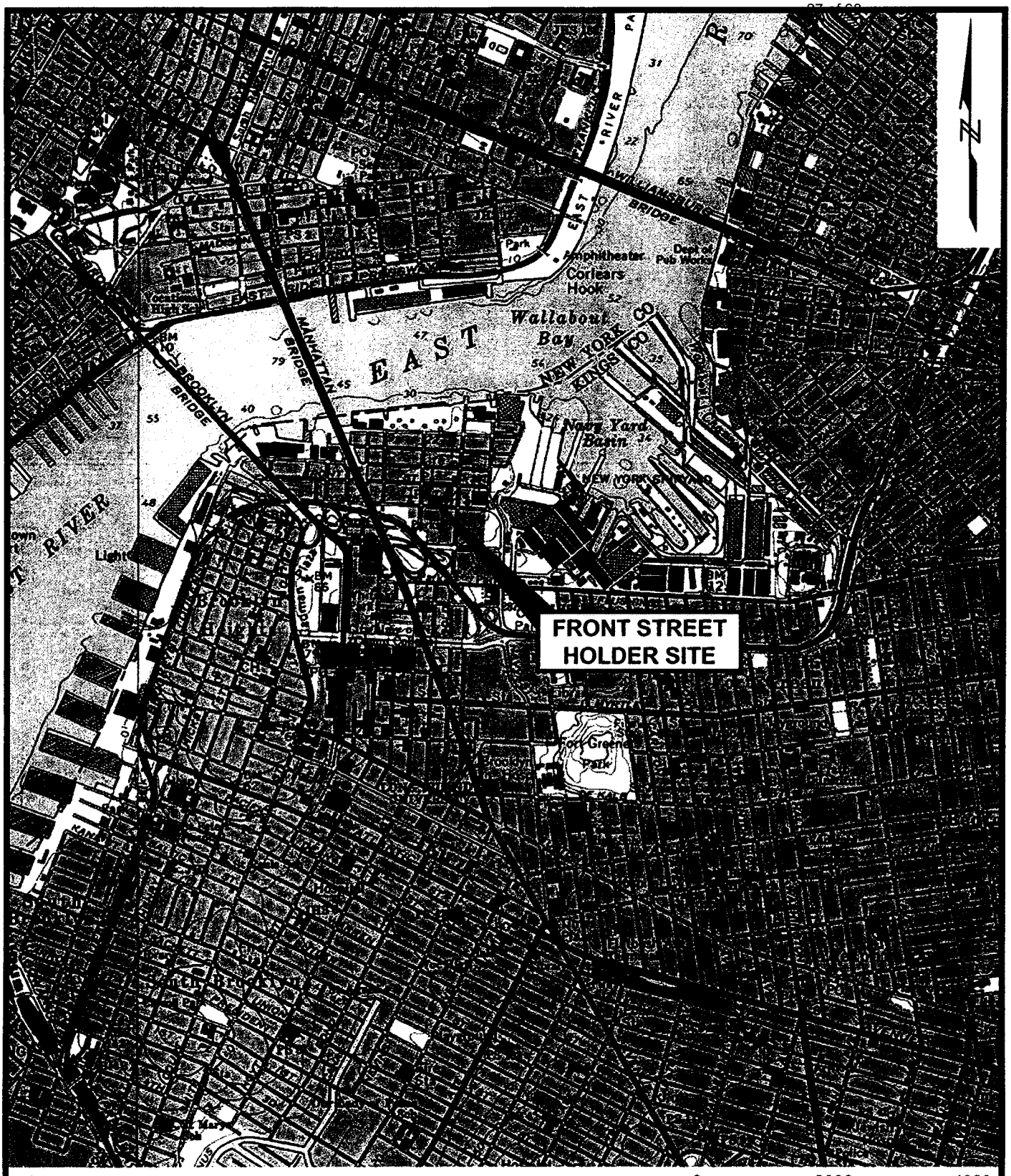
Descriptions of KeySpan MGP & Holder Sites & Maps of Sites included in Paragraphs 4.A and 4.B

Exhibit "A" - Table 1
KeySpan
Former MGP Sites & Holder Stations

Note: Sites in **BOLD CAPS** are entirely owned by KeySpan.
Sites in **Bold** are partially owned by KeySpan
All other sites are not owned by KeySpan

#	KeySpan Sites	Work Plan Type	Proposed Start	Site #	Location	County
1	Front Street Holder Station	SC	3/2007	224063	206 & 218 Front Street B Brooklyn, NY 11202	Kings
2	Rutledge Street Holder Station	SC	3/2007	224066	18 to 36 Rutledge Street & Heyward Street Brooklyn, 11211	Kings
3	Skillman Street Holder Station	SC	3/2007	224068	7 & 39 Skillman Street Brooklyn, NY 11205	Kings
4	Keap Street Holder Station	SC	3/2007	224064	18 Hooper Street (or 46 Keap St.) Brooklyn, NY 11221	Kings
5	Far Rockaway MGP Site	RI	6/2007	241032	Brunswick Avenue Queens, NY 11691	Queens
6	**Patchogue MGP Site & Hortonsphere**	RI	**9/2007**	**152182**	**Village of Patchogue Brookhaven, NY 11772**	**Suffolk**
7	Babylon MGP Site	RI	12/2007	152180	29 Evergreen Street Babylon, NY 11704	Suffolk
8	Dangman Park MGP	SC	3/2008	224047	486 Neptune Avenue Brooklyn, NY 11224	Kings
9	Peoples Works MGP Site	SC	3/2008	224053	Kent Ave & William St. Brooklyn, NY 11211	Kings
10	Jamaica Gas Light Co. MGP	SC	6/2008	241063	58-18 Beaver Road Jamaica, NY 11433	Queens
11	**Long Beach Holder Station**	SC	**6/2008**	130122	**Int. of: Riverside Blvd, Water St & Park Pl Long Beach, 11561**	**Nassau**
12	Jamaica Holder Station	SC	9/2008	241062	Liberty Ave. Bet 158 & 159 St Queens, NY	Queens
13	**Bay Ridge Holder Stations A & B**	SC	**9/2008**	224058	**A: 837 64th Street Brooklyn B:829-884 65th Street Brooklyn**	**Kings**
14	East Garden City (Stewart Avenue) Former Holder Station	SC	12/2008	130120	600-620 Stewart Avenue E. Garden City, NY 11530	Nassau

Note: Proposed Start = submittal of work plan



SOURCE: Map created with TOPO! ® ©2001 National Geographic
(www.nationalgeographic.com/topo)

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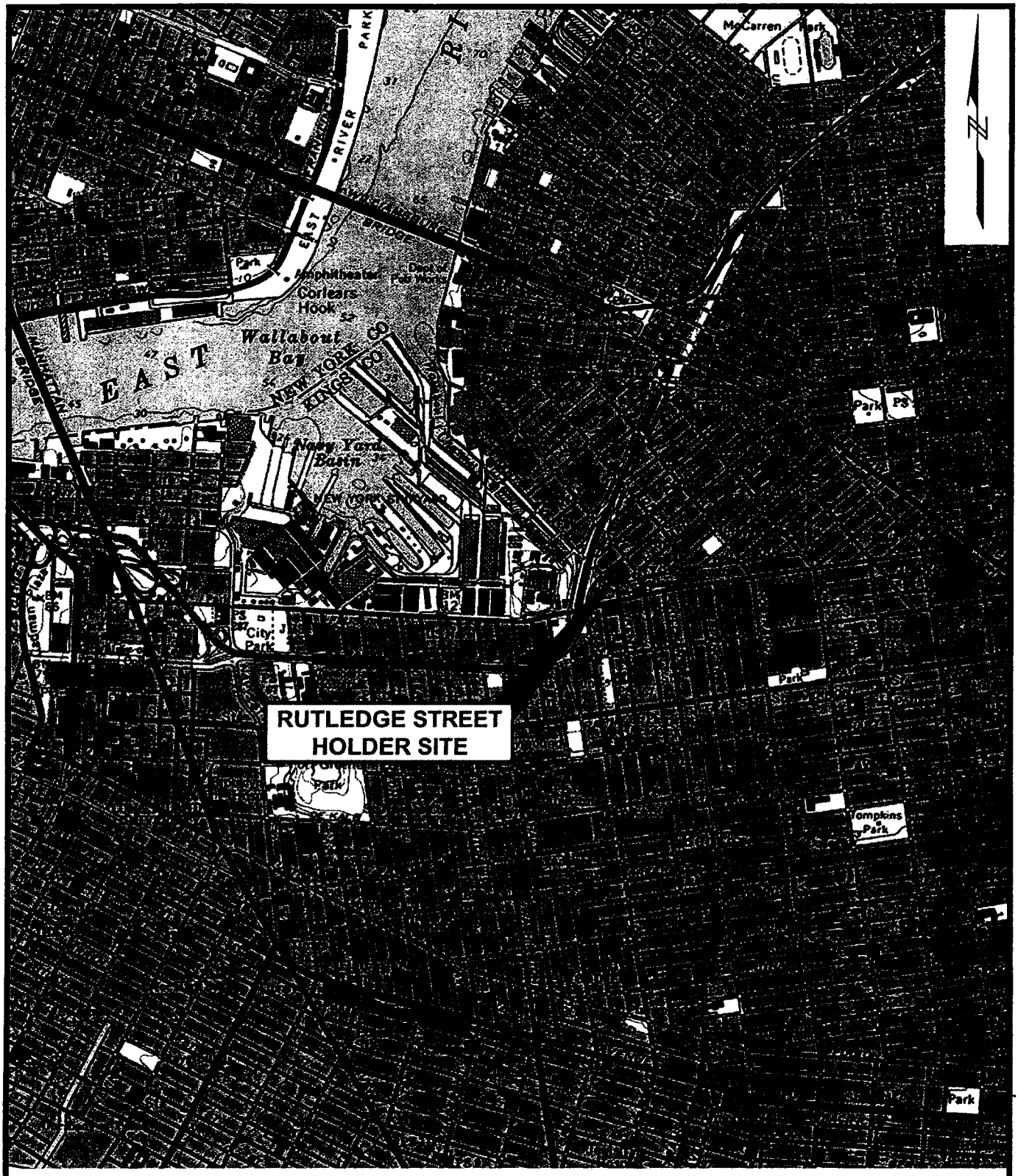
NYSDEC BCP APPLICATION FRONT STREET HOLDER BROOKLYN, NEW YORK	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



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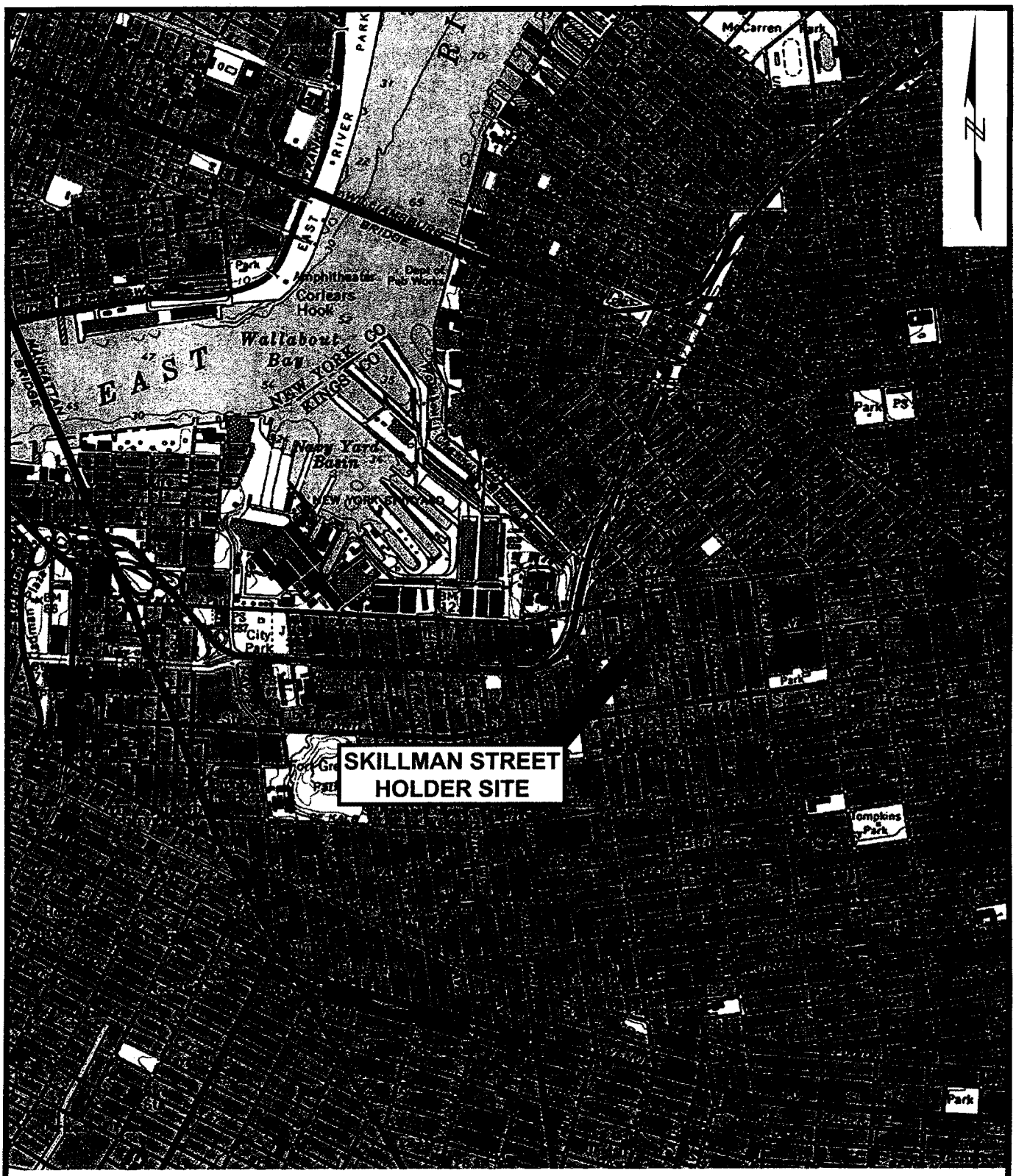
NYSDEC BCP APPLICATION RUTLEDGE STREET HOLDER STATION BROOKLYN, NEW YORK	GEI Consultants	SITE LOCATION MAP
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



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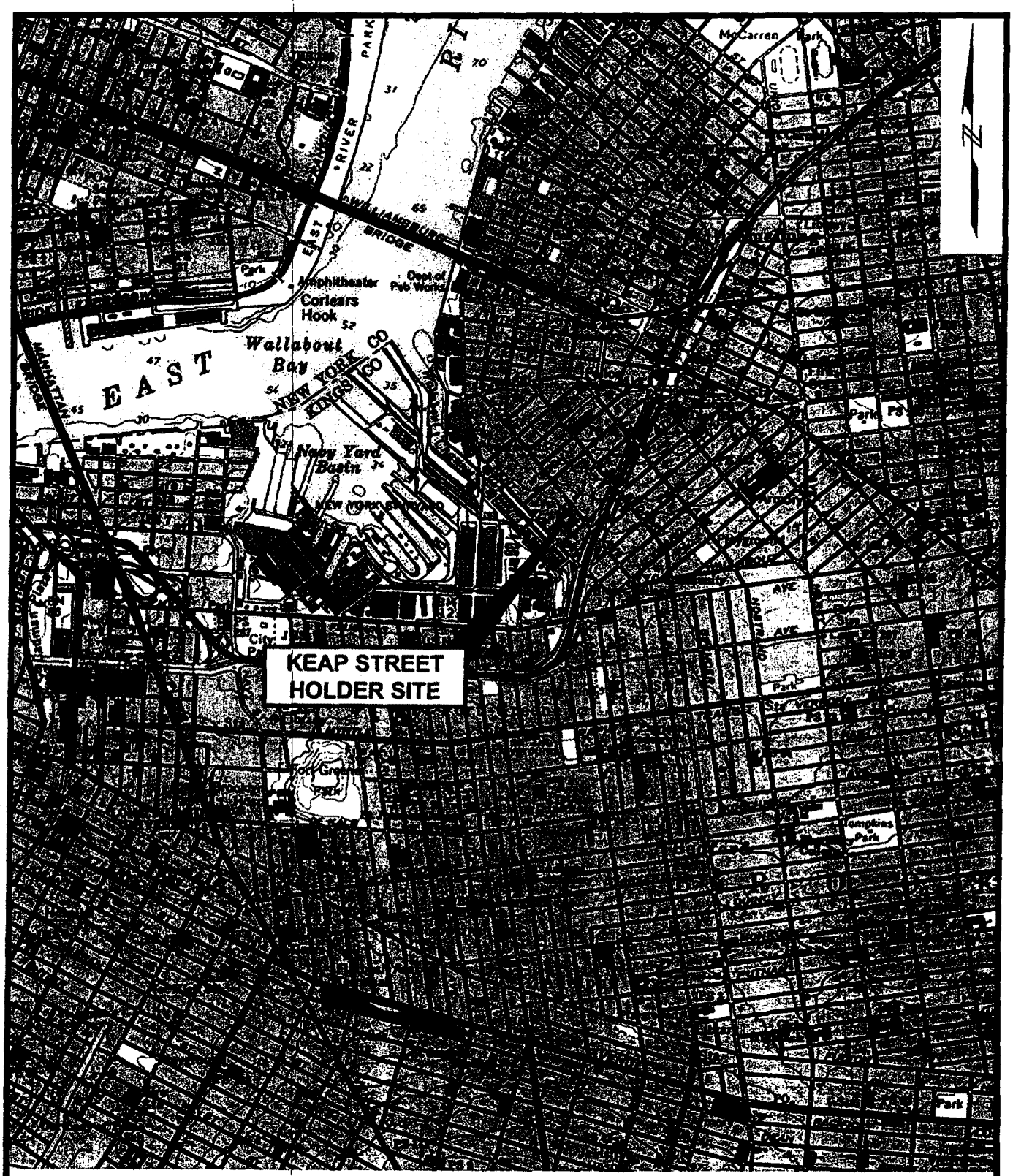
SCALE, FEET

NYSDEC BCP APPLICATION SKILLMAN STREET HOLDER STATION BROOKLYN, NEW YORK	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



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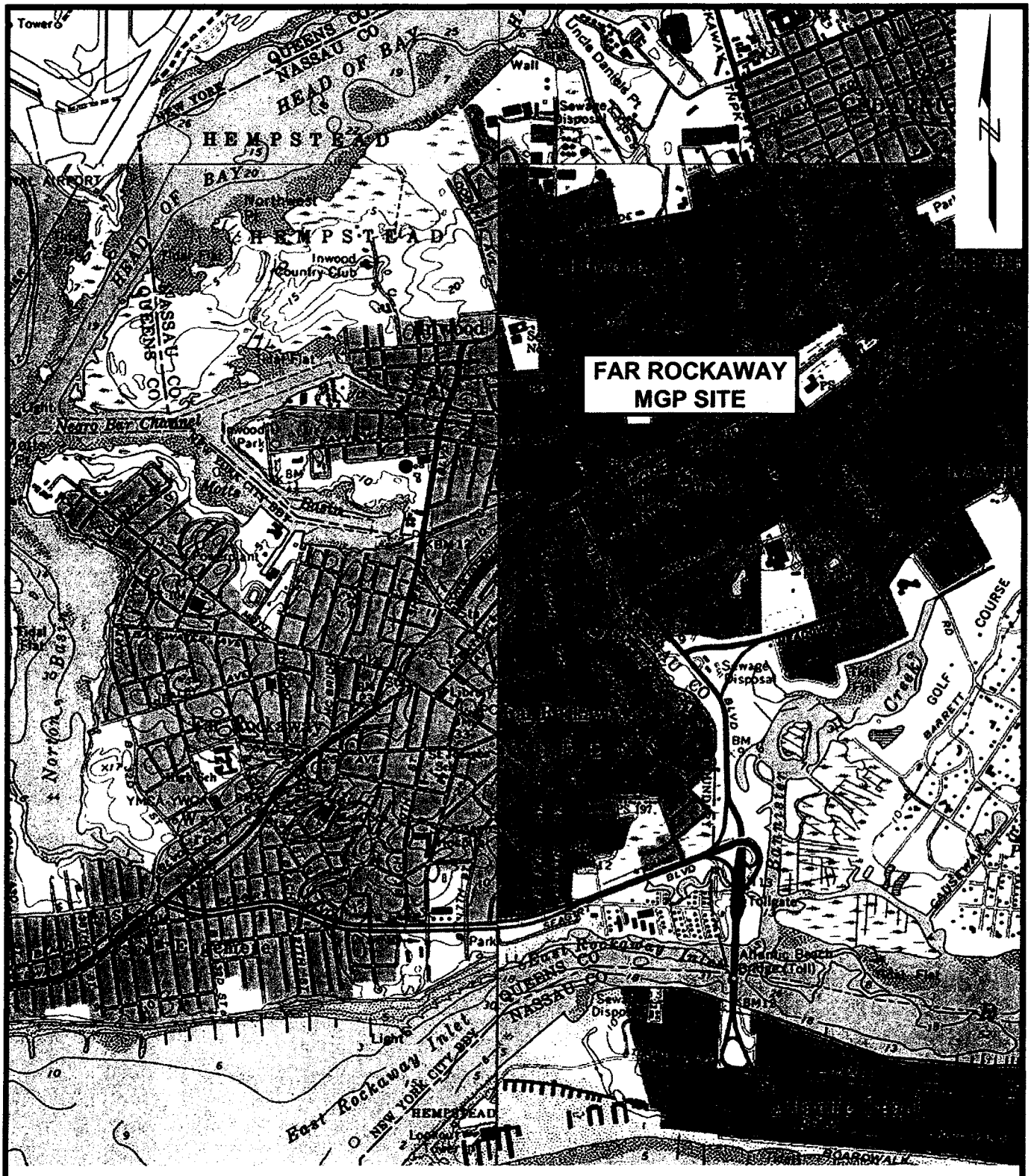
NYSDEC BCP APPLICATION KEAP STREET HOLDER STATION BROOKLYN, NEW YORK	GEI Consultants	SITE LOCATION MAP
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

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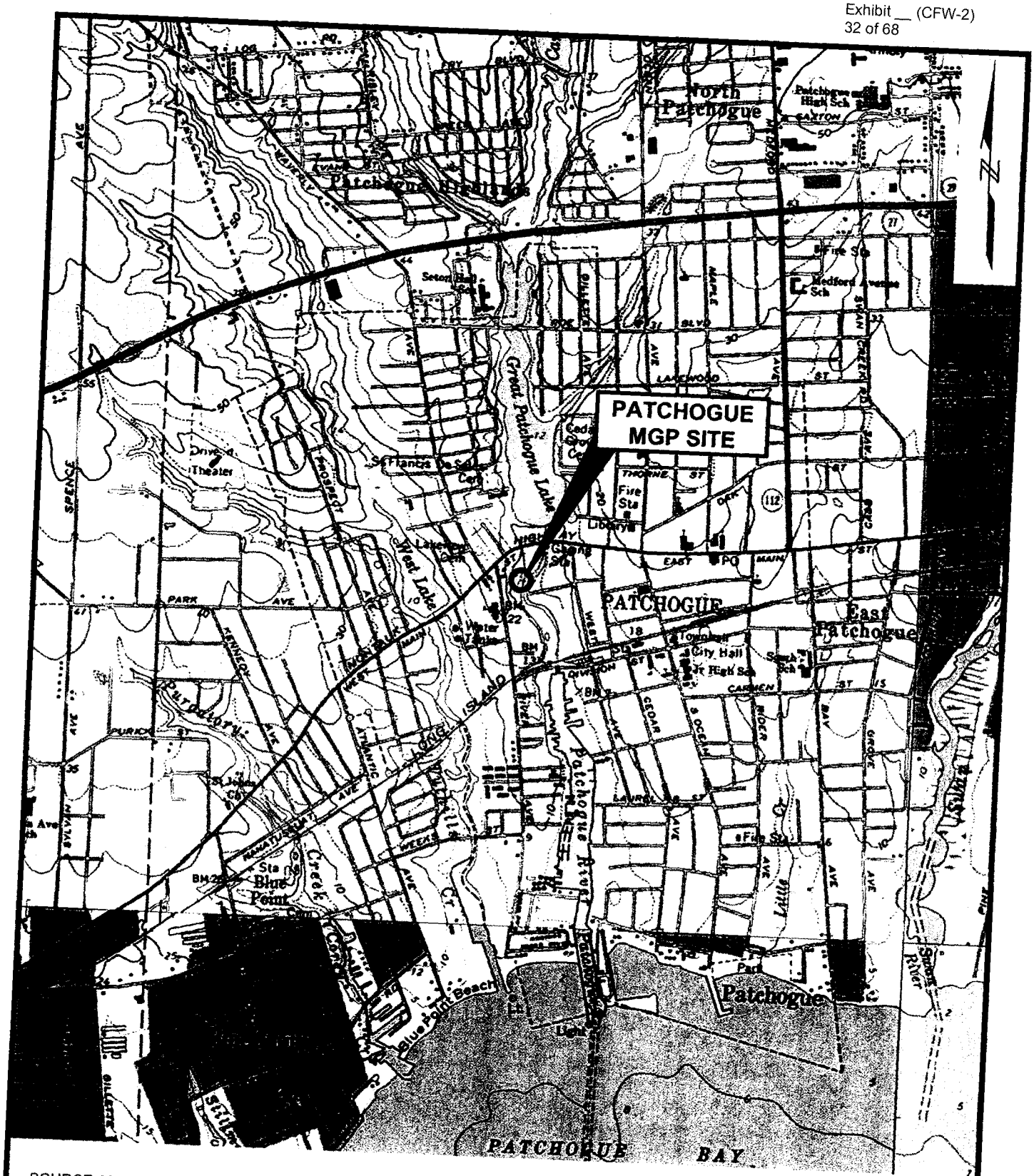
NYSDEC BCP APPLICATION FAR ROCKAWAY MGP QUEENS, NEW YORK	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

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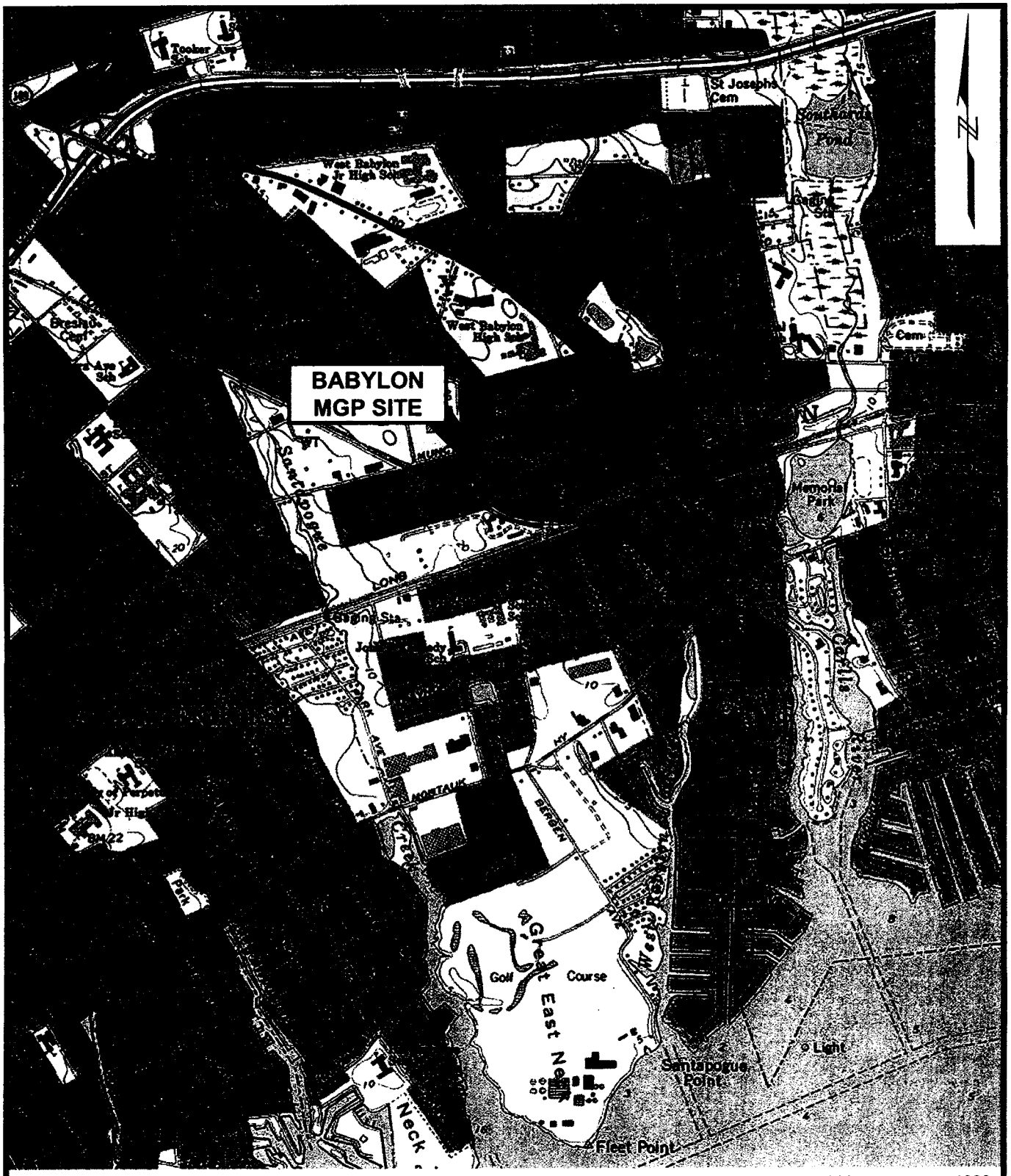
SCALE, FEET

NYSDEC BCP APPLICATION PATCHOGUE MGP PATCHOGUE, NEW YORK	GEI Consultants	SITE LOCATION MAP
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT

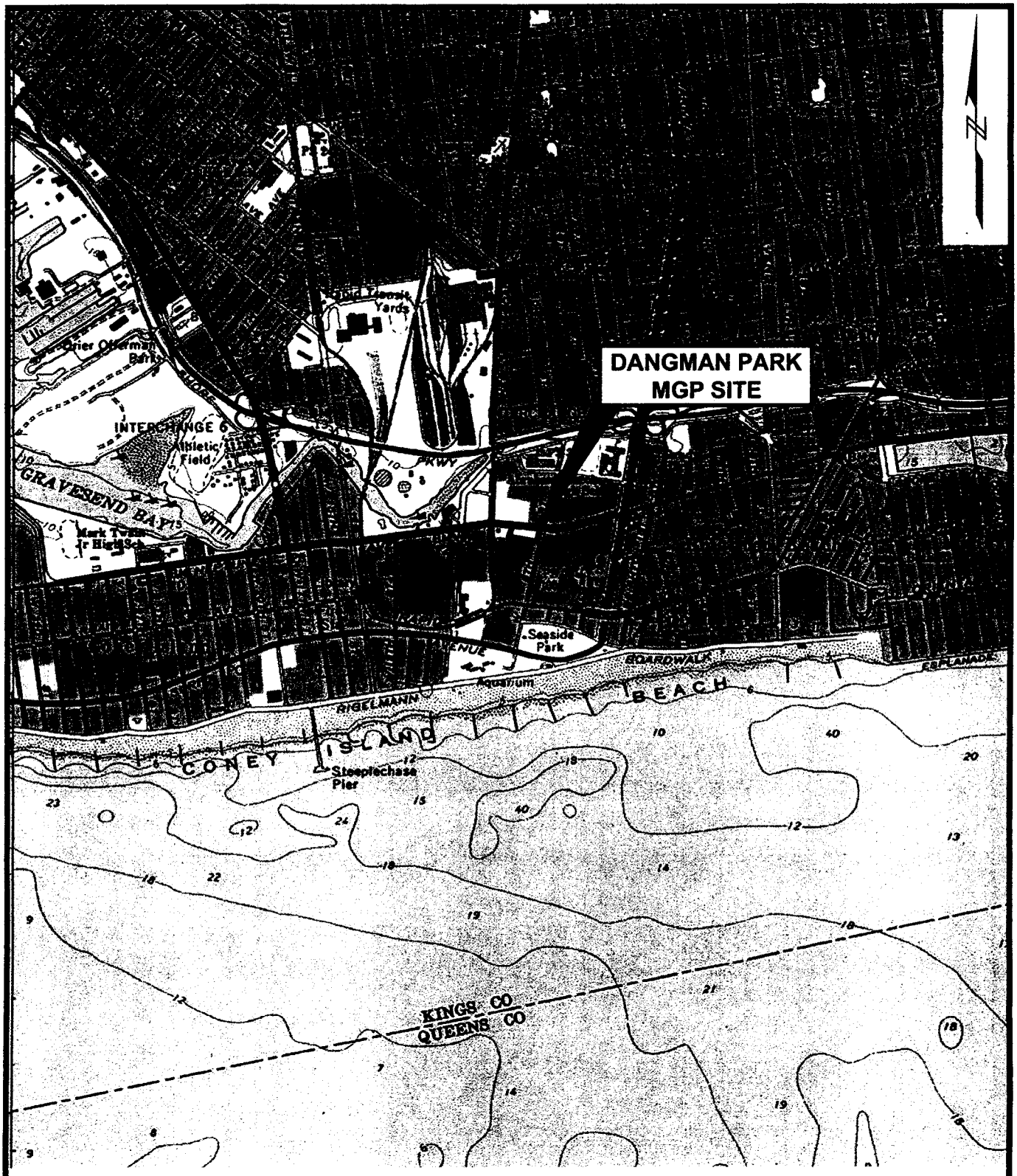


SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

SCALE, FEET

NYSDEC BCP APPLICATION **BABYLON MGP** **BABYLON, NEW YORK**	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

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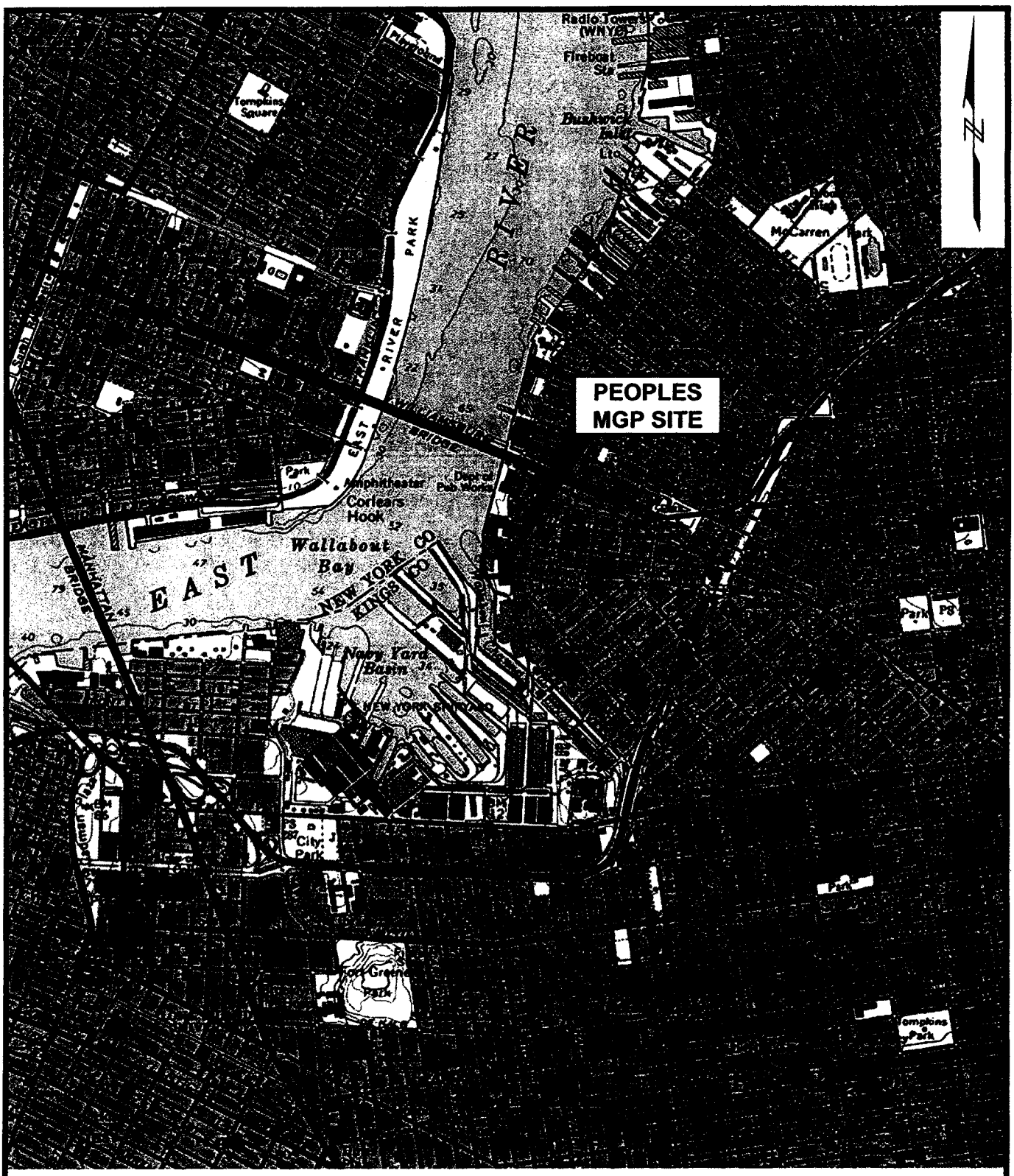
SCALE, FEET

NYSDEC BCP APPLICATION DANGMAN PARK MGP BROOKLYN, NEW YORK	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

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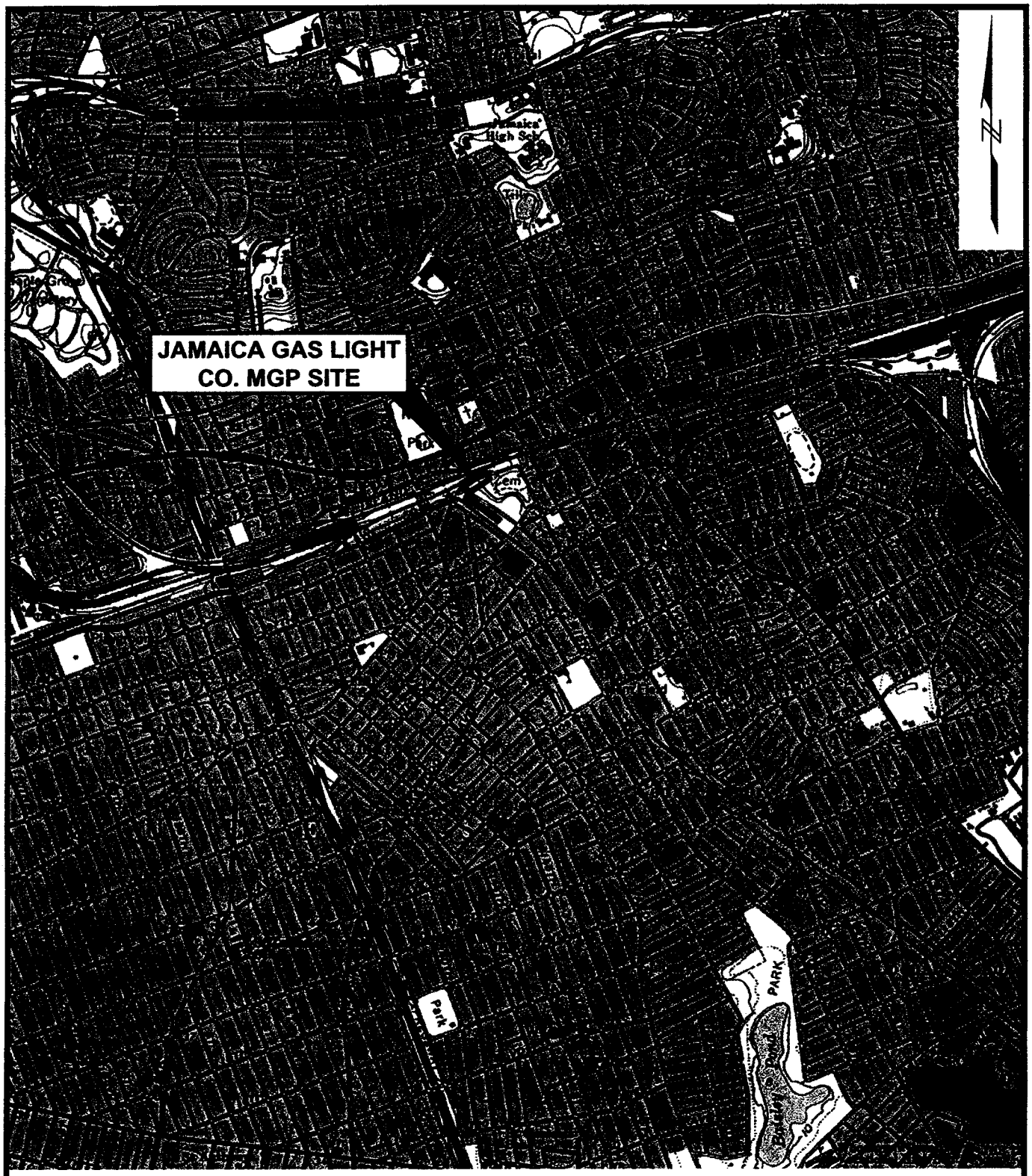
SCALE, FEET

NYSDEC BCP APPLICATION PEOPLES WORKS MGP BROOKLYN, NEW YORK	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



SOURCE: Map created with TOPO! ® ©2001 National Geographic
(www.nationalgeographic.com/topo)

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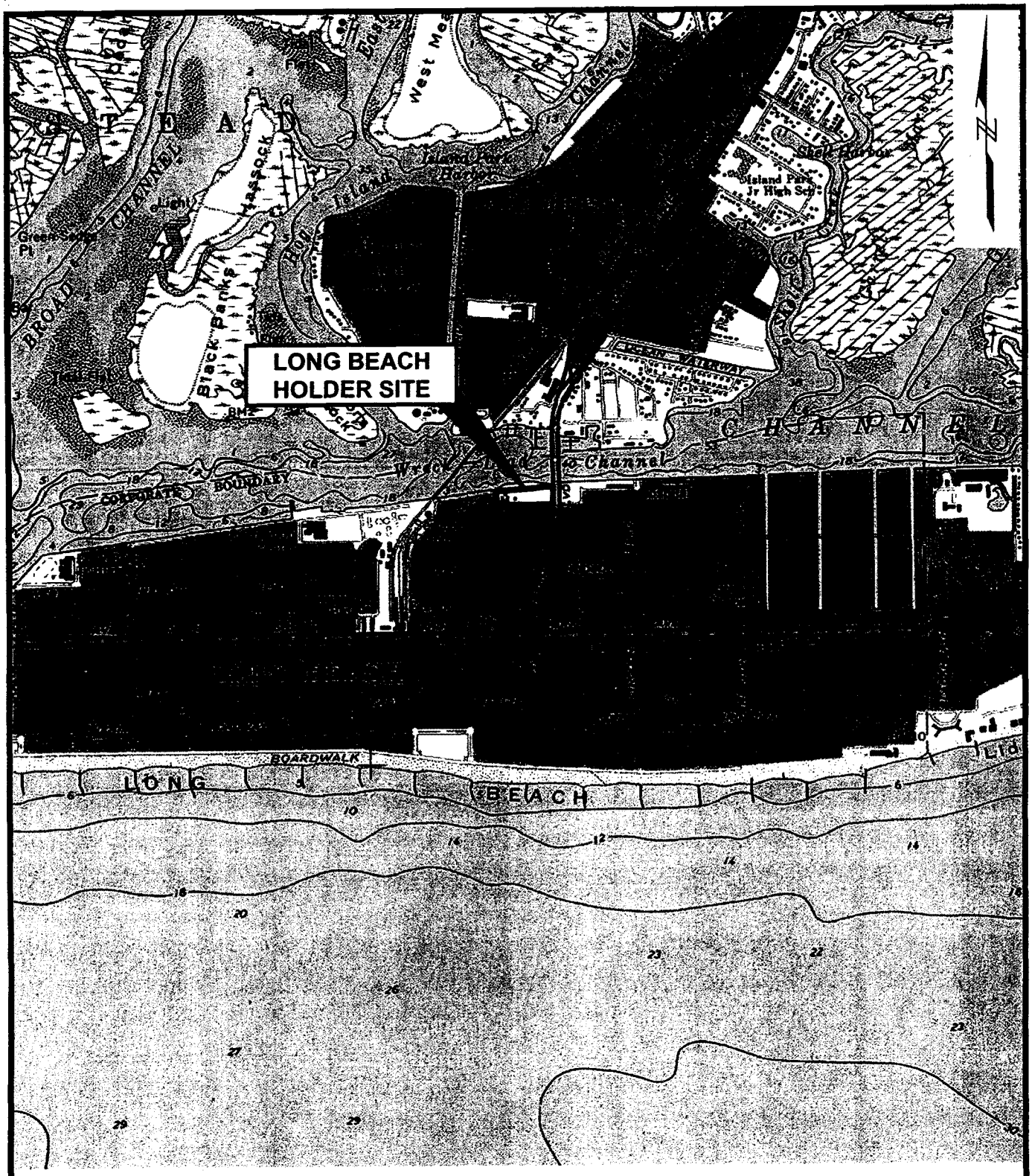
SCALE, FEET

NYSDEC BCP APPLICATION JAMAICA GAS LIGHT COMPANY MGP JAMAICA, NEW YORK	GEI Consultants	SITE LOCATION MAP
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1



SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

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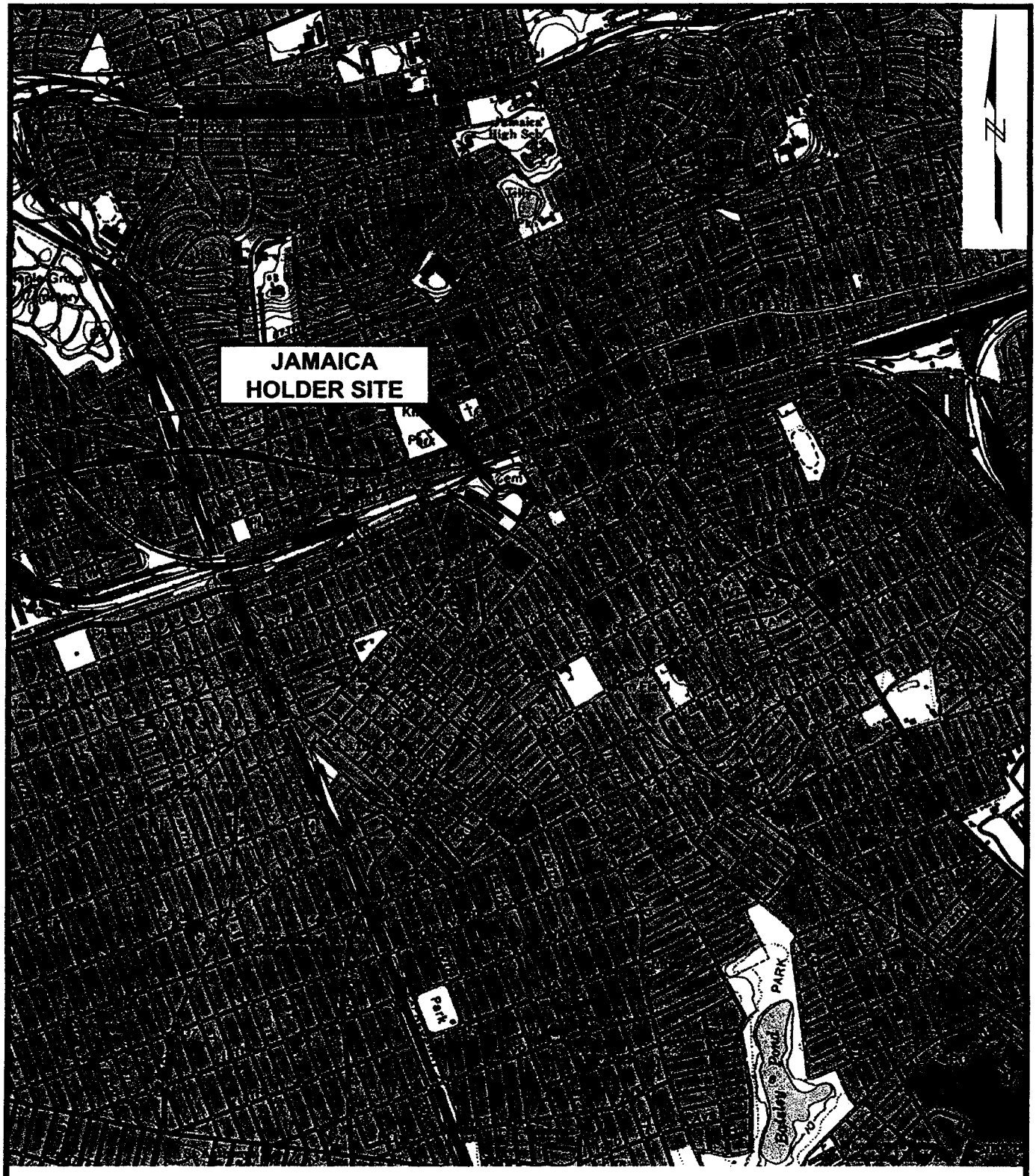
SCALE, FEET

NYSDEC BCP APPLICATION LONG BEACH HOLDER LONG BEACH, NEW YORK	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1



SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

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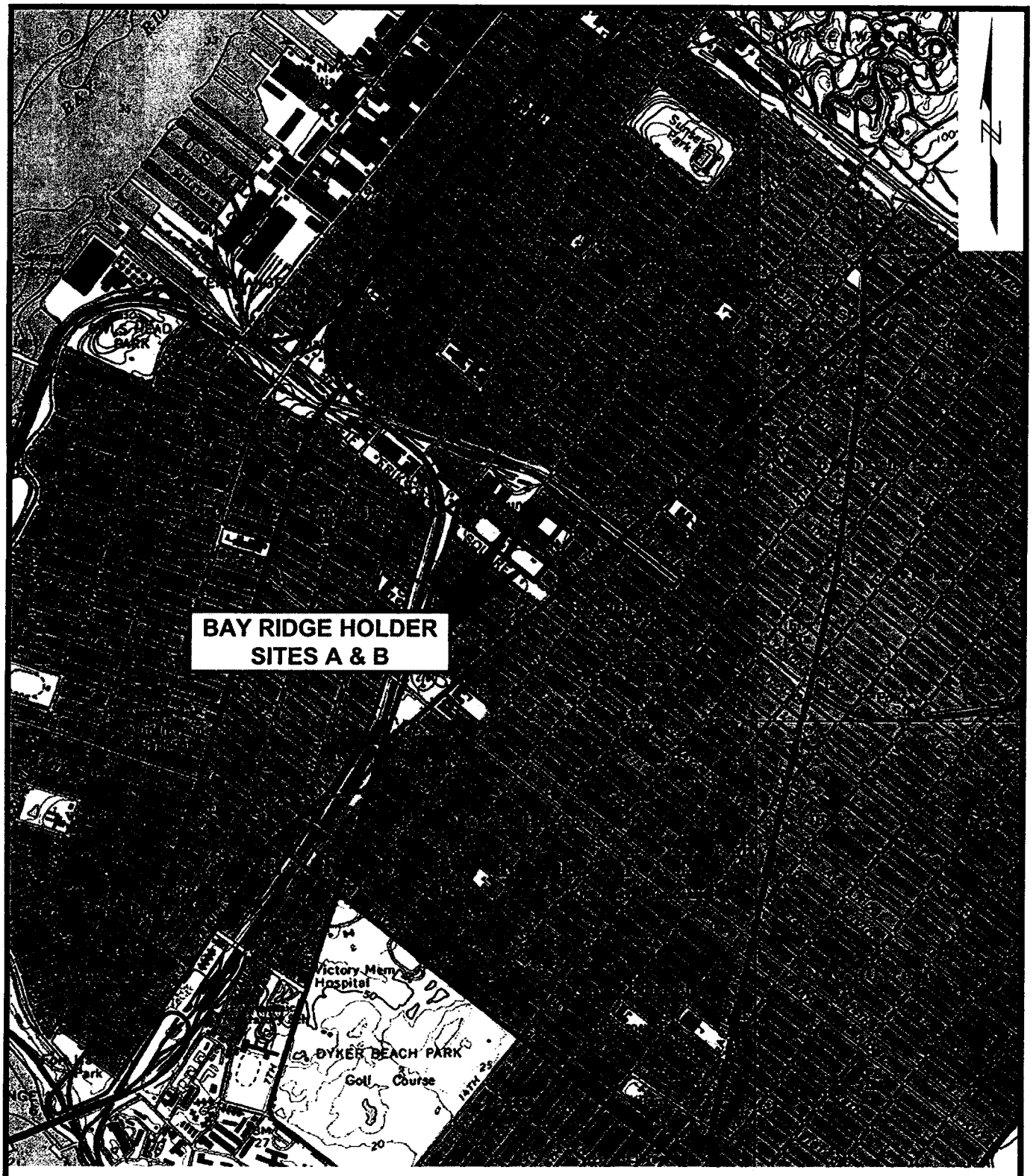
SCALE, FEET

NYSDEC BCP APPLICATION JAMAICA HOLDER STATION JAMAICA, NEW YORK	GEI Consultants	SITE LOCATION MAP
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

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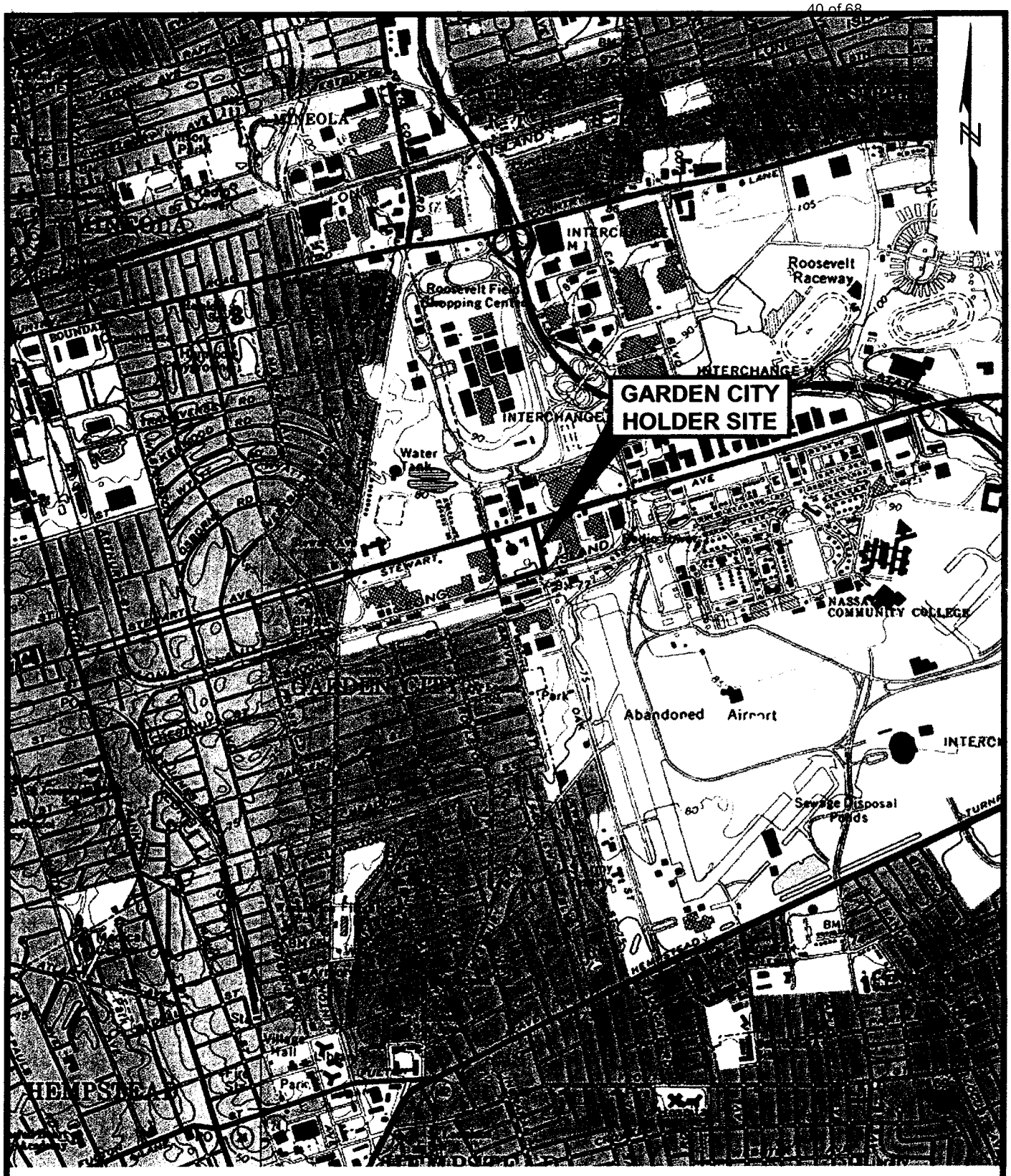
SCALE, FEET

NYSDEC BCP APPLICATION BAY RIDGE HOLDER STATIONS A & B BROOKLYN, NEW YORK	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT



SOURCE: Map created with TOPO! ® ©2001 National Geographic (www.nationalgeographic.com/topo)

0 2000 4000
SCALE, FEET

NYSDEC BCP APPLICATION GARDEN CITY (STEWART AVENUE) HOLDER HEMPSTEAD, NEW YORK	GEI Consultants	**SITE LOCATION MAP**
KEYSPAN CORPORATION BROOKLYN, NEW YORK	PROJECT 982482	AUGUST 2004 FIGURE 1

KEYSPAN\BCP-APPLICATION\FIGURES\BCPAPP-LOC.PPT

Exhibit "A" - Table 2
KeySpan MGP Related Sites
Subject to
Existing Orders and/or Agreements

A. **Order on Consent # D2-001-94-12**: Investigation and Interim Remedial Measures

Coney Island MGP Site Site # 224026 (Class 2)
Neptune Avenue
Brooklyn, NY 11224 County: Kings
Status: ROD for OU-1 signed in March 2001. ROD for Coney Island Creek, OU-2, signed in March 2002. Phase I of OU-1 completed in 2004. Phase II of OU-1 and OU-2 expected to begin in 2006.

B. **Order on Consent Index # D2-0001-98-04**: Investigation and Remediation

Clifton MGP Site Site # 243023
25 & 40 Willow Avenue
Staten Island, NY 10305 County: Richmond
Status: RI for 40 Willow parcel is complete - ROD issued in 2004. RI for 25 Willow Ave. parcel approved, PRAP issued 2006.

C. **Order on Consent Index # D1-0001-98-11**: Investigation and Remedial Response Program

Bay Shore MGP Site Site # 152172
Clinton Avenue
Town of Islip, NY 11706 County: Suffolk
Status: Several rounds of investigation were completed and RI and Final RI Reports issued in 2002 and 2003. Site was subsequently segregated into four operable units. IRMs were completed or are in progress in OU-2, 3 and 4. (See Watchogue Creek below.) Regarding OU-1, a RAP was issued in 2004 and RAWP in 2005. Pre-design construction studies were conducted and the remedial design is in progress.

Hempstead MGP Site Site # 130086
Intersection Street
Town of Hempstead NY 11530 County: Nassau
Status: Final RI was completed in 2005. The report is being finalized based on NYSDEC/NYSDOH comments.

Glen Cove MGP Site Site # 130089
Stanco Street
City of Glen Cove, NY 11542 County: Nassau
Status: Initial phase of RI field work was completed in the summer of 2004. A supplemental investigation to further define the extent of site contamination was conducted in 2005/2006 and RI to be issued in 2006.

Halesite MGP Site Site # 152173
40 New York Avenue
Town of Huntington, NY 11743 County: Suffolk
Status: Remedial Investigation was completed in 2004. Feasibility study to develop remedial alternatives was completed and RAP issued in 2006.

D. **Order on Consent Index # D1-0002-98-11**: Remedial Investigation/Feasibility Studies

Sag Harbor MGP Site Site # 152159 (Class 2)
Bridge Street
Town of Southampton, NY 11963 County: Suffolk
Status: The site was recently decommissioned as a natural gas distribution station. Completed investigations include a PSA (1995) and two phases RI (2002 and 2003). Supplemental data was collected and a report was generated in 2005. A PRAP and ROD were issued in 2006.

Rockaway Park MGP Site Site # 241029 (Class 2)
Beach Channel Drive & Beach 108th Street
Queens, NY 11694 County: Queens
Status: Completed investigations include a PSA (1997) and two RIs (2002 and 2004). A ROD was issued for the site in October, 2004 and a design to implement that decision is currently being developed.

E. **Order on Consent Index # D1-0001-99-05**: Preliminary Site Assessment

Patchogue MGP Site Site # 152182
River Ave & West Main Street
Town of Brookhaven, NY 11772 County: Suffolk
Status: Field work for the investigation has been completed. MGP waste was observed in test pits. To be included in this Order & Settlement Agreement (see Table 1)

Far Rockaway MGP Site Site # 241032
Beach 12th Street & Brunswick Avenue
Queens, NY 11691 County: Queens
Status: Originally started as a State Superfund Project, the site has been continued under a Consent Order signed by Keyspan in September of 1999. PSA Report, which indicated that further investigative work is necessary, has been reviewed by DEC and accepted. To be included in this Order & Settlement Agreement (see Table 1)

Clinton Road (Hempstead) MGP Site Site # 130106
Clinton Road
Town of Hempstead, NY County: Nassau
Status: No Further Action; PSA completed in 2003

Babylon MGP Site Site # 152181
Evergreen Street
Town of Babylon, NY County: Suffolk
Status: Originally started as a State Superfund Project, the site has been continued under a Consent Order signed by Keyspan Corp in September of 1999. PSA completed. To be included in this Order & Settlement Agreement (see Table 1)

Garden City MGP Site Site #: 130105
Hilton Avenue & 11th Street
Town of Hempstead, NY County: Nassau
Status: No Further Action; PSA completed in 2002.

F. **Agreement Index # D1-0002-99-05**: Investigation Program for MGP & Petroleum Derived Waste

Watchogue Creek
Near Smith Avenue & Mechanicsville Road
Bay Shore, NY County: Suffolk
Status: Investigation conducted in 2000. Creek sediment removal/channel realignment completed as IRM under Bay Shore OU-4.

G. **Voluntary Cleanup Agreement Index # R1-0001-01-01**: Remedial Response Program

Glenwood Landing Propane Plant and Compressor Station Site # V00351
Shore Road
Glenwood Landing, NY 11547 County: Nassau
Status: A VCA agreement was signed in March 2001. As part of the agreement, data from several previous investigations was accepted and a further RI commenced in March 2001. The investigations indicated that groundwater VOCs and soil hot spots for PCBs, SVOCs, and metals. RI completed in September 2001 - additional sampling and bench scale studies have continued. An RA of hot spot removal for Areas 1A, 2, and 3, cover of area 1B, and in-situ chemical oxidation for groundwater in Area 1 were proposed and accepted. Excavation work and capping were completed in January 2003 with approval pending additional information and the groundwater treatment. A bench scale treatability study for the ISCO is complete and final design of the injection is on-going.

H. **Voluntary Cleanup Agreement Index # A2-0460-0502**: RI/FS

Citizens Gas Works MGP Site Site # 224012 & V00360
(Includes Hoyt Street Holder Station)
Corner of 5th Street & Smith Street
Brooklyn, NY 11218 County: Kings
Status: First phase of the RI was completed in spring 2003; second phase of the RI, initiated in December 2004, has been approved. Supplementary RI including off-site soil borings is currently underway.

I. **Voluntary Cleanup Agreement Index # W2-1062-05-03**

Greenpoint Energy Center Site #: V00631
287 Maspeth Avenue
Brooklyn, NY 11211 County: Kings

Status: In August 2005, an IRM Completion Report was submitted for four non-contiguous parcels located within the larger Greenpoint Energy Center.

J. **Order on Consent and Administrative Settlement Index # A2-0523-0705**: RI Program

Gowanus Canal
Brooklyn, NY County: Kings
Status: Related to Citizens Gas, Fulton Works and Metropolitan Works former MGP Sites. Off-site Remedial Investigation of Gowanus Canal underway.

K. **Voluntary Cleanup Agreement Index # D2-0002-99-10:** Investigation and, if needed, Remediation

Newtown Holder Station Site #: V00406
78-01 57th Avenue
Elmhurst, NY 11373 County: Queens
Status: Remediated. Property donated to New York City to develop as a park and community center. KeySpan is negotiating with an adjacent property owner for easements/deed restrictions.

L. Order on Consent Index # W2-1090-06-06

Brooklyn Navy Yard Site #: 2-24-019A, OU-2
(Includes **Nassau Works MGP Site** as OU-2)
540 Kent Avenue
Brooklyn, NY 11205 County: Kings
Status: Several preliminary characterization studies have been performed, and some interim remedial actions have been undertaken. Currently, the NYC Department of Sanitation is preparing a work plan to address the majority of the site (OU-1), while Keyspan is investigating the Coal Gasification portion of the site (OU-2).

M. **Voluntary Cleanup Agreement Index # R2-0330-98-01:** In the Matter of the Implementation of a Response Program

LILCO - Edgemere Substation Site #: V00147
s/s Rockaway Beach Boulevard
w/o 52nd Street, Edgemere
Rockaway, NY County: Queens
Status: The site is a retired LILCO electrical substation with various types of fill material including fill from former gas manufacturing facilities. Remediation completed. Awaiting comments from DEC on the soil vapor monitoring plan NYSDOH required be performed on an adjacent property.

Exhibit "A" - Table 3
Additional KeySpan
MGP Sites, Holder Stations & Hortonsphere Sites

#	KeySpan Sites	Site #	Location	County
	MGP or Holder Sites			
1	Flatbush Holder Stations A & B	224061	Winthrop St,.& Clarkson Ave. Brooklyn, NY 11226	Kings
2	Fulton Municipal Works MGP & 3rd Ave Holder Station	224051	Brooklyn, NY 11217 (near Gowanus Canal)	Kings
3	Metropolitan Works MGP Site	224046	124-136 Second Avenue Brooklyn, NY 11215 (near Gowanus Canal)	Kings
4	Williamsburg Works MGP	224055	Kent Avenue Bet. N.11 & N.12 Street Brooklyn, NY 11211	Kings
5	Inwood Holder Site & LP Cracking/Reformation	130121	Inwood, NY 11096	Nassau
6	Wythe Ave (Berry Street) Holder Site	224069	No. 12 & 13 Berry St. Brooklyn, NY 11211	Kings
7	Equity Works MGP Site	224050	Maspeth Avenue Brooklyn, NY 11211	Kings
8	Greenpoint Mgp Site	224052	287 Maspeth Avenue Brooklyn, NY 11211	Kings
9	Kings County MGP Site	224056	5400-5600 1st Avenue Brooklyn, NY 11220	Kings
10	Belmont Holder Station	224060	290 Belmont Avenue Brooklyn, NY 11207	Kings
11	Brooklyn Gas Light Works MGP Site	224048	1 Hudson Avenue Brooklyn, NY 11201	Kings
12	Plymouth Street Holder Station	224065	49 Gold Street Brooklyn, NY 11211	Kings
13	Union Station (Citizens Branch) Holder Station	224054	2940 Atlantic Avenue Brooklyn, NY 11207	Kings

Exhibit "A" - Table 3
Additional KeySpan
MGP Sites, Holder Stations & Hortonsphere Sites

#	KeySpan Sites	Site #	Location	County
	MGP or Holder Sites (cont'd)			
14	Scholes Street Holder Station	224067	338 & 350 Scholes St. & 154 Bogart Street Brooklyn, NY 11206	Kings
15	Brooklyn MGP Site		61st Street & New Utrecht Ave Brooklyn, NY	Kings
16	Saltaire, acetylene gas production		Believed W/S Beacon Walk S/O Lighthouse Promenade Saltaire 11706	Suffolk
17	Southold, acetylene gas production		Believed W/S Hobart St & N/O Korn Rd	Suffolk
18	East Hampton, gasoline cracking		Buells Lane East Hampton, NY	Suffolk
	Former & Existing Hortonsphere Sites			
19	Bellmore		Newbridge Rd/Grand Ave N/O LIRR, S of Substation	Nassau
20	East Hampton (active)		S/S Railroad Ave btwn Race Lane& Fresno Pl East Hampton, NY 11937	Suffolk
21	Lynbrook		Ocean Ave & Merrick Rd	Nassau
22	Manhasset		High Street	Nassau
23	Oyster Bay		End of Willow Street	Nassau
24	Pinelawn/Farmingdale		E/S Broadhollow Rd, N/O Conklin St & S/O LIRR ROW East Farmingdale, NY 11735	Suffolk

EXHIBIT "B"

Department-Approved Work Plan(s)

EXHIBIT "C"

Cost Summary

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EXHIBIT "D"

NOTICE OF ORDER & SETTLEMENT AGREEMENT

_____ ("Respondent") is subject to an Order On Consent and Administrative Settlement (Index # _____) (the "Order & Settlement Agreement") issued by the Commissioner of the New York State Department of Environmental Conservation (the "Department"under Article 27, Title 13, and Article 71, Title 27 of the Environmental Conservation Law of the State of New York ("ECL") for a site located at _____, New York (the "Site").

[The Site has been designated by the Department as an inactive hazardous waste disposal site, as that term is defined at ECL Section 27-1301.2, and has been listed in the Registry of Inactive Hazardous Waste Disposal Sites in New York State as Site # _____. The Department has classified the Site as a Class "____" site pursuant to ECL Section 27-1305.4.b. This classification means that the Department has determined that the Site_____. The Site is more particularly described in the legal description that is attached hereto as Schedule "A."]**[Note: This paragraph can be changed, modified or deleted to set forth the status of the particular site.]**

The purpose of the Order & Settlement Agreement is to provide for the development and implementation of an inactive hazardous waste disposal site remedial program for the Site. The effective date of the Order & Settlement Agreement was _____. A copy of the Order & Settlement Agreement, as well as any and all Department-approved Work Plans under this Order & Settlement Agreement can be reviewed at the Department's _____ offices located at _____ by contacting _____.

This Notice of Order & Settlement Agreement is being filed with the _____recording officer in accordance with Paragraph IX of the Order & Settlement Agreement to give all parties who may acquire any interest in the Site notice of this Order & Settlement Agreement.

-or-

At the request of Respondent, and in accordance with Respondent's responsibilities pursuant to Paragraph IX of the Order & Settlement Agreement, to give all parties who may acquire any interest in the Site notice of this Order & Settlement Agreement, this Notice of Order and Settlement Agreement is being filed with the _____recording officer by _____, the fee Owner of the Site,

WHEREFORE, the undersigned has signed this Notice of Order & Settlement Agreement in compliance with the terms of the Order & Settlement Agreement.

_____
Respondent (or Owner)

By:_____

Title:_____

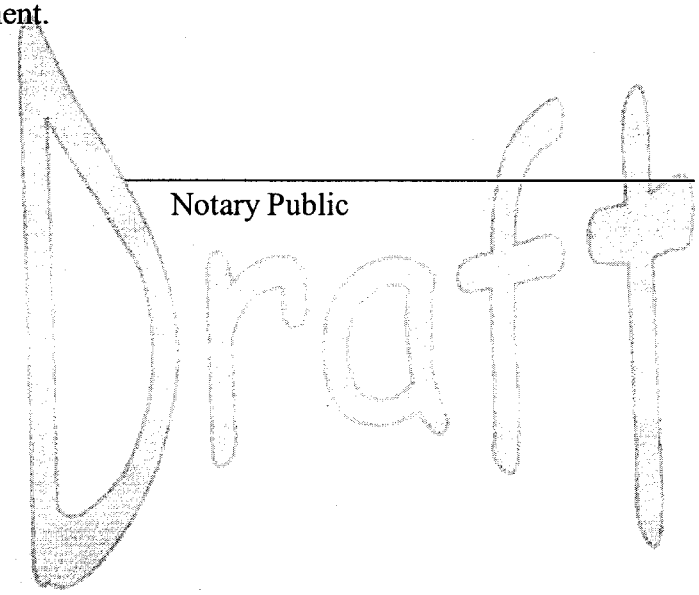
Date:_____

STATE OF NEW YORK

) ss.:

COUNTY OF_____

On the _____ day of _____ in the year _____ before me, the undersigned, a notary public in and for said State, personally appeared _____ _____ personally known to me or proved to me on the basis of satisfactory evidence to be the individual(s) whose name(s) is (are) subscribed to the within instrument and acknowledged to me that he/she/they executed the same in his/her/their capacity(ies), and that by his/her/their signature(s) on the instrument, the individuals) or the person upon behalf of which the individual(s) acted, executed this instrument.



Notary Public

Appendix "A"

(to Exhibit "D")

Map of the Property

EXHIBIT "E"

DRAFT ***DRAFT***

ENVIRONMENTAL EASEMENT

THIS INDENTURE made this ____day of _____, 200__, between ***Owner(s)*** _____ residing at (or having an office at) _____(the "Grantor"), and The People of the State of New York (the "Grantee."), acting through their Commissioner of the Department of Environmental Conservation (the "Commissioner", or "NYSDEC" or "Department" as the context requires) with its headquarters located at 625 Broadway, Albany, New York 12233,

WHEREAS, the Legislature of the State of New York has declared that it is in the public interest to encourage the remediation of abandoned and likely contaminated properties ("brownfield sites") that threaten the health and vitality of the communities they burden while at the same time ensuring the protection of public health and the environment; and

WHEREAS, the Legislature of the State of New York has declared that it is in the public interest to establish within the Department a statutory environmental remediation program that includes the use of environmental easements as an enforceable means of ensuring the performance of operation, maintenance, and/or monitoring requirements and of ensuring the potential restriction of future uses of the land, when an environmental remediation project leaves residual contamination at levels that have been determined to be safe for a specific use, but not all uses, or which includes engineered structures that must be maintained or protected against damage to perform properly and be effective, or which requires groundwater use or soil management restrictions; and

WHEREAS, the Legislature of the State of New York has declared that environmental easement shall mean an interest in real property, created under and subject to the provisions of Article 71, Title 36 of the New York State Environmental Conservation Law ("ECL") which contains a use restriction and/or a prohibition on the use of land in a manner inconsistent with engineering controls which are intended to ensure the long term effectiveness of a brownfield site remedial program or eliminate potential exposure pathways to hazardous waste or petroleum; and;

WHEREAS, Grantor, is the owner of real property located in the City/Town/Village of _____, _____County, New York known and designated on the tax map of the _____ of _____ as tax map parcel number_____, section ___block ___ lot____ , being the same as that property conveyed to Grantor by deed on _____, and recorded in the Land Records of the _____ County Clerk at page ____, liber _____ of Deeds, comprised of approximately ____ acres, and hereinafter more fully described in Schedule A attached hereto and made a part hereof (the " Controlled Property"); and;

Attach an adequate legal description of the property subject to the easement, or reference a recorded map. If the easement is on only a part of a parcel of land which is not subdivided into encumbered and unencumbered portions, a legal description needs to be

created by a survey bearing the seal and signature of a licensed land surveyor with reference to a metes and bounds description.

WHEREAS, the Commissioner does hereby acknowledge that the Department accepts this Environmental Easement in order to ensure the protection of human health and the environment and to achieve the requirements for remediation established at this Controlled Property until such time as this Environmental Easement is extinguished pursuant to ECL Article 71, Title 36;and

NOW THEREFORE, in consideration of the covenants and mutual promises contained herein and the terms and conditions of **Brownfield Cleanup Agreement Number**_____ /State **Assistance Contract Number**_____ **/Order on Consent Number**_____, Grantor grants, conveys and releases to Grantee a permanent Environmental Easement pursuant to Article 71, Title 36 of the ECL in, on, over, under, and upon the Controlled Property as more fully described herein ("Environmental Easement").

1. Purposes. Grantor and Grantee acknowledge that the Purposes of this Environmental Easement are: to convey to Grantee real property rights and interests that will run with the land in perpetuity in order to provide an effective and enforceable means of encouraging the reuse and redevelopment of this Controlled Property at a level that has been determined to be safe for a specific use while ensuring the performance of operation, maintenance, and/or monitoring requirements; and to ensure the potential restriction of future uses of the land that are inconsistent with the above-stated purpose.

2. Institutional and Engineering Controls. The following controls apply to the use of the Controlled Property, run with the land are binding on the Grantor and the Grantor's successors and assigns, and are enforceable in law or equity against any owner of the Controlled Property, any lessees, and any person using the Controlled Property:

A. The Controlled Property may be used for

residential
commercial
industrial
use as long as the following long-term engineering controls are employed:

B. The Controlled Property may not be used for a higher level of use such as **unrestricted/ residential / commercial** use and the above-stated engineering controls may not be discontinued without an amendment or extinguishment of this Environmental Easement.

C. Grantor covenants and agrees that until such time as the Environmental Easement is extinguished in accordance with the requirements of Article 71, Title 36 of the ECL, the property deed and all subsequent instruments of conveyance relating to the Controlled Property shall state in at least fifteen-point bold-faced type:

This property is subject to an environmental easement held by the New York State Department of Environmental Conservation pursuant of Title 36 to Article 71 of the Environmental Conservation Law.

C. Grantor covenants and agrees that this Environmental Easement shall be incorporated in full or by reference in any leases, licenses, or other instruments granting a right to use the Controlled Property.

D. Grantor covenants and agrees that it shall annually, or such time as NYSDEC may allow, submit to NYSDEC a written statement by an expert the NYSDEC may find acceptable certifying under penalty of perjury that the controls employed at the Controlled Property are unchanged from the previous certification or that any changes to the controls employed at the Controlled Property were approved by the NYSDEC, and that nothing has occurred that would impair the ability of such control to protect the public health and environment or constitute a violation or failure to comply with any Site Management Plan for such controls and giving access to such Controlled Property to evaluate continued maintenance of such controls.

3. Right to Enter and Inspect. Grantee, its agents, employees, or other representatives of the State may enter and inspect the Controlled Property in a reasonable manner and at reasonable times to assure compliance with the above-stated restrictions.

4. Reserved Grantor's Rights. Grantor reserves for itself, its assigns, representatives, and successors in interest with respect to the Property, all rights as fee owner of the Controlled Property, including:

1. Use of the Controlled Property for all purposes not inconsistent with, or limited by the terms of this Environmental Easement;

2. The right to give, sell, assign, or otherwise transfer the underlying fee interest to the Controlled Property by operation of law, by deed, or by indenture, subject and subordinate to this Environmental Easement;

5. Enforcement

A. This environmental easement is enforceable in law or equity in perpetuity by Grantor, Grantee, or any affected local government, as defined in ECL Section 71-3603, against the owner of the Property, any lessees, and any person using the land. Enforcement shall not be defeated because of any subsequent adverse possession, laches, estoppel, or waiver. It is not a defense in any action to enforce this environmental easement that: it is not appurtenant to an interest in real property; it is not of a character that has been recognized traditionally at common law; it imposes a negative burden; it imposes affirmative obligations upon the owner of any interest in the burdened property; the benefit does not touch or concern real property; there is no privity of estate or of contract; or it imposes an unreasonable restraint on alienation.

B. If any person intentionally violates this environmental easement, the Grantee may revoke the Certificate of Completion provided under ECL Article 27, Title 14, or the Satisfactory

Completion of Project provided under ECL Article 56, Title 5 with respect to the Controlled Property.

C. Grantee shall notify Grantor of a breach or suspected breach of any of the terms of this Environmental Easement. Such notice shall set forth how Grantor can cure such breach or suspected breach and give Grantor a reasonable amount of time from the date of receipt of notice in which to cure. At the expiration of such period of time to cure, or any extensions granted by Grantee, the Grantee shall notify Grantor of any failure to adequately cure the breach or suspected breach. Grantor shall then have a reasonable amount of time from receipt of such notice to cure. At the expiration of said second period, Grantee may commence any proceedings and take any other appropriate action reasonably necessary to remedy any breach of this Environmental Easement in accordance with applicable law to require compliance with the terms of this Environmental Easement.

D. The failure of Grantee to enforce any of the terms contained herein shall not be deemed a waiver of any such term nor bar its enforcement rights in the event of a subsequent breach of or noncompliance with any of the terms of this Environmental easement.

6. Notice. Whenever notice to the State (other than the annual certification) or approval from the State is required, the Party providing such notice or seeking such approval shall identify the Controlled Property by referencing its County tax map number or the Liber and Page or computerized system tracking/ identification number and address correspondence to:

Division of Environmental Enforcement
Office of General Counsel
New York State Department of Environmental Conservation
625 Broadway
Albany New York 12233-5500

Such correspondence shall be delivered by hand, or by registered mail or by Certified mail and return receipt requested. The Parties may provide for other means of receiving and communicating notices and responses to requests for approval.

7. Recordation. Grantor shall record this instrument, within thirty (30) days of execution of this instrument by the Commissioner or her/his authorized representative in the office of the recording officer for the county or counties where the Property is situated in the manner prescribed by Article 9 of the Real Property Law.

8. Amendment. This environmental easement may be amended only by an amendment executed by the Commissioner of the New York State Department of Environmental Conservation and filed with the office of the recording officer for the county or counties where the Property is situated in the manner prescribed by Article 9 of the Real Property Law.

9. Extinguishment. This environmental easement may be extinguished only by a release by the Commissioner of the New York State Department of Environmental Conservation and filed

with the office of the recording officer for the county or counties where the Property is situated in the manner prescribed by Article 9 of the Real Property Law.

10. Joint Obligation. If there are two or more parties identified as Grantor herein, the obligations imposed by this instrument upon them shall be joint and several.

IN WITNESS WHEREOF, Grantor has caused this instrument to be signed in its name.

Grantor's Name

By:_____

Title:_____

Date:_____

THIS ENVIRONMENTAL EASEMENT IS HEREBY ACCEPTED BY THE PEOPLE OF THE STATE OF NEW YORK, Acting By and Through the Department of Environmental Conservation

By: _____
Denise M. Sheehan, Commissioner

Draft

Grantor's Acknowledgment

STATE OF NEW YORK)
) ss:
COUNTY OF)

On the _____ day of _____, in the year 200_, before me, the undersigned, personally appeared _____, personally known to me or proved to me on the basis of satisfactory evidence to be the individual(s) whose name is (are) subscribed to the within instrument and acknowledged to me that he/she/they executed the same in his/her/their capacity(ies), and that by his/her/their signature(s) on the instrument, the individual(s), or the person upon behalf of which the individual(s) acted, executed the instrument.

_____
Notary Public - State of New York

Grantee's Acknowledgment

STATE OF NEW YORK)
) ss:
COUNTY OF)

On the _____ day of _____, in the year 200_, before me, the undersigned, personally appeared _____, personally known to me or proved to me on the basis of satisfactory evidence to be the individual(s) whose name is (are) subscribed to the within instrument and acknowledged to me that he/she/ executed the same in his/her/ capacity as Commissioner of the State of New York Department of Environmental Conservation, and that by his/her/ signature on the instrument, the individual, or the person upon behalf of which the individual acted, executed the instrument.

_____
Notary Public - State of New York

Draft

EXHIBIT "F"

RECORDS SEARCH REPORT

1. Detail all environmental data and information within Respondent's or Respondent's agents' or consultants' possession or control regarding environmental conditions at or emanating from the Site.

2. A comprehensive list of all existing relevant reports with titles, authors, and subject matter, as well as a description of the results of all previous investigations of the Site and of areas immediately surrounding the Site which are or might be affected by contamination at the Site, including all available topographic and property surveys, engineering studies, and aerial photographs.

3. A concise summary of information held by Respondent and Respondent's attorneys and consultants with respect to:

(i) a history and description of the Site, including the nature of operations;

(ii) the types, quantities, physical state, locations, methods, and dates of disposal or release of hazardous waste at or emanating from the Site;

(iii) a description of current Site security (i.e. fencing, posting, etc.); and

(iv) the names and addresses of all persons responsible for disposal of hazardous waste, including the dates of such disposal and any proof linking each such person responsible with the hazardous wastes identified.

EXHIBIT "G"

SC WORK PLAN REQUIREMENTS

The SC Work Plan shall include but not be limited to:

1. A chronological description of the anticipated SC activities together with a schedule for the performance of these activities.

2. A Sampling and Analysis Plan that shall include:

(i) A quality assurance project plan that describes the quality assurance and quality control protocols necessary to achieve the initial data quality objectives. This plan shall designate a data validation expert and must describe such individual's qualifications and experience;

(ii) A field sampling plan that defines sampling and data gathering methods in a manner consistent with the "Field Methods Compendium," OSWER Directive 9285.2-11 (draft June 1993), as supplemented by the Department; and

(iii) A health and safety plan to protect persons at and in the vicinity of the Site during the performance of the SC which shall be prepared in accordance with 29 CFR 1910 and all other applicable standards by a certified health and safety professional. Respondent shall add supplemental items to this plan necessary to ensure the health and safety of all persons at or in the vicinity of the Site during the performance of any work pursuant to this Order & Settlement Agreement.

3. The Work Plan shall incorporate all elements of an SC as set forth in Department technical and administrative guidance documents including, but not limited to, investigations of surface and subsurface soils, surface waters, ground water, and air.

4. The SC must be sufficiently comprehensive to allow the Department to determine whether a consequential amount of hazardous waste has been disposed at the Site and, if so, whether the contamination presents a significant threat to public health and/or the environment.

EXHIBIT "H"

RI/FS WORK PLAN REQUIREMENTS

The Investigation Work Plan shall include but not be limited to:

1. A chronological description of the anticipated RI/FS activities together with a schedule for the performance of these activities.

2. A Sampling and Analysis Plan that shall include:

(i) A quality assurance project plan that describes the quality assurance and quality control protocols necessary to achieve the initial data quality objectives. This plan shall designate a data validation expert and must describe such individual's qualifications and experience;

(ii) A field sampling plan that defines sampling and data gathering methods in a manner consistent with the "Field Methods Compendium," OSWER Directive 9285.2-11 (draft June 1993), as supplemented by the Department;

(iii) A health and safety plan to protect persons at and in the vicinity of the Site during the performance of the RI/FS which shall be prepared in accordance with 29 CFR 1910 and all other applicable standards by a certified health and safety professional. Respondent shall add supplemental items to this plan necessary to ensure the health and safety of all persons at or in the vicinity of the Site during the performance of any work pursuant to this Order & Settlement Agreement; and

(iv) A citizen participation plan that is, at a minimum, consistent with the Department's publication "Citizen Participation in New York's Hazardous Waste Site Remediation Program: A Guidebook," dated June 1998, any subsequent revisions thereto, and 6 NYCRR Part 375.

3. The Work Plan shall incorporate all elements of an RI/FS as set forth in CERCLA, as amended, the NCP, the USEPA guidance document entitled "Guidance for Conducting Remedial Investigations and Feasibility Studies under CERCLA," dated October 1988, and any subsequent revisions thereto in effect at the time the RI/FS Work Plan is submitted, and appropriate USEPA and Department technical and administrative guidance documents.

4. The Work Plan shall provide for an FS evaluating on-Site and off-Site remedial actions to restore the Site to pre-disposal conditions, to the extent feasible and authorized by law. At a minimum, alternatives shall evaluate the elimination or mitigation of all significant threats to the public health and to the environment presented by hazardous waste disposed at the Site through the proper application of scientific and engineering principals.

EXHIBIT "I"

IRM WORK PLAN REQUIREMENTS

The IRM Work Plan shall include, at a minimum, the following:

1. a summary of the data supporting the extent of the proposed IRM;

2. a chronological description of the anticipated IRM activities;

3. a schedule for performance of the IRM activities;

4. detailed documents and/or specifications prepared, signed, and sealed by a Professional Engineer providing sufficient detail to implement the Department-approved IRM, including, as appropriate, a description of soil and sediment erosion control, storm water management and monitoring, and dust, odor, and organic vapor control and monitoring procedures to be implemented during remedial activities, and a detailed description of confirmation sampling and site restoration plans;

5. a health and safety plan, including a community air monitoring plan;

6. a contingency plan, including a description of procedures for dismantling and removing remedial structures and equipment from the Site, if applicable;

7. a citizen participation plan, if required, that incorporates appropriate activities outlined in the Department's publication "Citizen Participation in New York's Hazardous Waste Site Remediation Program: A Guidebook," dated June 1998, any subsequent revisions thereto, and 6 NYCRR Part 375;

8. an OM&M Plan, if the performance of the Department-approved IRM results in a treatment system which is expected to operate for greater than 18 months. If the system will not operate for greater than 18 months, or if only monitoring is required, only a monitoring plan will be needed; and

9. a description of institutional controls to be implemented as well as written approval from the owner of the affected property if the remedy selected requires implementation of an institutional control at an off-Site location or if the person responsible for the remedy is not the Site owner.

EXHIBIT "J"

REMEDIATION WORK PLAN REQUIREMENTS

The Remediation ("RD/RA") Work Plan shall include the following:

1. A detailed description of the remedial objectives and the means by which each element of the selected remedial alternative will be implemented to achieve those objectives, including, but not limited to:

(i) the construction and operation of any structures;

(ii) the collection, destruction, treatment, and/or disposal of hazardous wastes and substances and their constituents and degradation products, and of any soil or other materials contaminated thereby;

(iii) the collection, destruction, treatment, and/or disposal of contaminated groundwater, leachate, and air;

(iv) physical security and posting of the Site;

(v) quality control and quality assurance procedures and protocols to be applied during implementation of the Remedial Construction; and

(vi) monitoring which integrates needs which are present on-Site and off-Site during implementation of the Department-selected remedial alternative.

2. A schedule for submission of "Biddable Quality" documents for the Remedial Design including, but not limited to, documents and specifications prepared, signed, and sealed by a Professional Engineer. These plans shall satisfy all applicable local, state, and federal laws, rules, and regulations;

3. A time schedule to implement the Remedial Design;

4. The parameters, conditions, procedures, and protocols to determine the effectiveness of the Remedial Design, including a schedule for periodic sampling of all media of concern, including groundwater monitoring wells on-Site and off-Site;

5. A description of operation, maintenance, and monitoring activities to be undertaken after the Department has approved construction of the Remedial Design, including the number of years during which such activities will be performed (where appropriate) and a specific description of the criteria to be used to decide when operation of such activities may be discontinued.

6. A contingency plan to be implemented if any element of the Remedial Design fails to achieve any of its objectives or otherwise fails to protect human health or the environment;

7. A health and safety plan for the protection of persons at and in the vicinity of the Site during and after construction. This plan shall be prepared in accordance with 29 CFR 1910 by a certified health and safety professional; and

8. A citizen participation plan which incorporates appropriate activities outlined in the Department's publication "Citizen Participation in New York's Hazardous Waste Site Remediation Program: A Guidebook," dated June 1998, any subsequent revisions thereto, and 6 NYCRR Part 375.

EXHIBIT "K"

SM WORK PLAN REQUIREMENTS

The SM Work Plan shall provide for:

1. Operation and maintenance of engineering controls and/or treatment systems;

2. Maintenance of institutional controls, where applicable;

3. Yearly certification by a Professional Engineer of the continued effectiveness of any institutional and/or engineering controls, where applicable. The certification must identify the required controls and evaluate whether the controls should remain in place and effective for the protection of public health and/or the environment;

4. A monitoring plan which describes the measures for monitoring the performance and effectiveness of the remedy at the Site;

5. A contingency plan which describes procedures which may be required to protect and/or maintain the operation of the remedy in the event of an emergency, such as a fire, spill, tank or drum overflow or rupture, severe weather, or vandalism;

6. A health and safety plan and a list of records and references;

7. Monitoring and reporting of the performance and effectiveness of the remedy, both short and long-term, by:

   (i) Assessing compliance with actual or equivalent discharge permit limits;

   (ii) Assessing achievement of the remedial performance criteria; and,

   (iii) Sampling and analysis of appropriate media.

8. A determination that the remedy is complete by demonstrating that the remedial action objectives have been achieved.

EXHIBIT "L"

RECORD OF DECISION

Glossary of Terms

The following terms shall have the following meanings:

"BPM Director": the Director of the Bureau of Program Management within the Division of Environmental Remediation.

"CERCLA": the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended, 42 U.S.C. 9601 et seq.

"Day": a calendar day. In computing any period of time under this Order & Settlement Agreement, where the last day would fall on a Saturday, Sunday or State holiday, the period shall run until the close of business of the next working day.

"Department": the New York State Department of Environmental Conservation.

"Director": the Division Director, Division of Environmental Remediation.

"ECL": the Environmental Conservation Law, Chapter 43-B of the Consolidated Laws of New York, as amended.

"Feasibility Study" or "FS": a study undertaken to develop and evaluate options for remedial action. The feasibility study emphasizes data analysis and is generally performed concurrently and in an interactive fashion with the remedial investigation, using data gathered during the remedial investigation. The term also refers to a report that describes the results of the study. (See 6 NYCRR Part 375)

"Force Majeure Event": an event which is brought on as a result of fire, lightning, earthquake, flood, adverse weather conditions, strike, shortages of labor and materials, war, riot, obstruction or interference by adjoining landowners, or any other fact or circumstance beyond Respondent's reasonable control.

"Inactive Hazardous Waste Disposal Site Remedial Program" or "Remedial Program": activities undertaken to eliminate, remove, abate, control, or monitor existing health hazards, existing environmental hazards, potential health hazards, and/or potential environmental hazards in connection with the Site and all activities to manage wastes and contaminated materials at or removed from the Site. (See ECL 27-1301(3) and 6 NYCRR Part 375)

"Interim Remedial Measure" or "IRM": a discrete set of activities, including removal activities, to address both emergency and non-emergency Site conditions, which can be undertaken without extensive investigation or evaluation, to prevent, mitigate, or remedy environmental damage or the consequences of environmental damage attributable to the Site. (See 6 NYCRR Part 375)

"National Contingency Plan" or "NCP": the National Oil and Hazardous Substances Pollution Contingency Plan promulgated pursuant to Section 105 of CERCLA, 42 U.S.C. 9605, and codified at 40 C.F.R. Part 300, and any amendments thereto.
"NL": the Navigation Law, Chapter 37 of the Consolidated Laws of New York, as amended.

"OH&M": the Office of Hearings and Mediation Services.

"Order & Settlement Agreement": this Order on Consent and Administrative Settlement and all exhibits attached hereto.

"Professional Engineer": an individual registered as a professional engineer in accordance with Article 145 of the New York State Education Law. If such individual is a member of a firm, that firm must be authorized to offer professional engineering services in the State of New York in accordance with Article 145 of the New York State Education Law.

"Record of Decision" or "ROD": the document reflecting the Department's selection of a remedy relative to the Site or any Operable Unit thereof. The ROD shall be attached to and made enforceable under this Order & Settlement Agreement as Exhibit "L."

"Remedial Action" or "RA": those activities, except for OM&M, to be undertaken under this Order & Settlement Agreement to implement the ROD.

"Remedial Investigation" or "RI": a process undertaken to determine the nature and extent of contamination. The remedial investigation emphasizes data collection and site characterization and generally is performed concurrently with the feasibility study. It includes sampling and monitoring, as necessary, and includes the gathering of sufficient information to determine the necessity for and the proposed extent of the program and to support the evaluation of proposed alternatives. (See 6 NYCRR Part 375)

"Site Characterization"or "SC": a process undertaken to allow the Department to determine whether a consequential amount of hazardous waste has been disposed at a Site and, if so, whether the contamination presents a significant threat to public health and/or the environment.

"Site Management" or "SM": the activities undertaken as the last phase of the remedial program at a site which continues until the remedial action objectives for the Site are met.
Site management is conducted in accordance with a site management plan, which identifies and implements the institutional and engineering controls required for a site, as well as any necessary monitoring and/or operation and maintenance of the remedy.

"Spill Fund": the New York State Environmental Protection and Spill Compensation Fund as established by Article 12, Part Three of the NL.

"State Costs": all the State's response expenses related to this Site, including, but not limited to, direct labor, fringe benefits, indirect costs, travel, analytical costs, and contractor costs incurred by the State of New York for negotiating, implementing, overseeing, administering, or enforcing this Order & Settlement Agreement, and any other response costs as defined under CERCLA. Approved agency fringe benefit and indirect cost rates will be applied.

"Termination Date": the date that this Order & Settlement Agreement is terminated pursuant to Paragraph XIII.

"USEPA": the United States Environmental Protection Agency.

Exhibit ___ (CFW-3)

Unilateral Administrative Order for the remedial design of the Gowanus Canal Superfund Site

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
REGION 2

IN THE MATTER OF THE GOWANUS CANAL SUPERFUND SITE	:	
Beam, Inc. Beazer East, Inc. Brink's Inc. Brooklyn Union Gas Co. d/b/a National Grid New York CBS Corp. Citigroup, Inc. Consolidated Edison Co. of New York, Inc. Dun and Bradstreet Corp. ExxonMobil Oil Corp. Hauck Manufacturing Co. Hess Corp. Honeywell International Inc. Kraft Foods Global, Inc. MCIZ Corp. and Fifteen Second Avenue LLC 36-2nd-J Corp. 107 Sixth Street LLC MRC Holdings, Inc. NL Industries, Inc. Northville Industries Corp. Patterson Fuel Oil Co., Inc. Phillips 66 Co. Puget Sound Commerce Center, Inc. Rexam Beverage Can Co. SPX Corp. Stauffer Management Company, LLC TDA Industries, Inc. The Brooklyn Improvement Co. The Union Oil Company of California Verizon New York Inc.	:	INDEX NO. CERCLA - 02-2014-2001 ADMINISTRATIVE ORDER FOR REMEDIAL DESIGN
Respondents.	:	
Proceeding under Section 106(a) of the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended, 42 U.S.C. § 9606(a).	:	

I. JURISDICTION AND GENERAL PROVISIONS

1. This Administrative Order ("Order") is issued to the above-captioned Respondents by the United States Environmental Protection Agency, Region 2 ("EPA") and requires Respondents to undertake a Remedial Design ("RD"), including various pre-RD investigations and analyses, to produce a set of biddable plans and specifications for the implementation of the remedy selected in EPA's September 27, 2013 Record of Decision ("ROD") for the Gowanus Canal Superfund Site ("Site"), other than the CSO controls and the cleanup and restoration of the former 1st Street turning basin, which EPA expects to be conducted by New York City.

2. This Order is issued to Respondents by EPA pursuant to the authority vested in the President of the United States under Section 106(a) of the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended ("CERCLA"), 42 U.S.C. § 9606(a), and delegated to the Administrator of EPA on January 23, 1987, by Executive Order No. 12580 (52 Federal Register 2926, January 29, 1987). This authority was further delegated to the EPA Regional Administrators by EPA Delegation Nos. 14-14-A and 14-14-B and to the Director of the Emergency and Remedial Response Division in Region 2 by Regional Delegation R-1200, dated November 23, 2004.

3. EPA has notified the New York State Department of Environmental Conservation ("NYSDEC") of this Order pursuant to Section 106(a) of CERCLA, 42 U.S.C. § 9606(a).

II. PARTIES BOUND

4. This Order shall apply to and be binding upon Respondents and their directors, officials, employees, agents, successors and assigns. No change in the status or control of Respondents shall alter Respondents' responsibilities under this Order. Respondents are jointly and severally responsible for carrying out all Work required by this Order.

5. Until EPA notifies Respondents under Paragraph 94 that the Work has been completed, Respondents shall provide a copy of this Order to any prospective purchaser or successor before a controlling interest in Respondents' assets or property rights are transferred to any successor.

III. DEFINITIONS

6. Unless otherwise expressly provided herein, terms used in this Order which are defined in CERCLA or in regulations promulgated under CERCLA shall have the meaning assigned to them in CERCLA or its implementing regulations. Whenever terms listed below are used in this Order or in an attachment to this Order, the following definitions shall apply:

a. "CERCLA" shall mean the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended, 42 U.S.C. §§ 9601-9675.

b. "Day" shall mean a calendar day. In computing any period of time under this Order, where the last day would fall on a Saturday, Sunday or federal holiday, the period shall run until the close of business of the next working day.

c. "Effective Date" shall be the date this Order goes into effect as provided in Subsection T (Opportunity to Confer, Effective Date).

d. "EPA" shall mean the United States Environmental Protection Agency and any successor departments or agencies of the United States.

e. "National Contingency Plan" or "NCP" shall mean the National Oil and Hazardous Substances Pollution Contingency Plan promulgated pursuant to Section 105 of CERCLA, 42 U.S.C. § 9605, and codified at 40 C.F.R. Part 300, including any amendments thereto.

f. "NYSDEC" shall mean the New York State Department of Environmental Conservation and any successor departments or agencies of the State.

g. "Order" shall mean this Administrative Order and all appendices attached hereto. In the event of conflict between this Order and any appendix, this Order shall control.

h. "Paragraph" shall mean a portion of this Order identified by an Arabic numeral.

i. "Performance Standards" shall mean the cleanup standards and Remedial Action Objectives and other measures of achievement of the goals of the remedy set forth in the ROD and Section II of the Remedial Design Statement of Work ("RD SOW") attached hereto as Appendix A.

j. "Pre-Remedial Design Work Plan" or "Pre-RD Work Plan" shall mean the document describing the preliminary fieldwork activities to be undertaken by Respondents to gather the information necessary to fully develop the Remedial Design. A draft Pre-RD Work Plan was prepared by Respondent Brooklyn Union Gas Co. d/b/a National Grid New York ("National Grid") pursuant to a January 24, 2014 Amendment to Administrative Order and Settlement Agreement, Index Number CERCLA-02-2010-2009 ("National Grid Amended Settlement Agreement"), attached hereto as Appendix B, and which, following approval by EPA, shall be incorporated into and made an enforceable part of this Order, as well as any amendments thereto.

k. "Record of Decision" or "ROD" shall mean the EPA Record of Decision relating to the Site signed on September 27, 2013 by the Director of the Emergency Remedial Response Division, EPA Region 2, including all attachments thereto, attached hereto as Appendix C.

l. "Remedial Design" or "RD" shall mean those activities to be undertaken by Respondents to develop the final plans and specifications for the Remedial Action pursuant to the Remedial Design Work Plan.

m. "Remedial Design Statement of Work" or "RD SOW" shall mean the Statement of Work attached hereto as Appendix A.

n. "Remedial Design Work Plan" or "RD Work Plan" shall mean the document developed by Respondent National Grid pursuant to the National Grid Amended Settlement Agreement, a draft of which is attached hereto as Appendix D and which, following approval by EPA, shall be incorporated into and made an enforceable part of this Order, as well as any amendments thereto.

o. "Respondents" shall mean:

1. Beam, Inc.
2. Beazer East, Inc.
3. Brink's Inc.
4. National Grid
5. CBS Corp.
6. Citigroup, Inc.
7. Consolidated Edison Co. of New York, Inc.
8. Dun and Bradstreet Corp.
9. ExxonMobil Oil Corp.
10. Hauck Manufacturing Co.
11. Hess Corp.
12. Honeywell International Inc.
13. Kraft Foods Global, Inc.
14. MCIZ Corp. and affiliated entities:
    a. Fifteen Second Avenue LLC
    b. 36-2nd-J Corp.
    c. 107 Sixth Street LLC
15. MRC Holdings, Inc.
16. National Grid
17. NL Industries, Inc.
18. Northville Industries Corp.
19. Patterson Fuel Oil Co., Inc.
20. Phillips 66 Co.
21. Puget Sound Commerce Center, Inc.
22. Rexam Beverage Can Co.
23. SPX Corp.
24. Stauffer Management Company, LLC
25. TDA Industries, Inc.
26. The Brooklyn Improvement Co.
27. The Union Oil Company of California

28. Verizon New York Inc.

p. "Section" shall mean a portion of this Order identified by an upper-case Roman numeral and includes one or more Paragraphs.

q. "Site" shall mean the Gowanus Canal Superfund Site, an approximately 100-foot wide, 1.8-mile-long canal located in the New York City borough of Brooklyn, Kings County, New York, and also includes any areas which are sources of contamination to the Canal, areas where contamination has migrated from the Canal, and/or suitable areas in very close proximity to the contamination which are necessary for implementation of the Work. The Site is depicted generally on the map attached as Appendix E.

r. "State" shall mean the State of New York.

s. "United States" shall mean the United States of America.

t. "Waste Material" shall mean (i) any "hazardous substance" under Section 101(14) of CERCLA, 42 U.S.C. § 9601(14); (ii) any "pollutant or contaminant" under Section 101(33) of CERCLA, 42 U.S.C. § 9601(33); and (iii) any "solid waste" under Section 1004(27) of the Solid Waste Disposal Act, as amended, 42 U.S.C. § 6903(27).

u. "Work" means all activities Respondents are required to perform pursuant to this Order, except those required by Paragraph 69, below.

IV. FINDINGS OF FACT AND CONCLUSIONS OF LAW

7. Gowanus Canal ("Canal") is a brackish, tidal arm of the New York–New Jersey Harbor Estuary, extending for approximately 1.8 miles through Brooklyn, New York. The approximately 100–foot–wide Canal runs southwest from Butler Street to Gowanus Bay and Upper New York Bay. The adjacent waterfront is primarily commercial and industrial, currently including concrete plants, warehouses and parking lots, and the Site is near several residential neighborhoods.

8. In 1849, the State authorized construction of the Canal to open the area to barge traffic, flush away sewage, receive storm water and fill the adjacent lowlands for development.

9. The Canal was constructed by bulkheading and dredging a tidal creek and wetland. Additional fill was utilized to raise the grade of the surrounding land. The authorizing legislation and the initial canal designs had recognized the likelihood that the Canal would be stagnant, creating pollution problems. As a result, various flushing solutions were contemplated. However, these were not implemented as part of its initial construction.

10. Following the construction of the Canal, infrastructure was added at various times, including combined sanitary and storm sewers and direct discharge pipes, all of which deposited into the Canal from the surrounding watershed, which is approximately 1,758 acres in size.

11. After completion of construction in the 1860s, the Canal quickly became one of the nation's busiest industrial waterways, home to heavy industry including gas works (*i.e.*, manufactured gas plants), coal yards, cement makers, soap makers, tanneries, paint and ink factories, machine shops, chemical plants and oil refineries.

12. As a result of decades of direct and indirect discharges of hazardous substances generated by industrial and other activity, the Canal is a repository for untreated industrial wastes, raw sewage, and runoff causing it to be one of New York's most polluted waterways.

13. The Canal was first declared a public nuisance in 1877 due to discharge of sanitary and industrial waste, in combination with stagnant water conditions. Subsequent studies and commissions have repeatedly examined methods of addressing the contamination. A series of unsuccessful solutions were implemented between 1891 and 1904, including directing additional sewage discharges to the Canal in order to improve flow.

14. A "Flushing Tunnel" began operating in 1911 as the next attempt to address the Canal's pollution problems. Designed to improve circulation and flush pollutants from the Canal, the Flushing Tunnel consists of a one mile long, 12-foot diameter tunnel stretching from New York Bay near Governors Island to the head of the Canal. Originally using a large ship propeller-type pump system, it could pump water in either direction. It operated with mixed results until the mid-1960s when it fell into disrepair.

15. Periodic infrastructure improvements have gradually reduced direct and indirect discharges to the Canal. The Owl's Head Waste Water Treatment Works ("WWTW") was completed in 1952, serving portions of the Park Slope area. The Red Hook WWTW was completed in 1987. The Second Avenue pump station was completed in 1990, eliminating the last area of dry weather discharges along the Canal. However, Combined Sewer Overflow ("CSO") discharges continue to the present date at an estimated volume of 377 million gallons per year. CSO discharges contain CERCLA hazardous substances from a range of sources, including but not limited to household and industrial discharges to the sanitary sewers and contaminated stormwater captured by storm drains.

16. Throughout the period described in Paragraph 15, above, depending on location relative to infrastructure improvements, facilities located directly adjacent to the Canal discharged untreated industrial and sanitary waste directly into the Canal in both dry weather and wet weather conditions due to the lack of infrastructure necessary to divert discharges from the Canal to upgradient sewer lines that discharged into New York Harbor. During this era, facilities not directly adjacent to the Canal but downgradient of main sewer lines also discharged untreated industrial and sanitary waste indirectly into the Canal in both dry weather and wet weather conditions. Facilities upgradient of the main sewer lines also discharged untreated industrial and sanitary waste indirectly into the Canal in wet weather conditions.

17. Hazardous substances, pollutants and contaminants have entered and continue to enter the Canal via several transport pathways or mechanisms, including spillage during product shipping and handling, direct disposal or discharge, contaminated groundwater discharge, surface water runoff, storm water discharge (including CSO events) and contaminated soil erosion.

18. Much of the heavy industrial activity along the Canal has ceased, although many upland areas adjacent to the Canal remain zoned as manufacturing districts. Land uses along and near certain portions of the Canal are in the process of transitioning from heavy industrial to light industrial, commercial, and residential uses. The Canal is currently used by some for recreational purposes such as boating, diving, and catching fish for consumption. The Canal and New York City harbor are subject to New York State fishing advisories for various ingestion risks, including polychlorinated biphenyls ("PCBs").

19. The Site was placed on the National Priorities List pursuant to Section 105 of CERCLA, 42 U.S.C. § 9605, on March 2, 2010.

20. A Remedial Investigation ("RI") report was completed by EPA in January 2011 and a Feasibility Study ("FS") report was completed by EPA in December 2011. An FS addendum report was issued by EPA in December 2012, together with a Proposed Plan. The Proposed Plan described the remedial alternatives considered to address the contamination in the Canal and identified the preferred remedy with the rationale for this preference.

21. Sampling results from the RI/FS document the presence of a wide range of hazardous substances in the groundwater, soil, and Canal sediments at the Site. These include polycyclic aromatic hydrocarbons ("PAHs"), PCBs, pesticides (such as methoxychlor and DDT), metals (such as barium, cadmium, copper, lead, mercury, nickel and silver), as well as volatile organic compounds ("VOCs") (such as benzene, toluene, ethylbenzene and xylene). The contamination in the sediments extends the entire length of the Canal. The contamination is present in both the sediment which has accumulated above the native sediments (referred to as "soft sediments"), and in the native sediment below the original bed of the Canal. Some of the hazardous substances are present at high levels. The soft sediment layer ranges in thickness from approximately 1 foot to greater than 20 feet, with an average thickness of about 10 feet. For example, total PAH concentrations in surface sediment (defined as the top 6 inches of the soft sediments, where potential exposure is more likely to occur) range up to 8,001,000 ug/kg . PCBs in surface sediment were detected up to 3,400 ug/kg. In the subsurface (i.e., deeper than 6 inches), total PAH concentrations in the soft sediment ranged up to 45,000,000 ug/kg. Total PAH concentrations in the native sediment were detected up to 47,500,000 ug/kg. In the subsurface, total PCB concentrations in the soft sediment were detected up to 50,700 ug/kg. In the native sediments, total PCBs were detected up to 2,610 ug/kg.

22. Based on the results of the RI/FS, chemical contamination in the Canal sediments presents an unacceptable ecological and human health risk, primarily due to exposure to PAHs, PCBs, and metals (barium, cadmium, copper, lead, mercury, nickel and silver) in surface water and sediment, and from ingesting fish and crabs from the Canal.

23. PCBs and PAHs have been demonstrated to cause a variety of adverse health effects. PCBs have been shown to cause cancer in test animals. PCBs have also been shown to cause a number of serious non-cancer health effects in animals, including effects on the immune system, reproductive system, nervous system, endocrine system and other health effects. Studies in humans provide supportive evidence for potential carcinogenic and non-carcinogenic effects of PCBs. The toxicity of PAHs can vary from being nontoxic to extremely toxic. EPA has classified seven PAH compounds as probable human carcinogens: benz[a]anthracene, benzo[a]pyrene, benzo[b]fluoranthene, benzo[k]fluoranthene, chrysene, dibenz(a,h)anthracene, and indeno(1,2,3-cd)pyrene. PAHs known for their carcinogenic, mutagenic, and teratogenic properties are benz[a]anthracene and chrysene, benzo[b]fluoranthene, benzo[j]fluoranthene, benzo[k]fluoranthene, benzo[a]pyrene, benzo[ghi]perylene, coronene, dibenz(a,h)anthracene, indeno(1,2,3-cd)pyrene, and ovalene. High prenatal exposure to PAHs is associated with lower IQ and childhood asthma. The Center for Children's Environmental Health reports studies that demonstrate that exposure to PAH pollution during pregnancy is related to adverse birth outcomes including low birth weight, premature delivery, and heart malformations. Cord blood in cases of prenatal exposure shows DNA damage that has been linked to cancer. Follow-up studies show increased developmental delays at age three, and lower scores on IQ tests and increased behavioral problems at ages six and eight.

24. EPA's ecological risk assessment of the Site determined that PAHs, PCBs and metals in the sediment are toxic to benthic organisms. PAHs were detected in sediment at the highest concentrations relative to their ecological screening benchmarks and represent the greatest site-related risk to the benthic community. PCBs and seven metals (barium, cadmium, copper, lead, mercury, nickel and silver) were also detected at concentrations above their ecological screening benchmarks and at concentrations significantly higher than those detected in reference area sediments and also represent a potential site-related risk to the benthic community. PAHs were found to be a potential risk to aquatic herbivores (represented by the black duck) and mercury was found to be a potential risk to avian omnivores (represented by the heron).

25. On September 27, 2013, EPA issued a ROD for the Site which includes the following response actions: 1) Dredging of the entire column of hazardous substance-contaminated soft sediments in the upper and mid-reaches of the canal; 2) in-situ stabilization of those native sediments in select areas in the upper and mid-reaches of the canal contaminated with high levels of nonaqueous phase liquid ("NAPL"); 3) construction of a multilayered cap in the upper and mid-reaches of the canal to isolate and prevent the migration of PAHs and residual NAPL from native sediments; 4) dredging of the entire soft sediment column in the lower reach of the canal; 5) construction of a multilayer cap to isolate and prevent the migration of PAHs from native sediments in the lower reach of the canal; 6) off-Site treatment with thermal desorption of the NAPL-impacted sediments dredged from the upper and mid-reaches of the canal, followed by beneficial reuse off-Site (*e.g.*, landfill daily cover) if possible; 7) off-Site stabilization of the less contaminated sediments dredged from the lower reach of the canal and the sediments in the other reaches not impacted by NAPL, followed by beneficial reuse off-Site; 8) excavation and restoration of approximately 475 feet of the filled-in former 1st Street turning basin; 9) excavation and restoration of the portion of the 5th Street turning basin beginning underneath the

3rd Avenue bridge and extending approximately 25 feet to the east and the installation of a barrier or interception system at the eastern boundary of the excavation; 10) implementation of institutional controls incorporating the existing fish consumption advisories (modified as needed), as well as other controls to protect the integrity of the cap; 11) periodic maintenance of the cap and long-term monitoring to insure that the remedy continues to function effectively; and 12) CSO controls to significantly reduce overall contaminated solid discharges to the canal, which shall include a) construction of in-line sewage/stormwater retention tanks to retain stormwater which currently discharges through outfalls RH-034 and OH-007; and b) implementation of appropriate engineering controls to ensure that hazardous substances and solids from separated stormwater, including from future upland development projects, are not discharged to the canal.

26. In 2009, EPA began the investigation of potentially responsible parties ("PRPs") for the Site. EPA began issuing letters notifying parties of their potential liability, and thus their status as PRPs, in August 2009.

27. In March 2012, EPA convened a meeting of all of the PRPs that had received a notice letter as of that date for the purpose of providing a technical and enforcement briefing regarding the Site, and to encourage the PRPs to begin preparations for future settlement negotiations.

28. On September 30, 2013, EPA issued a Notice for the Commencement of Remedial Design Negotiations and Demand for Past Costs ("Notice and Demand") to each of the Respondents named herein. The Notice and Demand sought $5 million in partial reimbursement of EPA's outstanding past costs for the Site, and execution of an RD consent order, a draft of which was included therein, by December 13, 2013, which deadline was determined by EPA to be necessary in order to ensure RD fieldwork could begin in spring 2014.

29. To facilitate settlement discussions between and among the Agency and the PRPs, EPA convened a meeting of the PRPs on November 7, 2013. To provide further time for negotiations, EPA extended the time for Respondents to enter into the RD consent order from December 13, 2013 to January 31, 2014 and then later to February 14, 2014.

30. To prevent a delay in implementing the RD during the negotiation extension period, on January 24, 2014, National Grid and EPA entered into the National Grid Amended Settlement Agreement. Pursuant to the National Grid Amended Settlement Agreement, National Grid developed and submitted for EPA approval the Pre-RD and RD Work Plans, on January 29, 2014 and February 27, 2014, respectively, attached hereto as Appendices B and D. On February 3, 2014, National Grid also paid EPA $1 million in partial reimbursement of EPA's outstanding past response costs.

31. EPA is currently conducting separate consent order negotiations with New York City for that portion of the RD that involves the siting and design of the CSO retention tanks and the design for the cleanup and restoration of the former 1st Street turning basin. EPA is also negotiating an administrative order for a removal action with Bayside Fuel Oil Corp. and its affiliates, Bayside Fuel Oil Depot Corp., Sackett Street Properties, LLC, Smith Street Properties

LLC, OAA Realty LLC, LAA Realty LLC and Victor Allegretti Credit Shelter Trust, which requires, under EPA supervision, implementation of bulkhead upgrades on Bayside's property to EPA's remedial standards, as well as the coordination and cooperation with Respondents regarding the RD.

32. The Site constitutes a "facility" within the meaning of Section 101(9) of CERCLA, 42 U.S.C. § 9601(9).

33. The contamination found at the Site, as identified in the Findings of Fact above, includes hazardous substances as defined in Section 101(14) of CERCLA, 42 U.S.C. § 9601(14).

34. Each Respondent is a responsible party with respect to the Site pursuant to Section 107(a) of CERCLA, 42 U.S.C. §9607(a), for reasons including but not limited to, each Respondent's status as the current owner/operator of a Facility at the Site and/or the owner/operator of a Facility at the Site at a time of disposal of one or more hazardous substances, and/or a successor in interest thereto.

35. Each Respondent is a corporation and/or a limited liability company, and therefore is a "person" within the meaning of Section 101(21) of CERCLA, 42 U.S.C. § 9601(21).

36. The conditions described in these Findings of Fact constitute an actual or threatened "release" of one or more a hazardous substances from a facility as defined by Section 101(22) of CERCLA, 42 U.S.C. § 9601(22). Such actual or threatened releases include, but are not limited to, the discharge of high levels of PAHs, PCBs, pesticides, metals, and VOCs into the Canal, and/or the indirect discharge of such hazardous substances into the Canal through sewer or other pipes and/or the soil and/or groundwater at the Site, as well as the potential for future migration of hazardous substances at and from the Site.

37. Each Respondent was given an opportunity to enter into a settlement agreement, either individually or collectively, for the Work required by this Order. No party consented to perform the Work.

V. DETERMINATIONS

38. Based on the Findings of Fact and Conclusions of Law set forth above and the entirety of the administrative record, EPA has determined that the release or threatened release of hazardous substances from the Site may present an imminent and substantial endangerment to the public health or welfare or the environment within the meaning of Section 106(a) of CERCLA, 42 U.S.C. § 9606(a).

39. The actions required by this Order are necessary to protect the public health or welfare or the environment, are in the public interest, and are consistent with CERCLA and the NCP.

VI. ORDER

40. Based upon the foregoing Findings of Fact, Conclusions of Law and Determinations, and the administrative record supporting the Record of Decision for this Site, it is hereby ordered that Respondents comply with all requirements of this Order including, but not limited to, performance of the Remedial Design at the Site in accordance with Subsection A herein (Description of Work), the RD SOW, and following approval by EPA, the Pre-RD and RD Work Plans.

41. Respondents are jointly and severally responsible for carrying out all activities required by this Order, the RD SOW, and following approval by EPA, the Pre-RD and RD Work Plans. Any failure to perform, in whole or in part, any requirement of this Order by any Respondent hereto shall not relieve Respondents of their obligation to perform each and every requirement of this Order. No Respondent shall interfere in any way with the performance of Work in accordance with this Order by any other Respondent.

A. Description of Work

42. Respondents shall perform the following:

a. Obligation to Cooperate and Coordinate: Respondents shall make best efforts to coordinate in the performance of the Work required by this Order with any person not a party to this Order who is directed by EPA and who makes good-faith offers to perform or, in lieu of performance to pay for, in whole or in part, the Work required by this Order. Best efforts to coordinate shall include, at a minimum:

i. replying in writing within a reasonable period of time to good-faith offers to perform or pay for the Work required by this Order;

ii. engaging in good-faith negotiations with any person not a party to this Order who makes good-faith offers to perform or pay for the Work required by this Order; and

iii. good-faith consideration of good-faith offers to perform or pay for the Work required by this Order.

Upon request of EPA and subject to any claims of applicable privileges(s), Respondents shall submit to EPA (1) any offer to perform or pay for, or (2) all documentation relating to the performance of or payment for, the Work required by this Order by any non--respondent to this Order.

Nothing in this Paragraph shall be construed to require or permit Respondents to delay implementing the Pre-RD and/or RD Work Plan, following EPA approval, or for otherwise complying with the terms of this Order.

b. Work: Respondents shall conduct the Work required hereunder in accordance with CERCLA, the NCP, the ROD, the Performance Standards, the RD SOW, and, following EPA approval, the Pre-RD Work Plan, and RD Work Plan, as well as applicable provisions of the following guidance documents, (and of other guidance documents referenced therein) as they may be amended or modified by EPA: *Uniform Federal Policy for Implementing Quality Systems* (UFP-QS), EPA-505-F-03-001, March 2005, *Uniform Federal Policy for Quality Assurance Project Plans* (UFP-QAPP), Parts 1, 2 and 3, EPA-505-B-04-900A, B and C, March 2005, *EPA Region 2's "Clean and Green Policy"* which may be found at http://epa.gov/region2/superfund/greenremediation/policy.html, *Guidance for Scoping the Remedial Design* (EPA 540/R-95/025, March 1995), and *Guide to Management of Investigation-Derived Wastes* (OSWER Publication 9345.3-03FS, January 1992). The tasks that Respondents must perform (including future deliverables) and the scope of such Work are identified in this Order and the RD SOW which is incorporated into and is an enforceable part of this Order. Each deliverable submitted pursuant to this Settlement Agreement shall be deemed incorporated into and an enforceable part of this Settlement Agreement upon its approval by EPA.

43. Respondents shall assure that all field personnel used by Respondents are properly trained in the use of field equipment and in chain-of-custody procedures.

B. Designation Of Contractor and Designated Project Coordinator

44. Within twenty-one (21) days after the Effective Date, Respondents shall select a coordinator to be known as the Project Coordinator and shall submit the name, address, qualifications, and telephone number of the Project Coordinator to EPA. The Project Coordinator shall be responsible on behalf of Respondents for oversight of the implementation of the Work to be carried out under this Order. The Project Coordinator shall not be an attorney engaged in the practice of law. He or she shall have the technical expertise sufficient to adequately oversee all aspects of the Work contemplated by this Order. The Project Coordinator shall be knowledgeable at all times about all matters relating to the Work being performed under this Order.

45. Selection of the Project Coordinator shall be subject to approval by EPA in writing. If EPA disapproves a proposed Project Coordinator, Respondents shall propose a different person and notify EPA of that person's name, address, telephone number and qualifications within seven (7) days following EPA's disapproval. Respondents may change their Project Coordinator provided that EPA has received written notice at least seven (7) days prior to the desired change. All changes of the Project Coordinator shall be subject to EPA approval.

46. EPA correspondence related to this Order will be sent to the Project Coordinator. Notice by EPA in writing to the Project Coordinator shall be deemed notice to Respondents for all matters relating to the Work under this Order and shall be effective upon receipt. To the extent possible, the Project Coordinator shall be present on-Site or readily available for EPA to contact

during all working days and be retained by Respondents at all times until EPA issues a notice of termination of this Order upon the completion of the Work in accordance with Paragraph 94.

47. Within twenty-one (21) days after the Effective Date, Respondents shall select a Supervising Contractor and shall submit the name, address, qualifications, and telephone number of the Supervising Contractor to EPA. The Supervising Contractor may be the same person as the Project Coordinator. Respondents shall also notify EPA of the name and qualifications of any other contractor or subcontractor proposed to perform Work under this Order at least ten (10) days prior to commencement of such Work.

48. All activities required of Respondents under the terms of this Order shall be performed only by well-qualified persons possessing all necessary permits, licenses, and other authorizations required by Federal, State and/or local governments consistent with Section 121 of CERCLA, 42 U.S.C. § 9621, and all Work conducted pursuant to this Order shall be performed in accordance with prevailing professional standards. All plans and specifications shall be prepared under the supervision of, and signed/certified by, a licensed New York professional engineer.

49. EPA retains the right to disapprove any or all of the contractors and/or subcontractors proposed by Respondents to conduct the Work. If EPA disapproves in writing of any of Respondents' proposed contractors to conduct the Work, Respondents shall propose a different contractor within seven (7) days of receipt of EPA's disapproval.

50. Respondents shall provide a copy of this Order to each contractor and subcontractor approved and retained to perform the Work required by this Order. Respondents shall include in all contracts or subcontracts entered into for Work required under this Order provisions stating that such contractors or subcontractors, including their agents and employees, shall perform activities required by such contracts or subcontracts in compliance with this Order and all applicable laws and regulations. Respondents shall be responsible for ensuring that their contractors and subcontractors perform the Work contemplated herein in accordance with this Order.

C. EPA Remedial Project Manager, Other Personnel and Modification to EPA-Approved Pre-RD and RD Work Plans

51. EPA has designated Christos Tsiamis of the New York Remediation Branch, Emergency and Remedial Response Division, EPA Region 2, as its Remedial Project Manager ("RPM") for the Site. Except as otherwise provided in this Settlement Agreement, Respondents shall direct all submissions required by this Settlement Agreement to the RPM via e-mail at tsiamis.christos@epa.gov and by regular mail, at U.S. EPA, Region 2, 290 Broadway, 20th Floor, New York, NY 10007.

52. EPA, including the RPM, or his authorized representative, will conduct oversight of the implementation of this Order. The RPM shall have the authority vested in an RPM by the NCP, including the authority to halt, conduct or direct any Work required by this Order, or to direct

any other response action undertaken by EPA or Respondents at the Site consistent with this Order. Absence of the RPM from the Site shall not be cause for stoppage of Work unless specifically directed by the RPM.

53. As appropriate during the course of implementation of the actions required of Respondents pursuant to this Order, Respondents or their consultants or contractors, acting through the Designated Project Coordinator, may confer with EPA concerning the required actions. Based upon new circumstances or new information not in the possession of EPA on the Effective Date of this Order, the Project Coordinator may request in writing EPA approval of modification(s) to the EPA-approved Pre-RD Work Plan and RD Work Plan. In addition, Respondents may propose other additional investigations, studies, and response actions and, upon EPA approval of the same, Respondents shall conduct such actions pursuant to this Order. Only modifications approved by EPA in writing shall be deemed effective.

D. EPA Review of Submissions

54. After review of any deliverable, plan, report or other item which is required to be submitted for review and approval pursuant to this Order, EPA may: (a) approve the submission; (b) approve the submission with modifications; (c) disapprove the submission and direct Respondents to re-submit the document after incorporating EPA's comments; or (d) disapprove the submission and assume responsibility for performing all or any part of the response action. As used in this Order, the terms ”approval by EPA,” “ EPA approval,” or a similar term means the action described in subparagraphs (a) or (b) of this Paragraph.

55. In the event of approval or approval with modifications by EPA, Respondents shall proceed to take any action required by the plan, report or other item, as approved or modified by EPA.

56. Upon receipt of a notice of disapproval or a direction for a modification, Respondents shall correct the deficiencies and resubmit the plan, report or other item for approval within thirty (30) days or such other time as may be specified by EPA in its notice of disapproval or request for modification. Notwithstanding the notice of disapproval or approval with modifications, Respondents shall proceed, at the direction of EPA, to take any action required by any non-deficient portion of the submission.

57. If any plan, report or other item required to be submitted to EPA for approval pursuant to this Order is disapproved by EPA, even after being resubmitted following Respondents’ receipt of EPA's comments on the initial submittal, Respondents shall be deemed to be out of compliance with this Order. If any resubmitted plan, report or other item, or portion thereof, is disapproved by EPA, EPA may again direct Respondents to make the necessary modifications thereto, and/or EPA may amend or develop the item(s) and recover the costs of doing so from Respondents. Respondents shall implement any such item(s) as amended or developed by EPA.

58. EPA shall be the final arbiter regarding the sufficiency or acceptability of all documents submitted and all activities performed pursuant to this Order. EPA may modify those documents and/or perform or require the performance of additional work unilaterally.

59. All plans, reports and other submittals required to be submitted to EPA under this Order shall, upon approval by EPA, be deemed to be incorporated in and an enforceable part of this Order. In the event EPA approves a portion of a plan, report or other item required to be submitted to EPA under this Order, the approved portion shall be deemed to be incorporated in and an enforceable part of this Order.

E. Reporting Requirements

60. Reporting

a. Respondents shall submit written progress reports to EPA concerning actions undertaken pursuant to this Order every thirtieth (30^{th}) day after the date of receipt of EPA's approval of the RD Work Plan until termination of this Order, unless otherwise directed in writing by EPA. These reports shall describe all significant developments during the preceding period, including the actions performed and any problems encountered, analytical data received during the reporting period, and the developments anticipated during the next reporting period, including a schedule of actions to be performed, anticipated problems and planned resolutions of past or anticipated problems.

b. Respondents shall submit copies of all plans, reports or other submissions required by this Order, the RD SOW or any approved work plan as set forth below. Any electronic submissions must be in a format that is compatible with EPA software and in database files and sizes to be specified by EPA. Reports should be submitted to the following:

4 copies: (2 bound, 1 unbound, 1 electronic)	Remedial Project Manager – Gowanus Canal Site Emergency and Remedial Response Division United States Environmental Protection Agency, Region 2 290 Broadway, 20^{th} Floor New York, New York 10007-1866
1 copy:	Chief, New York/Caribbean Superfund Branch Office of Regional Counsel United States Environmental Protection Agency, Region 2 290 Broadway, 17^{th} Floor New York, New York 10007-1866 Attn: Gowanus Canal Superfund Site Attorney
3 copies: (2 unbound, 1 electronic)	Director, Division of Environmental Remediation New York State Department of Environmental Conservation 625 Broadway, 12^{th} Floor Albany, New York 12233-7011

Attn: Gowanus Canal Superfund Site

F. Oversight

61. During the implementation of the requirements of this Order, Respondents and their contractor(s) and subcontractors shall be available for such conferences with EPA and inspections by EPA or its authorized representatives as EPA may determine are necessary to adequately oversee the Work being carried out or to be carried out by Respondents, including inspections at the Site and at laboratories where analytical work is being done hereunder.

62. Respondents and their employees, agents, contractor(s) and consultant(s) shall cooperate with EPA in its efforts to oversee Respondents' implementation of this Order.

G. Community Relations

63. Respondents shall cooperate with EPA in providing information relating to the Work required hereunder to the public. As requested by EPA, Respondents shall participate in the preparation of all appropriate information disseminated to the public; participate in public meetings which may be held or sponsored by EPA to explain activities at or concerning the Site; and provide a suitable location for public meetings, as needed.

H. Access to Property and Information

64. EPA, NYSDEC and their designated representatives, including, but not limited to, employees, agents, contractor(s) and consultant(s) thereof, shall be permitted to observe the Work carried out pursuant to this Order. Respondents shall at all times permit EPA, NYSDEC, and their designated representatives full access to and freedom of movement at the Site and any other premises where Work under this Order is to be performed for purposes of inspecting or observing Respondents' progress in implementing the requirements of this Order, verifying the information submitted to EPA by Respondents, conducting investigations relating to contamination at the Site or for any other purpose EPA determines to be reasonably related to EPA oversight of the implementation of this Order.

65. In the event that action under this Order is to be performed in areas owned by or in possession of a person other than Respondents, Respondents shall use their best efforts to obtain access agreements from such persons within forty-five (45) working days of the Effective Date of this Order for purposes of implementing the requirements of this Order. Such agreements shall provide access not only for Respondents, but also for EPA and its designated representatives or agents, as well as NYSDEC and its designated representatives or agents. Such agreements shall specify that Respondents are not EPA's representative with respect to liability associated with Site activities. If such access agreements are not obtained by Respondents within the time period specified herein, Respondents shall immediately notify EPA of their failure to obtain access and shall include in that notification a summary of the steps Respondents have taken to attempt to obtain access. Subject to the United States' non-reviewable discretion, EPA may use its legal authorities to obtain access for Respondents, may

perform those response actions with EPA contractors at the property in question, or may terminate the Order if Respondents cannot obtain access agreements. If EPA performs those tasks or activities with EPA contractors and does not terminate the Order, Respondents shall perform all other activities not requiring access to that property. Respondents shall integrate the results of any such tasks undertaken by EPA into its reports and deliverables.

66. Upon request, Respondents shall provide EPA with access to all records and documentation related to the conditions at the Site, hazardous substances found at or released from the Site and the actions conducted pursuant to this Order except for those items, if any, subject to the attorney-client or work product privilege. Nothing herein shall preclude Respondents from asserting a business confidentiality claim pursuant to 40 CFR Part 2, Subpart B. All data, information and records created, maintained or received by Respondents or their contractor(s) or consultant(s) in connection with implementation of the Work under this Order, including, but not limited to, contractual documents, invoices, receipts, work orders and disposal records shall, without delay, be made available to EPA upon request, subject to the same privileges specified above in this paragraph. EPA shall be permitted to copy all such documents. Respondents shall submit to EPA upon receipt the results of all sampling or tests and all other technical data generated by Respondents or their contractor(s), or on the Respondents' behalf, in connection with the implementation of this Order.

67. Upon request by EPA, Respondents shall provide EPA or its designated representatives with duplicate and/or split samples of any material sampled in connection with the implementation of this Order.

68. Notwithstanding any other provision of this Order, EPA hereby retains all of its information gathering, access, and inspection authority under CERCLA, the Resource Conservation and Recovery Act ("RCRA"), 42 U.S.C §6901, *et seq.* and any other applicable statutes or regulations.

I. Record Retention, Documentation, Availability of Information

69. Respondents shall preserve all documents and information relating to Work performed under this Order, or relating to Waste Materials found on or released from the Site, for ten (10) years after completion of the Work required by this Order. At the end of the ten (10) year period, Respondents shall notify EPA at least thirty (30) days before any such document or information is destroyed that such documents and information are available for inspection. Upon request, Respondents shall provide EPA with the originals or copies of such documents and information.

70. All documents submitted by Respondents to EPA in the course of implementing this Order shall be available to the public unless identified as confidential by Respondents pursuant to 40 CFR Part 2, Subpart B, and determined by EPA to merit treatment as confidential business information in accordance with applicable law. In addition, EPA may release all non-confidential documents to NYSDEC, and NYSDEC may make those documents available to

the public unless Respondents conform with applicable New York State law and regulations regarding confidentiality. Respondents shall not assert a claim of confidentiality regarding any monitoring or hydrogeological data, any information specified under Section 104(e)(7)(F) of CERCLA, or any other chemical, scientific or engineering data relating to the Work performed hereunder.

J. Off-Site Shipments

71. All hazardous substances, pollutants, or contaminants removed from the Site pursuant to this Order for off-Site treatment, storage or disposal shall be treated, stored or disposed of in compliance with (a) Section 121(d)(3) of CERCLA, 42 U.S.C. § 9621(d)(3), (b) Section 300.440 of the NCP,(c) RCRA, (d) the Toxic Substances Control Act, 15 U.S.C. § 2601, et seq. and (e) all other applicable federal and New York State requirements.

72. If hazardous substances from the Site are to be shipped outside of the State of New York, Respondents shall provide prior notification of such Waste Material shipments to the RPM at the address set forth in Paragraph 60 and in accordance with the EPA Memorandum entitled "Notification of Out-of-State Shipments of Superfund Site Wastes" (OSWER Directive 9330.2-07, September 14, 1989). At least five (5) working days prior to such Waste Material shipments, Respondents shall notify the environmental agency of the accepting state of the following: (a) the name and location of the facility to which the Waste Materials are to be shipped; (b) the type and quantity of Waste Material to be shipped; (c) the expected schedule for the Waste Material shipments; (d) the method of transportation and name of transporter; and (e) the treatment and/or disposal method of the Waste Material streams.

73. Certificates of destruction must be provided to EPA upon Respondents' receipt of such. These certificates must be included in the monthly progress reports and in the Final Report.

K. Compliance With Other Laws

74. Respondents shall undertake all action that this Order requires in accordance with the requirements of all applicable local, state and federal laws and regulations, unless an exemption from such requirements is specifically provided by law or in this Order.

75. Except as provided in Section 121(e)(1) of CERCLA, 42 U.S.C. § 9621(e)(1), and the NCP, no permit shall be required for any portion of the Work required hereunder that is conducted entirely on-Site. Where any portion of the Work requires a federal or New York State permit or approval, Respondent shall submit timely applications and shall take all other actions necessary to obtain and to comply with all such permits or approvals. This Order is not, nor shall it be construed to be, a permit issued pursuant to any federal or New York State statute or regulation.

L. Emergency Response and Notification of Releases

76. Upon the occurrence of any event during performance of the Work required hereunder which, pursuant to Section 103 of CERCLA, 42 U.S.C. § 9603, requires reporting to the National Response Center (800) 424-8802, Respondents shall then immediately orally notify the Chief of the Removal Action Branch of the Emergency and Remedial Response Division of EPA, Region 2, at (732) 321-6658, of the incident or Site conditions. Respondent shall also submit a written report to EPA within seven (7) days after the onset of such an event, setting forth the events that occurred and the measures taken or to be taken, if any, to mitigate any release or endangerment caused or threatened by the release and to prevent the reoccurrence of such a release. The reporting requirements of this paragraph are in addition to, not in lieu of, reporting under CERCLA Section 103, 42 U.S.C. § 9603, and Section 304 of the Emergency Planning and Community Right-To-Know Act of 1986, 42 U.S.C. § 11004.

77. In the event of any action or occurrence during Respondents' performance of the requirements of this Order which causes or threatens to cause a release of a hazardous substance or which may present an immediate threat to public health or welfare or the environment, Respondents shall immediately take all appropriate action to prevent, abate or minimize the threat and shall immediately notify EPA as provided in the preceding paragraph. Respondents shall take such action in accordance with applicable provisions of this Order including, but not limited to, the Health and Safety Plan required to be submitted pursuant to Section IV.E. of the RD SOW. In the event that EPA determines that (a) the activities performed pursuant to this Order, (b) significant changes in conditions at the Site or (c) emergency circumstances occurring at the Site pose a threat to human health or the environment, EPA may direct Respondents to stop further implementation of any actions pursuant to this Order or to take other and further actions reasonably necessary to abate the threat.

78. Nothing in the preceding paragraph shall be deemed to limit any authority of the United States to take, direct, or order all appropriate action to protect human health and the environment or to prevent, abate, or minimize an actual or threatened release of hazardous substances on, at, or from the Site.

M. Modifications

79. No informal advice, guidance, suggestion, or comment by EPA regarding reports, plans, specifications, schedules or any other writing submitted by Respondents shall relieve Respondents of their obligation to obtain such formal approval as may be required by this Order and to comply with all requirements of this Order.

N. Delay in Performance

80. Any delay in performance of the Work under this Order that, in EPA's judgment, is not properly justified by Respondents under the terms of Paragraph 81 below, shall be considered a

violation of this Order. Any delay in performance of this Order shall not affect Respondents' obligations to perform all obligations fully under the terms and conditions of this Order.

81. Respondents shall notify EPA of any delay or anticipated delay in performing any requirement of this Order. Such notification shall be made by telephone to EPA's RPM as soon as Respondents know that a delay might occur. Respondents shall adopt all reasonable measures to avoid or minimize any such delay. Within two (2) days after notifying EPA by telephone, Respondents shall provide written notification fully describing the nature of the delay, any justification for the delay, any reason why Respondents should not be held strictly accountable for failing to comply with any relevant requirements of this Order, the measures planned and taken to minimize the delay and a schedule for implementing the measures that have been or will be taken to mitigate the effect of the delay. Increased cost or expense associated with the implementation of the activities called for in this Order is not a justification for any delay in performance.

O. Enforcement and Reservation of Rights

82. Notwithstanding any other provision of this Order, failure of any Respondent to comply with any provision of this Order may subject such Respondent to civil penalties of up to thirty-seven thousand five hundred dollars ($37,500) per violation per day, as provided in Section 106(b)(1) of CERCLA, 42 U.S.C. § 9606(b)(1), the Debt Collection and Improvement Act of 1996 (see Civil Monetary Penalty Inflation Adjustment Rule, 73 Fed. Reg. 75340 (December 11, 2008) and 40 CFR Part 19. Respondents also may be subject to punitive damages in an amount at least equal to but not more than three times the amount of any costs incurred by the United States as a result of such failure to comply with this Order, as provided in Section 107(c)(3) of CERCLA, 42 U.S.C. § 9607(c)(3). Should Respondents violate this Order or any portion thereof, EPA may carry out the required actions unilaterally, pursuant to Section 104 of CERCLA, 42 U.S.C. § 9604, and/or may seek judicial enforcement of this Order pursuant to Section 106 of CERCLA, 42 U.S.C. § 9606.

83. Nothing herein shall limit the power and authority of EPA or the United States to take, direct or order all actions necessary to protect public health, welfare, or the environment or to prevent, abate or minimize an actual or threatened release of hazardous substances, pollutants or contaminants or hazardous or solid waste on, at or from the Site. Further, nothing herein shall prevent EPA from seeking legal or equitable relief to enforce the terms of this Order, from taking other legal or equitable action as it deems appropriate, or from requiring Respondents in the future to perform additional activities pursuant to CERCLA or any other applicable law. EPA reserves the right to bring an action against Respondent under Section 107 of CERCLA, 42 U.S.C. § 9607, for recovery of any response costs incurred by the United States related to this Order or the Site.

P. Other Claims

84. By issuance of this Order, the United States and EPA assume no liability for injuries or damages to persons or property resulting from any acts or omissions of Respondent or

Respondent's employees, agents, contractors, or consultants in carrying out any action or activity pursuant to this Order. The United States or EPA shall not be held out as or deemed a party to any contract entered into by Respondents or their directors, officers, employees, agents, successors, representatives, assigns, contractors or consultants in carrying out actions pursuant to this Order.

85. Nothing in this Order constitutes or shall be construed as a satisfaction of or release from any claim or cause of action against Respondents or any person not a party to this Order for any liability that Respondents or other persons may have under CERCLA, other statutes, or the common law, including but not limited to any claims of the United States for injunctive relief, costs, damages and interest under Sections 106(a) and 107 of CERCLA, 42 U.S.C. §§ 9606(a) and 9607. Nothing herein shall constitute a finding that Respondents are the only responsible parties with respect to the release and threatened release of hazardous substances at and from the Site.

86. Nothing in this Order shall affect any right, claim, interest, defense, or cause of action of any party hereto with respect to third parties.

87. Nothing in this Order shall be construed to constitute preauthorization under Section 111(a)(2) of CERCLA, 42 U.S.C. § 9611(a)(2), and 40 CFR § 300.700(d).

Q. Insurance

88. At least five (5) days prior to commencing any on-Site Work under this Order, Respondents shall secure and shall maintain for the duration of this Order comprehensive general liability insurance and automobile insurance with limits of $5 million dollars, combined single limit, naming the EPA as an additional insured. Within the same period, Respondents shall provide EPA with certificates of such insurance and a copy of each insurance policy. Respondents shall submit such certificates and copies of policies each year on the anniversary of the Effective Date. In addition, for the duration of the Order, Respondents shall satisfy, or shall ensure that their contractors or subcontractors satisfy, all applicable laws and regulations regarding the provision of worker's compensation insurance for all persons performing the Work on behalf of Respondents in furtherance of this Order. If Respondents demonstrate by evidence satisfactory to EPA that any contractor or subcontractor maintains insurance equivalent to that described above, or insurance covering some or all of the same risks but in an equal or lesser amount, then Respondents need provide only that portion of the insurance described above that is not maintained by such contractor or subcontractor.

R. Financial Assurance

89. Within thirty (30) days of the Effective Date of this Order, Respondents shall demonstrate their financial ability to complete the Work by submitting to EPA copies of one or more of Respondents' most recent Annual Reports. Such Annual Reports shall demonstrate that Respondents have sufficient assets to perform the Work, which is valued by EPA at $35,000,000. Each year thereafter, until the completion of the work, Respondents shall submit one or more

Respondents' most recent Annual Reports to EPA within thirty (30) days of publication of such reports. In the event that EPA determines at any time that the financial assurances provided by the Annual Reports do not demonstrate Respondents' ability to complete the Work, then Respondents shall establish and maintain financial security in the amount needed to complete the Work, in one or more of the following forms:

a. A surety bond unconditionally guaranteeing payment and/or performance of the Work that is issued by a surety company among those listed as acceptable sureties on Federal bonds as set forth in Circular 570 of the U.S. Department of the Treasury;

b. One or more irrevocable letters of credit, payable to or at the direction of EPA, that is issued by one or more financial institution(s) (i) that has the authority to issue letters of credit and (ii) whose letter-of-credit operations are regulated and examined by a U.S. Federal or State agency;

c. A trust fund established for the benefit of EPA that is administered by a trustee (i) that has the authority to act as a trustee and (ii) whose trust operations are regulated and examined by a U.S. Federal or State agency;

d. A policy of insurance that (i) provides EPA with acceptable rights as a beneficiary thereof; and (ii) is issued by an insurance carrier (a) that has the authority to issue insurance policies in the applicable jurisdiction(s) and (b) whose insurance operations are regulated and examined by a State agency;

e. A demonstration by one or more Respondents that such Respondent(s) meets the financial test criteria of 40 C.F.R. § 264.143(f) with respect to the Estimated Cost of the Work, provided that all other requirements of 40 C.F.R. § 264.143(f) are satisfied; and

f. A written guarantee to fund or perform the Work executed in favor of EPA by one or both of the following: (i) a direct or indirect parent company of a Respondent or (ii) a company that has a "substantial business relationship" (as defined in 40 C.F.R. § 264.141(h)) with at least one Respondent; provided, however, that any company providing such a guarantee must demonstrate to the satisfaction of EPA that it satisfies the financial test requirements of 40 C.F.R. § 264.143(f) with respect to the Estimated Cost of the Work that it proposes to guarantee hereunder.

90. Any and all financial assurance instruments provided pursuant to this Section shall be in form and substance satisfactory to EPA, determined in EPA's sole discretion. In the event that EPA determines at any time that the financial assurances provided pursuant to this Section (including, without limitation, the instrument(s) evidencing such assurances) are inadequate, Respondents shall, within thirty (30) days of receipt of notice of EPA's determination, obtain and present to EPA for approval one of the other forms of financial assurance listed in Paragraph 89, above. In addition, if at any time EPA notifies Respondents that the anticipated cost of completing the Work has increased, then, within thirty (30) days of such notification, Respondents shall obtain and present to EPA for approval a revised form of financial assurance

(otherwise acceptable under this Section) that reflects such cost increase. Respondents' inability to demonstrate financial ability to complete the Work shall in no way excuse performance of any activities required under this Order.

91. If Respondents seek to ensure completion of the Work through a guarantee pursuant to Subparagraph 89(e) or 89(f) of this Order, Respondents shall: (i) demonstrate to EPA's satisfaction that the guarantor satisfies the requirements of 40 C.F.R. Part 264.143(f); and (ii) resubmit sworn statements conveying the information required by 40 C.F.R. Part 264.143(f) annually, on the anniversary of the Effective Date, to EPA. For the purposes of this Order, wherever 40 C.F.R. Part 264.143(f) references "sum of current closure and post-closure costs estimates and the current plugging and abandonment costs estimates," the current EPA cost estimate of $35,000,000 for the Work at the Site shall be used in relevant financial test calculations.

92. If, after the Effective Date, Respondents can show that the estimated cost to complete the remaining Work has diminished below the amount set forth in Paragraph 89 of this Section, Respondents may, on any anniversary date of the Effective Date, or at any other time agreed to by EPA and Respondents, reduce the amount of the financial security provided under this Section to the estimated cost of the remaining Work to be performed. Respondents shall submit a proposal for such reduction to EPA, in accordance with the requirements of this Section, and may reduce the amount of the security after receiving written approval from EPA.

93. Respondents may change the form of financial assurance provided under this Section at any time, upon notice to and prior written approval by EPA, provided that EPA determines that the new form of assurance meets the requirements of this Section.

S. Termination and Satisfaction

94. Upon a determination by EPA that the Work required pursuant to this Order has been fully carried out in accordance with this Order, EPA will so notify Respondents in writing.

T. Opportunity to Confer, Effective Date

95. This Order shall be effective ten (10) days after receipt by Respondents, unless a conference is timely requested pursuant to Paragraph 96 below. If such a conference is timely requested, this Order shall become effective three (3) days following the date the conference is held, unless the effective date is modified by EPA. All times for performance of ordered activities shall be calculated from this Effective Date.

96. Respondents may, within ten (10) days after receipt of this Order, request a conference with EPA to discuss this Order. If requested, the conference shall occur within seven (7) days of Respondents' request for a conference. The conference may occur in person or telephonically.

97. The purpose and scope of the conference is to discuss issues involving the implementation of the Work required by this Order and the extent to which Respondents intend

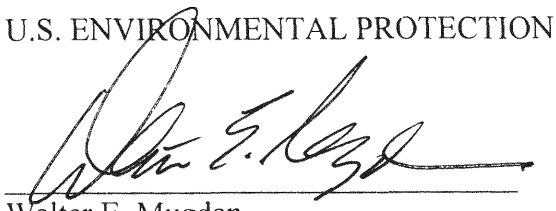
to comply with this Order. The conference is not intended to be a forum for discussing liability issues or whether the Order should have been issued. This conference is not an evidentiary hearing and does not constitute a proceeding to challenge this Order. It does not give Respondents a right to seek review of this Order or to seek resolution of potential liability, and no official stenographic record of the conference will be made. At any conference held pursuant to Respondents' request, Respondents may appear in person or by an attorney or other representative.

98. A request for a conference must be made by telephone to Brian E. Carr, Assistant Regional Counsel, Office of Regional Counsel, EPA Region II, telephone (212) 637-3170, followed by written confirmation emailed that day to Mr. Carr at carr.brian@epa.gov.

U. Notice of Intent to Comply

99. Each Respondent shall provide, not later than five (5) days after the Effective Date, written notice to EPA stating whether it will comply with the terms of this Order. The notice should state the manner in which the Respondent intends to comply. If a Respondent does not unequivocally commit to perform the work required by this Order, it shall be deemed to have violated this Order and to have failed or refused to comply with this Order. Each Respondent's written notice shall describe, using facts that exist on or prior to the effective date of this Order, any "sufficient cause" defenses asserted by Respondent under Sections 106(b) and 107(c)(3) of CERCLA. Each Respondent's written notice shall be sent to the EPA addressees listed in Paragraph 60 above. The absence of a response by EPA to the notice required by this paragraph shall not be deemed to be an acceptance of a Respondent's assertions.

U.S. ENVIRONMENTAL PROTECTION AGENCY



Walter E. Mugdan
Director
Emergency and Remedial Response Division
U.S. Environmental Protection Agency
Region 2

March 20, 2014
Date of Issuance

APPENDICES

Available Electronically on EPA's Gowanus Webpage:
http://www.epa.gov/region02/superfund/npl/gowanus/additionaldocs.html

Appendix A – Statement of Work

Appendix B – Draft Pre-Design Work Plan

Appendix C – Record of Decision

Appendix D - Draft Remedial Design Work Plan

Appendix E – Site Map

Exhibit ___ (CFW-3A)

Unilateral Administrative Order 2 on the Gowanus Canal Superfund Site

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
REGION 2

IN THE MATTER OF THE GOWANUS CANAL SUPERFUND SITE	:	
Beazer East, Inc.	:	INDEX NO.
Brink's Inc.	:	CERCLA-02-2019-2010
Brooklyn Union Gas Co. d/b/a National Grid New York	:	
Citigroup, Inc.	:	SECOND AMENDMENT TO
City of New York	:	ADMINISTRATIVE ORDER
Consolidated Edison Co. of New York, Inc.	:	FOR REMOVAL ACTION
Dun and Bradstreet Corp.	:	
ExxonMobil Oil Corp.	:	
Hauck Manufacturing Co.	:	
Hess Corp.	:	
Honeywell International Inc.	:	
Kraft Foods Global, Inc.	:	
MCIZ Corp. and	:	
Fifteen Second Avenue LLC	:	
36-2nd-J Corp.	:	
107 Sixth Street LLC	:	
MRC Holdings, Inc.	:	
NL Industries, Inc.	:	
Northville Industries Corp.	:	
Phillips 66 Co.	:	
Puget Sound Commerce Center, Inc.	:	
Rexam Beverage Can Co.	:	
SPX Corp.	:	
Stauffer Management Company, LLC	:	
TDA Industries, Inc.	:	
The Brooklyn Improvement Co.	:	
The Union Oil Company of California	:	
Verizon New York Inc.	:	
Respondents,	:	
Proceeding under Section 106 of the Comprehensive Environmental Response, Compensation, and Liability Act, as amended, 42 U.S.C. § 9606.	:	

I. JURISDICTION AND GENERAL PROVISIONS

1. This second amendment ("Second Amendment") to the Administrative Order for Removal Action, Index Number CERCLA-02-2019-2010 ("Order"), dated April 11, 2019, is issued to the above-captioned Respondents by the United States Environmental Protection Agency, Region 2 ("EPA") and requires Respondents to undertake a Removal Action to perform the tasks necessary to provide structural support for bulkheads and infrastructure in preparation for implementation of the dredging and capping remedy selected in EPA's September 27, 2013 Record of Decision ("ROD") for the Gowanus Canal Superfund Site ("Site"), Brooklyn, New York. This Second Amendment adds tasks related to two additional properties where upgraded bulkheads are needed.

2. This Second Amendment is issued to Respondents by EPA pursuant to the authority vested in the President of the United States under Section 106(a) of the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended ("CERCLA"), 42 U.S.C. § 9606(a), and delegated to the Administrator of EPA on January 23, 1987, by Executive Order No. 12580 (52 Federal Register 2926, January 29, 1987). This authority was further delegated to the EPA Regional Administrators by EPA Delegation Nos. 14-14-A and 14-14-B and to the Director of the Emergency and Remedial Response Division in Region 2 by Regional Delegation R-1200, dated January 19, 2017. Effective April 28, 2019, the Emergency and Remedial Response Division has been renamed the Superfund and Emergency Management Division. All delegations to the Director of the Emergency and Remedial Response Division were conferred upon the Director of the Superfund and Emergency Management Division in a memorandum by the EPA Regional Administrator dated March 27, 2019.

3. As revised by this Second Amendment, the Order remains in full force and effect.

4. EPA has notified the New York State Department of Environmental Conservation ("NYSDEC") of this Second Amendment pursuant to Section 106(a) of CERCLA, 42 U.S.C. § 9606(a).

II. AMENDMENT

5. Paragraph 62 of Section VIII. (Work to be Performed) of the Order is amended by the addition of the following subparagraph at the end of the paragraph:

RTA 2 Bulkhead Construction

l. Following EPA approval, pursuant to the Dredging RD UAO and City RD UAO, of the bulkhead structural support designs and construction schedule for Block 990, Lot 369 (76 Sixth Street), and Block 990, Lot 50 (37 Ninth Street), Respondents shall construct such bulkhead structural supports. In lieu of Respondents' performance of the construction of the

bulkhead structural support at 76 Sixth Street and/or 37 Ninth Street, Respondents may submit to EPA for review and approval a proposed PRP funding agreement with the owner(s) of 76 Sixth Street and/or 37 Ninth Street providing for such owner's performance of such construction and consent for EPA oversight under this Order. Such agreement(s) shall not relieve Respondents of the obligation to ensure performance of these tasks under this Order.

III. OPPORTUNITY TO CONFER, EFFECTIVE DATE

6. This Second Amendment shall be effective five (5) days after receipt by Respondents ("Second Amendment Effective Date"), unless a conference is timely requested pursuant to Paragraph 7 below. If such a conference is timely requested, this Amendment shall become effective three (3) days following the date the conference is held, unless the Second Amendment Effective Date is modified by EPA. All times for performance of ordered activities shall be calculated from the Second Amendment Effective Date.

7. Respondents may, within fifteen (15) days after receipt of this Second Amendment, request a conference with EPA to discuss this Second Amendment. If requested, the conference shall occur within fourteen (14) days of Respondents' request for a conference. The conference may be held in person or by telephone or videoconference.

IV. NOTICE OF INTENT TO COMPLY

9. Each Respondent shall provide, not later than fourteen (14) days after the Second Amendment Effective Date, written notice to EPA stating whether it will comply with the terms of this Second Amendment. If a Respondent does not unequivocally commit to perform the work required by this Second Amendment, it shall be deemed to have violated this Second Amendment and to have failed or refused to comply with this Second Amendment. Each Respondent's written notice shall describe, using facts that exist on or prior to the Second Amendment Effective Date, any "sufficient cause" defenses asserted by Respondent under Sections 106(b) and 107(c)(3) of CERCLA. Each Respondent's written notice shall be sent to Brian Carr, Assistant Regional Counsel, Office of Regional Counsel, EPA Region 2, 290 Broadway, 17th floor, New York, New York 10007-1866, or by email to carr.brian@epa.gov. The absence of a response by EPA to the notice required by this paragraph shall not be deemed to be an acceptance of a Respondent's assertions.

By: Evangelista, Pat Digitally signed by Evangelista, Pat Date: 2020.09.18 09:34:36 -04'00'

PAT EVANGELISTA
Director
Superfund and Emergency Management Division
U.S. Environmental Protection Agency
Region 2

Date

Exhibit ___ (CFW-3B)

Unilateral Administrative Order 2 Amendment on the Gowanus Canal Superfund Site

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
REGION 2

IN THE MATTER OF THE GOWANUS CANAL SUPERFUND SITE	
Beazer East, Inc. Brink's Inc. Brooklyn Union Gas Co. d/b/a National Grid New York Citigroup, Inc. City of New York Consolidated Edison Co. of New York, Inc. Dun and Bradstreet Corp. ExxonMobil Oil Corp. Hauck Manufacturing Co. Hess Corp. Honeywell International Inc. Kraft Foods Global, Inc. MCIZ Corp. and Fifteen Second Avenue LLC 36-2nd-J Corp. 107 Sixth Street LLC MRC Holdings, Inc. NL Industries, Inc. Northville Industries Corp. Phillips 66 Co. Puget Sound Commerce Center, Inc. Rexam Beverage Can Co. SPX Corp. Stauffer Management Company, LLC TDA Industries, Inc. The Brooklyn Improvement Co. The Union Oil Company of California Verizon New York Inc. Respondents,	INDEX NO. CERCLA-02-2019-2010 AMENDMENT TO ADMINISTRATIVE ORDER FOR REMOVAL ACTION
Proceeding under Section 106 of the Comprehensive Environmental Response, Compensation, and Liability Act, as amended, 42 U.S.C. § 9606.	

I. JURISDICTION AND GENERAL PROVISIONS

1. This amendment ("Amendment") to the Administrative Order for Removal Action, Index Number CERCLA-02-2019-2010 ("Order"), dated April 11, 2019, is issued to the above-captioned Respondents by the United States Environmental Protection Agency, Region 2 ("EPA") and requires Respondents to undertake a Removal Action to perform the tasks necessary to provide structural support for bulkheads and infrastructure in preparation for implementation of the dredging and capping remedy selected in EPA's September 27, 2013 Record of Decision ("ROD") for the Gowanus Canal Superfund Site ("Site"), Brooklyn, New York. This Amendment adds tasks related to an additional property where an upgraded bulkhead is needed.

2. This Amendment is issued to Respondents by EPA pursuant to the authority vested in the President of the United States under Section 106(a) of the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended ("CERCLA"), 42 U.S.C. § 9606(a), and delegated to the Administrator of EPA on January 23, 1987, by Executive Order No. 12580 (52 Federal Register 2926, January 29, 1987). This authority was further delegated to the EPA Regional Administrators by EPA Delegation Nos. 14-14-A and 14-14-B and to the Director of the Emergency and Remedial Response Division in Region 2 by Regional Delegation R-1200, dated January 19, 2017. Effective April 28, 2019, the Emergency and Remedial Response Division has been renamed the Superfund and Emergency Management Division. All delegations to the Director of the Emergency and Remedial Response Division were conferred upon the Director of the Superfund and Emergency Management Division in a memorandum by the EPA Regional Administrator dated March 27, 2019.

3. As revised by this Amendment, the Order remains in full force and effect.

4. EPA has notified the New York State Department of Environmental Conservation ("NYSDEC") of this Amendment pursuant to Section 106(a) of CERCLA, 42 U.S.C. § 9606(a).

II. AMENDMENT

5. Paragraph 31 of Section IV. (Findings of Fact and Conclusions of Law) of the Order is deleted and replaced with the following:

> In preparation for the RTA 1 dredging, EPA has also entered into administrative consent orders for the performance of removal actions requiring the construction of upgraded bulkheads at various upland parcels in RTA 1. The respondents for these bulkhead orders have generally been non-PRP volunteers who are in the process of redeveloping the parcels. Such bulkhead upgrades have been completed at 363 Bond Street, 365 Bond Street and 155 3rd Street. EPA has entered into another such order with the owners of 319-323-325-327 Bond Street, as well as a bulkhead upgrade design order with the owner of 322 3rd Avenue.

EPA is in the process of negotiating, or has negotiated, orders for bulkhead work with the owners of 495-498-510 Sackett Street/267 Bond Street, 420 Carroll Street, 450 Union Street, and 175-225 3rd Street. Those orders, in conjunction with this Order, will provide for completion of all upgraded bulkheads in RTA 1 in advance of the scheduled start of RTA 1 dredging. With regard to property located at 450 Union Street, designated on the Kings County Tax Map as Block 438, Lot 7, construction of the EPA-approved bulkhead design at 450 Union Street in advance of the scheduled start of RTA 1 dredging must be addressed under this Order in order for that work to be timely performed or funded.

6. Section VIII. (Work to be Performed) of the Order is amended as follows:

a. Paragraph 62.a. is amended to add the following sub-paragraph:

v. 450 Union Street.

b. Paragraph 62.c.i. is deleted and replaced with the following:

In lieu of Respondents' performance of the construction of the bulkhead structural support at 479 Degraw Street and/or 450 Union Street, Respondents may submit to EPA for review and approval a proposed PRP funding agreement with the owner of 479 Degraw Street and/or 450 Union Street providing for the respective owner's performance of such construction and consent for EPA oversight under this Order. Such agreement shall not relieve Respondents of the obligation to ensure performance of this task under this Order.

7. Appendix B is amended to add the following schedule requirement:

Construction of the EPA-approved bulkhead design for 450 Union Street shall be completed by July 31, 2020.

III. OPPORTUNITY TO CONFER, EFFECTIVE DATE

8. This Amendment shall be effective five (5) days after receipt by Respondents ("Amendment Effective Date"), unless a conference is timely requested pursuant to Paragraph 9 below. If such a conference is timely requested, this Amendment shall become effective three (3) days following the date the conference is held, unless the Amendment Effective Date is modified by EPA. All times for performance of ordered activities shall be calculated from the Amendment Effective Date.

9. Respondents may, within ten (10) days after receipt of this Amendment, request a conference with EPA to discuss this Amendment. If requested, the conference shall occur

within fourteen (14) days of Respondents' request for a conference. The conference may be held in person or by telephone or videoconference.

IV. NOTICE OF INTENT TO COMPLY

9. Each Respondent shall provide, not later than ten (10) days after the Amendment Effective Date, written notice to EPA stating whether it will comply with the terms of this Amendment. If a Respondent does not unequivocally commit to perform the work required by this Amendment, it shall be deemed to have violated this Amendment and to have failed or refused to comply with this Amendment. Each Respondent's written notice shall describe, using facts that exist on or prior to the Amendment Effective Date, any "sufficient cause" defenses asserted by Respondent under Sections 106(b) and 107(c)(3) of CERCLA. Each Respondent's written notice shall be sent to Brian Carr, Assistant Regional Counsel, Office of Regional Counsel, EPA Region 2, 290 Broadway, 17th floor, New York, New York 10007-1866, or by email to carr.brian@epa.gov. The absence of a response by EPA to the notice required by this paragraph shall not be deemed to be an acceptance of a Respondent's assertions.

By: _____ 3/3/2020
ERIC J. WILSON Date
Acting Director
Superfund and Emergency Management Division
U.S. Environmental Protection Agency
Region 2

Exhibit ___ (CFW-3C)

Unilateral Administrative Order 3 on the Gowanus Canal Superfund Site

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
REGION 2

IN THE MATTER OF THE GOWANUS CANAL SUPERFUND SITE	:	
Brooklyn Union Gas Co. d/b/a National Grid New York City of New York Consolidated Edison Co. of New York, Inc. Hess Corp. Honeywell International Inc. The Brooklyn Improvement Co.,	:	INDEX NO. CERCLA-02-2020-2003 ADMINISTRATIVE ORDER FOR REMEDIAL ACTION
Respondents,	:	
Proceeding under Section 106 of the Comprehensive Environmental Response, Compensation, and Liability Act, as amended, 42 U.S.C. § 9606.	:	

I. JURISDICTION AND GENERAL PROVISIONS

1. This administrative order ("Order") is issued to the above-captioned Respondents by the United States Environmental Protection Agency, Region 2 ("EPA") and requires Respondents to undertake the Remedial Action ("RA") to implement a portion of the dredging and capping remedy selected in EPA's September 27, 2013 Record of Decision ("ROD") for the Gowanus Canal Superfund Site ("Site"), Brooklyn, New York. For purposes of the planning and implementation of the ROD, EPA divided the Gowanus Canal ("Canal") into three Remediation Target Areas ("RTAs"). RTA 1 extends from Butler Street to 3rd Street. This Order requires implementation of the RA for RTA 1 and the procurement of a staging area for future RTA 2 dredging and capping work. Implementation of the balance of the RA will be the subject of one or more future enforcement documents.

2. This Order is issued to Respondents by EPA pursuant to the authority vested in the President of the United States under Section 106(a) of the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended ("CERCLA"), 42 U.S.C. § 9606(a), and delegated to the Administrator of EPA on January 23, 1987, by Executive Order No. 12580, 52 Federal Register 2926 (January 29, 1987), and was further delegated to the EPA Regional Administrators by EPA Delegation Nos. 14-14-A and 14-14-B, dated May 11, 1994, and to the EPA Region 2 Director of the Emergency and Remedial Response Division by Regional Delegation R-1200, dated January 19, 2017. Effective April 28, 2019, the Emergency and Remedial Response Division has been renamed the Superfund and Emergency Management Division. All delegations to the Director of the Emergency and Remedial Response Division were conferred upon the Director of the Superfund and Emergency Management Division in a memorandum by the EPA Regional Administrator dated March 27, 2019.

3. EPA has notified the New York State Department of Environmental Conservation ("NYSDEC") of this Order pursuant to Section 106(a) of CERCLA, 42 U.S.C. § 9606(a).

II. PARTIES BOUND

4. This Order applies to and is binding upon Respondents and their successors and assigns. Any change in ownership or control of the Site or change in corporate or partnership status of a Respondent, including, but not limited to, any transfer of assets or real or personal property, shall not alter Respondents' responsibilities under this Order. Respondents are jointly and severally responsible for carrying out all Work required by this Order. Compliance or noncompliance by any Respondent with any provision of this Order shall not excuse or justify noncompliance by any other Respondent. No Respondent shall interfere in any way with performance of the Work in accordance with this Order by any other Respondent. In the event of the insolvency or other failure of any Respondent to implement the requirements of this Order, the remaining Respondents shall complete all such requirements.

5. Until EPA notifies Respondents under Paragraph 101 that the Work has been completed, Respondents shall provide a copy of this Order to any successors before a controlling interest in such Respondents' assets or property rights are transferred to any successor.

III. DEFINITIONS

6. Unless otherwise expressly provided herein, terms used in this Order that are defined in CERCLA or in regulations promulgated under CERCLA shall have the meaning assigned to them in CERCLA or its implementing regulations. Whenever terms listed below are used in this Order or in the appendices attached hereto, or incorporated by reference into this Order, the following definitions shall apply:

a. "Bulkhead Removal Order" shall mean the Administrative Order, Index Number CERCLA-02-2019-2010, issued by EPA on April 11, 2019 to 28 parties requiring, among other things, the performance of bulkhead and bridge structural support construction and access dredging in RTA 1 at the Site.

b. "CERCLA" shall mean the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended, 42 U.S.C. §§ 9601-9675.

c. "Day" shall mean a calendar day. In computing any period under this Order, where the last day would fall on a Saturday, Sunday, or federal or State holiday, the period shall run until the close of business of the next working day.

d. "Effective Date" means the date specified in Paragraph 102.

e. "EPA" shall mean the United States Environmental Protection Agency and any successor departments or agencies of the United States.

f. "NCP" shall mean the National Oil and Hazardous Substances Pollution Contingency Plan promulgated pursuant to Section 105 of CERCLA, 42 U.S.C. § 9605, and codified at 40 C.F.R. Part 300, including any amendments thereto.

g. "Order" shall mean this Unilateral Administrative Order and all appendices attached hereto. In the event of conflict between this Order and any appendix, this Order shall control.

h. "Paragraph" shall mean a portion of this Order identified by an Arabic numeral.

i. "Record of Decision" or "ROD" shall mean the EPA Record of Decision relating to the Site signed on September 27, 2013 by the Director of the Emergency

Remedial Response Division, EPA Region 2, including all attachments thereto, attached as Appendix A.

j. "Remedial Design" or "RD" shall mean those activities being undertaken by Respondents, among others, pursuant to two RD administrative orders issued by EPA in 2014, Index Numbers CERCLA-02-2014-2001 and CERCLA-02-2014-2019.

k. "Respondents" shall mean:

1. Brooklyn Union Gas Co. d/b/a National Grid New York
2. City of New York
3. Consolidated Edison Co. of New York, Inc.
4. Hess Corp.
5. Honeywell International Inc.
6. The Brooklyn Improvement Co.

l. "Site" shall mean the Gowanus Canal Superfund Site, an approximately 100-foot wide, 1.8-mile-long canal located in the New York City borough of Brooklyn, Kings County, New York, and also includes any areas which are sources of contamination to the Gowanus Canal ("Canal"), including any areas where contamination has migrated from the Canal, and/or suitable areas in very close proximity to the contamination which are necessary for implementation of the Work.

m. "United States" shall mean the United States of America.

n. "Waste Material" shall mean i) any "hazardous substance" under Section 101(14) of CERCLA, 42 U.S.C. § 9601(14); ii) any "pollutant or contaminant" under Section 101(33) of CERCLA, 42 U.S.C. § 9601(33); and iii) any "solid waste" under Section 1004(27) of the Solid Waste Disposal Act, as amended, 42 U.S.C. § 6903(27).

o. "Work" shall mean all activities Respondents are required to perform under this Order, except those required by Paragraph 72 (Record Retention).

IV. FINDINGS OF FACT AND CONCLUSIONS OF LAW

7. The Gowanus Canal ("Canal") is a brackish, tidal arm of the New York–New Jersey Harbor Estuary, extending for approximately 1.8 miles through Brooklyn, New York. The approximately 100–foot–wide Canal runs southwest from Butler Street to Gowanus Bay and Upper New York Bay. The adjacent waterfront is primarily commercial and industrial, currently

including concrete plants, warehouses, and parking lots, and the Site is near several residential neighborhoods.

8. In 1849, State legislation authorized construction of the Canal by the City of Brooklyn to open the area to barge traffic, flush away sewage, receive storm water and fill the adjacent lowlands for development. Pursuant to the 1849 legislation, the City of New York ("City"), successor in interest to the City of Brooklyn, has been the owner and operator of the Canal since its construction. The Canal bottomlands are owned by the City.

9. The Canal was constructed by bulkheading and dredging a tidal creek and wetland. Additional fill was utilized to raise the grade of the surrounding land. The authorizing legislation and the initial Canal design had recognized the likelihood that the Canal would be stagnant, creating pollution problems. To prevent this, various flushing solutions were contemplated. However, none were implemented as part of its initial construction.

10. After completion of construction in the 1860s, the Canal quickly became one of the nation's busiest industrial waterways, home to heavy industry including manufactured gas plants ("MGPs"), coal yards, cement makers, soap makers, tanneries, paint and ink factories, machine shops, chemical and asphalt plants, oil refineries and shipyards.

11. A tunnel was constructed by the City (the "Flushing Tunnel") and began operating in 1911 in an effort to address the Canal's pollution problems. Designed to improve circulation and flush pollutants from the Canal, the Flushing Tunnel consists of a one mile long, 12-foot diameter tunnel stretching from New York Bay near Governors Island to the head of the Canal. Originally using a large ship propeller-type pump system, the Flushing Tunnel could pump water in either direction. The Flushing Tunnel was operated by the City with mixed results until the mid-1960s when it fell into disrepair. The Flushing Tunnel was re-started in 1999 after a partial rehabilitation, before being shut down in 2010 due to equipment problems. Following a full-scale rehabilitation by the City, the Flushing Tunnel began operating again in 2014.

12. Hazardous substances, pollutants and contaminants have entered the Canal via several transport pathways or mechanisms, including spillage during product shipping and handling, direct disposal or discharge, contaminated groundwater discharge, surface water runoff, storm water discharge (including combined sewer overflow ("CSO")) and contaminated soil erosion. Because of decades of direct and indirect discharges of hazardous substances generated by industrial and other activity, the Canal became a repository for untreated industrial wastes, raw sewage, and runoff, causing it to become one of New York's most polluted waterways.

13. Several lateral extensions of the Canal, known as "turning basins," were constructed as part of the Canal. Portions of several turning basins have been filled over time, covering unremediated areas of contamination which are hydraulically connected to the Canal.

14. The contaminated former 1st Street turning basin was originally utilized to deliver coal via barges to an adjacent electric generating station (the "Power House") originally built to provide power to the former Brooklyn Rapid Transit Authority ("BRT") subway system. The Power House began operations in 1904. During operations, it consumed large quantities of coal, fed from coal piles which surrounded the building and were located adjacent to the Canal. The 1st Street turning basin was filled in between 1954 and 1966 after the Power House became obsolete and was removed from service.

15. Much of the heavy industrial activity along the Canal has ceased, although many upland areas adjacent to the Canal remained zoned as manufacturing districts. Land uses along and near certain portions of the Canal are in the process of transitioning from heavy industrial to light industrial, commercial, and residential uses. The Canal is currently used by some for recreational purposes such as boating, diving, and catching fish for consumption. The Canal and New York City harbor are subject to New York State fishing advisories.

16. The Site was placed on the National Priorities List pursuant to Section 105 of CERCLA, 42 U.S.C. § 9605, on March 2, 2010.

17. In April 2010, EPA entered into administrative consent orders with the City and The Brooklyn Union Gas Co. d/b/a National Grid New York ("National Grid") to perform work in support of EPA's remedial investigation/feasibility study ("RI/FS") at the Site. An RI report was completed in January 2011 and an FS report was completed in December 2011. An FS addendum report was completed in December 2012.

18. Sampling results from the RI/FS document the presence of a wide range of hazardous substances in the groundwater, soil, and Canal sediments at the Site. These include polycyclic aromatic hydrocarbons ("PAHs"), polychlorinated biphenyls ("PCBs"), pesticides (such as methoxychlor and DDT), metals (such as barium, cadmium, copper, lead, mercury, nickel and silver), as well as volatile organic compounds ("VOCs") (such as benzene, toluene, ethylbenzene, and xylene). The contamination in the sediments extends the entire length of the Canal. The contamination is present in both the sediment that has accumulated above the native sediments (referred to as "soft sediments") and in the native sediment. Some of the hazardous substances are present at high levels. The soft sediment layer ranges in thickness from approximately 1 foot to greater than 20 feet, with an average thickness of about 10 feet. For example, total PAH concentrations in surface sediment (defined as the top 6 inches of the soft sediments, where potential exposure is more likely to occur) range up to 8,001,000 micrograms per kilogram (µg/kg). PCBs in surface sediment were detected up to 3,400 µg/kg. In the subsurface (*i.e.,* deeper than 6 inches), total PAH concentrations in the soft sediment ranged up to 45,000,000 µg/kg. Total PAH concentrations in the native sediment were detected up to 47,500,000 µg/kg. In the subsurface, total PCB concentrations in the soft sediment were detected up to 50,700 µg/kg. In the native sediments, total PCBs were detected up to 2,610 µg/kg.

19. Based on the results of the RI/FS, it has been concluded that chemical contamination in the Canal sediments present an unacceptable ecological and human health risk, primarily due to exposure to PAHs, PCBs and metals (barium, cadmium, copper, lead, mercury, nickel and silver).

20. PCBs and PAHs have been demonstrated to cause a variety of adverse health effects. PCBs have been shown to cause cancer in test animals. PCBs have also been shown to cause a number of serious non-cancer health effects in animals, including effects on the immune system, reproductive system, nervous system, endocrine system and other health effects. Studies in humans provide supportive evidence for potential carcinogenic and non-carcinogenic effects of PCBs. The toxicity of PAHs can vary from being nontoxic to extremely toxic. EPA has classified seven PAH compounds as probable human carcinogens, all of which are found at the Site: benz[a]anthracene; benzo[a]pyrene; benzo[b]fluoranthene; benzo[k]fluoranthene; chrysene; dibenz(a,h)anthracene; and indeno(1,2,3-cd)pyrene. PAHs known for their carcinogenic, mutagenic, and teratogenic properties, and found at the Site, include: benz[a]anthracene, chrysene, benzo[b]fluoranthene, benzo[k]fluoranthene, benzo[a]pyrene, benzo[ghi]perylene, dibenz(a,h)anthracene, and indeno(1,2,3-cd)pyrene. High prenatal exposure to PAHs is associated with lower IQ and childhood asthma.

21. EPA's ecological risk assessment for the Site determined that PAHs, PCBs and metals in the sediment are toxic to benthic organisms. PAHs were detected in sediment at the highest concentrations relative to their ecological screening benchmarks and represent the greatest Site-related risk to the benthic community. PCBs and seven metals (barium, cadmium, copper, lead, mercury, nickel and silver) were also detected at concentrations above their ecological screening benchmarks and at concentrations significantly higher than those detected in reference area sediments and also represent a potential Site-related risk to the benthic community. PAHs were found to be a potential risk to aquatic herbivores (represented by the black duck) and mercury was found to be a potential risk to avian omnivores (represented by the heron).

22. On September 27, 2013, EPA issued a ROD for the Site that selected the following response actions: 1) Dredging of the entire column of hazardous substance-contaminated soft sediments which have accumulated above the native sediments in the upper and mid-reaches of the Canal; 2) in-situ stabilization ("ISS") of those native sediments in select areas in the upper and mid-reaches of the Canal contaminated with high levels of nonaqueous phase liquid ("NAPL"); 3) construction of a multilayered cap in the upper and mid-reaches of the Canal to isolate and prevent the migration of PAHs and residual NAPL from native sediments; 4) dredging of the entire soft sediment column in the lower reach of the Canal; 5) construction of a multilayer cap to isolate and prevent the migration of PAHs from native sediments in the lower reach of the Canal; 6) off-Site treatment of the NAPL-impacted sediments dredged from the upper and mid-reaches of the Canal with thermal desorption, followed by beneficial reuse off-Site (*e.g.,* landfill daily cover) if possible; 7) off-Site stabilization of the less contaminated sediments dredged from the lower reach of the Canal and the sediments in the other reaches not

impacted by NAPL, followed by beneficial reuse off-Site; 8) excavation and restoration of approximately 475 feet of the filled-in former 1st Street turning basin; 9) excavation and restoration of the portion of the 5th Street turning basin beginning underneath the 3rd Avenue bridge and extending approximately 25 feet to the east and the installation of a barrier or interception system at the eastern boundary of the excavation; 10) implementation of institutional controls incorporating the existing fish consumption advisories (modified, as needed), as well as other controls to protect the integrity of the cap; 11) periodic maintenance of the cap and long-term monitoring to insure that the remedy continues to function effectively; and 12) CSOs controls to significantly reduce overall contaminated solid discharges to the Canal, which include a) construction of two retention tanks to retain discharges from outfalls RH-034 and OH-007; and b) implementation of appropriate engineering controls to ensure that hazardous substances and solids from separated stormwater, including from future upland development projects, are not discharged to the Canal.

23. On March 21, 2014, EPA issued an administrative order for remedial design ("RD"), Index Number CERCLA-02-2014-2001 (the "Dredging RD UAO"), to 31 parties requiring the performance of various pre-RD investigations and analyses, the preparation of biddable plans and specifications for the implementation of the remedy selected in EPA's September 27, 2013 ROD for the Site, other than the CSO controls, and the cleanup and restoration of the former 1st Street turning basin.

24. On May 28, 2014, EPA issued an administrative order for RD, Index Number CERCLA-02-2014-2019 (the "City RD UAO"), to the City requiring the preparation of the RD of the two CSO retention tanks and of the cleanup and restoration of the City-owned former 1st Street turning basin, as well as the City's coordination and participation in the Dredging RD UAO.

25. Pursuant to the City RD UAO, the City performed the RD for the cleanup and restoration of the former 1st Street turning basin. Sampling conducted within the former 1st Street turning basin by the City as part of the RD confirmed the presence of a range of hazardous substances, including naphthalene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, chrysene, dibenzo(a,h)anthracene, indeno(1,2,3-cd)pyrene, naphthalene, phenanthrene, metals and PCBs. The PCBs appear to have been released from the adjacent Powerhouse. The 100% RD for the 1st Street turning basin cleanup and restoration was approved by EPA on June 12, 2019.

26. On April 11, 2019, EPA issued an administrative order requiring a removal action, Index Number CERCLA-02-2019-2010 (the "Bulkhead Removal UAO"), to 28 parties requiring, among other things, the performance of bulkhead and bridge structural support construction and access dredging at the Site in preparation for the RTA 1 dredging and capping. The Bulkhead Removal Order and several additional administrative orders issued by EPA provide for the construction of bulkhead structural support necessary to conduct the RTA 1 dredging and capping.

27. The structural support options available to reinforce bulkhead and bridges at the Canal are limited. Typical reinforcement measures involve installation of new, deeper sheet piling along the existing structure's face, resulting in incremental encroachment on the Canal. NYSDEC seeks to limit and mitigate such encroachment when approving such repairs and upgrades along the Canal. In an effort to mitigate such encroachment, EPA's ROD requires the creation of a wetlands mitigation area for offsetting such encroachment when it cannot be accomplished at the parcel where the structural support is constructed. Such a wetlands mitigation area will be primarily achieved by means of the excavation and restoration of the contaminated, filled-in former 1st Street turning basin.

28. The work required by the Dredging RD UAO, the Bulkhead Removal UAO, and dredging and capping-related requirements of the City RD UAO, is currently being conducted by a group of potentially responsible parties ("PRPs") ("Performing PRPs"). Geosyntec Consultants, Inc. is the EPA-approved primary consultant for the Performing PRPs. The Performing PRPs have retained Jay Cashman, Inc., together with other sub-contractors, to perform certain RD, removal and pre-RA work.

29. On September 30, 2019, the Performing PRPs submitted to EPA the draft 90% design for the RTA 1 dredging and capping. This Order is being issued in advance of EPA's final approval of the 100% design for the RTA 1 dredging and capping to facilitate the technical, legal, contracting, and procurement transition from RD to RA.

30. RD work for the RTA 2 and 3 dredging and capping RA is currently being conducted. EPA intends to complete the oversight, review and approval of the RTA 2 and 3 dredging and capping design so that the remediation of the rest of the Canal can continue seamlessly following the RTA 1 dredging and capping work. In order to ensure that the dredging and capping can continue in a such a manner, this Order requires the Respondents to obtain such staging area(s), by access agreement, lease or acquisition, as are necessary to conduct the dredging and capping for RTA 1 and RTA 2 in a continuous manner.

31. The Operation and Maintenance ("O&M") of the RTA 1 dredging and capping, including the establishment of Institutional Controls ("ICs") is not included in this Order. EPA currently anticipates that such O&M and ICs will be addressed as part of the completed RD and implemented through a later enforcement instrument.

32. The New York City Department of Environmental Protection ("NYCDEP") manages the Flushing Tunnel and Gowanus Pump Station facilities located at Bond and Butler Streets in RTA 1. These facilities are critical infrastructure for the City. To implement the RTA 1 dredging and capping required by this Order at the Flushing Tunnel/Gowanus Pump Station property, EPA has concluded that it may be necessary to temporarily suspend operation of the Flushing Tunnel for a short period, or periods, of time. Consistent with the provisions of this Order and the RTA 1 dredging and capping design, when approved by EPA, EPA will oversee the implementation of

such Work so as to minimize impacts to current water quality in the Canal. Similarly, EPA has concluded that it may be necessary to take steps to modify the operation of the RH-034 outfall during implementation of the Work required by this Order.

33. The New York City Department of Transportation ("NYCDOT") manages the six bridges that cross the Canal, all of which will be required to operate in a timely and coordinated manner to implement the Work required by this Order.

34. The Site includes a "facility" within the meaning of Section 101(9) of CERCLA, 42 U.S.C. § 9601(9).

35. The contamination found at the Site, as identified in these Findings of Fact above, includes hazardous substances as defined in Section 101(14) of CERCLA, 42 U.S.C. § 9601(14).

36. The conditions described in these Findings of Fact above constitute an actual or threatened "release" of a hazardous substance from a facility as defined by Section 101(22) of CERCLA, 42 U.S.C. § 9601(22). Such releases include, but are not limited to, PAHs, PCBs, pesticides, metals, and volatile organic compounds that were discharged into the soil, surface water, groundwater and sediments at the Site.

37. Respondent City of New York is a municipal corporation chartered by the State of New York and is a "person" within the meaning of Section 101(21) of CERCLA, 42 U.S.C. § 9601(21).

38. Each of the other Respondents is a corporation and is a "person" within the meaning of Section 101(21) of CERCLA, 42 U.S.C. § 9601(21).

39. Each Respondent is a liable party with respect to the Site within the meaning of Section 107(a) of CERCLA, 42 U.S.C. § 9607(a).

V. DETERMINATIONS

40. Based on the Findings of Fact and Conclusions of Law set forth above, as well as the administrative record, the actual or threatened release of hazardous substances from the Site may present an imminent and substantial endangerment to the public health, welfare, or the environment within the meaning of Section 106(a) of CERCLA, 42 U.S.C. §9606(a).

41. Solely for purposes of Section 113(j) of CERCLA, 42 U.S.C. § 9613(j), the remedy set forth in the ROD and the Work to be performed by Respondents shall constitute a response action taken or ordered by the President for which judicial review shall be limited to the administrative record.

42. Based upon the Findings of Fact and Conclusions of Law set forth above, as well as the administrative record, the actions required by this Order are necessary to protect the public health or welfare or the environment.

VI. ORDER

43. Based on the Findings of Fact, Conclusions of Law, and Determinations set forth above, and the administrative record, Respondents are hereby ordered to comply with this Order and any modifications to this Order, including, but not limited to, all appendices and all documents incorporated by reference into this Order.

VII. DESIGNATED EPA PROJECT MANAGER AND RESPONDENTS' PROJECT COORDINATOR

44. Respondents have previously designated and EPA has previously approved Respondents' Project Coordinator pursuant to the Dredging RD UAO. Respondents' approved Project Coordinator for the Work under this Order is:

David Himmelheber
Geosyntec Consultants, Inc.
7 Graphics Drive, Suite 106
Ewing, New Jersey 08628
609-493-9012
dhimmelheber@geosyntec.com

The Project Coordinator shall be responsible, on behalf of Respondents, for the implementation of the Work being performed under this Order. The Project Coordinator shall be knowledgeable about all matters relating to the Work being performed under this Order. To the extent possible, the Project Coordinator shall be present on-Site or readily available for EPA to contact during all working days. Notice by EPA in writing to the Project Coordinator shall be deemed notice to Respondents for all matters relating to the Work under this Order and shall be effective upon receipt. Respondents shall retain a Project Coordinator until EPA issues a Termination and Satisfaction in accordance with Paragraph 101.

45. If Respondents change their Project Coordinator, Respondents shall submit the name, address, qualifications, email address and telephone number of the new Project Coordinator to the EPA Remedial Project Manager ("RPM") specified in Paragraph 51, below. The new Project Coordinator shall not be an attorney engaged in the practice of law. He or she shall have the technical expertise sufficient to adequately oversee all aspects of the Work contemplated by this Order.

46. Selection of a new Project Coordinator shall be subject to approval by EPA in writing. If EPA disapproves a proposed Project Coordinator, Respondents shall propose a different person and notify EPA of that person's name, address, telephone number and qualifications within seven (7) days following EPA's disapproval. Respondents may change their Project Coordinator if EPA has received written notice at least ten (10) days prior to the desired change. The initial notification may be made orally, but shall be promptly followed by a written notice. All changes of the Project Coordinator shall be subject to EPA approval.

47. Respondents have previously retained, and EPA has previously approved of, Geosyntec Consultants, Inc. as their Supervising Contractor pursuant to the Dredging RD UAO, and this contractor shall serve as Supervising Contractor for the Work hereunder. Respondents shall notify EPA of the name and qualifications of any change to the Supervising Contractor under this Order, as well as any other contractor or subcontractor proposed to perform Work under this Order, at least ten (10) days prior to commencement of Work by the proposed Supervising Contractor.

48. All activities required of Respondents under the terms of this Order shall be performed only by well-qualified persons possessing all necessary permits, licenses, and other authorizations required by federal, state, and/or local governments consistent with Section 121 of CERCLA, 42 U.S.C. § 9621, and all Work conducted pursuant to this Order shall be performed in accordance with prevailing professional standards. All plans and specifications shall be prepared under the supervision of, and signed/certified by, a licensed New York professional engineer.

49. EPA retains the right to disapprove any or all of the contractors and/or subcontractors proposed by Respondents to conduct the Work. Within ten (10) days of the Effective Date, Respondents shall notify EPA of the name and qualifications of all contractors and subcontractor proposed to perform Work under this Order. If EPA disapproves in writing any of Respondents' proposed contractors or subcontractors, Respondents shall propose a different contractor within seven (7) days of receipt of EPA's disapproval.

50. Respondents shall provide a copy of this Order to each contractor, subcontractor, laboratory, or consultant approved and retained to perform any Work under this Order, within five (5) days after the Effective Date or on the date such services are retained, whichever date occurs later. Respondents shall also provide a copy of this Order to each person representing Respondents with respect to the Site or the Work and shall condition all contracts and subcontracts entered into hereunder upon performance of the Work in conformity with the terms of this Order and in compliance with all applicable laws and regulations. Notwithstanding the terms of any contract, Respondents are responsible for compliance with this Order and for ensuring that its contractors, subcontractors and agents comply with this Order, and perform any Work in accordance with this Order.

51. EPA has designated Christos Tsiamis of the New York Remediation Branch, Superfund and Emergency Management Division, EPA Region 2, as its RPM for the Site. Except as otherwise provided in this Order, Respondents shall direct all submissions required by this Order to the RPM via e-mail at tsiamis.christos@epa.gov and by regular mail, at U.S. EPA, Region 2, 290 Broadway, 19th Floor, New York, NY 10007. EPA will notify Respondent's Project Coordinator if EPA designates a different RPM for the Order.

52. EPA's RPM shall have the authority lawfully vested in an RPM by the NCP. In addition, EPA's RPM shall have the authority, consistent with the NCP, to halt any Work required by this Order and to take any necessary response action when the RPM determines that conditions at the Site may present an immediate endangerment to public health, welfare or the environment. The absence of the RPM from the area under study pursuant to this Order shall not be cause for the stoppage or delay of Work.

VIII. WORK TO BE PERFORMED

53. Respondents shall perform all actions necessary to complete the Work set forth below:

a. Respondents shall implement the RTA 1 dredging and capping in accordance with the RTA 1 100% design within the time frames set forth in the EPA-approved work schedule set forth therein.

b. For purposes of implementing the RTA 1 dredging and capping Work, the City shall ensure that NYCDOT complies with all opening, closing and use restrictions for the City-owned and operated bridges as may be determined by EPA to be needed for the Work such that there is no delay in the scheduled implementation of such Work.

c. For purposes of implementing the RTA 1 dredging and capping Work, the City shall ensure that NYCDEP temporarily suspends operation of the Flushing Tunnel for such period, or periods, of time as may be determined by EPA to be needed for the Work in the vicinity of NYCDEP's Flushing Tunnel/Gowanus Pump Station property at Bond and Butler Streets. The City shall also ensure that NYCDEP promptly facilitates the implementation of any measures as may be required for the Work in the vicinity of the RH-034 CSO outfall or other potentially affected City infrastructure.

d. With respect to implementation of the RTA 1 dredging and capping Work as set forth above, the City shall ensure that NYCDEP and NYCDOT provide such construction support resources as are necessary and appropriate for protection of the affected critical infrastructure and to ensure the proper and prompt completion of such Work.

e. Respondents shall implement the EPA-approved 100% design for the cleanup and restoration of the former 1st Street turning basin, attached hereto as Appendix B. Within sixty (60) days of the Effective Date of this Order, Respondents shall submit to EPA, for review and approval, a proposed schedule for implementing the Work required by this subparagraph.

f. In preparation for RTA 1 and 2 dredging and capping Work, Respondents shall, with EPA approval and oversight, obtain such staging areas, by access agreement, lease or acquisition, as are necessary to conduct the dredging and capping for RTA 1 and RTA 2 in a continuous manner.

g. Respondents to this Order shall coordinate and cooperate with respondents to any administrative orders issued by EPA for the construction of any bulkheads in RTA 1 which are necessary for implementation of the Work and the ROD.

54. Health and Safety Plan. Respondents shall submit for EPA review and comment a Health and Safety Plan ("HASP") that ensures the protection of on-Site workers and the public during the performance of on-Site Work under this Order. This HASP shall be prepared in accordance with "OSWER Integrated Health and Safety Program Operating Practices for OSWER Field Activities," Pub. 9285.0-OlC (Nov. 2002), available on the National Service Center for Environmental Publications database at https://www.epa.gov/nscep, and "EPA's Emergency Responder Health and Safety Manual," OSWER Directive 9285.3-12 (July 2005 and updates), available at https://www.epaosc.org/_HealthSafetyManual/manual-index.htm. In addition, the HASP shall comply with all currently applicable Occupational Safety and Health Administration ("OSHA") regulations found at 29 C.F.R. Part 1910. If EPA determines that it is appropriate, the HASP shall also include contingency planning. Respondents shall incorporate all changes to the HASP recommended by EPA and shall implement the HASP during the pendency of the Work. If any of the Work performed by Respondents pursuant to this Order requires alteration of the HASP, Respondents shall submit to EPA for review such amendments to the HASP prior to the performance of such Work.

55. Quality Assurance, Sampling, and Data Analysis. Respondents shall use Quality Assurance/Quality Control ("QA/QC") and other technical activities and chain of custody procedures for all samples consistent with "EPA Requirements for Quality Assurance Project Plans (QA/R5)" (EPA/240/B-01/003, March 2001, reissued May 2006), "Guidance for Quality Assurance Project Plans (QA/G-5)" (EPA/240/R-02/009, December 2002), and subsequent amendments to such guidelines. Amendments to such guidelines shall apply only to procedures conducted after such notification.

Prior to the commencement of any data collection or other monitoring under this Order, Respondents shall submit to EPA for approval a Quality Assurance Project Plan ("QAPP") that is consistent with EPA-approved plans, the NCP, the Uniform Federal Policy for Implementing

Quality Systems ("UFP-QS"), EPA-505-F-03-001, March 2005; Uniform Federal Policy for Quality Assurance Project Plans ("UFP-QAPP"), Parts 1, 2, and 3, EPA-505-B-04-900A, B, and C, March 2005 and March 2012 or newer, and other guidance documents referenced in the aforementioned guidance documents as well as http://www2.epa.gov/fedfac/assuring-quality-federal-cleanups. Respondents shall ensure that EPA personnel and EPA's authorized representatives are allowed access at reasonable times to all laboratories utilized by Respondents in implementing this Order. In addition, Respondents shall ensure that such laboratories shall analyze all samples submitted by EPA pursuant to the QAPP for quality assurance, quality control, and technical activities that will satisfy the stated performance criteria as specified in the QAPP and that sampling and field activities are conducted in accordance with "EPA QA Field Activities Procedure" (https://www.epa.gov/irmpoli8/epa-qa-field-activities-procedures). Respondents shall ensure that the laboratories it utilizes for the analysis of samples taken pursuant to this Order meet the competency requirements set forth in EPA's "Policy to Assure Competency of Laboratories, Field Sampling, and Other Organizations Generating Environmental Measurement Data under Agency-Funded Acquisitions" (http://www.epa.gov/fem/pdfs/fem-lab-competency-policy.pdf) and that the laboratories perform all analyses according to accepted EPA methods. Accepted EPA methods consist of, but are not limited to, methods that are documented in the EPA's Contract Laboratory Program (http://www.epa.gov/superfund/programs/clp/), SW 846 "Test Methods for Evaluating Solid Waste, Physical/Chemical Methods" (http://www.epa.gov/epawaste/hazard/testmethods/sw846/online/index.htm), "Standard Methods for the Examination of Water and Wastewater" (http://www.standardmethods.org/), 40 C.F.R. Part 136, "Air Toxics - Monitoring Methods" (http://www.epa.gov/ttnamti1/airtox.html)," and any amendments made thereto during the course of the implementation of this Order. However, upon approval by EPA, Respondents may use other appropriate analytical method(s), as long as (a) QA/QC criteria are contained in the method(s) and the method(s) are included in the QAPP, (b) the analytical method(s) are at least as stringent as the methods listed above, and (c) the method(s) have been approved for use by a nationally recognized organization responsible for verification and publication of analytical methods, *e.g.,* EPA, American Society for Testing and Materials, National Institute for Occupational Safety and Health, OSHA, etc. Respondents shall ensure that all laboratories it uses for the analysis of samples taken pursuant to this Order have a documented Quality System that complies with ANSI/ASQC E-4-2004, "Quality Systems for Environmental Data and Technology Programs: Requirements with Guidance for Use" (American National Standard, 2004, and "EPA Requirements for Quality Management Plans (QA/R-2)" (EPA/240/B-01/002, March 2001, reissued May 2006), or equivalent documentation as determined by EPA. EPA may consider Environmental Response Laboratory Network laboratories, laboratories accredited under the National Environmental Laboratory Accreditation Program, or laboratories that meet International Standardization Organization (ISO 17025) standards or other nationally recognized programs (http://www.epa.gov/fem/accredit.htm) as meeting the Quality System requirements. Respondents shall ensure that all field methodologies utilized in collecting samples for subsequent analysis pursuant to this Order are conducted in accordance with the procedures set forth in the QAPP approved by EPA.

Upon request, Respondents shall provide split or duplicate samples to EPA or its authorized representatives. Respondents shall notify EPA not less than seven (7) days in advance of any sample collection activity unless shorter notice is agreed to by EPA. In addition, EPA has the right to collect any additional samples that EPA deems necessary. Upon request, EPA will provide to Respondents split or duplicate samples of any samples it collects as part of EPA's oversight of Respondents' implementation of the Work.

Respondents shall submit to EPA the results of all sampling and/or tests or other data obtained or generated by or on behalf of Respondents with respect to the implementation of this Order.

56. Respondents shall assure that field personnel used by Respondents are properly trained in the use of field equipment and in chain-of-custody procedures.

57. Respondents shall conduct all Work under this Order in compliance with CERCLA, the NCP, and all applicable EPA guidances, policies, and procedures, including, without limitation, EPA Region 2's "Clean and Green Policy" which may be found at: https://www.epa.gov/greenercleanups/epa-region-2-clean-and-green-policy

58. Off-Site Shipments

a. Respondents may ship hazardous substances, pollutants, and contaminants from the Site to an off-Site facility only if they comply with Section 121(d)(3) of CERCLA, 42 U.S.C. § 9621(d)(3), and 40 C.F.R. § 300.440. Respondents will be deemed to be in compliance with CERCLA § 121(d)(3) and 40 C.F.R. § 300.440 regarding a shipment if Respondents obtain a prior determination from EPA that the proposed receiving facility for such shipment is acceptable under the criteria of 40 C.F.R. § 300.440(b).

b. Respondents may ship Waste Material from the Site to an out-of-state waste management facility only if, prior to any shipment, they provide written notice to the appropriate state environmental official in the receiving facility's state and to the OSC. This notice requirement will not apply to any off-Site shipments when the total quantity of all such shipments will not exceed ten cubic yards. The written notice must include the following information, if available: 1) the name and location of the receiving facility; 2) the type and quantity of Waste Material to be shipped; 3) the schedule for the shipment; and 4) the method of transportation. Respondents shall also notify the state environmental official referenced above and the OSC of any major changes in the shipment plan, such as a decision to ship the Waste Material to a different out-of-state facility. Respondents shall provide the notice after the award of the contract for the response action and before the Waste Material is shipped.

c. Respondents may ship Investigation Derived Waste ("IDW") from the Site to an

off-Site facility only if they comply with Section 121(d)(3) of CERCLA, 42 U.S.C. § 9621(d)(3), 40 C.F.R. § 300.440, EPA's "Guide to Management of Investigation Derived Waste," OSWER 9345.3-03FS (Jan. 1992), and any IDW-specific requirements contained in the 100% design. Wastes shipped off-Site to a laboratory for characterization, and RCRA hazardous wastes that meet the requirements for an exemption from RCRA under 40 C.F.R. § 261.4(e) shipped off-Site for treatability studies, are not subject to 40 C.F.R. § 300.440.

59. As appropriate during the course of implementation of the actions required of Respondents pursuant to this Order, Respondents or their consultants or contractors, acting through the Project Coordinator, may confer with EPA concerning the required actions. Based upon new circumstances or new information not in the possession of EPA on the Effective Date of this Order, the Project Coordinator may request, in writing, EPA approval of modification(s) to the EPA-approved plans. Only modifications approved by EPA in writing shall be deemed effective. Upon approval by EPA, such modifications shall be deemed incorporated in this Order and shall be implemented by Respondents.

XI. EPA APPROVAL OF PLANS AND OTHER SUBMISSIONS

60. If EPA disapproves or otherwise requires any modifications to any plan, report or other item required to be submitted to EPA for approval pursuant to this Order, Respondents shall have fourteen (14) days from the receipt of notice of such disapproval or the required modifications to correct any deficiencies and resubmit the plan, report, or other written document to EPA for approval, unless a shorter or longer period is specified in the notice. Any notice of disapproval will include an explanation of why the plan, report, or other item is being disapproved. Respondents shall address each of the comments and resubmit the plan, report, or other item with the required changes within the time stated above. At such time as EPA determines that the plan, report, or other item is acceptable, EPA will transmit to Respondents a written statement to that effect.

61. If any plan, report, or other item required to be submitted to EPA for approval pursuant to this Order is disapproved by EPA, even after being resubmitted following Respondents' receipt of EPA's comments on the initial submittal, Respondents shall be deemed to be out of compliance with this Order. If any resubmitted plan, report, or other item, or portion thereof, is disapproved by EPA, EPA may again direct Respondents to make the necessary modifications thereto, and/or EPA may amend or develop the item(s) and recover the costs of doing so from Respondents. Respondents shall implement any such item(s) as amended or developed by EPA.

62. EPA shall be the final arbiter in any dispute regarding the sufficiency or acceptability of all documents submitted and all activities performed pursuant to this Order. EPA may modify those documents and/or perform or require the performance of additional work unilaterally.

63. All plans, reports and other submittals required to be submitted to EPA pursuant to this Order, upon approval by EPA, shall be deemed to be incorporated in and an enforceable part of this Order.

X. SUBMISSION OF PLANS AND REPORTING REQUIREMENTS

64. Reporting.

Commencing on the thirtieth (30th) day after the Effective Date, unless there is field work at the Site, Respondents shall provide monthly, or as otherwise requested by EPA, progress reports. Whenever, during the implementation of this Order, Respondents are engaged in active field work, Respondents shall provide, via email to EPA's Project Coordinator, at least one (1) week advance notice of all field activities. During active field work, Respondents shall provide EPA with written progress reports, including photo documentation, every seven (7) days beginning from the date of commencement of field work. After active field work has been completed, Respondents shall resume monthly written progress reports, commencing thirty (30) days after the submission of the last weekly written progress report. All progress reports shall fully describe all actions and activities undertaken pursuant to this Order. Such progress reports shall, among other things: a) describe the actions taken toward achieving compliance with this Order during the previous week; b) include all results of sampling and tests and all other data received by Respondents after the most recent progress report submitted to EPA; c) describe all actions which are scheduled for the next week; d) provide other information relating to the progress of Work as is customary in the industry and e) include information regarding percentage of completion, all delays encountered or anticipated that may affect the future schedule for completion of the Work required hereunder, and a description of all efforts made to mitigate those delays or anticipated delays.

b. Respondents shall submit copies of all plans, reports or other submissions required by this Order or any approved work plan as set forth below. Any electronic submissions must be in a format that is compatible with EPA software and in database files and sizes to be specified by EPA. All deliverables shall be submitted to the following:

2 copies: (1 bound, 1 electronic)	Remedial Project Manager - Gowanus Canal Site Superfund and Emergency Management Division U.S. Environmental Protection Agency, Region 2 290 Broadway, 19th Floor New York, New York 10007-1866 tsiamis.christos@epa.gov
1 copy: (via email or electronic)	Chief, New York/Caribbean Superfund Branch Office of Regional Counsel United States Environmental Protection Agency, Region 2

290 Broadway, 17th Floor
New York, New York 10007-1866
Attn: Gowanus Canal Superfund Site Attorney
carr.brian@epa.gov

1 copy each: (via email)
N.Y.S. Department of Environmental Conservation
patrick.foster@dec.ny.gov
aaron.fischer@dec.ny.gov
gerard.burke@dec.ny.gov
heidi.dudek@dec.ny.gov

XI. OVERSIGHT

65. During the implementation of the requirements of this Order, Respondents and their contractor(s) and subcontractors shall be available for such conferences with EPA and inspections by EPA or its authorized representatives as EPA may determine are necessary to adequately oversee the Work being carried out or to be carried out by Respondents, including inspections at the Site and at laboratories where analytical work is being done hereunder.

66. Respondents and their employees, agents, contractor(s), and consultant(s) shall cooperate with EPA in its efforts to oversee Respondents' implementation of this Order.

XII. COMMUNITY RELATIONS

67. Respondents shall cooperate with EPA in providing information relating to the Work required hereunder to the public. As requested by EPA, Respondents shall participate in the preparation of all appropriate information disseminated to the public; participate in public meetings that may be held or sponsored by EPA to explain activities at or concerning the Property; and provide a suitable location for public meetings, as needed.

XIII. SITE ACCESS AND ACCESS TO INFORMATION

68. EPA, NYSDEC and their designated representatives, including, but not limited to, employees, agents, contractor(s) and consultant(s) thereof, shall be permitted to observe the Work carried out pursuant to this Order. Respondents shall at all times permit EPA, NYSDEC and their designated representatives full access to and freedom of movement at the Site and any other premises where Work under this Order is to be performed for purposes of inspecting or observing Respondents' progress in implementing the requirements of this Order, verifying the information submitted to EPA by Respondents, conducting investigations relating to contamination at the Site or for any other purpose EPA determines to be reasonably related to EPA oversight of the implementation of this Order.

69. In the event that action under this Order is to be performed in areas owned by or in possession of a person other than Respondents, Respondents shall use best efforts to obtain access agreements from such persons within thirty (30) days of the Effective Date of this Order for purposes of implementing the requirements of this Order. Such agreements shall provide access not only for Respondents, but also for EPA and its designated representatives or agents, as well as NYSDEC and its designated representatives or agents. Such agreements shall specify that Respondents are not EPA's representative with respect to liability associated with Site activities. If such access agreements are not obtained by Respondents within the time period specified herein, Respondents shall immediately notify EPA of the failure to obtain access and shall include in that notification a summary of the steps Respondents have taken to attempt to obtain access. For purposes of this Paragraph, "best efforts" include the payment of reasonable sums of money in consideration of access. EPA may use its legal authorities to obtain access for Respondents, may perform those response actions with EPA contractors at the property in question or may terminate the Order if Respondents cannot obtain access agreements. If EPA performs those tasks or activities with EPA contractors and does not terminate the Order, Respondents shall perform all other activities not requiring access to that property. Respondents shall integrate the results of any such tasks undertaken by EPA into its reports and deliverables.

70. Upon request, Respondents shall provide EPA with access to all records and documentation related to the conditions at the Site, hazardous substances found at or released from the Site, and the actions conducted pursuant to this Order except for those items, if any, subject to the attorney-client or work product privilege. All data, information and records created, maintained, or received by Respondents or their contractor(s) or consultant(s) in connection with implementation of the Work under this Order, including, but not limited to, contractual documents, invoices, receipts, work orders and disposal records shall, without delay, be made available to EPA upon request, subject to the same privileges specified above in this Paragraph. EPA shall be permitted to copy all such documents. Respondents shall submit to EPA upon receipt the results of all sampling or tests and all other technical data generated by Respondents or their contractor(s), or on Respondents' behalf, in connection with the implementation of this Order.

71. Notwithstanding any other provision of this Order, EPA hereby retains all its information gathering, access, and inspection authority under CERCLA, the Resource Conservation and Recovery Act and any other applicable statutes or regulations.

XIV. RECORD RETENTION

72. During the pendency of this Order and for a minimum of 10 years after EPA provides Notice of Termination and Satisfaction pursuant to Paragraph 101, each Respondent shall preserve and retain all non-identical copies of Records (including Records in electronic form) now in its possession or control or that come into its possession or control that relate in any

manner to liability under CERCLA with respect to the Site. Each Respondent must also retain, and instruct its contractors and agents to preserve, for the same period of time specified above, all non-identical copies of the last draft or final version of any Records (including Records in electronic form) now in its possession or control or that come into its possession or control that relate in any manner to the performance of the Work, provided, however, that each Respondent (and its contractor and agents) must retain, in addition, copies of all data generated during performance of the Work and not contained in the aforementioned Records to be retained. Each of the above record retention requirements shall apply regardless of any corporate retention policy to the contrary.

73. At the end of the ten (10)-year period referred to in Paragraph 72, Respondents shall notify EPA at least thirty (30) days before any such document or information is destroyed that such documents and information are available for inspection. Upon request, Respondents shall provide EPA with the originals or copies of such documents and information.

74. All documents submitted by Respondents to EPA during the implementation of this Order shall be available to the public unless claimed as privileged or confidential pursuant to applicable law. In addition, EPA may release all such documents to NYSDEC, and NYSDEC may make those documents available to the public unless Respondents conform to applicable New York State law and regulations regarding confidentiality. Respondents shall not assert a claim of privilege or confidentiality regarding any monitoring or hydrogeologic data, any information specified under Section 104(e)(7)(F) of CERCLA, or any other chemical, scientific or engineering data relating to the Work performed hereunder.

XV. COMPLIANCE WITH OTHER LAWS

75. Respondents shall undertake all action that this Order requires in accordance with the requirements of all applicable local, state, and federal laws and regulations, unless an exemption from such requirements is specifically provided by law or in this Order. The activities conducted pursuant to this Order, if approved by EPA, shall be considered consistent with the NCP.

76. As provided in Section 121(e) of CERCLA, 42 U.S.C. § 9621(e), and the NCP, no permit shall be required for any portion of the Work conducted entirely on-Site, including temporary modifications to the operation of the Flushing Tunnel and RH-034 outfall. Where any portion of the Work requires a federal or state permit or approval, Respondents shall submit timely applications and take all other actions necessary to obtain and to comply with all such permits or approvals.

77. This Order is not, and shall not be construed to be, a permit issued pursuant to any federal or state statute or regulation.

XVI. EMERGENCY RESPONSE AND NOTIFICATION OF RELEASES

78. Upon the occurrence of any event during performance of the Work required hereunder which, pursuant to Section 103 of CERCLA, 42 U.S.C. §9603, requires reporting to the National Response Center (telephone number 800-424-8802), Respondents shall immediately orally notify the Chief of the Response and Prevention Branch of the Superfund and Emergency Management Division of EPA, Region 2, at 732-321-6656 of the incident or Site conditions. Respondents shall also submit a written report to EPA within seven (7) days after the onset of such an event, setting forth the events that occurred and the measures taken or to be taken, if any, to mitigate any release or endangerment caused or threatened by the release and to prevent the reoccurrence of such a release. The reporting requirements of this Paragraph are in addition to, not in lieu of, reporting under CERCLA Section 103, 42 U.S.C. §9603, and Section 304 of the Emergency Planning and Community Right-To-Know Act of 1986, 42 U.S.C. §11004.

79. In the event of any action or occurrence during Respondents' performance of the requirements of this Order that causes or threatens to cause a release of a hazardous substance or which may present an immediate threat to public health, welfare or the environment, or cause potential or actual adverse impacts to upland structures, Respondents shall immediately take all appropriate action to prevent, abate or minimize the threat and shall immediately notify EPA as provided in the preceding paragraph. Respondents shall take such action in accordance with applicable provisions of this Order. In the event that EPA determines that a) the activities performed pursuant to this Order, b) significant changes in conditions at the Site or c) emergency circumstances occurring at the Site pose a threat to human health or the environment, EPA may direct Respondents to stop further implementation of any actions pursuant to this Order or to take other and further actions reasonably necessary to abate the threat.

80. Nothing in the preceding paragraph shall be deemed to limit any authority of the United States to take, direct, or order all appropriate action to protect human health and the environment or to prevent, abate, or minimize an actual or threatened release of hazardous substances on, at, or from the Site.

XVII. MODIFICATIONS

81. No informal advice, guidance, suggestion or comment by EPA regarding reports, plans, specifications, schedules or any other writing submitted by Respondents shall relieve Respondents of their obligation to obtain such formal approval as may be required by this Order and to comply with all requirements of this Order.

XVIII. DELAY IN PERFORMANCE

82. Any delay in performance of the Work under this Order that, in EPA's judgment, is not properly justified by Respondents under the terms of the paragraph below, shall be considered a violation of this Order. Any delay in performance of this Order shall not affect Respondents' obligations to perform all obligations fully under the terms and conditions of this Order.

83. Respondents shall notify EPA of any delay or anticipated delay in performing any requirement of this Order. Such notification shall be made by telephone to EPA's Project Coordinator as soon as Respondents know that a delay might occur. Respondents shall adopt all reasonable measures to avoid or minimize any such delay. Within two (2) days after notifying EPA by telephone, Respondents shall provide written notification fully describing the nature of the delay, any justification for the delay, any reason why Respondents should not be held strictly accountable for failing to comply with any relevant requirements of this Order, the measures planned and taken to minimize the delay, and a schedule for implementing the measures that have been or will be taken to mitigate the effect of the delay. Increased cost or expense associated with the implementation of the activities called for in this Order is not a justification for any delay in performance.

XIX. ENFORCEMENT/WORK TAKEOVER AND RESERVATION OF RIGHTS

84. Any willful violation of, or failure or refusal to comply with any provision of this Order may subject Respondents to civil penalties up to the maximum amount authorized by law. CERCLA § 106(b)(1), 42 U.S.C. § 9606(b)(1). As of the date of issuance of this Order, the statutory maximum amount is fifty-eight thousand three hundred twenty-eight dollars ($58,328) per violation per day. This maximum amount may increase in the future, as EPA amends its civil penalty amounts through rulemaking pursuant to the 1990 Federal Civil Penalties Inflation Adjustment Act (Public Law 101-410, codified at 28 U.S.C. § 2461), as amended by the 2015 Federal Civil Penalties Inflation Adjustment Act Improvement Act (Section 701 of Public Law 114-74)). The maximum amount to be applied to this violation may be set as the most recent maximum amount set forth in 40 CFR Section 19.4 as of the date that the U.S. District Court assesses any such penalty. In the event of such willful violation, or failure or refusal to comply, EPA may unilaterally carry out the actions required by this Order, pursuant to Section 104 of CERCLA, 42 U.S.C. § 9604, and/or may seek judicial enforcement of this Order pursuant to Section 106 of CERCLA, 42 U.S.C. § 9606. Nothing herein shall limit the power and authority of EPA or the United States to take, direct, or order all actions necessary to protect public health, welfare, or the environment or to prevent, abate, or minimize an actual or threatened release of hazardous substances, pollutants or contaminants, or hazardous or solid waste on, at, or from the Site. Further, nothing herein shall prevent EPA from seeking legal or equitable relief to enforce the terms of this Order, from taking other legal or equitable action as it deems appropriate, or from requiring Respondents in the future to perform additional activities pursuant to CERCLA or any other applicable law. EPA reserves

the right to bring an action against Respondents under Section 107 of CERCLA, 42 U.S.C. § 9607, for recovery of any response costs incurred by the United States related to this Order or the Site. In addition, nothing in this Order shall limit EPA's authority under Section XXI (Financial Assurance). Respondents may also be subject to punitive damages at least equal to but not more than three times the amount of any cost incurred by the United States as a result of such failure to comply, as provided in Section 107(c)(3) of CERCLA, 42 U.S.C. § 9607(c)(3).

XX. OTHER CLAIMS

85. By issuance of this Order, the United States and EPA assume no liability for injuries or damages to persons or property resulting from any acts or omissions of Respondents or Respondents' employees, agents, contractors, or consultants in carrying out any action or activity pursuant to this Order. The United States or EPA shall not be held out as or deemed a party to any contract entered into by Respondents or their directors, officers, employees, agents, successors, representatives, assigns, contractors, or consultants in carrying out actions pursuant to this Order.

86. Nothing in this Order constitutes or shall be construed as a satisfaction of or release from any claim or cause of action against Respondents or any person not a party to this Order for any liability that Respondents or other persons may have under CERCLA, other statutes, or the common law, including but not limited to any claims of the United States for injunctive relief, costs, damages, and interest under Sections 106(a) and 107 of CERCLA, 42 U.S.C. §§ 9606(a) and 9607. Nothing herein shall constitute a finding that any Respondent is the only responsible party with respect to the release and threatened release of hazardous substances at and from the Site.

87. Nothing in this Order shall affect any right, claim, interest, defense, or cause of action of any party hereto with respect to third parties.

88. Nothing in this Order shall be construed to constitute preauthorization under Section 111(a)(2) of CERCLA, 42 U.S.C. §9611(a)(2), and 40 CFR § 300.700(d).

XXI. FINANCIAL ASSURANCE

89. In order to ensure completion of the Work, Respondents shall secure financial assurance, initially in the amount of $125,000,000 ("Estimated Cost of the Work"). The financial assurance must be one or more of the mechanisms listed below, in a form substantially identical to the relevant sample documents available from EPA or under the "Financial Assurance - Orders" category on the Cleanup Enforcement Model Language and Sample Documents Database at https://cfpub.epa.gov/compliance/models/, and satisfactory to EPA. Respondents may use multiple mechanisms if they are limited to trust funds, surety bonds guaranteeing payment, and/or letters of credit.

a. A trust fund: 1) established to ensure that funds will be available as and when needed for performance of the Work; 2) administered by a trustee that has the authority to act as a trustee and whose trust operations are regulated and examined by a federal or state agency; and 3) governed by an agreement that requires the trustee to make payments from the fund only when the Director of the Superfund and Emergency Management Divisions advises the trustee in writing that: i) payments are necessary to fulfill the affected Respondents' obligations under the Order; or ii) funds held in trust are in excess of the funds that are necessary to complete the performance of Work in accordance with this Order;

b. A surety bond, issued by a surety company among those listed as acceptable sureties on federal bonds as set forth in Circular 570 of the U.S. Department of the Treasury, guaranteeing payment or performance in accordance with Paragraph 95 (Access to Financial Assurance);

c. An irrevocable letter of credit, issued by an entity that has the authority to issue letters of credit and whose letter-of-credit operations are regulated and examined by a federal or state agency, guaranteeing payment in accordance with Paragraph 95 (Access to Financial Assurance);

d. A demonstration by a Respondent that it meets the relevant financial test criteria of Paragraph 92; or

e. A guarantee to fund or perform the Work executed by a company 1) that is a direct or indirect parent company of a Respondent or has a "substantial business relationship" (as defined in 40 C.F.R. § 264.141(h)) with a Respondent; and 2) can demonstrate to EPA's satisfaction that it meets the financial test criteria of Paragraph 92.

90. **Standby Trust**. If Respondents seek to establish financial assurance by using a surety bond, a letter of credit, or a corporate guarantee, Respondents shall at the same time establish and thereafter maintain a standby trust fund, which must meet the requirements specified in Paragraph 89.a., and into which payments from the other financial assurance mechanism can be deposited if the financial assurance provider is directed to do so by EPA pursuant to Paragraph 95 (Access to Financial Assurance). An originally signed duplicate of the standby trust agreement must be submitted, with the other financial mechanism, to EPA in accordance with Paragraph 91. Until the standby trust fund is funded pursuant to Paragraph 95 (Access to Financial Assurance), neither payments into the standby trust fund nor annual valuations are required.

91. Within thirty (30) days after the Effective Date, Respondents shall submit to EPA proposed financial assurance mechanisms in draft form in accordance with Paragraph 89 for EPA's review. Within sixty (60) days after the Effective Date, or 30 days after EPA's approval of the form and substance of Respondents' financial assurance, whichever is later, Respondents shall secure all executed and/or otherwise finalized mechanisms or other documents consistent

with the EPA-approved form of financial assurance and shall submit such mechanisms and documents to the EPA contacts set forth in Paragraph 64.b. and to:

Chief, Resource Management/Cost Recovery Section
Program Support Branch
Superfund and Emergency Management Division
U.S. Environmental Protection Agency, Region 2
290 Broadway, 18th Floor
New York, New York 10007-1866
keating.robert@epa.gov.

92. Respondents seeking to provide financial assurance by means of a demonstration or guarantee under Paragraph 89.d. or Paragraph 89.e. must, within sixty (60) days after the Effective Date:

a. Demonstrate that:

(1) the affected Respondent or guarantor has:

i. Two of the following three ratios: a ratio of total liabilities to net worth less than 2.0; a ratio of the sum of net income plus depreciation, depletion, and amortization to total liabilities greater than 0.1; and a ratio of current assets to current liabilities greater than 1.5;

ii. Net working capital and tangible net worth each at least six times the sum of the Estimated Cost of the Work and the amounts, if any, of other federal, state, or tribal environmental obligations financially assured through the use of a financial test or guarantee;

iii. Tangible net worth of at least $10 million; and

iv. Assets located in the United States amounting to at least 90 percent of total assets or at least six times the sum of the Estimated Cost of the Work and the amounts, if any, of other federal, state, or tribal environmental obligations financially assured through the use of a financial test or guarantee; or

(2) The affected Respondent or guarantor has:

i. A current rating for its senior unsecured debt of AAA, AA, A, or BBB as issued by Standard and Poor's or Aaa, Aa, A or Baa as issued by Moody's;

v. Tangible net worth at least six times the sum of the Estimated Cost of the Work and the amounts, if any, of other federal, state, or tribal environmental obligations financially assured through the use of a financial test or guarantee;

vi. Tangible net worth of at least $10 million; and

vii. Assets located in the United States amounting to at least 90 percent of total assets or at least six times the sum of the Estimated Cost of the Work and the amounts, if any, of other federal, state, or tribal environmental obligations financially assured through the use of a financial test or guarantee; and

b. Submit to EPA for the affected Respondent or guarantor: 1) a copy of an independent certified public accountant's report of the entity's financial statements for the latest completed fiscal year, which must not express an adverse opinion or disclaimer of opinion; and 2) a letter from its chief financial officer and a report from an independent certified public accountant substantially identical to the sample letter and reports available from EPA or under the "Financial Assurance – Orders" subject list category on the Cleanup Enforcement Model Language and Sample Documents Database at https://cfpub.epa.gov/compliance/models/.

93. Respondents shall diligently monitor the adequacy of the financial assurance. If any Respondent becomes aware of any information indicating that the financial assurance provided under this Section is inadequate or otherwise no longer satisfies the requirements of this Section, such Respondent shall notify EPA of such information within 30 days. If EPA determines that the financial assurance provided under this Section is inadequate or otherwise no longer satisfies the requirements of this Section, EPA will notify the affected Respondent of such determination. Respondents shall, within 30 days after notifying EPA or receiving notice from EPA under this Paragraph, secure and submit to EPA for approval a proposal for a revised or alternative financial assurance mechanism that satisfies the requirements of this Section. Respondents shall follow the procedures of Paragraph 96 (Modification of Amount, Form, or Terms of Financial Assurance) in seeking approval of, and submitting documentation for, the revised or alternative financial assurance mechanism. Respondents' inability to secure financial assurance in accordance with this Section does not excuse performance of any other obligation under this Order.

94. Respondents providing financial assurance by means of a demonstration or guarantee under Paragraph 89.d. or Paragraph 89.e. must also:

a. Annually resubmit the documents described in Paragraph 92.b. within 90 days after the close of the affected Respondent's or guarantor's fiscal year;

b. Notify EPA within 30 days after the affected Respondent or guarantor determines that it no longer satisfies the relevant financial test criteria and requirements set forth in this Section; and

c. Provide to EPA, within 30 days of EPA's request, reports of the financial condition of the affected Respondent or guarantor in addition to those specified in Paragraph 92.b.; EPA may make such a request at any time based on a belief that the affected Respondent or guarantor may no longer meet the financial test requirements of this Section.

95. **Access to Financial Assurance**

a. If EPA determines that Respondents 1) have ceased implementation of any portion of the Work, 2) are seriously or repeatedly deficient or late in their performance of the Work, or 3) are implementing the Work in a manner that may cause an endangerment to human health or the environment, EPA may issue a written notice ("Performance Failure Notice") to both Respondents and the financial assurance provider regarding the affected Respondents' failure to perform. Any Performance Failure Notice issued by EPA will specify the grounds upon which such notice was issued and will provide Respondents a period of 10 days within which to remedy the circumstances giving rise to EPA's issuance of such notice. If, after expiration of the 10-day period specified in this Paragraph, Respondents have not remedied to EPA's satisfaction the circumstances giving rise to EPA's issuance of the relevant Performance Failure Notice, then, in accordance with any applicable financial assurance mechanism, EPA may at any time thereafter direct the financial assurance provider to immediately: i) deposit any funds assured pursuant to this Section into the standby trust fund; or ii) arrange for performance of the Work in accordance with this Order.

b. If EPA is notified by the provider of a financial assurance mechanism that it intends to cancel the mechanism, and the affected Respondent fails to provide an alternative financial assurance mechanism in accordance with this Section at least 30 days prior to the cancellation date, EPA may, prior to cancellation, direct the financial assurance provider to deposit any funds guaranteed under such mechanism into the standby trust fund for use consistent with this Section.

96. **Modification of Amount, Form, or Terms of Financial Assurance**. Respondents may submit, on any anniversary of the Effective Date or following Respondents' request for, and EPA's approval of, another date, a request to reduce the amount, or change the form or terms, of the financial assurance mechanism. Any such request must be submitted to the EPA individual(s) referenced in Paragraph 91, and must include an estimate of the cost of the remaining Work, an explanation of the bases for the cost calculation, a description of the

proposed changes, if any, to the form or terms of the financial assurance, and any newly proposed financial assurance documentation in accordance with the requirements of Paragraphs 89 and 90 (Standby Trust). EPA will notify Respondents of its decision to approve or disapprove a requested reduction or change. Respondents may reduce the amount or change the form or terms of the financial assurance only in accordance with EPA's approval. Within 30 days after receipt of EPA's approval of the requested modifications pursuant to this Paragraph, Respondents shall submit to the EPA individual(s) referenced in Paragraph 91 all executed and/or otherwise finalized documentation relating to the amended, reduced, or alternative financial assurance mechanism. Upon EPA's approval, the Estimated Cost of the Work shall be deemed to be the estimate of the cost of the remaining Work in the approved proposal.

97. **Release, Cancellation, or Discontinuation of Financial Assurance**. Respondents may release, cancel, or discontinue any financial assurance provided under this Section only: a) after receipt of documentation issued by EPA certifying completion of the Work; or b) in accordance with EPA's written approval of such release, cancellation, or discontinuation.

XXI. INSURANCE

98. At least fourteen (14) days prior to commencing any on-site Work under this Order, Respondents shall secure, and shall maintain for the duration the Work being performed pursuant to this Order, commercial general liability insurance with limits of liability of $1 million per occurrence, automobile liability insurance with limits of liability of $1 million per accident, and umbrella liability insurance with limits of liability of $5 million in excess of the required commercial general liability and automobile liability limits, naming EPA as an additional insured with respect to all liability arising out of the activities performed by or on behalf of Respondents pursuant to this Order. Within the same period, Respondents shall provide EPA with certificates of such insurance and, if requested, a copy of each insurance policy. Respondents shall submit such certificates and, if requested, copies of policies each year on the anniversary of the Effective Date. In addition, for the duration of the Work being performed pursuant to Order, Respondents shall satisfy, or shall ensure that their contractors or subcontractors satisfy, all applicable laws and regulations regarding the provision of worker's compensation insurance for all persons performing the Work on behalf of Respondents in furtherance of this Order. If Respondents demonstrate by evidence satisfactory to EPA that any contractor or subcontractor maintains insurance equivalent to that described above, or insurance covering some or all of the same risks but in an equal or lesser amount, then Respondents need provide only that portion of the insurance described above which is not maintained by such contractor or subcontractor.

XXIII. INTEGRATION/APPENDICES

99. In the event of a conflict between any provision of this Order and the provisions of any document attached to this Order or submitted or approved pursuant to this Order, the provisions of this Order shall control.

100. The following documents are attached to and incorporated into this Order:

Appendix A is the ROD (electronic)

Appendix B is the EPA-approved 100% RD for the 1st Street turning basin (electronic)

XXIV. TERMINATION AND SATISFACTION

101. Upon a determination by EPA that the Work required pursuant to this Order has been fully carried out in accordance with this Order, EPA will so notify Respondents in writing.

XXV. OPPORTUNITY TO CONFER, EFFECTIVE DATE

102. This Order shall be effective fourteen (14) days after receipt by Respondents, unless a conference is timely requested pursuant to Paragraph 103 below. If such a conference is timely requested, this Order shall become effective seven (7) days following the date the conference is held, unless the Effective Date is modified by EPA. All times for performance of ordered activities shall be calculated from this Effective Date.

103. Respondents may, within thirteen (13) days after receipt of this Order, request a conference with EPA to discuss this Order. If requested, the conference shall occur within fourteen (14) days of Respondents' request for a conference. The conference may be held in person or by telephone or videoconference.

104. If a conference is held, Respondents may present any information, arguments, or comments regarding this Order. This conference is not an evidentiary hearing, and does not constitute a proceeding to challenge this Order. It does not give Respondents a right to seek review of this Order or to seek resolution of potential liability, and no official stenographic record of the conference will be made. At any conference held pursuant to Respondents' request, Respondents may appear in person or by an attorney or other representative.

105. A request for a conference must be made by telephone to Brian Carr, Assistant Regional Counsel, Office of Regional Counsel, EPA Region 2, telephone (212) 637-3170, followed by written confirmation emailed that day to Mr. Carr at carr.brian@epa.gov.

XXVI. NOTICE OF INTENT TO COMPLY

106. Each Respondent shall provide, not later than seven (7) days after the Effective Date, written notice to EPA stating whether it will comply with the terms of this Order. If a Respondent does not unequivocally commit to perform the work required by this Order, it shall be deemed to have violated this Order and to have failed or refused to comply with this Order. Each Respondent's written notice shall describe, using facts that exist on or prior to the Effective Date, any "sufficient cause" defenses asserted by Respondent under Sections 106(b) and 107(c)(3) of CERCLA. Each Respondent's written notice shall be sent to Brian Carr, Assistant Regional Counsel, Office of Regional Counsel, EPA Region 2, 290 Broadway, 17th floor, New York, New York 10007-1866, or by email to carr.brian@epa.gov. The absence of a response by EPA to the notice required by this Paragraph shall not be deemed to be an acceptance of a Respondent's assertions.

By: _____ January 28, 2020
ERIC WILSON Date
For Acting Director
Superfund and Emergency Management Division
U.S. Environmental Protection Agency
Region 2

Exhibit ___ (CFW-4)

Administrative Settlement Agreement and Order on Consent for the remedial investigation and feasibility study of the Newtown Creek Superfund Site

UNITED STATES
ENVIRONMENTAL PROTECTION AGENCY
REGION 2

IN THE MATTER OF: The Newtown Creek Superfund Site Kings County and Queens County, New York City, New York Phelps Dodge Refining Corporation, Texaco, Inc., BP Products North America Inc., The Brooklyn Union Gas Company d/b/a National Grid NY, ExxonMobil Oil Corporation, and The City of New York Respondents.	ADMINISTRATIVE SETTLEMENT AGREEMENT AND ORDER ON CONSENT FOR REMEDIAL INVESTIGATION/FEASIBILITY STUDY U.S. EPA Region 2 CERCLA Docket No. CERCLA-02-2011-2011 Proceeding Under Sections 104, 107 and 122 of the Comprehensive Environmental Response, Compensation, and Liability Act, as amended, 42 U.S.C. §§ 9604, 9607 and 9622.

ADMINISTRATIVE SETTLEMENT AGREEMENT AND ORDER ON CONSENT FOR REMEDIAL INVESTIGATION/FEASIBILITY STUDY

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ADMINISTRATIVE SETTLEMENT AGREEMENT AND ORDER ON CONSENT
FOR REMEDIAL INVESTIGATION/FEASIBILITY STUDY

I. JURISDICTION AND GENERAL PROVISIONS

1. This Administrative Settlement Agreement and Order on Consent is entered into voluntarily by the United States Environmental Protection Agency ("EPA") and Phelps Dodge Refining Corporation, Texaco, Inc., BP Products North America Inc., The Brooklyn Union Gas Company d/b/a National Grid NY, ExxonMobil Oil Corporation, and The City of New York. The Settlement Agreement concerns the preparation and performance of a remedial investigation and feasibility study ("RI/FS") at or in connection with the Newtown Creek Study Area portion of the Newtown Creek Superfund Site. The Newtown Creek Study Area includes a facility encompassing the body of water known as the Newtown Creek which is situated at the border of the boroughs of Brooklyn (Kings County) and Queens (Queens County) in the City of New York, State of New York, roughly centered at the geographic coordinates of 40° 42' 54.69" north latitude (40.715192°) and 73° 55' 50.74" west longitude (-73.930762°), having an approximate 3.8-mile reach, including Newtown Creek proper and its five branches (or tributaries) which are known, respectively, as Dutch Kills, Maspeth Creek, Whale Creek, East Branch and English Kills, as well as the sediments below the water, and the water column above the sediments, up to and including the landward edge of the shoreline, and including also any bulkheads or riprap containing the water body, except where no bulkhead or riprap exists, then the Study Area shall extend to the ordinary high water mark, as defined in 33 C.F.R. §328.3(e), of Newtown Creek, and the areal extent of the contamination from such area, but the Study Area does not include the upland areas beyond the landward edge of the shoreline. The Settlement Agreement also concerns the payment by Respondents of Future Response Costs and Past Response Costs.

2. This Settlement Agreement is issued under the authority vested in the President of the United States by Sections 104, 107 and 122 of the Comprehensive Environmental Response, Compensation, and Liability Act, as amended, 42 U.S.C. §§ 9604, 9607 and 9622 ("CERCLA"). This authority was delegated to the Administrator of EPA on January 23, 1987, by Executive Order 12580, 52 Fed. Reg. 2926 (Jan. 29, 1987), and further delegated to Regional Administrators on May 11, 1994, by EPA Delegation Nos. 14-14-C and 14-14-D. This authority was further redelegated by the Regional Administrator of EPA Region 2 to the Director of the Emergency and Remedial Response Division on November 23, 2004 by Region 2 Redelegation R-1200.

3. In accordance with Sections 104(b)(2) and 122(j)(1) of CERCLA, 42 U.S.C. §§ 9604(b)(2) and 9622(j)(1), EPA notified the relevant Federal and State natural resource trustees, U.S. Fish and Wildlife Service, the National Oceanic and Atmospheric Administration, and NYSDEC, on May 4, 2010, of negotiations with potentially responsible parties ("PRPs") regarding the release of hazardous substances that may have resulted in injury to the natural resources under Federal and/or State trusteeship.

4. Other than the RI/FS Work Plan activities required to be performed by Respondents by this Settlement Agreement, and except to the extent otherwise provided in Paragraphs 39 or 44 of this Settlement Agreement, Respondents shall not be required by this Settlement Agreement to fund or perform any remedial or removal actions selected by EPA for the Study Area or the Site. Any remedial or removal actions selected for the Study Area or the Site may be the subject of settlement agreements between EPA, Respondents, and/or other persons not a party to this Settlement Agreement or may otherwise be the subject of separate enforcement actions by EPA.

5. EPA and Respondents recognize that this Settlement Agreement has been negotiated in good faith and that the actions undertaken by Respondents in accordance with this Settlement Agreement do not constitute an admission of any liability. Respondents do not admit, and retain the right to controvert in any subsequent proceedings other than proceedings to implement or enforce this Settlement Agreement, the validity of the findings of fact, conclusions of law and determinations in Sections V and VI of this Settlement Agreement. Respondents agree to comply with and be bound by the terms of this Settlement Agreement and further agree that they will not contest the basis or validity of this Settlement Agreement or its terms in any action to enforce its provisions.

6. It is EPA's present intention to identify, and consistent with standard practice and in its unreviewable discretion, take measures to obtain the participation of additional parties in the Work and other activities required by this Settlement Agreement as the RI/FS proceeds. Such measures may include taking enforcement actions pursuant to Sections 106(a) and 107(a) of CERCLA, 42 U.S.C. § 9606(a) and § 9607(a). This does not alter the Respondents' obligations to perform the Work under this Settlement Agreement. EPA and Respondents agree to cooperate in the identification of newly identified parties and in seeking to obtain their participation in this Settlement Agreement. EPA and the Respondents acknowledge that this Settlement Agreement may be amended, upon mutually acceptable terms and conditions, to include additional parties who consent to become Respondents after this Settlement Agreement becomes effective, provided, however, that if Respondents' Liaison Counsel certifies in writing to EPA that a newly identified party has entered into an agreement with Respondents to participate and cooperate with Respondents in the performance or funding of the Work and all other activities required pursuant to this Settlement Agreement, then any amendment to this Settlement Agreement that merely adds such newly identified party as an additional Respondent shall require the signatures of only such additional Respondent and EPA.

II. PARTIES BOUND

7. This Settlement Agreement applies to and is binding upon EPA and upon Respondents and their successors and assigns. Any change in ownership or corporate status of a Respondent including, but not limited to, any transfer of assets or real or personal

property shall not alter such Respondent's responsibilities under this Settlement Agreement.

8. Respondents are jointly and severally liable for carrying out all activities required by this Settlement Agreement. In the event of the insolvency or other failure of any one or more of the Respondents to implement the requirements of this Settlement Agreement, the remaining Respondents shall complete all such requirements. Notwithstanding the foregoing, it is intended that Work Respondents shall perform all of the Work and pay Past Response Costs and Future Response Costs under this Settlement Agreement and that, as long as the Work Respondents do not fail to perform all obligations under this Settlement Agreement, Funding Respondents shall participate only to the extent that they shall make any payments of money to the Work Respondents in amounts and on terms agreed to by and among Respondents by separate agreement in order to fund such activities.

9. Respondents shall ensure that their contractors, subcontractors, and representatives receive a copy of and comply with this Settlement Agreement. Respondents shall be responsible for any noncompliance with this Settlement Agreement by Respondents, their contractors, subcontractors, or representatives.

10. Each undersigned representative of Respondents certifies that he or she is fully authorized to enter into the terms and conditions of this Settlement Agreement and to execute and legally bind Respondents to this Settlement Agreement.

III. STATEMENT OF PURPOSE

11. In entering into this Settlement Agreement, the objectives of EPA and Respondents are: (a) to determine the nature and extent of contamination and any threat to the public health, welfare, or the environment caused by the release or threatened release of hazardous substances, pollutants or contaminants at or from the Study Area, by conducting a Remedial Investigation as more specifically set forth in the Creekwide Remedial Investigation/Feasibility Study Work Plan dated June 2011 ("RI/FS Work Plan"), a copy of which is attached as Appendix A to this Settlement Agreement; (b) to identify and evaluate remedial alternatives to prevent, mitigate or otherwise respond to or address any release or threatened release of hazardous substances, pollutants, or contaminants at or from the Study Area, by conducting a Feasibility Study as more specifically set forth in the RI/FS Work Plan; and (c) to recover response and oversight costs incurred by EPA with respect to this Settlement Agreement as well as Past Response Costs.

12. The Work conducted under this Settlement Agreement is subject to approval by EPA and shall provide all appropriate and necessary information to assess conditions affecting the Study Area and evaluate alternatives to the extent necessary to select a remedy that will be consistent with CERCLA and the National Oil and Hazardous Substances Pollution

Contingency Plan, 40 C.F.R. Part 300 ("NCP"). Respondents shall conduct all Work under this Settlement Agreement in compliance with CERCLA, the NCP, and all applicable EPA guidances, policies, and procedures, including, without limitation, EPA Region 2's "Clean and Green Policy" which may be found at: http://www.epa.gov/region02/superfund/green_remediation.

IV. DEFINITIONS

13. Unless otherwise expressly provided in this Settlement Agreement, terms used in this Settlement Agreement that are defined in CERCLA or in regulations promulgated under CERCLA shall have the meaning assigned to them in CERCLA or in such regulations. Whenever terms listed below are used in this Settlement Agreement or in the appendices attached hereto and incorporated hereunder, the following definitions shall apply:

a. "CERCLA" shall mean the Comprehensive Environmental Response, Compensation, and Liability Act of 1980, as amended, 42 U.S.C. §§ 9601, *et seq.*

b. "Day" shall mean a calendar day. In computing any period of time under this Settlement Agreement, where the last day would fall on a Saturday, Sunday, or federal holiday, the period shall run until the close of business of the next Working Day.

c. "Effective Date" shall be the effective date of this Settlement Agreement as provided in Section XXXI.

d. "EPA" shall mean the United States Environmental Protection Agency and any successor departments or agencies of the United States.

e. "Future Response Costs" shall mean all costs not inconsistent with the NCP, including, but not limited to, direct and indirect costs, that the United States incurs in reviewing or developing plans, reports and other deliverables pursuant to this Settlement Agreement, in overseeing implementation of the Work, or otherwise implementing, overseeing, or enforcing this Settlement Agreement, including, but not limited to, payroll costs, contractor costs, travel costs, laboratory costs, Agency for Toxic Substances and Disease Registry ("ATSDR") costs, the costs incurred pursuant to Paragraph 58.a (including, but not limited to, costs and attorneys fees and any monies paid to secure access, including, but not limited to, the amount of just compensation), Paragraph 44 (emergency response), and Paragraph 87 (Work Takeover).

f. "Institutional controls" shall mean non-engineered instruments, such as administrative and/or legal controls, that help to minimize the potential for human exposure to contamination and/or protect the integrity of a remedy by limiting land and/or resource use. Examples of institutional controls include easements and covenants, zoning restrictions, special building permit requirements, and well drilling prohibitions.

g. "Interest" shall mean interest at the rate specified for interest on investments of the EPA Hazardous Substance Superfund established by 26 U.S.C. § 9507,

compounded annually on October 1 of each year, in accordance with 42 U.S.C. § 9607(a). The applicable rate of interest shall be the rate in effect at the time the interest accrues. The rate of interest is subject to change on October 1 of each year.

h. "NCP" shall mean the National Oil and Hazardous Substances Pollution Contingency Plan promulgated pursuant to Section 105 of CERCLA, 42 U.S.C. § 9605, codified at 40 C.F.R. Part 300, and any amendments thereto.

i. "Newtown Creek" shall mean Newtown Creek proper and its five branches (or tributaries) known as Dutch Kills, Maspeth Creek, Whale Creek, East Branch and English Kills, as well as the sediments below the water, and the water column above the sediments.

j. "Newtown Creek Superfund Site" or "Site" shall mean the Study Area and the areal extent of the contamination associated with the Study Area including facilities upland of the Study Area that are sources of contamination to the Study Area.

k. "NYSDEC" shall mean the New York State Department of Environmental Conservation and any successor departments or agencies of the State.

l. "Paragraph" shall mean a portion of this Settlement Agreement identified by an Arabic numeral or an upper case or lower case letter.

m. "Parties" shall mean EPA and Respondents.

n. "Past Response Costs" shall mean the sum of Seven Hundred and Fifty Thousand ($750,000) Dollars of the costs (including, but not limited to, direct and indirect costs) that the United States has paid at or in connection with the Study Area prior to the Effective Date.

o. "RCRA" shall mean the Resource Conservation and Recovery Act, also known as the Solid Waste Disposal Act, as amended, 42 U.S.C. §§ 6901, *et seq.*

p. "Respondents" shall mean those signatories to this Settlement Agreement identified in Appendix C, as may be amended from time to time.

q. "Work Respondents" shall mean those Respondents identified in Appendix D, as may be amended from time to time.

r. "Funding Respondents" shall mean those Respondents identified in Appendix E as may be amended from time to time.

s. "RI/FS Work Plan" shall mean the EPA-approved Creekwide Remedial Investigation/Feasibility Study Work Plan dated June 2011, a copy of which is attached as Appendix A, and including all additional work plans to be developed hereafter pursuant to the RI/FS Work Plan, so as to constitute the full work plan for the complete RI/FS for the Study Area, which subsequent work plans, upon EPA approval thereof, shall be deemed incorporated within the RI/FS Work Plan and added to Appendix A.

t. "Section" shall mean a portion of this Settlement Agreement identified by a Roman numeral.

u. "Settlement Agreement" shall mean this Administrative Settlement Agreement and Order on Consent including, the RI/FS Work Plan and any other appendix attached

hereto (listed in Section XXIX) and all documents incorporated by reference into this document including, without limitation, EPA-approved submissions. EPA-approved submissions (other than progress reports) are incorporated into and become a part of the Settlement Agreement upon approval by EPA. In the event of conflict between this Settlement Agreement and any appendix or other incorporated documents, this Settlement Agreement shall control.

v. "Study Area" shall mean the portion of the Newtown Creek Superfund Site that encompasses the body of water known as Newtown Creek, situated at the border of the boroughs of Brooklyn (Kings County) and Queens (Queens County) in the City of New York and the State of New York, roughly centered at the geographic coordinates of 40° 42' 54.69" north latitude (40.715192°) and 73° 55' 50.74" west longitude (-73.930762°), having an approximate 3.8-mile reach, including Newtown Creek proper and its five branches (or tributaries) known respectively as Dutch Kills, Maspeth Creek, Whale Creek, East Branch and English Kills, as well as the sediments below the water, and the water column above the sediments, up to and including the landward edge of the shoreline, and including also any bulkheads or riprap containing the water body, except where no bulkhead or riprap exists, then the Study Area shall extend to the ordinary high water mark, as defined in 33 C.F.R. §328.3(e), of Newtown Creek, and the areal extent of the contamination from such area, but not including upland areas beyond the landward edge of the shoreline (notwithstanding that such upland areas may subsequently be identified as sources of contamination to the water body and its sediments or that such upland areas may be included within the scope of the Newtown Creek Superfund Site as listed pursuant to Section 105(a)(8) of CERCLA). The Study Area is depicted generally on the map attached as Appendix B.

w. "State" shall mean the State of New York.

x. "United States" shall mean the United States of America and each department, agency, and instrumentality of the United States, including EPA.

y. "Waste Material" shall mean (1) any "hazardous substance" under Section 101(14) of CERCLA, 42 U.S.C. § 9601(14); (2) any pollutant or contaminant under Section 101(33) of CERCLA, 42 U.S.C. § 9601(33); and (3) any "solid waste" under Section 1004(27) of RCRA, 42 U.S.C. § 6903(27).

z. "Work" shall mean all activities Respondents are required to perform under this Settlement Agreement, except those required by Section XIV (Retention of Records).

aa. "Working Day" shall mean any day of the week excluding Saturdays, Sundays and federal holidays as defined at 5 U.S.C. § 6103.

V. FINDINGS OF FACT

14. The Site includes a body of water located in Kings County and Queens County in the City of New York and the State of New York roughly centered at the geographic coordinates of 40° 42' 54.69" north latitude (40.715192°) and 73° 55' 50.74 74". west longitude (-73.930762°). Newtown Creek is a tidal arm of the New York-New Jersey Harbor Estuary that forms the northwestern-most border between the New York City boroughs of Brooklyn and Queens. It is a tributary of the East River and it itself includes five branches (or tributaries) along its 3.8-mile reach. Historically, Newtown Creek drained the uplands of western Long Island and flowed through wetlands and marshes. However, due to heavy industrial development and governmental activities dating from the 1800's, formerly wet areas have been filled, Newtown Creek has been channelized, and its banks have been stabilized with bulkheads and rip rap. The historic development has resulted in changes in the nature of Newtown Creek from a natural drainage condition to one that is governed largely by engineered and institutional systems.

15. The area around Newtown Creek has a history of extensive industrial development dating back to the 1800's. By the 1850's, the area surrounding and adjacent to Newtown Creek had become highly industrialized. By 1870, more than 50 petroleum refineries were located along its banks, and by the close of the 19th century, the Creek was lined with oil refineries and petrochemical plants, fertilizer and glue factories, copper-smelting and fat-rendering plants, shipbuilders, sugar refineries, hide tanning plants, canneries, governmental facilities, sawmills, paint works, and lumber and coal yards. During its industrial past, Newtown Creek was crowded with commercial vessels, including large boats bringing in raw materials and fuel and taking out oil, fat, varnish, chemicals, and metals. The majority of the shoreline area of Newtown Creek is currently zoned for heavy manufacturing and industrial use with a small portion, near the mouth of the Creek, zoned for residential use. Currently, the predominant land use around Newtown Creek includes industrial facilities, petroleum bulk storage facilities, manufacturing facilities, warehouses, transportation and transportation infrastructure including rail and highways, utility facilities and a large municipal Water Pollution Control Plant. The majority of the upland area bordering Newtown Creek is within one of three Industrial Business Zones of the City of New York, and the majority of Newtown Creek has also been designated by the City of New York as one of the City's six Significant Maritime and Industrial Areas. Comprehensive plans developed by the City of New York for areas within one-quarter mile of Newtown Creek call for development of approximately 7,000 potential units of housing and nearly 8 acres of public open space along shoreline on or adjacent to Newtown Creek in the Greenpoint-Williamsburg area of Brooklyn that was rezoned to residential in 2005, and for development of approximately 5,000 potential units of housing units, 11 acres of public open space and waterfront access along shoreline that is on or adjacent to Newtown Creek on land that was rezoned to residential in 2008 in the Hunters Point South area of Queens.

16. Numerous upland facilities within the Newtown Creek watershed have been identified by NYSDEC for inclusion under various New York State programs, including the Brownfield Cleanup Program, the Environmental Restoration Program, the State Superfund Program, and the Voluntary Cleanup Program. In addition, there are 15 facilities within the Newtown Creek watershed that are listed as RCRA Large Quantity Generator facilities including landfills, solid waste and hazardous waste facilities and facilities with underground storage tanks holding chemicals or petroleum products.

17. EPA conducted an Expanded Site Investigation ("ESI") of Newtown Creek in 2009 as part of the Hazard Ranking System scoring process. Based on the ESI, which was focused on Newtown Creek itself and not its tributaries, EPA concluded that metals, volatile organic compounds, and semi-volatile organic compounds (including polycyclic aromatic hydrocarbons and polychlorinated biphenyls) were present in Creek sediments at concentrations above levels in nearby locations in the Atlantic basin. The variety and distribution of the detected constituents suggests that they originated from a variety of sources. Previous environmental investigations of Newtown Creek, or specific portions of the Creek (including investigations performed by certain of the Respondents under oversight of NYSDEC and investigations performed by the City of New York in connection with proposed navigational dredging) also disclosed that sediments in Newtown Creek are contaminated by a wide variety of hazardous substances. Environmental investigations of certain upland parcels adjacent to or nearby the Creek have disclosed contamination of those parcels by hazardous substances similar to hazardous substances found in sediments in Newtown Creek.

18. Potential sources of hazardous substances to Newtown Creek include a wide range of current and historical industrial and municipal discharges, including historic and potentially ongoing industrial discharges from industries along the banks and within the Newtown Creek watershed, incidental releases or other discharges during loading or unloading of barges servicing Creekside industrial or governmental facilities, historic placement of fill by both industry and municipalities along the banks of Newtown Creek for disposal or for filling marshland or swamps, historic placement of dredge material during dredging and channelization activities, historic discharge of sanitary sewage and industrial wastes, historic and ongoing releases of combined storm, sanitary and industrial discharges through combined sewage overflow systems into Newtown Creek, and releases from navigational shipping and maritime traffic.

19. The water in Newtown Creek is currently classified by the NYSDEC as Class SD saline surface water with a protected use of fish survival only. The Creek does not presently meet parameters for that protected use, e.g., dissolved oxygen. NYSDEC defines class SD as, "...water that, because of natural or manmade conditions, cannot meet the requirements for primary and secondary contact recreation and fish propagation." As such, the waters of Newtown Creek are not suitable for primary (e.g., swimming) or secondary (e.g., boating or fishing) direct contact recreation. However, current recreational uses of

the Newtown Creek include kayaking and there are existing and planned waterfront access points. Human exposure to contaminated sediments and waters of Newtown Creek through recreational use, use of the shoreline areas by workers, residents, recreationists or trespassers or through consumption of fish or other biota impacted by contamination associated with Newtown Creek, could cause adverse health effects that will be evaluated in the Remedial Investigation ("RI"), and effects of exposure of contaminated sediments and water to flora and fauna will also be evaluated.

20. The Newtown Creek Superfund Site (which includes the Study Area) was proposed for inclusion on the National Priorities List ("NPL") pursuant to CERCLA Section 105, 42 U.S.C. § 9605, on September 23, 2009 by publication in the Federal Register at 74 Federal Register 48511, and the Site was added to the NPL by rule published in the Federal Register on September 29, 2010 at 75 Federal Register 59983.

21. Each Respondent currently owns or operates, or, at the time of disposal of hazardous substances, owned or operated a facility or facilities at the Study Area or within other areas at the Site from which there was a release or there is a threat of release of hazardous substances to the Study Area, or is a person who arranged for disposal of hazardous substances at a facility within the Site from which facility hazardous substances were released or threaten to be released into the Study Area, or is a person who transported hazardous substances to a facility, selected by such person, within the Site, from which facility hazardous substances were released or threaten to be released into the Study Area. Hazardous substances of the type released from such facilities have been identified in the sediments of Newtown Creek.

22. EPA has evaluated information obtained in connection with the Site and has determined on the basis of such information that hazardous substances were released and/or there is a threat of release from facilities that are currently owned or operated or that were, at the time of disposal of hazardous substances owned or operated by each of the Respondents or by predecessor companies or predecessor municipalities of Respondents, or that a Respondent is a person who arranged for disposal of hazardous substances at the Site from which hazardous substances were released or threaten to be released into the Study Area, or is a person who transported hazardous substances to a facility selected by such person within the Site from which hazardous substances were released or threaten to be released into the Study Area. Hazardous substances of the type released from such facilities have been identified in the sediments of Newtown Creek.

VI. CONCLUSIONS OF LAW AND DETERMINATIONS

Based on the Findings of Fact set forth above, EPA has determined that:

23. The Study Area is a "facility" as defined in Section 101(9) of CERCLA, 42 U.S.C. § 9601(9).

24. The contamination found at the Study Area, as identified in the Findings of Fact, above, includes "hazardous substances" as defined in Section 101(14) of CERCLA, 42 U.S.C. § 9601(14).

25. The conditions described in the Findings of Fact, above, constitute an actual and/or threatened "release" of a hazardous substance from a facility as defined in Section 101(22) of CERCLA, 42 U.S.C. § 9601(22).

26. Each Respondent is a "person" as defined in Section 101(21) of CERCLA, 42 U.S.C. § 9601(21).

27. Each Respondent is a person who currently, or at the time of disposal of hazardous substances, owned or operated facilities within the Site, or is a person who arranged for disposal of hazardous substances at the Site or is a person who transported hazardous substances to a facility selected by such person within the Site. Each Respondent therefore may be liable under Section 107(a) of CERCLA, 42 U.S.C. § 9607(a).

28. Respondents are responsible parties for the purposes of Sections 104, 107 and 122 of CERCLA, 42 U.S.C. §§ 9604, 9607 and 9622.

29. The actions required by this Settlement Agreement are necessary to protect the public health, welfare and/or the environment, are in the public interest, are consistent with CERCLA and the NCP, and will expedite effective remedial action and minimize litigation, 42 U.S.C. §§ 9604(a)(1) and 9622(a).

30. EPA has determined that Respondents are qualified to conduct the RI/FS within the meaning of Section 104(a) of CERCLA, 42 U.S.C. § 9604(a), and will carry out the Work properly and promptly, in accordance with Sections 104(a) and 122(a) of CERCLA, 42 U.S.C. §§ 9604(a) and 9622(a), if Respondents comply with the terms of this Settlement Agreement.

VII. SETTLEMENT AGREEMENT AND ORDER

31. Based upon the foregoing Findings of Fact and Conclusions of Law and Determinations, it is hereby ordered and agreed that Respondents shall comply with all provisions of this Settlement Agreement, including, but not limited to, all appendices to this Settlement Agreement and all documents incorporated by reference into this Settlement Agreement.

VIII. DESIGNATION OF CONTRACTORS AND PROJECT COORDINATORS

32. Selection of Contractors, Personnel. All Work performed under this Settlement Agreement shall be under the direction and supervision of qualified personnel. Within

thirty days after the Effective Date of this Settlement Agreement, and before the Work outlined below begins, Respondents shall notify EPA in writing of the names, titles, and qualifications of the key personnel, including contractors, subcontractors, consultants and laboratories to be used in carrying out such Work. With respect to any proposed contractor, Respondents shall demonstrate that the proposed contractor has a quality system which complies with ANSI/ASQC E4-1994, "Specifications and Guidelines for Quality Systems for Environmental Data Collection and Environmental Technology Programs" (American National Standard, January 5, 1995, or most recent version), by submitting a copy of the proposed contractor's Quality Management Plan ("QMP"). The QMP should be prepared in accordance with "EPA Requirements for Quality Management Plans (QA/R-2)," (EPA/240/B-01/002, March 2001 (Reissued May 2006)) or equivalent documentation as determined by EPA. The qualifications of the key personnel undertaking the Work for Respondents shall be subject to EPA's review, for verification that such persons meet minimum technical background and experience requirements. This Settlement Agreement is contingent on Respondents' demonstration to EPA's satisfaction that Respondents are qualified to perform properly and promptly the actions set forth in this Settlement Agreement. If EPA disapproves in writing of any key person's technical qualifications, Respondents shall notify EPA of the identity and qualifications of the replacement(s) within thirty days of the written notice. If EPA subsequently disapproves of the replacement, EPA reserves the right to terminate this Settlement Agreement and to conduct a complete RI/FS, and to seek reimbursement for costs and penalties from Respondents. During the course of the RI/FS, Respondents shall notify EPA in writing of any changes or additions to the key personnel used to carry out such Work, providing their names, titles, and qualifications. EPA shall have the same right to disapprove changes and additions to personnel as it has hereunder regarding the initial notification of identified personnel by Respondents.

33. Within fourteen days after the Effective Date, Respondents shall designate a Project Coordinator who shall be responsible for administration of all actions by Respondents required by this Settlement Agreement and shall submit to EPA the designated Project Coordinator's name, address, telephone number, and qualifications. To the greatest extent possible, the Project Coordinator shall be present at the Study Area or readily available during performance of Work at the Study Area. EPA retains the right to disapprove of the designated Project Coordinator. If EPA disapproves of the designated Project Coordinator, Respondents shall retain a different Project Coordinator and shall notify EPA of that person's name, address, telephone number and qualifications within fourteen days following EPA's disapproval. Respondents shall have the right to change their Project Coordinator, subject to EPA's right to disapprove. Respondents shall notify EPA fourteen days before such a change is made. The initial notification may be made orally, but shall be promptly followed by a written notification. Receipt by Respondents' Project Coordinator of any notice or communication from EPA relating to this Settlement Agreement shall constitute receipt by Respondents.

34. EPA has designated the following individual as its Project Coordinator for the Site:

Caroline Kwan
Remedial Project Manager
Special Projects Branch
Emergency and Remedial Response Division
United States Environmental Protection Agency, Region 2
290 Broadway, 20th Floor
New York, NY 10007-1866
212-637-4275
kwan.caroline@epa.gov

EPA will notify Respondents of a change of its designated Project Coordinator.

35. EPA's Project Coordinator shall have the authority lawfully vested in a Remedial Project Manager ("RPM") and On-Scene Coordinator ("OSC") by the NCP. In addition, EPA's Project Coordinator shall have the authority consistent with the NCP, to halt any Work required by this Settlement Agreement, and to take any necessary response action when she determines that conditions at the Study Area may present an immediate endangerment to public health or welfare or the environment. The absence of the EPA Project Coordinator from the area under study pursuant to this Settlement Agreement shall not be cause for the stoppage or delay of Work.

36. EPA shall arrange for a qualified person to assist in its oversight and review of the conduct of the RI/FS, as required by Section 104(a) of CERCLA, 42 U.S.C. § 9604(a). Such person shall have the authority to observe Work and make inquiries in the absence of EPA, but not to modify the RI/FS Work Plan.

37. Respondents' Liaison Counsel. Within fifteen days after the Effective Date, Respondents shall designate an attorney to act as liaison counsel and shall identify such counsel ("Liaison Counsel") to EPA by name, address, telephone number and e-mail address. During the course of the RI/FS, Respondents shall notify EPA in writing of changes to their Liaison Counsel. Liaison Counsel shall provide the written certifications, if any, pursuant to Paragraph 6 of this Settlement Agreement.

IX. WORK TO BE PERFORMED

38. Respondents shall conduct the RI/FS in accordance with the provisions of this Settlement Agreement, the RI/FS Work Plan, CERCLA, the NCP and EPA guidance, including, but not limited to the "Interim Final Guidance for Conducting Remedial Investigations and Feasibility Studies under CERCLA" (OSWER Directive # 9355.3-01, October 1988 or subsequently issued guidance), "Guidance for Data Usability in Risk Assessment" (OSWER Directive #9285.7-05, October 1990 or subsequently issued guidance), EPA Contaminated Sediment Guidance (OSWER Directive #9355.0-85, December 2005 or subsequently issued guidance) and guidance referenced therein, and guidances referenced

in the RI/FS Work Plan, as may be amended or modified by EPA. The RI shall consist of collecting data to characterize conditions at the Study Area, determining the nature and extent of the contamination at or from the Study Area, including identification and characterization of sources within the Study Area as provided in the RI/FS Work Plan, assessing risk to human health and the environment and conducting treatability testing as necessary to evaluate the potential performance and cost of the treatment technologies that are being considered. The Feasibility Study ("FS") shall determine and evaluate (based on treatability testing, where appropriate) alternatives for remedial action to prevent, mitigate or otherwise respond to or remedy the release or threatened release of hazardous substances, pollutants, or contaminants at or from the Study Area. The alternatives evaluated must include, but shall not be limited to, the range of alternatives described in the NCP, and shall include remedial actions that utilize permanent solutions and alternative treatment technologies or resource recovery technologies to the maximum extent practicable. In evaluating the alternatives, Respondents shall address the factors required to be taken into account by Section 121 of CERCLA, 42 U.S.C. § 9621, and Section 300.430(e) of the NCP, 40 C.F.R. § 300.430(e).

39. Modification of the RI/FS Work Plan.

a. If at any time during the RI/FS process, Respondents identify a need for modifications to the RI/FS Work Plan, Respondents shall submit a memorandum documenting the need for proposed modifications to the EPA Project Coordinator within ten days of identification. In addition, if at any time during the RI/FS process, Respondents identify a need for additional data, Respondents shall submit a memorandum documenting a need for additional data to the EPA Project coordinator within ten days of identification. EPA in its discretion will determine whether the additional data will be collected by Respondents and whether it will be incorporated into plans, reports and other deliverables. Notwithstanding the foregoing, Respondents retain the right to collect and submit data for any other purpose.

b. In the event of unanticipated or changed circumstances at the Study Area, Respondents shall notify the EPA Project Coordinator by telephone within 24 hours of discovery of the unanticipated or changed circumstances. In the event that EPA determines that the unanticipated or changed circumstances warrant changes in the RI/FS Work Plan, EPA shall modify or amend the RI/FS Work Plan in writing accordingly. Respondents shall perform the RI/FS Work Plan as modified or amended.

c. EPA may determine that in addition to tasks defined in the initially approved RI/FS Work Plan, other additional Work may be necessary to accomplish the objectives of the RI/FS consistent with CERCLA and the NCP. Respondents agree to perform these RI/FS Work activities in addition to those required by the initially approved RI/FS Work Plan, including any modifications approved by EPA, if EPA determines that such actions are necessary for a complete RI/FS.

d. Respondents shall confirm their willingness to perform the additional Work in writing to EPA within fifteen days of receipt of the EPA request. If Respondents object to any modification determined by EPA to be necessary pursuant to this Paragraph, Respondents may seek dispute resolution pursuant to Section XV (Dispute Resolution). The RI/FS Work Plan shall be modified in accordance with the final resolution of the dispute.

e. Respondents shall complete the additional Work according to the standards, specifications, and schedule set forth or approved by EPA in a written modification to the RI/FS Work Plan or written RI/FS Work Plan supplement. EPA reserves the right to conduct the Work itself at any point, to seek reimbursement from Respondents, and/or to seek any other appropriate relief.

f. Nothing in this Paragraph shall be construed to limit EPA's authority, in separate enforcement actions, to require performance of further response actions at the Study Area or the Site.

40. Off-Site Shipment of Waste Material. Respondents shall, prior to any off-site shipment of Waste Material from the Study Area to an out-of-state waste management facility, provide written notification of such shipment of Waste Material to the appropriate state environmental official in the receiving facility's state and to EPA's designated Project Coordinator. However, this notification requirement shall not apply to any off-site shipments when the total volume of all such shipments will not exceed 10 cubic yards.

a. Respondents shall include in the written notification the following information: (1) the name and location of the facility to which the Waste Material is to be shipped; (2) the type and quantity of the Waste Material to be shipped; (3) the expected schedule for the shipment of the Waste Material; and (4) the method of transportation. Respondents shall notify the state in which the planned receiving facility is located of major changes in the shipment plan, such as a decision to ship the Waste Material to another facility within the same state or to a facility in another state.

b. The identity of the receiving facility and state will be determined by Respondents following the award of the contract for the RI/FS. Respondents shall provide the information required by Paragraphs 40.a and 40.c as soon as practicable after the award of the contract and before the Waste Material is actually shipped.

c. Before shipping any hazardous substances, pollutants, or contaminants from the Study Area to an off-site location, Respondents shall obtain EPA's certification that the proposed receiving facility is operating in compliance with the requirements of CERCLA Section 121(d)(3), 42 U.S.C. § 9621(d)(3), and 40 C.F.R. § 300.440. Respondents shall only send hazardous substances, pollutants, or contaminants from the Study Area to an off-site facility that complies with the requirements of the statutory provision and regulation cited in the preceding sentence.

41. Meetings. Respondents shall make presentations at, and participate in, meetings as requested by EPA during the initiation, conduct, and completion of the RI/FS. In addition to discussion of the technical aspects of and substantive issues relating to the RI/FS, topics may include anticipated problems or new issues. Meetings will be scheduled at EPA's discretion. Respondents may also request such meetings.

42. Progress Reports. In addition to the plans, reports and other deliverables required to be submitted under this Settlement Agreement, Respondents shall provide to EPA monthly progress reports by the tenth day of each month. At a minimum, with respect to the preceding month, these progress reports shall (1) describe the actions which have been taken to comply with this Settlement Agreement, (2) include all results of sampling and tests and all other data received by Respondents, (3) describe Work planned for the next two months with schedules relating such Work to the overall project schedule for RI/FS completion, and (4) describe all problems encountered and any anticipated problems, any actual or anticipated delays, and solutions developed and implemented to address any actual or anticipated problems or delays.

43. Respondents shall submit copies of all plans, reports, or other submissions required by this Settlement Agreement, or any approved plan, as set forth below. Upon request by EPA, Respondents shall submit in electronic form all portions of any plan, report or other deliverable Respondents are required to submit pursuant to provisions of this Settlement Agreement. Any electronic submissions must be in a format that is compatible with EPA software and in database files and sizes to be specified by EPA. Reports should be submitted to the following:

a. Four copies (2 bound, 1 unbound and 1 electronic) to:

United States Environmental Protection Agency, Region 2
Emergency and Remedial Response Division
Special Projects Branch: Newtown Creek Superfund Site
290 Broadway, 20th Floor
New York, NY 10007-1866
Attention: Newtown Creek Superfund Site Remedial Project Manager

b. One copy (bound), to:

Chief, New York/Caribbean Superfund Branch
Office of Regional Counsel
United States Environmental Protection Agency, Region 2
290 Broadway, 17th Floor
New York, New York 10007-1866
Attention: Newtown Creek Superfund Site Attorney

c. Three copies (2 unbound, 1 electronic), to:

Director, Division of Environmental Remediation
N.Y. State Department of Environmental Conservation
625 Broadway, 12th Floor
Albany, New York 12233-7011
Attention: Newtown Creek Superfund Site Project Manager

44. Emergency Response and Notification of Releases.

a. In the event of any action or occurrence caused by or otherwise resulting from the performance of Work by Respondents, which results in a release or threat of release of Waste Material from the Study Area that constitutes an emergency situation or which may present an immediate threat to public health or welfare or the environment, Respondents shall immediately take all appropriate action. Respondents shall take these actions in accordance with all applicable provisions of this Settlement Agreement, including, but not limited to, the Health and Safety Plan, in order to prevent, abate or minimize such release or endangerment caused or threatened by the release. Respondents shall also immediately notify the Chief of the Response and Prevention Branch of the Emergency and Remedial Response Division of EPA, Region 2, at (732) 321-6656 of the incident or Site conditions. In the event that Respondents fail to take appropriate response action as required by this Paragraph, and EPA takes such action instead, then EPA's costs of such action shall constitute Future Response Costs and Respondents shall reimburse EPA pursuant to Section XVIII (Payment of Response Costs) for the costs of the response action which are not inconsistent with the NCP.

b. In addition, in the event of any release of a hazardous substance from the Study Area, Respondents shall immediately notify the National Response Center at (800) 424-8802 as well as the EPA Project Coordinator or, in the event of his/her unavailability, the Chief of the Special Projects Branch currently at 212-637-4435. Respondents shall submit a written report to EPA within seven days after any such release, setting forth the events that occurred and the measures taken or to be taken to mitigate any release or endangerment caused or threatened by the release and to prevent the reoccurrence of such a release.

c. This foregoing reporting requirement is in addition to, and not in lieu of, reporting under Section 103(c) of CERCLA, 42 U.S.C. § 9603(c), and Section 304 of the Emergency Planning and Community Right-To-Know Act of 1986, 42 U.S.C. § 11004, *et seq.*

X. EPA APPROVAL OF PLANS AND OTHER SUBMISSIONS

45. After review of any plan, report or other item that is required to be submitted for approval pursuant to this Settlement Agreement, EPA shall notify Respondents that it either: (a) approves, in whole or in part, the submission; (b) approves the submission upon specified conditions; (c) modifies the submission to cure the deficiencies; (d) disapproves, in whole

or in part, the submission, directing that Respondents modify the submission; or (e) any combination of the above. However, EPA shall not modify a submission without first providing Respondents at least one notice of deficiency and an opportunity to cure within 21 days, except where to do so would cause serious disruption to the Work or where previous submission(s) have been disapproved due to material defects, or greater than 21 days if mutually agreed to in writing by EPA and the Respondents.

46. In the event of approval, approval upon conditions, and/or modification by EPA, pursuant to Paragraph 45.a, 45.b, 45.c, or 45.e, Respondents shall proceed to take any action required by the plan, report or other deliverable, as approved or modified by EPA subject only to their right to invoke the Dispute Resolution procedures set forth in Section XV (Dispute Resolution) with respect to the modifications or conditions made by EPA. Following EPA approval or modification of a submission or portion thereof, Respondents shall not thereafter alter or amend such submission or portion thereof unless directed by EPA. In the event that EPA modifies the submission to cure the deficiencies pursuant to Paragraph 45.c and the submission had a material defect, EPA retains the right to seek stipulated penalties, as provided in Section XVI (Stipulated Penalties).

47. Resubmission.

a. Upon receipt of a notice of disapproval, Respondents shall, within thirty days or such longer time as specified by EPA in such notice, correct the deficiencies and resubmit the plan, report, or other deliverable for approval. Any stipulated penalties applicable to the submission, as provided in Section XVI, shall accrue during the thirty-day period or otherwise specified period but shall not be payable unless the resubmission is disapproved or modified due to a material defect as provided in Paragraphs 48 and 49.

b. Notwithstanding the receipt of a notice of disapproval, Respondents shall proceed to take any action required by any non-deficient portion of the submission, unless otherwise directed by EPA. Implementation of any non-deficient portion of a submission shall not relieve Respondents of any liability for stipulated penalties under Section XVI (Stipulated Penalties).

c. Respondents shall not proceed further with subsequent activities or tasks until receiving EPA approval, approval on condition or modification of the following deliverables: each subsequent work plan or work plan addendum for the RI/FS that are required to be developed pursuant to the RI/FS Work Plan, Sampling and Analysis Plan(s), Quality Assurance Project Plan(s), Draft Remedial Investigation Report(s), Draft Pathways Analysis Report(s), Draft Human Health Risk Assessment Report, Draft Screening Level Ecological Risk Assessment, Draft Baseline Ecological Risk Assessment Report and Treatability Testing Work Plan(s) and associated Sampling and Analysis Plan(s) and Draft Feasibility Study Report(s). While awaiting EPA approval, approval on condition or modification of these deliverables, Respondents shall proceed with all other tasks and activities

which may be conducted independently of these deliverables, in accordance with the schedule set forth in the RI/FS Work Plan or as otherwise approved by EPA.

d. For all remaining deliverables not listed above in Paragraph 47.c., Respondents shall proceed with all subsequent tasks, activities and deliverables without awaiting EPA approval on a submitted deliverable. EPA reserves its right to stop Respondents from proceeding further, either temporarily or permanently, on any task, activity or deliverable at any point during the RI/FS, and Respondents agree to cease any task, activity or deliverable if so directed by EPA.

48. If EPA disapproves a resubmitted plan, report or other deliverable, or portion thereof, EPA may again direct Respondents to correct the deficiencies. EPA shall also retain the right to modify or develop the plan, report, or other deliverable. Respondents shall implement any such plan, report, or deliverable as corrected, modified or developed by EPA, subject only to Respondents' right to invoke the procedures set forth in Section XV (Dispute Resolution).

49. If upon resubmission, a plan, report, or other deliverable is disapproved or modified by EPA due to a material defect, Respondents shall be deemed to have failed to submit such plan, report, or other deliverable timely and adequately unless Respondents invoke the dispute resolution procedures in accordance with Section XV (Dispute Resolution) and EPA's action is revoked or substantially modified pursuant to a Dispute Resolution decision issued by EPA or superceded by an agreement reached pursuant to that Section. The provisions of Section XV (Dispute Resolution) and Section XVI (Stipulated Penalties) shall govern the implementation of the Work and accrual and payment of any stipulated penalties during Dispute Resolution. If EPA's disapproval or modification is not otherwise revoked, substantially modified or superceded as a result of a decision or agreement reached pursuant to the Dispute Resolution process set forth in Section XV, stipulated penalties shall accrue for such violation from the date on which the initial submission was originally required, as provided in Section XVI.

50. In the event that EPA takes over some of the tasks, but not the preparation of the RI Report or the FS Report, Respondents shall incorporate and integrate information supplied by EPA into the final reports.

51. All plans, reports, and other deliverables submitted to EPA under this Settlement Agreement shall, upon approval or modification by EPA, be incorporated into and enforceable under this Settlement Agreement. In the event EPA approves or modifies a portion of a plan, report, or other deliverable submitted to EPA under this Settlement Agreement, the approved or modified portion shall be incorporated into and enforceable under this Settlement Agreement.

52. Neither failure of EPA to expressly approve or disapprove of Respondents' submissions within a specified time period, nor the absence of comments, shall be construed as

approval by EPA. Whether or not EPA gives express approval for Respondents' deliverables, Respondents are responsible for preparing deliverables acceptable to EPA.

XI. QUALITY ASSURANCE, SAMPLING, AND ACCESS TO INFORMATION

53. Quality Assurance. Respondents shall assure that Work performed, samples taken and analyses conducted conform to the requirements of the RI/FS Work Plan, the EPA-approved Quality Assurance Project Plan ("QAPP") and guidances identified therein. Respondents will assure that field personnel used by Respondents are properly trained in the use of field equipment and in chain of custody procedures. Respondents shall only use laboratories which have a documented quality system that complies with "EPA Requirements for Quality Management Plans (QA/R-2)" (EPA/240/B-01/002, March 2001, reissued May 2006) or equivalent documentation as determined by EPA.

54. Sampling.

a. All results of sampling, tests, modeling or other data (including raw data) generated by Respondents, or on Respondents' behalf, during the performance of the Work required under this Settlement Agreement, shall be submitted to EPA in the next monthly progress report after they are generated as described in Paragraph 42 of this Settlement Agreement. EPA will make available to Respondents validated data generated by EPA, as well as all QA/QC data, unless it is exempt from disclosure by any federal or state law or regulation.

b. Respondents shall verbally notify EPA at least 14 days prior to conducting significant field events as described in the RI/FS Work Plan or Sampling and Analysis Plan, provided, however, for any sampling events during wet weather/storm flows or tides, Respondents shall verbally notify EPA as soon as practicable, but not less than one Working Day prior to such sampling event. At EPA's verbal or written request, or the request of EPA's oversight assistant, Respondents shall allow split or duplicate samples to be taken by EPA (and its authorized representatives) of any samples collected in implementing the Work under this Settlement Agreement. All split samples analyzed by EPA shall be analyzed by the methods identified in the QAPP.

c. Circumstances permitting, EPA shall verbally notify Respondents at least 14 days prior to conducting significant field events relating to the Study Area, other than emergency response actions, activities conducted pursuant to Paragraph 86 (Work Takeover), or any enforcement-related events that require confidentiality. At Respondents' verbal or written request, EPA shall allow split or duplicate samples to be taken by Respondents (and their authorized representatives) of any samples collected in overseeing this Settlement Agreement or otherwise investigating the Site. All split samples shall be analyzed by the methods identified in the QAPP.

d. Any Respondent that has or independently produces as a requirement under any other regulatory program (i) any reports relating to Waste Materials in the Study Area, (ii) any reports relating to Waste Materials from direct or indirect discharges

into the Study Area, or (iii) any other reports relating to the Study Area or related to direct or indirect discharges into the Study Area, shall disclose such reports to EPA and, upon request of EPA or the Respondents, shall provide a copy of such report to EPA and the Respondents.

e. To the extent provided by the RI/FS Work Plan, if EPA or the Respondents propose to install a groundwater monitoring well on a facility which is situated upland of Newtown Creek and which is owned by or under the control of any of the Respondents, then such Respondent shall install such groundwater monitoring well.

55. Access to Information.

a. Respondents shall provide to EPA, upon request, copies of all documents and information within their possession or control or that of their contractors or agents relating to implementing the RI/FS Work Plan Activities or additional Work as required hereunder at the Site or to the implementation of this Settlement Agreement, including, but not limited to, sampling, analysis, chain of custody records, manifests, trucking logs, receipts, reports, sample traffic routing, correspondence, or other documents or information related to the Work. Respondents shall also make available to EPA, for purposes of investigation, information gathering, or testimony, their employees, agents, or representatives with knowledge of relevant facts concerning the performance of the Work.

b. Respondents may assert business confidentiality claims covering part or all of the documents or information submitted to EPA under this Settlement Agreement to the extent permitted by and in accordance with Section 104(e)(7) of CERCLA, 42 U.S.C. § 9604(e)(7), and 40 C.F.R. § 2.203(b). Documents or information determined to be confidential by EPA will be afforded the protection specified in 40 C.F.R. Part 2, Subpart B. If no claim of confidentiality accompanies documents or information when it is submitted to EPA, or if EPA has notified Respondents that the documents or information are not confidential under the standards of Section 104(e)(7) of CERCLA or 40 C.F.R. Part 2, Subpart B, the public may be given access to such documents or information without further notice to Respondents. Respondents shall segregate and clearly identify all documents or information submitted under this Settlement Agreement for which Respondents assert business confidentiality claims.

c. Respondents may assert that certain documents, records and other information are privileged under the attorney-client privilege or any other privilege recognized by federal law. If the Respondents assert such a privilege in lieu of providing documents, they shall provide EPA with the following: (1) the title of the document, record, or information; (2) the date of the document, record, or information; (3) the name and title of the author of the document, record, or information; (4) the name and title of each addressee and recipient; (5) a description of the contents of the document, record, or information; and (6) the privilege asserted by Respondents. However, no documents, reports or other

information required to be created or generated by this Settlement Agreement shall be withheld on the grounds that they are privileged, provided that Respondents may assert such privileges with respect to internal draft documents that have not been disclosed to persons other than Respondents, their counsel, contractors or agents.

d. No claim of confidentiality shall be made with respect to any data, including, but not limited to, all sampling, analytical, monitoring, hydrogeologic, scientific, chemical, or engineering data, or any other documents or information evidencing conditions at or around the Site to the extent that EPA or Respondents believe that such data relates to the Study Area.

56. In entering into this Settlement Agreement, Respondents waive any objections to any data gathered, generated, or evaluated by EPA, the State or Respondents in the performance or oversight of the Work that has been verified according to the quality assurance/quality control ("QA/QC") procedures required by the Settlement Agreement or the RI/FS Work Plan or Sampling and Analysis Plans. If Respondents object to any other data relating to the RI/FS, Respondents shall submit to EPA a report that specifically identifies and explains their objections, describes the acceptable uses of the data, if any, and identifies any limitations to the use of the data. Such report must be submitted to EPA within fifteen days of the monthly progress report containing the data.

XII. SITE ACCESS

57. If any portion of the Study Area or Site, or any other property where access is needed to implement this Settlement Agreement, is owned or controlled by any of Respondents, such Respondents shall, commencing on the Effective Date, provide EPA and the State, and their representatives, including contractors, with access at all reasonable times to such property, including without limitation, any vessels, where Respondents are conducting any Work, for the purpose of conducting or overseeing any activity related to this Settlement Agreement.

58. Access to Areas Not Owned by a Respondent.

a. Where any action under this Settlement Agreement is to be performed in areas owned by or in possession of someone other than Respondents, Respondents shall use their best efforts to obtain all necessary access agreements from the present owners within forty-five days of the later of i) the Effective Date of this Settlement Agreement, or ii) the date as it is determined by the EPA Project Manager that access to such other properties is needed for performance of Work. Respondents shall immediately notify EPA if after using their best efforts they are unable to obtain such agreements. For purposes of this Paragraph, "best efforts" includes the payment of reasonable sums of money in consideration of access, unless EPA has identified the property owner as a PRP under CERCLA in connection with the Study Area. Respondents shall describe in writing their efforts to obtain access. If Respondents cannot obtain access agreements, EPA may either (i) obtain access

for Respondents or assist Respondents in gaining access, to the extent necessary to effectuate the response actions described in this Settlement Agreement, using such means as EPA deems appropriate; or (ii) perform those tasks or activities with EPA contractors. Respondents shall reimburse EPA for all costs and attorney's fees incurred by the United States in obtaining such access, in accordance with the procedures in Section XVIII (Payment of Response Costs). If EPA performs those tasks or activities with EPA contractors and does not terminate the Settlement Agreement, Respondents shall perform all other tasks or activities not requiring access to that property, and Respondents shall reimburse EPA for all costs incurred in performing such tasks or activities. Respondents shall integrate the results of any such tasks or activities undertaken by EPA into its plans, reports and other deliverables.

b. Notwithstanding Paragraph 58.a, if access is sought to any upland facility situated outside of the Study Area which is not owned by or under the control of a Respondent, but which is suspected to be a significant source of groundwater impact or discharge to the Study Area such that EPA or Respondents propose to install a groundwater well or take other action in connection with the RI/FS, Respondents will identify the current facility owner or operator to EPA for further agency action. If the facility owner cannot be identified, is no longer in existence, or fails or refuses to install any requested upland groundwater well or to take such other actions, then the Respondents may, but will not be required to, install such monitoring well or perform such action under this Settlement Agreement.

59. Notwithstanding any provision of this Settlement Agreement, EPA retains all of its access authorities and rights, including enforcement authorities related thereto, under CERCLA, RCRA, and any other applicable statutes or regulations.

XIII. COMPLIANCE WITH OTHER LAWS

60. Respondents shall comply with all applicable local, state, and federal laws and regulations when performing the RI/FS. No local, state, or federal permit shall be required for any portion of any action conducted entirely on-site, including studies, if the action is selected and carried out in compliance with CERCLA and the NCP, provided that Respondents shall comply with the substantive requirements that would otherwise be included in such permits. Where any portion of the Work is to be conducted off-site and requires a federal, state or local permit or approval, Respondents shall submit timely and complete applications and take all other actions necessary to obtain and to comply with all such permits or approvals. Respondents may seek relief under the provisions of Section XVII (Force Majeure) of this Settlement Agreement for any delay in performing the Work due to a failure to obtain, or a delay in obtaining, any permit required for the Work, provided that Respondents have made proper, timely, and complete application for such permit and have taken all other actions necessary to obtain and to comply with all such permits or approvals. This Settlement Agreement is not, and shall not be construed to be, a permit issued pursuant to any federal or state statute or regulation.

XIV. RETENTION OF RECORDS

61. During the pendency of this Settlement Agreement and for a minimum of 10 years after commencement of construction of any remedial action, each Respondent shall preserve and retain all non-identical copies of documents, records, and other information (including documents, records, or other information in electronic form) now in its possession or control or which come into its possession or control that relate in any manner to the performance of the Work or the liability of any person under CERCLA with respect to the Study Area, regardless of any corporate or governmental retention policy to the contrary. Until 10 years after commencement of construction of any remedial action, Respondents shall also instruct their contractors and agents to preserve all documents, records, and other information of whatever kind, nature, or description relating to performance of the Work.

62. At the conclusion of this document retention period, Respondents shall notify EPA at least ninety days prior to the destruction of any such documents, records, or other information, and, upon request by EPA, Respondents shall deliver any such documents, records, or other information to EPA. Respondents may assert that certain documents, records, and other information are privileged under the attorney-client privilege or any other privilege recognized by federal law. If Respondents assert such a privilege, they shall provide EPA with the information set forth in Paragraph 55.c (1)-(6). Except as otherwise provided in Paragraph 55.c., no documents, records or other information required to be created or generated pursuant to this Settlement Agreement shall be withheld on the grounds that they are privileged.

63. Each Respondent hereby certifies individually that to the best of its knowledge and belief, after thorough inquiry, it has not altered, mutilated, discarded, destroyed or otherwise disposed of any records, documents or other information (other than identical copies) relating to its potential liability regarding the Study Area, since notification of potential liability by EPA or the filing of suit against it regarding the Study Area, and that it has fully complied with any and all EPA requests for information pursuant to Sections 104(e) and 122(e) of CERCLA, 42 U.S.C. §§ 9604(e) and 9622(e), and Section 3007 of RCRA, 42 U.S.C. § 6927.

XV. DISPUTE RESOLUTION

64. Unless otherwise expressly provided for in this Settlement Agreement, the dispute resolution procedures of this Section shall be the exclusive mechanism for resolving disputes arising under this Settlement Agreement. The Parties shall attempt to resolve any disagreements concerning this Settlement Agreement expeditiously and informally.

65. If Respondents object to any EPA action taken pursuant to this Settlement Agreement, including billings for Future Response Costs, they shall notify EPA in writing of their

objection(s) within 14 days of such action, unless the objection(s) has/have been resolved informally. EPA and Respondents shall have 30 days from EPA's receipt of Respondents' written objection(s) to resolve the dispute (the "Negotiation Period"). The Negotiation Period may be extended at the sole discretion of EPA. Such extension may be granted verbally but must be confirmed in writing.

66. Any agreement reached by the Parties pursuant to this Section shall be in writing and shall, upon signature by the Parties, be incorporated into and become an enforceable part of this Settlement Agreement. If the Parties are unable to reach an agreement within the Negotiation Period, the Chief of the Special Projects Branch of the Emergency and Remedial Response Division, EPA Region 2, will issue a written decision. When feasible, Respondents shall be given an opportunity to meet with the Chief of the Special Projects Branch of the Emergency and Remedial Response Division, EPA Region 2, before the decision on the dispute is made. The administrative record of the dispute shall be maintained by EPA and will include all correspondence and material exchanged between EPA and Respondents during the dispute resolution process. EPA's decision shall be incorporated into and become an enforceable part of this Settlement Agreement. Respondents' obligations under this Settlement Agreement shall not be tolled by submission of any objection for dispute resolution under this Section. Following resolution of the dispute, as provided by this Section, Respondents shall fulfill the requirement that was the subject of the dispute in accordance with the agreement reached or with EPA's decision, whichever occurs, and regardless of whether Respondents agree with the decision.

XVI. STIPULATED PENALTIES

67. Respondents shall be liable to EPA for stipulated penalties in the amounts set forth in Paragraphs 68.a and 68.b for failure to comply with any of the requirements of this Settlement Agreement as specified below unless excused by EPA under Section XVII (Force Majeure) or otherwise. "Compliance" by Respondents shall include completion of the Work required under this Settlement Agreement including any activities contemplated under any RI/FS Work Plan or other plan approved under this Settlement Agreement as identified below, in accordance with all applicable requirements of law, this Settlement Agreement, the RI/FS Work Plan, and any plans or other documents approved by EPA pursuant to this Settlement Agreement and within the specified time schedules established by and approved under this Settlement Agreement.

68. Stipulated Penalty Amounts

a. The following stipulated penalties shall accrue per day for any non-compliance identified in Paragraph 68.a:

Penalty Per Violation Per Day	Period of Noncompliance

$1,500	1st through 14th day
$3,000	15th through 30th day
$6,000	31st day and beyond

Stipulated penalties as specified in Paragraph 68.a shall accrue per violation per day if Respondents fail to timely submit the following major deliverables required under the RI/FS Work Plan: The RI/FS Work Plan attached hereto as Appendix A and including all additional work plans to be developed after the Effective Date pursuant to the RI/FS Work Plan and also including all supporting plans (FSAP, QAPP, HASP, DMP and CIP); Phase I RI Interim Data Report; Phase I RI Work Plan Addendum; Phase I RI Data Summary Report; Phase II RI Work Plan including supporting plans (specifically FSAP, QAPP, HASP and DMP); RI Report; BERA Problem Formulation Report; BERA Work Plan; PAR-HHRA; BERA Report; HHRA Report; FS Work Plan; FS Field Program Work Plan (if needed); and FS Report (including FS Field Program Report to be attached thereto).

All of these major deliverables shall be completed and submitted by Respondents in accordance with the RI/FS Work Plan Estimated Program Schedule set forth in Section 3.5 of the RI/FS Work Plan or as otherwise modified by agreement of EPA and Respondents.

b. For all other violations of this Settlement Agreement, including, without limitation, Respondents failure to provide timely or adequate certificates of insurance pursuant to Paragraph 101 or periodic progress reports required under Paragraph 42, or failure to timely pay Past Response Costs or Future Response Costs in accordance with Section XVIII, stipulated penalties shall accrue per violation per day as follows:

Penalty Per Violation Per Day	Period of Noncompliance
$ 750	1st through 14th day
$1,500	15th through 30th day
$3,000	31st day and beyond

69. In the event that EPA assumes performance of all or a substantial portion of the Work pursuant to Paragraph 87 of Section XX (Reservation of Rights by EPA), Respondents shall be liable for a stipulated penalty in the amount of five million dollars ($5-million). EPA agrees that any penalty assessed against Respondents under this Paragraph shall be reduced by the percentage of Work completed by the Respondents as of the time of the

assessment. This Paragraph shall not apply to circumstances described in Paragraph 58.b in which EPA performs work because Respondents are unable to obtain access.

70. All penalties shall begin to accrue on the day after the complete performance is due or the day a violation occurs, and they shall continue to accrue through the final day of the correction of the noncompliance or completion of the activity. However, stipulated penalties shall not accrue: (1) with respect to a deficient submission under Section X (EPA Approval of Plans and Other Submissions), during the period, if any, beginning on the 31st day after EPA's receipt of such submission until the date that EPA notifies Respondents of any deficiency; and (2) with respect to a decision by the Chief of the Special Projects Branch as provided by Paragraph 66 of Section XV (Dispute Resolution), during the period, if any, beginning on the 31st day after the Negotiation Period begins until the date that the Chief of the Special Projects Branch issues a final decision regarding such dispute. Nothing in this Settlement Agreement shall prevent the simultaneous accrual of separate penalties for separate violations of this Settlement Agreement.

71. Following EPA's determination that Respondents have failed to comply with a requirement of this Settlement Agreement, EPA may give Respondents written notification of the same and describe the noncompliance. EPA may send Respondents a written demand for the payment of the penalties. However, penalties shall accrue as provided in the preceding Paragraph regardless of whether EPA has notified Respondents of a violation.

72. All penalties accruing under this Section shall be due and payable to EPA within thirty days of Respondents' receipt from EPA of a demand for payment of the penalties, unless Respondents invoke the dispute resolution procedures under Section XVI (Dispute Resolution). All payments to EPA under this Paragraph shall be identified as "stipulated penalties" and shall be made by Fedwire Electronic Funds Transfer in the manner set forth in Paragraph 80.b, below.

73. The payment of penalties shall not alter in any way Respondents' obligation to complete performance of the Work required under this Settlement Agreement.

74. Penalties shall continue to accrue as provided in Paragraph 70 during any dispute resolution period, but need not be paid until fifteen days after the dispute is resolved by agreement or by receipt of EPA's decision.

75. If Respondents fail to pay stipulated penalties when due, EPA may institute proceedings to collect the penalties, as well as Interest. Respondents shall pay Interest on the unpaid balance, which shall begin to accrue on the date of demand made pursuant to Paragraph 72.

76. Nothing in this Settlement Agreement shall be construed as prohibiting, altering, or in any way limiting the ability of EPA to seek any other remedies or sanctions available by virtue

of Respondents' violation of this Settlement Agreement or of the statutes and regulations upon which it is based, including, but not limited to, penalties pursuant to Section 122(l) of CERCLA, 42 U.S.C. § 9622(l), and punitive damages pursuant to Section 107(c)(3) of CERCLA, 42 U.S.C. § 9607(c)(3). However, EPA shall not seek civil penalties pursuant to Section 122(l) of CERCLA or punitive damages pursuant to Section 107(c)(3) of CERCLA for any violation for which a stipulated penalty is provided in this Settlement Agreement, except in the case of willful violation of this Settlement Agreement or in the event that EPA assumes performance of a portion or all of the Work pursuant to Section XX (Reservation of Rights by EPA), Paragraph 87. Notwithstanding any other provision of this Section, EPA may, in its unreviewable discretion, waive any portion of stipulated penalties that have accrued pursuant to this Settlement Agreement.

XVII. FORCE MAJEURE

77. Respondents agree to perform all requirements of this Settlement Agreement within the time limits established under this Settlement Agreement, unless the performance is delayed by a *force majeure*. For purposes of this Settlement Agreement, *force majeure* is defined as any event arising from causes beyond the control of Respondents or of any entity controlled by Respondents, including but not limited to their contractors and subcontractors, which delays or prevents performance of any obligation under this Settlement Agreement despite Respondents' best efforts to fulfill the obligation. *Force majeure* does not include financial inability to complete the Work or increased cost of performance.

78. If any event occurs or has occurred that may delay the performance of any obligation under this Settlement Agreement, whether or not caused by a *force majeure* event, Respondents shall notify the EPA Project Coordinator (or, in her absence, the Chief of the Special Projects Branch of the Emergency and Remedial Response Division of EPA Region 2 currently at (212) 637-4435) within two Working Days of when Respondents or their agents, contractors or representatives knew or should have known that the event might cause a delay. Within ten Working Days thereafter, Respondents shall provide to EPA in writing an explanation and description of the reasons for the delay; the anticipated duration of the delay; all actions taken or to be taken to prevent or minimize the delay; a schedule for implementation of any measures to be taken to prevent or mitigate the delay or the effect of the delay; Respondents' rationale for attributing such delay to a *force majeure* event if they intend to assert such a claim; and a statement as to whether, in the opinion of Respondents, such event may cause or contribute to an endangerment to public health, welfare or the environment. Failure to comply with the above requirements shall preclude Respondents from asserting any claim of *force majeure* for that event for the period of time of such failure to comply and for any additional delay caused by such failure.

79. If EPA agrees that the delay or anticipated delay is attributable to *a force majeure* event, the time for performance of the obligations under this Settlement Agreement that are

affected by the *force majeure* event will be extended by EPA for such time as is necessary to complete those obligations. If EPA agrees that the delay is attributable to a *force majeure* event, EPA will notify Respondents in writing of the length of the extension, if any, for performance of the obligations affected by the *force majeure* event. An extension of the time for performance of the obligations affected by the *force majeure* event shall not, of itself, extend the time for performance of any other obligation. If EPA does not agree that the delay or anticipated delay has been or will be caused by a *force majeure* event, EPA will notify Respondents in writing of its decision.

XVIII. PAYMENT OF RESPONSE COSTS

80. Payment of Past Response Costs.

a. Within thirty days after the Effective Date, Respondents shall pay to EPA $750,000 for Past Response Costs in the manner set forth in Paragraph 80.b.

b. Payments shall be made to EPA by Fedwire Electronic Funds Transfer to:

Federal Reserve Bank of New York
ABA: 021030004
Account: 68010727
SWIFT address: FRNYUS33
Street Address: 33 Liberty Street, New York NY 10045
Field Tag 4200 of the Fedwire message should read "D 68010727 Environmental Protection Agency"
Amount: Specify the amount of the payment
Respondents' Names: Specify Respondents' Names
EPA Index Number: CERCLA-02-2011-2011
Site/spill identifier: A206

At the time of payment, Respondent shall send notice that such payment has been made by email to acctsreceivable.cinwd@epa.gov, and by mail to the addressees designated in Paragraph 43.a and Paragraph 43.b, above as well as to the following addressee:

U.S. Environmental Protection Agency
Attn: Richard Rice
26 W. Martin Luther King Drive
Cincinnati Finance Center, MS: NWD
Cincinnati, Ohio 45268
e-mail: rice.richard@epa.gov

c. The total amount to be paid by Respondents pursuant to Paragraph 80.a. and Paragraph 82 shall be deposited in the Newtown Creek Superfund Site Special

Account within the EPA Hazardous Substance Superfund to be retained and used to conduct or finance response actions at or in connection with the Site, or to be transferred by EPA to the EPA Hazardous Substance Superfund.

81. Payments of Future Response Costs.

a. Respondents shall pay EPA all Future Response Costs. On a periodic basis, EPA will send Respondents a bill requiring payment that includes a SCORPIOS Report, which includes direct and indirect costs incurred by EPA and its contractors, and, as applicable, a DOJ Cost Report, which includes costs incurred by DOJ and its contractors. Respondents shall make all payments within thirty days of receipt of each bill requiring payment, except as otherwise provided in Paragraph 82 of this Settlement Agreement, and payment shall be made in the manner set forth in Paragraph 80.b.

b. The total amount to be paid by Respondents pursuant to Paragraph 81.a. and Paragraph 82 shall be deposited in the Newtown Creek Superfund Site Special Account within the EPA Hazardous Substance Superfund to be retained and used to conduct or finance response actions at or in connection with the Site, or to be transferred by EPA to the EPA Hazardous Substance Superfund.

82. If Respondents fail to pay Past Response Costs within 30 days after the Effective Date or Future Response Costs within 30 days of Respondents' receipt of a bill, Respondents shall pay Interest on the unpaid balance of Past Response Costs or Future Response Costs. The Interest on unpaid Past Response Costs or Future Response Costs shall begin to accrue on the Effective Date or on the date of the bill, as applicable, and shall continue to accrue until the date of payment. If EPA receives a partial payment, Interest shall accrue on any unpaid balance. Payments of Interest made under this Paragraph shall be in addition to such other remedies or sanctions available to the United States by virtue of Respondents' failure to make timely payments under this Section, including but not limited to, payments of stipulated penalties pursuant to Section XVI. Respondents shall make all payments required by this Paragraph in the manner described in Paragraph 80.b.

83. Respondents may contest payment of any Future Response Costs under Paragraph 81 if they determine that EPA has made a mathematical error or included a cost item that is not within the definition of Future Response Costs, or if they believe EPA incurred excess costs as a direct result of an EPA action that was inconsistent with a specific provision or specific provisions of the NCP. Such objection shall be made in writing within 30 days of receipt of the bill and must be sent to the EPA Project Coordinator. Any such objection shall specifically identify the contested Future Response Costs and the basis for objection. In the event of an objection, Respondents shall within the 30 day period pay all uncontested Future Response Costs to EPA in the manner described in Paragraph 81. Simultaneously, Respondents shall establish, in a duly chartered bank or trust company, an interest-bearing escrow account that is insured by the Federal Deposit Insurance

Corporation, and remit to that escrow account funds equivalent to the amount of the contested Future Response Costs. Respondents shall send to the EPA Project Coordinator a copy of the transmittal letter and check paying the uncontested Future Response Costs, and a copy of the correspondence that establishes and funds the escrow account, including, but not limited to, information containing the identity of the bank and bank account under which the escrow account is established as well as a bank statement showing the initial balance of the escrow account. Simultaneously with establishment of the escrow account, Respondents shall initiate the Dispute Resolution procedures in Section XV (Dispute Resolution). If EPA prevails in the dispute, within 5 days of the resolution of the dispute, Respondents shall pay the sums due (with accrued interest) to EPA in the manner described in Paragraph 81. If Respondents prevail concerning any aspect of the contested costs, Respondents shall pay that portion of the costs (plus associated accrued interest) for which they did not prevail to EPA in the manner described in Paragraph 81. Respondents shall be disbursed any balance of the escrow account. The dispute resolution procedures set forth in this Paragraph in conjunction with the procedures set forth in Section XV (Dispute Resolution) shall be the exclusive mechanisms for resolving disputes regarding Respondents' obligation to pay EPA for its Future Response Costs.

XIX. COVENANT NOT TO SUE BY EPA

84. In consideration of the actions that will be performed and the payments that will be made by Respondents under the terms of this Settlement Agreement, and except as otherwise specifically provided in this Settlement Agreement, EPA covenants not to sue or to take administrative action against Respondents pursuant to Sections 106 and 107(a) of CERCLA, 42 U.S.C. §§ 9606 and 9607(a), for the Work, Past Response Costs, and Future Response Costs. This covenant not to sue shall take effect upon receipt by EPA of the Past Response Costs due under Section XVIII of this Settlement Agreement and any Interest or Stipulated Penalties due for failure to pay Past Response Costs as required by Sections XVIII and XVI of this Settlement Agreement. This covenant not to sue is conditioned upon the complete and satisfactory performance by Respondents of their respective obligations under this Settlement Agreement, including, but not limited to, payment of Future Response Costs pursuant to Section XVIII. This covenant not to sue extends only to Respondents and does not extend to any other person.

XX. RESERVATIONS OF RIGHTS BY EPA

85. Except as specifically provided in this Settlement Agreement, nothing in this Settlement Agreement shall limit the power and authority of EPA or the United States to take, direct, or order all actions necessary to protect public health, welfare, or the environment or to prevent, abate, or minimize an actual or threatened release of hazardous substances, pollutants, or contaminants, or hazardous or solid waste on, at, or from the Site. Further, nothing herein shall prevent EPA from seeking legal or equitable relief to enforce the terms of this Settlement Agreement, from taking other legal or equitable action as it deems

appropriate and necessary, or from requiring Respondents in the future to perform additional activities pursuant to CERCLA or any other applicable law.

86. The covenant not to sue set forth in Section XIX, above, does not pertain to any matters other than those expressly identified therein. EPA reserves, and this Settlement Agreement is without prejudice to, all rights against Respondents with respect to all other matters, including, but not limited to:

a. claims based on a failure by Respondents to meet a requirement of this Settlement Agreement;
b. liability for costs not included within the definition of Past Response Costs or Future Response Costs;
c. liability for performance of response activities other than the Work;
d. criminal liability;
e. liability for damages for injury to, destruction of, or loss of natural resources, and for the costs of any natural resource damage assessments;
f. liability arising from the past, present, or future disposal, release, or threat of release of Waste Materials outside of the Study Area;
g. liability for costs incurred or to be incurred by the Agency for Toxic Substances and Disease Registry related to the Study Area not paid as Future Response Costs under this Settlement Agreement.

87. Work Takeover.

a. In the event EPA determines that Respondents have (i) ceased implementation of any portion of the Work, (ii) are seriously or repeatedly deficient or late in their performance of the Work, or (iii) are implementing the Work in a manner which may cause an endangerment to human health or the environment, EPA may issue a written notice ("Work Takeover Notice") to the Respondents. Any Work Takeover Notice issued by EPA will specify the grounds upon which such notice was issued and will provide Respondents a period of 21 days within which to remedy the circumstances giving rise to EPA's issuance of such notice.
b. If, after expiration of the 21-day notice period specified in Paragraph 87.a, Respondents have not remedied to EPA's satisfaction the circumstances giving rise to EPA's issuance of the relevant Work Takeover Notice, EPA may at any time thereafter assume the performance of all or any portions of the Work as EPA deems necessary ("Work Takeover"). EPA shall notify Respondents in writing if EPA determines that implementation of a Work Takeover is warranted under this Paragraph 87.b.
c. Respondents may invoke the procedures set forth in Section XV (Dispute Resolution) to dispute EPA's implementation of a Work Takeover under Paragraph 87.b. However, notwithstanding Respondents' invocation of such dispute resolution procedures, and during the pendency of any such

dispute, EPA may in its sole discretion commence and continue a Work Takeover under Paragraph 87.b until the earlier of (i) the date that Respondents remedy, to EPA's satisfaction, the circumstances giving rise to EPA's issuance of the relevant Work Takeover Notice or (ii) the date that a final decision is rendered in accordance with Section XV (Dispute Resolution), requiring EPA to terminate such Work Takeover.

XXI. RESERVATION OF RIGHTS BY RESPONDENTS

88. Except as otherwise provided in this Settlement Agreement, Respondents expressly reserve all rights and affirmative defenses under statute, including CERCLA, or common law.

XXII. COVENANT NOT TO SUE BY RESPONDENTS

89. Respondents covenant not to sue and agree not to assert any claims or causes of action against the United States, or its contractors or employees, with respect to the Work, Past Response Costs, Future Response Costs, or this Settlement Agreement, including, but not limited to:

a. any direct or indirect claim for reimbursement from the Hazardous Substance Superfund established by 26 U.S.C. § 9507, based on Sections 106(b)(2), 107, 111, 112, or 113 of CERCLA, 42 U.S.C. §§ 9606(b)(2), 9607, 9611, 9612, or 9613, or any other provision of law;

b. any claim arising out of the Work or arising out of the response actions for which the Past Response Costs or Future Response Costs have been or will be incurred, respectively, including any claim under the United States Constitution, the New York State Constitution, the Tucker Act, 28 U.S.C. § 1491, the Equal Access to Justice Act, 28 U.S.C. § 2412, as amended, or at common law; or

c. any claim against the United States pursuant to Sections 107 and 113 of CERCLA, 42 U.S.C. §§ 9607 and 9613, relating to the Work or payment of Past Response Costs or Future Response Costs provided however, that this Settlement Agreement shall not have any effect on claims or causes of action in contribution that Respondents have or may have pursuant to CERCLA against the United States or any of its agencies or departments, other than EPA, as a responsible party under Section 107(a) of CERCLA, 42 U.S.C. § 9607(a), relating to the Site.

90. Except as expressly provided in Paragraph 92 (*De Minimis*/Ability to Pay Waivers) of this Section XXII (Covenant Not to Sue by Respondents), these covenants not to sue shall not apply in the event the United States brings a cause of action or issues an order pursuant to the reservations set forth in Section XX (Reservations of Rights by EPA), other than in Paragraph 86.a (claims for failure to meet a requirement of the Settlement Agreement) or

86.d (criminal liability), but only to the extent that Respondents' claims arise from the same response action, response costs, or damages that the United States is seeking pursuant to the applicable reservation.

91. Nothing in this Agreement shall be deemed to constitute approval or preauthorization of a claim within the meaning of Section 111 of CERCLA, 42 U.S.C. § 9611, or 40 C.F.R. § 300.700(d).

92. *De Minimis*/Ability to Pay Waivers. Respondents agree not to assert any claims and agree to waive all claims or causes of action (including but not limited to claims or causes of action under Sections 107(a) or 113 of CERCLA) that they may have for all matters relating to the Site, against any person who has settled with EPA, or who hereafter settles with EPA with respect to the Site, in a final Section 122(g) *de minimis* settlement or a final settlement based on such person's limited ability to pay, as of the Effective Date. This waiver shall not apply with respect to any defense, claim, or cause of action that a Respondent may have against any person if such person asserts a claim or cause of action relating to the Site against such Respondent.

XXIII. OTHER CLAIMS

93. By issuance of this Settlement Agreement, the United States and EPA assume no liability for injuries or damages to persons or property resulting from any acts or omissions of Respondents.

94. Except as expressly provided in Section XXII (Covenant Not to Sue by Respondents), Paragraph 92 (*De Minimis*/Ability to Pay Waivers), and Section XIX (Covenant Not to Sue by EPA), nothing in this Settlement Agreement constitutes a satisfaction of or release from any claim or cause of action against Respondents or any person not a Party to this Settlement Agreement, for any liability such person may have under CERCLA, other statutes, or common law, including but not limited to any claims of the United States for costs, damages and interest under Sections 106 and 107 of CERCLA, 42 U.S.C. §§ 9606 and 9607.

95. No action or decision by EPA pursuant to this Settlement Agreement shall give rise to any right to judicial review except as set forth in Section 113(h) of CERCLA, 42 U.S.C. § 9613(h).

XXIV. CONTRIBUTION PROTECTION

96. Contribution Protection.

a. The Parties agree that this Settlement Agreement constitutes an administrative settlement for purposes of Section 113(f)(2) of CERCLA, 42 U.S.C. § 9613(f)(2) and that Respondents are entitled, as of the Effective Date, to protection from

contribution actions or claims as provided by Section 113(f)(2) of CERCLA, 42 U.S.C. § 9613(f)(2) or as may be otherwise provided by law, for "matters addressed" in this Settlement Agreement. The "matters addressed" in this Settlement Agreement are the Work, Past Response Costs, and Future Response Costs. Notwithstanding such protections against contribution actions or claims by non-parties to this Settlement Agreement, the Respondents agree that they may allocate or reallocate any and all response costs incurred in connection with this Settlement Agreement among themselves.

b. The Parties agree that this Settlement Agreement constitutes an administrative settlement for purposes of Section 113(f)(3)(B) of CERCLA, 42 U.S.C. § 9613(f)(3)(B), pursuant to which Respondents have, as of the Effective Date, resolved their liability to the United States for the Work, Past Response Costs, and Future Response Costs.

c. Except as provided in Section XXII (Covenant Not to Sue by Respondents), Paragraph 92 (*De Minimis*/Ability to Pay Waivers), nothing in this Settlement Agreement shall be construed to create any rights in, or grant any cause of action to, any person not a Party to this Settlement Agreement. Except as provided in Paragraph 92 (De Minimis/Ability to Pay Waivers), each of the Parties expressly reserves any and all rights (including, but not limited to, pursuant to Section 113 of CERCLA, 42 U.S.C. § 9613), defenses, claims, demands, and causes of action which each Party may have with respect to any matter, transaction, or occurrence relating in any way to the Site against any person not a Party to this Settlement Agreement. Nothing in this Settlement Agreement diminishes the right of the United States, pursuant to Section 113(f)(2) and (3) of CERCLA, 42 U.S.C. § 9613(f)(2) and (3), to pursue any such persons to obtain additional response costs or response action and to enter into settlements that give rise to contribution protection pursuant to Section 113(f)(2).

97. Each Respondent shall, with respect to any suit or claim brought by it for matters related to this Settlement Agreement, notify EPA in writing no later than 60 days prior to the initiation of such suit or claim. Each Respondent also shall, with respect to any suit or claim brought against it for matters related to this Settlement Agreement, notify EPA in writing within 10 Working Days of service of the complaint or claim upon it. In addition, for matters related to this Settlement Agreement, each Respondent shall notify EPA within 10 Working Days of service or receipt of any Motion for Summary Judgment and within 10 Working Days of receipt of any order from a court setting a case for trial.

XXV. INDEMNIFICATION

98. Respondents shall indemnify, save and hold harmless the United States, its officials, agents, contractors, subcontractors, employees and representatives from any and all claims or causes of action arising from, or on account of negligent or other wrongful acts or omissions of Respondents, their officers, directors, employees, agents, contractors, or subcontractors, in carrying out actions pursuant to this Settlement Agreement. In

addition, Respondents agree to pay the United States all costs incurred by the United States, including but not limited to attorneys fees and other expenses of litigation and settlement, arising from or on account of claims made against the United States based on negligent or other wrongful acts or omissions of Respondents, their officers, directors, employees, agents, contractors, subcontractors and any persons acting on their behalf or under their control, in carrying out activities pursuant to this Settlement Agreement. The United States shall not be held out as a party to any contract entered into by or on behalf of Respondents in carrying out activities pursuant to this Settlement Agreement. Neither Respondents nor any such contractor shall be considered an agent of the United States.

99. The United States shall give Respondents notice of any claim for which the United States plans to seek indemnification pursuant to this Section and shall consult with Respondents prior to settling such claim.

100. Respondents waive all claims against the United States for damages or reimbursement or for set-off of any payments made or to be made to the United States arising from or on account of any contract, agreement, or arrangement between any one or more of Respondents and any person for performance of Work at or relating to the Study Area. In addition, Respondents shall indemnify and hold harmless the United States with respect to any and all claims for damages or reimbursement arising from or on account of any contract, agreement, or arrangement between any one or more of Respondents and any person for performance of Work at or relating to the Study Area.

XXVI. INSURANCE

101. At least fourteen days prior to commencing any on-site Work under this Settlement Agreement, Respondents shall secure, and shall maintain for the duration the Work being performed pursuant to this Settlement Agreement, comprehensive general liability insurance and automobile insurance with limits of $5-million, combined single limit, naming EPA as an additional insured. Within the same period, Respondents shall provide EPA with certificates of such insurance and, if requested, a copy of each insurance policy. Respondents shall submit such certificates and, if requested, copies of policies each year on the anniversary of the Effective Date. In addition, for the duration of the Work being performed pursuant to Settlement Agreement, Respondents shall satisfy, or shall ensure that their contractors or subcontractors satisfy, all applicable laws and regulations regarding the provision of worker's compensation insurance for all persons performing the Work on behalf of Respondents in furtherance of this Settlement Agreement. If Respondents demonstrate by evidence satisfactory to EPA that any contractor or subcontractor maintains insurance equivalent to that described above, or insurance covering some or all of the same risks but in an equal or lesser amount, then Respondents need provide only that portion of the insurance described above which is not maintained by such contractor or subcontractor.

XXVII. FINANCIAL ASSURANCE

102. Within 30 days of the Effective Date, Respondents shall establish and maintain financial security for the benefit of EPA in the amount of $25-million in one or more of the following forms, in order to secure the full and final completion of Work by Respondents:

a. a surety bond unconditionally guaranteeing payment and/or performance of the Work;
b. one or more irrevocable letters of credit, payable to or at the direction of EPA, issued by financial institution(s) acceptable in all respects to EPA equaling the total estimated cost of the Work;
c. a trust fund administered by a trustee acceptable in all respects to EPA;
d. a policy of insurance issued by an insurance carrier acceptable in all respects to EPA, which ensures the payment and/or performance of the Work;
e. a written guarantee to pay for or perform the Work provided by one or more parent companies of a corporate Respondent, or by one or more unrelated companies that have a substantial business relationship with at least one of Respondents, including a demonstration that any such guarantor company satisfies the financial test requirements of 40 C.F.R. Part 264.143(f); and/or
f. a demonstration of sufficient financial resources to pay for the Work made by one or more of Respondents, which shall consist of a demonstration that any such Respondent satisfies the requirements of 40 C.F.R. Part 264.143(f), provided, however that a demonstration made by a local government shall be required to satisfy the requirements of 40 C.F.R. Part 258.74(f).

103. Any and all financial assurance instruments provided pursuant to this Section shall be in form and substance satisfactory to EPA, determined in EPA's sole discretion. In the event that EPA determines at any time that the financial assurances provided pursuant to this Section (including, without limitation, the instrument(s) evidencing such assurances) are inadequate, Respondents shall, within 30 days of receipt of notice of EPA's determination, obtain and present to EPA for approval one of the other forms of financial assurance listed in Paragraph 102, above. In addition, if at any time EPA notifies Respondents that the anticipated cost of completing the Work has increased, then, within 30 days of such notification, Respondents shall obtain and present to EPA for approval a revised form of financial assurance (otherwise acceptable under this Section) that reflects such cost increase. Respondents' inability to demonstrate financial ability to complete the Work shall in no way excuse performance of any activities required under this Settlement Agreement.

104. If Respondents seek to ensure completion of the Work through a guarantee pursuant to Paragraph 102.e. or 102.f. of this Settlement Agreement, Respondents shall (i)

demonstrate to EPA's satisfaction that the guarantor satisfies the requirements of 40 C.F.R. Part 264.143(f); and (ii) resubmit to EPA sworn statements conveying the information required by 40 C.F.R. Part 264.143(f) annually, on the anniversary of the Effective Date, or such other date as agreed by EPA. For the purposes of this Settlement Agreement, wherever 40 C.F.R. Part 264.143(f) references "sum of current closure and post-closure costs estimates and the current plugging and abandonment costs estimates," the dollar amount to be used in the relevant financial test calculations shall be the current cost estimate of $25 million for the Work at the Study Area plus any other RCRA, CERCLA, TSCA, or other federal environmental obligations financially assured by the relevant Respondent or guarantor to EPA by means of passing a financial test.

105. If, after the Effective Date, Respondents can show that the estimated cost to complete the remaining Work has diminished below the amount set forth in Paragraph 102 of this Section, Respondents may, on any anniversary date of the Effective Date, or at any other time agreed to by the Parties, request that the amount of the financial security provided under this Section be reduced to the estimated cost of the remaining Work to be performed. Respondents shall submit a proposal for such reduction to EPA, in accordance with the requirements of this Section, and may reduce the amount of the security after receiving written approval from EPA. In the event of a dispute, Respondents may seek dispute resolution pursuant to Section XV (Dispute Resolution). Respondents may reduce the amount of security by agreement among the Parties resolving the dispute or, failing such agreement, by EPA's written decision resolving the dispute.

106. Respondents may change the form of financial assurance provided under this Section at any time, upon notice to and prior written approval by EPA, provided that EPA determines that the new form of assurance meets the requirements of this Section. In the event of a dispute, Respondents may change the form of the financial assurance only in accordance with an agreement among the Parties resolving the dispute or, failing such agreement, EPA's written decision resolving the dispute.

XXVIII. CITY OF NEW YORK TRUST FUND ACCOUNT

107. The City of New York shall establish a trust fund account for the purpose of enabling the City of New York to fund a share of the Work and to enter into the Settlement Agreement as a Respondent. Within 30 days after the Effective Date of this Settlement Agreement, the City of New York shall enter into a trust fund agreement (City of New York Trust Fund Agreement) in substantially the same form as that set forth in Appendix F hereto in order to establish the Newtown Creek City of New York Special Trust Fund Account ("City of New York Trust Fund").

108. The City of New York shall make payments into the City of New York Trust Fund in such amounts and at such times as has been agreed to by the City and the other Respondents. The City of New York shall notify EPA of each such payment within 7 days after such payment has been made. The City of New York Trust Fund shall be used by the City for

the sole purpose of funding performance of the Work required by this Settlement Agreement and administering the City of New York Trust Fund as provided in the City of New York Trust Fund Agreement, and shall not be used for any other purpose. Payments from the City of New York Trust Fund shall be made in accordance with the City of New York Trust Fund Agreement and has been agreed to by the City and the other Respondents. The Parties acknowledge that funds placed into the City of New York Trust Fund remain the property of the City of New York until expended pursuant to the Trust Fund Agreement and that unexpended funds may be withdrawn by the City of New York at the time of termination of the City of New York Trust Fund Agreement.

109. Nothing in this Section XXVIII (City of New York Trust Fund Account) shall reduce, diminish or otherwise affect the joint and several obligations of the City of New York and each other Respondent pursuant to Paragraph 8 of this Settlement Agreement.

XXIX. INTEGRATION/APPENDICES

110. This Settlement Agreement and its appendices and any deliverables, technical memoranda, specifications, schedules, documents, plans, reports (other than progress reports), etc. that will be developed pursuant to this Settlement Agreement and become incorporated into and enforceable under this Settlement Agreement constitute the final, complete and exclusive agreement and understanding among the Parties with respect to the settlement embodied in this Settlement Agreement. The Parties acknowledge that there are no representations, agreements or understandings relating to the settlement other than those expressly contained in this Settlement Agreement. The following appendices are attached to and incorporated into this Settlement Agreement:

"Appendix A" is the RI/FS Work Plan.

"Appendix B" is a map generally depicting the Study Area.

"Appendix C" is the list of Respondents.

"Appendix D" is the list of Work Respondents.

"Appendix E" is the list of Funding Respondents.

"Appendix F " is the City of New York Trust Fund Agreement.

XXX. ADMINISTRATIVE RECORD

111. EPA will, pursuant to CERCLA and the NCP, determine the contents of the administrative record file for selection of any remedial action. Respondents shall submit to EPA documents developed during the course of the RI/FS upon which selection of a response action may be based. Upon request of EPA, Respondents shall provide copies of

plans, task memoranda for further action, quality assurance memoranda and audits, raw data, field notes, laboratory analytical reports and other reports. Upon request of EPA, Respondents shall additionally submit any previous studies conducted under state, local or other federal authorities relating to selection of the response action, and all communications between Respondents and state, local or other federal authorities concerning selection of a response action. At EPA's discretion, Respondents shall establish a community information repository at or near the Study Area, to house one copy of the administrative record.

XXXI. EFFECTIVE DATE AND SUBSEQUENT MODIFICATION

112. This Settlement Agreement shall be effective three days after it has been signed by EPA by its delegated official and delivered to Liaison Counsel.

113. This Settlement Agreement may be amended as provided in Paragraph 6 to add additional Respondents, or may be otherwise amended by mutual agreement of EPA and Respondents. Amendments shall be in writing and shall be effective when signed by EPA by its delegated official.

114. No informal advice, guidance, suggestion, or comment by the EPA Project Coordinator or other EPA representatives regarding reports, plans, specifications, schedules, or any other writing submitted by Respondents shall relieve Respondents of their obligation to obtain any formal approval required by this Settlement Agreement, or to comply with all requirements of this Settlement Agreement, unless it is formally modified. The EPA Project Coordinator may extend any deadline under this Settlement Agreement, provided that any such extension shall be in writing (which may include electronic mail).

XXXII. NOTICE OF COMPLETION OF WORK

115. If Respondents believe that all Work and all other activities have been fully performed in accordance with this Settlement Agreement, with the exception of any continuing obligations, such as Respondents' obligation to retain records pursuant to Section XIV of this Settlement Agreement, Respondents may request that EPA provide them with written notice of completion of the Work. Upon receipt of such a request, EPA will either provide written notice of completion to Respondents or, if EPA determines that any Work has not been completed in accordance with this Settlement Agreement, EPA will so notify Respondents, provide a list of the deficiencies, and require that Respondents correct such deficiencies. Failure by Respondents to correct such deficiencies shall be a violation of

this Settlement Agreement.

It is so ORDERED AND AGREED this 7th day of July, 2011.

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY

BY: _____ DATE: 7/7/2011

WALTER E. MUGDAN, Director
Emergency and Remedial Response Division
United States Environmental Protection Agency
Region 2

In the Matter of the Newtown Creek Superfund Site
Administrative Settlement Agreement and Order on Consent for Remedial Investigation/Feasibility Study, CERCLA Docket No. CERCLA-02-2011-2011

FOR RESPONDENT __Phelps Dodge Refining Corporation_____

_____ 6/24/11
Signature Date

L. Richards McMillan, II

_____
Senior Vice President

In the Matter of the Newtown Creek Superfund Site
Administrative Settlement Agreement and Order on Consent for Remedial Investigation/Feasibility Study, CERCLA Docket No. CERCLA-02-2011-2011

FOR RESPONDENT Texaco Inc.

Frank G. Soler — 14 June 2011

Signature — Date

Frank G. Soler

Printed Name

Vice President and Secretary

Title

In the Matter of the Newtown Creek Superfund Site
Administrative Settlement Agreement and Order on Consent for Remedial Investigation/Feasibility Study, CERCLA Docket No. CERCLA-02-2011-2011

FOR RESPONDENT BP PRODUCTS NORTH AMERICA INC.

Patricia Gallery
Signature

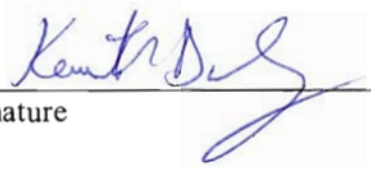
27 MAY 2011
Date

PATRICIA GALLERY
Printed Name

GLOBAL PORTFOLIO MANAGER R&M/AE
Title

In the Matter of the Newtown Creek Superfund Site
Administrative Settlement Agreement and Order on Consent for Remedial Investigation/Feasibility Study, CERCLA Docket No. CERCLA-02-2011-2011

FOR RESPONDENT **The Brooklyn Union Gas Company d/b/a National Grid NY**



Signature

June 16, 2011
Date

Kenneth D. Daly
Printed Name

Chief Financial Officer and Treasurer
Title

In the Matter of the Newtown Creek Superfund Site
Administrative Settlement Agreement and Order on Consent for Remedial Investigation/Feasibility Study, CERCLA Docket No. CERCLA-02-2011-2011

FOR RESPONDENT ExxonMobil Oil Corporation

C Pearson
Signature

6/27/2011
Date

Clifford L Pearson
Printed Name

Manager, Major Projects
Title

ExxonMobil Environmental Services Co.

In the Matter of the Newtown Creek Superfund Site
Administrative Settlement Agreement and Order on Consent for Remedial Investigation/Feasibility Study, CERCLA Docket No. CERCLA-02-2011-2011

FOR RESPONDENT CITY OF NEW YORK

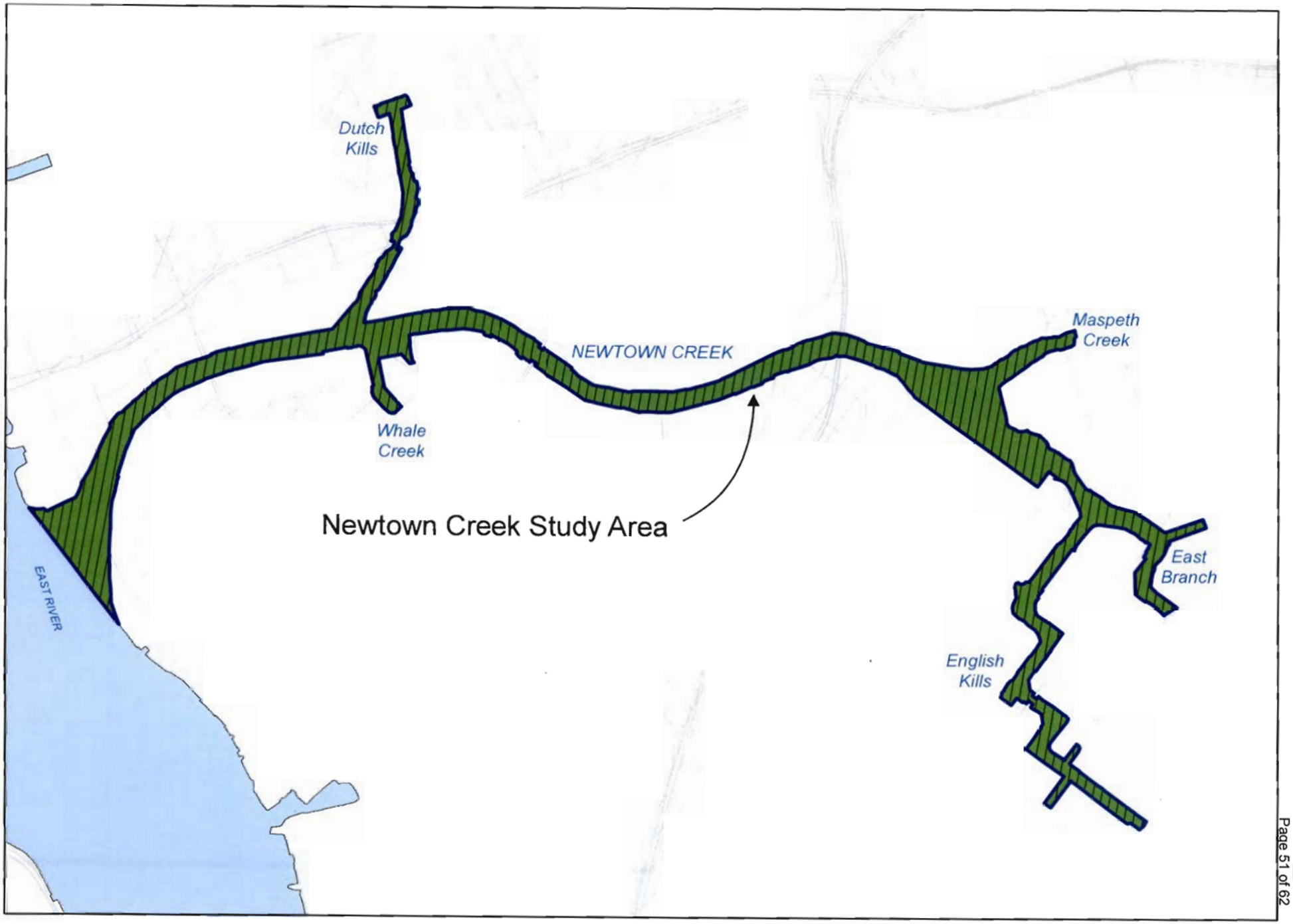
Susan E Amron
Signature

June 30, 2011
Date

Susan E. Amron
Printed Name

Chief, Environmental Law Division, New York
City Law Department
Title

Appendix A
RI/FS Work Plan



J:\indi_Service\Project Files\Newtown Creek\10.0 Data\GIS\Projects\Work Plan February 15\Figure 1.mxd

AECOM

0 1,000 Feet

Source:
ESRI 9.3 Streetmap Data Layer.

Appendix B
Newtown Creek Study Area

Exhibit __ (CFW-4)
Page 51 of 62

Appendix C
List of Respondents
As of the Effective Date

Phelps Dodge Refining Corporation (a New York corporation)
Texaco Inc. (a Delaware corporation)
BP Products North America Inc. (a Delaware corporation)
The Brooklyn Union Gas Company d/b/a National Grid NY (a New York corporation)
ExxonMobil Oil Corporation (a New York corporation)
The City of New York (a municipal corporation organized and existing under the laws of the State of New York)

Appendix D
List of Work Respondents
As of the Effective Date

Phelps Dodge Refining Corporation (a New York corporation)
Texaco Inc. (a Delaware corporation)
BP Products North America Inc. (a Delaware corporation)
The Brooklyn Union Gas Company d/b/a National Grid NY (a New York corporation)
ExxonMobil Oil Corporation (a New York corporation)
The City of New York (a municipal corporation organized and existing under the laws of the State of New York)

Appendix E
List of Funding Respondents
As of the Effective Date

There are no Funding Respondents as of the Effective Date

Appendix F
City of New York Trust Fund Agreement

Attached

TRUST AGREEMENT

For City of New York Payment Obligations at the Newtown Creek Study Area
Dated: July ___, 2011

This Trust Agreement (this "Agreement") is entered into as of _____ by and between the City of New York, a municipal corporation organized and existing under the laws of the State of New York (the "Grantor"), and JP Morgan Chase Bank, a corporation organized and existing under the laws of the State of New York (the "Trustee").

Whereas, the United States Environmental Protection Agency ("EPA"), an agency of the United States federal government, and the Grantor have entered into an Administrative Settlement Agreement and Order on Consent, In the Matter of the Newtown Creek Superfund Site, Kings County and Queens County, New York, New York CERCLA Docket No. CERCLA-02-2011-2011, (hereinafter the "Settlement Agreement");

Whereas, the Settlement Agreement provides that the Grantor together with other Work Respondents, as hereinafter defined, will perform Work required by the Settlement Agreement;

Whereas, in order to provide a mechanism for Grantor to sign the Settlement Agreement as a Respondent and contribute to the funding of such Work, Grantor has agreed to establish and fund the trust created by this Agreement; and

Whereas, the Grantor, acting through its duly authorized officers, has selected the Trustee to be the trustee under this Agreement, and the Trustee has agreed to act as trustee hereunder.

Now, therefore, the Grantor and the Trustee agree as follows:

Section 1. Definitions. As used in this Agreement:

(a) The term "Beneficiary" shall have the meaning assigned thereto in Section 3 of this Agreement.

(b) The term "Business Day " means any day, other than a Saturday or a Sunday, that banks are open for business in New York, New York, USA.

(c) The term "Claim Certificate" shall have the meaning assigned thereto in Section 4(a) of this Agreement.

(d) The term "Fund" shall have the meaning assigned thereto in Section 3 of this Agreement.

(e) The term "Grantor" shall have the meaning assigned thereto in the first paragraph of this Agreement.

(f) The term "Study Area" shall have the meaning assigned thereto in paragraph 13.v of the Settlement Agreement.

(g) The term "Trust" shall have the meaning assigned thereto in Section 3 of this Agreement.

(h) The term "Trustee" shall mean the trustee identified in the first paragraph of this Agreement, along with any successor trustee appointed pursuant to the terms of this Agreement.

(i) The term "Work" shall have the meaning assigned thereto in paragraph 13.z of the Settlement Agreement.

(j) The term "Work Respondents" shall have the meaning assigned thereto in paragraph 13.q of the Settlement Agreement.

Section 2. Identification of Facilities and Costs. This Agreement pertains to costs, including related administrative costs, for Work required at the Newtown Creek Study Area in Kings County and Queens County, New York, New York (the "Study Area"), pursuant to the above referenced Settlement Agreement.

Section 3. Establishment of Trust Fund. The Grantor and the Trustee hereby establish a trust (the "Trust"), for the benefit of EPA (the "Beneficiary"), to enable the Grantor to sign the Settlement Agreement as a respondent and contribute to payment for performance of the Work required by, and in accordance with the terms of, the Settlement Agreement. The Grantor and the Trustee intend that no third party shall have access to monies or other property in the Trust except as expressly provided herein. The Trust is established initially as consisting of funds in the amount of [_____] U.S. Dollars ($ _____). Such funds, along with any other monies and/or other property hereafter deposited into the Trust, and together with all earnings and profits thereon, are referred to herein collectively as the "Fund." The Fund shall be held by the Trustee, IN TRUST, as hereinafter provided. The Trustee shall not be responsible nor shall it undertake any responsibility for the amount or adequacy of, nor any duty to collect from the Grantor, any payments necessary to discharge any liabilities of the Grantor.

Section 4. Payment for Work Required Under the Settlement Agreement. The Trustee shall make payments from the Fund in accordance with the following procedures.

(a) From time to time, the Grantor may request that the Trustee make payment from the Fund for Work performed under the Settlement Agreement, or for related administrative costs, by delivering to the Trustee a written invoice and certificate (together, a "Claim Certificate") signed by a duly authorized representative of the Grantor and certifying:

(i) that the invoice is for Work performed at the Study Area in accordance with the Settlement Agreement or for administrative costs incurred in connection with the Work or Settlement Agreement; and

(ii) a description of the Work that has been performed, the amount of the claim, and the identity of the payee(s);

(b) Upon receipt of a Claim Certificate, the Trustee shall promptly make the payment from the Fund requested in such Claim Certificate.

(c) While this Agreement is in effect, disbursements from the Fund are governed exclusively by the express terms of this Agreement.

Section 5. Trust Management. The Trustee shall invest and reinvest the principal and income of the Fund and keep the Fund invested as a single fund, without distinction between principal and income, in accordance with directions which the Grantor may communicate in writing to the Trustee from time to time, except that:

(a) securities, notes, and other obligations of any person or entity shall not be acquired or held by the Trustee with monies comprising the Fund, unless they are securities, notes, or other obligations of the U.S. federal government or any U.S. state government or as otherwise permitted in writing by the Grantor;

(b) the Trustee is authorized to invest the Fund in time or demand deposits of the Trustee, to the extent such deposits are insured by an agency of the U. S. federal or any U.S. state government; and

(c) the Trustee is authorized to hold cash awaiting investment or distribution uninvested for a reasonable time and without liability for the payment of interest thereon.

Section 6. Express Powers of Trustee. Without in any way limiting the powers and discretion conferred upon the Trustee by the other provisions of this Agreement or by law, the Trustee is expressly authorized and empowered:

(a) to make, execute, acknowledge, and deliver any and all documents of transfer and conveyance and any and all other instruments that may be necessary or appropriate to carry out the powers herein granted;

(b) to register any securities held in the Fund in its own name or in the name of a nominee and to hold any security in book entry, or to deposit or arrange for the deposit of such securities in a qualified central depositary even though, when so deposited, such securities may be merged and held in bulk in the name of the nominee of such depositary with other securities deposited therein by another person, or to deposit or arrange for the deposit of any securities issued by the U.S. federal government or any U.S. state government, or any agency or instrumentality thereof, with a Federal Reserve bank having net capital of not less than $50 million, but the books and records of the Trustee shall at all times show that all such securities are part of the Fund; and

(c) to deposit any cash in the Fund in interest-bearing accounts maintained or savings certificates issued by the Trustee, in its separate corporate capacity, or in any other banking institution affiliated with the Trustee, to the extent insured by an agency of the U.S. federal government.

Section 7. Taxes and Expenses. All taxes of any kind that may be assessed or levied against or in respect of the Fund and all expenses and charges incurred by the Trustee in connection with the administration of the Fund and this Trust shall be paid from the Fund.

Section 8. Annual Valuation. The Trustee shall provide monthly statements to the Grantor. The Trustee upon the direction of the Grantor shall make records available at reasonable times to any entity which the Grantor shall notify the Trustee is authorized to conduct an audit of the Fund. The Trustee shall also provide such information concerning the Fund and this Trust as EPA may request from time to time.

Section 9. Advice of Counsel. The Trustee may from time to time consult with counsel with respect to any question arising as to the construction of this Agreement or any action to be taken hereunder; provided, however, that any counsel retained by the Trustee for such purposes may not, during the period of its representation of the Trustee, serve as counsel to the Grantor.

Section 10. Trustee Compensation. The Trustee shall be entitled to reasonable compensation for its services as agreed upon in writing from time to time with the Grantor and as notified in writing to the Beneficiary.

Section 11. Trustee and Successor Trustee. The Trustee and any replacement Trustee must not be affiliated with the Grantor. The Trustee may resign or the Grantor may replace the Trustee, but such resignation or replacement shall not be effective until the Grantor has appointed a successor trustee and this successor accepts such appointment. The successor trustee shall have the same powers and duties as those conferred upon the Trustee hereunder. Upon the successor trustee's acceptance of the appointment, the Trustee shall assign, transfer, and pay over to the successor trustee the funds and properties then constituting the Fund. If for any reason the Grantor cannot or does not act in the event of the resignation of the Trustee, the Trustee may apply to a court of competent jurisdiction for the appointment of a successor trustee or for instructions. The successor trustee shall specify the date on which it assumes administration of the Fund and the Trust in a writing sent to the Grantor, the Beneficiary, and the present Trustee by certified mail no less than 10 days before such change becomes effective. Any expenses incurred by the Trustee as a result of any of the acts contemplated by this Section shall be paid as provided in Section 7.

Section 12. Instructions to the Trustee. All instructions to the Trustee shall be in writing, signed by such persons as are empowered to act on behalf of the entity giving such instructions. The Trustee shall be fully protected in acting without inquiry on such written instructions given in accordance with the terms of this Agreement. The Trustee shall have no duty to act in the absence of such written instructions, except as expressly provided for herein.

Section 13. Amendment of Agreement. This Agreement may be amended only by an instrument in writing executed by the Grantor and the Trustee, and with the prior written consent of EPA.

Section 14. Irrevocability and Termination. Subject to the right to amend this Agreement as provided in Section 13, this Trust shall be irrevocable and shall continue until terminated upon the earlier to occur of (a) the written direction of the Grantor to terminate which shall include a certification from the Grantor that the Grantor has provided prior written notice to EPA and to the Work Respondents of the Grantor's intent to terminate the Trust and (b) the complete exhaustion of the Fund comprising the Trust as certified in writing by the Trustee to

EPA and the Grantor. Upon termination of the Trust pursuant to Section 14(a), all remaining trust property (if any), less final trust administration expenses, shall be delivered to the Grantor.

Section 15. Immunity and Indemnification. The Trustee shall not incur personal liability of any nature in connection with any act or omission, made in good faith, in the administration of this Trust, or in carrying out any directions by the Grantor or the EPA issued in accordance with this Agreement. Grantor shall indemnify and hold harmless the Trustee from and against any personal liability to which the Trustee may be subjected by reason of any act or conduct made by the Trustee in its official capacity by reason of its execution or performance of this Agreement, except those resulting from the Trustee's gross negligence or willful misconduct, including all expenses reasonably incurred in its defense in the event the Grantor fails to provide such defense.

Section 16. Choice of Law. This Agreement shall be administered, construed, and enforced according to the laws of the State of New York.

Section 17. Interpretation. As used in this Agreement, words in the singular include the plural and words in the plural include the singular. The descriptive headings for each Section of this Agreement shall not affect the interpretation or the legal efficacy of this Agreement.

Section 18. Notices. All notices and other communications given under this agreement shall be in writing and shall be addressed to the parties as follows or to such other address as the parties shall by written notice designate:

(a) If to the Grantor, to [_____].

(b) If to the Trustee, to [_____].

(c) If to EPA, to EPA Region 2, Remedial Project Manger for the Study Area and EPA Region 2, Office of Regional Counsel contact for the Study Area, at [_____].

(d) If to Work Respondents, to [_____]

[Remainder of page left blank intentionally.]

In Witness Whereof, the parties hereto have caused this Agreement to be executed by their respective officers duly authorized and attested as of the date first above written:

GRANTOR

[Signature of Grantor]
[Name and Title]

State of _____
County of _____

On this [date], before me personally came [name of Grantor official], to me known, who, being by me duly sworn, did depose and say that she/he is [title] of [corporation], the corporation described in and which executed the above instrument; and that she/he signed her/his name thereto.

[Signature of Notary Public]

TRUSTEE

[Signature of Trustee]
[Name and Title]

State of _____
County of _____

On this [date], before me personally came [name of Trustee official], to me known, who, being by me duly sworn, did depose and say that she/he is [title] of [corporation], the corporation described in and which executed the above instrument; and that she/he signed her/his name thereto.

[Signature of Notary Public]

APPROVED AS TO FORM AND CERTIFIED
AS TO LEGAL AUTHORITY

_____

Acting Corporation Counsel of
The City of New York

Exhibit ___ (CFW-5)

Value Engineering examples

KEDNY Examples of Value Engineering that led to Changes to DEC and EPA Remedy Decisions and associated Cost Savings

From 2019 through 2022, KEDNY has documented over 100 opportunities where value engineering and project optimizations have reduced costs or avoided additional costs to the projects. Below are examples of some of these reductions.

Citizens Works Former MGP Site - Remedial Action of Parcels I, II and III

Redesigned cost-effective approach to support of excavation

KEDNY successfully developed an alternative obstruction removal approach along the Citizen's Parcel 1 and 3 soil excavation support of excavation alignments. This approach relied upon utilizing an excavator to demolish subsurface obstructions to the vertical limits of the equipment (as deep as 20 feet below grade) versus the prior approach which included drilling through these shallower obstructions with a sonic rig. Utilizing the alternate approach resulted in savings of $688,000. In addition, KEDNY was able to re-evaluate the support of excavation construction sequencing and design to avoid coring and additional pre-drilling to remove obstructions. This resulted in an additional $870,000 in savings.

Developed a lower cost temporary sound attenuation barrier

KEDNY successfully developed a lower cost temporary sound attenuation barrier (using Cargo Containers instead of a fence supported curtain) to help reduce noise complaints. This improvised design was successful at mitigating noise complaints from the local community around the site and resulted in $400,000 in savings over the original approach.

Hydraulic Relief System Design Change

EPA provided comments on the hydraulic relief system in fall of 2019, which required the system to be redesigned. The Contractor provided a lump sum price to install the system, which the project team thought was higher than expected. We renegotiated an alternate pricing structure for installing the majority of the system on a time and materials basis and the section around the gas shaft on a lump sum basis. The revised pricing approach resulted in an approximate savings of $338,000 compared to the contractor's initial pricing proposal.

Change from Asphalt to Crushed Stone

KEDNY evaluated the restoration materials from the Parcel II and III Remedial Design (originally asphalt) towards the end of the project and determined that crushed stone is sufficient to prevent soil erosion prior to redevelopment of the properties. The estimated savings to restore Parcels II and III with crushed stone instead of pavement was $320,000 in savings.

Elimination of Temporary Fabric Structure

The previous excavations were completed with a tent (temporary fabric structure - TFS) to protect the community from fugitive dust and odors. In order to reduce the cost and duration to complete the additional DEC required Area 12 excavation, KEDNY developed a phased excavation approach that avoided the need for a TFS. This alternative concept was presented and accepted by the contractor, and the DEC, and resulted in a savings of $460,000.

Fulton Municipal Works MGP Site

Negotiated savings in cost agreement with third party

KEDNY successfully negotiated the Fulton Parcel 3 cost sharing agreement with the property owner over the course of several years for a cost avoidance or offset of $2.7 million.

Gowanus Canal Superfund Site

Negotiations resulted in long-term lease reductions

Negotiations from January 2022 through September 2022 led to a successful negotiation of the long term (15-year) property lease for project Site Support. Negotiations led by KEDNY on behalf of the Trust resulted in approximately $21 million in cost avoidance and future savings on the lease.

Revised capping design

Design Team realized cost avoidance/savings thru 90% design submission. The Team negotiated capping treatment layer design to optimize placement of treatment layers instead of a more expensive, equally protective EPA cap placement design. Reduced cap thickness to 4 feet overall versus upwards of 8 feet in areas of EPA design. This resulted in future project savings of $19 million.

Reduced dredge volume in RTA-3

Design Team optimized dredge volume for RTA-3 from 631,400 CY to 431,123 CY. Project team identified a method to quantify the additional volumes requested by the US Army Corps of Engineers (ACOE) to create safety buffer between the cap and ACOE maintenance dredge vessels which require an additional 5 feet of sediment. Making more precise measurements to meet the request reduced the cost for stabilization and transport & disposal by $39 million.

Greenpoint Energy Center Former MGP

Negotiated disposal cost with bidder

Disposal costs that the contractor presented during initial bid were very high. We collected estimates from various disposal facilities and summarized what it cost to dispose soils and sediments for other sites. Negotiated with the bidder and based on the initial estimated quantities, the cost was reduced by $1.6 million.

Greenpoint – Equity Works

Negotiated relocation of tenant

Our Greenpoint and Equity teams worked together to find space for current Equity site tenant on Greenpoint Energy Center in order to facilitate remediation of the Equity site. Due to shortage of suitable industrial space that met the tenant requirements, the cost could have been well over

$4 million. The Greenpoint and Equity teams collaborated with Facilities, Real Estate, Legal, Engineering, LNG Operations, CNG Station team, MRI-V team, Security, Warehousing, Environmental, Damage Prevention, Permitting, Community Relations, Process Safety, as well as NYSDEC to make the two distinct areas at Greenpoint available for the use for Equity tenant and remediation contractor.

Peoples Works

Negotiated transfer of site to the Brownfield Cleanup Program (BCP)

After several years of regulatory push-back on nature of and responsibility for impacts at the site, we successfullyterminatedit's the site consent order, allowing it to be transferred to the Brownfields Cleanup Program in February 2022. Transfer of the site into the BCP results in the avoidance of remedial costs which will now be assumed by the site developer. The cost savings totals approximately $5 million.

Exhibit ___ (CFW-6)

DEC work schedules for fiscal years ("FY") 2022, 2023 and 2024

National Grid – Former New York City Service Territory MGP Project Schedules

Site No.	Site name		DEC FY 2021 NG FY 2022 4/1/21 to 3/31/2022				DEC FY 2022 NG FY 2023 4/1/2022 to 3/31/2023				DEC FY 2023 NG FY 2024 4/1/2023 to 3/31/2024			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
224058	Bay Ridge Holder Station A,B	OU-1	NO FURTHER ACTION				NO FURTHER ACTION				NO FURTHER ACTION			
		OU-2	OM&M/ISMP[1]				OM&M/ISMP[1]				OM&M/ISMP[1]			
224060	Belmont Holder Station		SC[4]				SC[4]				SC[4]			
224048	Brooklyn Gas Light Works		SC[4]				SC[4]				SC[4]			
224012	Citizens Gas Works MGP		RA				RA				OM&M / SMP[1]			
243023	Clifton Works	OU-1	OM&M / SMP[1]				OM&M / SMP[1]				OM&M / SMP[1]			
		OU-2	OM&M / SMP[1]				OM&M / SMP[1]				OM&M / SMP[1]			
224026	Coney Island MGP		OM&M / SMP[1]				OM&M / SMP[1]				OM&M / SMP[1]			
224047	Dangman Park MGP	OU1	ISMP[1]				ISMP[1]				ISMP[1]			
		OU2	RI				IRM				ISMP[1]			
224050	Equity Works MGP		IRM				IRM				IRM			
224061	Flatbush Holder Station A, B		SC				SC	OM&M / SMP[1]			OM&M / SMP[1]			
224063	Front Street Holder Station		TBD[5]			ISMP[2]	ISMP[1]				ISMP[1]			
224051	Fulton Municipal Works MGP		RA				RA				RA			
	Gowanus Canal Superfund Site		RD RTAs 2/3 & RA RTA 1				RD RTAs 2/3 & RA RTA 1				RD RTAs 2/3 & RA RTA 1			
224052	Greenpoint Energy Center		RI/FS				RI/FS				FS			
241063	Jamaica Gas Light Company MGP		ON HOLD – NO ACCESS				ON HOLD – NO ACCESS				ON HOLD – NO ACCESS			
241062	Jamaica Holder Station		NO FURTHER ACTION				NO FURTHER ACTION				NO FURTHER ACTION			
224064	Keap Street Holder Station		NO FURTHER ACTION				NO FURTHER ACTION				NO FURTHER ACTION			
224056	Kings County Works MGP		OM&M				OM&M / ISMP[1]				OM&M / ISMP[1]			
224046	Metropolitan MGP		RI/FS				RD				RA			
224019A	Nassau Works MGP		FS		RD		RD				RA			
V00406	Newtown Creek Superfund Site		RI/FS				FS				FS			
224053	Peoples Works MGP		TBD		Not Currently Under Order		Not Currently Under Order				Not Currently Under Order			
224065	Plymouth Street Holder Station		SC[4]				SC[4]				SC[4]			
224066	Rutledge Street Holder Station		NO FURTHER ACTION				NO FURTHER ACTION				NO FURTHER ACTION			
224067	Scholes Street Holder Station		ISMP[1]				ISMP[1]				ISMP[1]			
224068	Skillman Street Holder Station		ISMP[1]				ISMP[1]				ISMP[1]			
224054	Union Street Holder Station		OM&M / SMP[1]				OM&M / SMP[1]				OM&M / SMP[1]			
224055	Williamsburg Works MGP		RI				FS(AA)				IRM			
224069	Wythe Avenue Holder Station	OU-1	ISMP[1]				ISMP[1]				ISMP[1]			
		OU-2	ISMP[1]				ISMP[1]				ISMP[1]			

National Grid – Former New York City Service Territory MGP Project Schedules

Color Key and Header Notes		
SC	Site Characterization	DEC FY – NYSDEC Fiscal Year
RI/FS	Remedial Investigation & Feasibility Study	NG FY – National Grid Fiscal Year
RD	Remedial Design	SMP[1] – Requires Periodic Review Report and management of intrusive work
RA	Remedial Action	SMP[2] – Requires inspection only
IRM	Interim Remedial Measure	XXX[3] – Access constraints are limiting the range of activities which could be performed
OM&M	Operation, Monitoring & Maintenance / Site Management Plan	XXX[4] – Site activities on hold per schedule agreement between NG and NYSDEC
		TBD[5] – Future phase(s) to be determined (TBD) based on redevelopment status

Exhibit ___ (CFW-7)

KEDNY's compliance with existing timetables and DEC requirements

Compliance with Existing Timetables

Site Name	Current Status Q4 CY22	NYSDEC Approved Schedule for CY (2022) Q1	Q2	Q3	Q4	Variance Q4 CY22 (Y/N)	Explanation of variance
Bay Ridge	OMM	OMM	OMM	OMM	OMM	N	
Belmont Holder Station	SC	SC	SC	SC	SC	N	
Brooklyn Gas Light Works	SC	SC	SC	SC	SC	N	
Citizens Works (Carroll Gardens)	RA	RA	RA	RA	RA	N	
Clifton Works	OMM	OMM	OMM	OMM	OMM	N	
Coney Island MGP	OMM	OMM	OMM	OMM	OMM	N	
Dangman Park	RI/RP	RI	RA	RA	RA	Y	Delays resulting from regulatory approval of ROD and constraints with obtaining access to OU2.
Equity Works	RP	RA	RA	RA	RA	Y	NAPL IRM inprogress. Delays to implement ISS IRM resulting from site access constraints.
Flatbush Holder Station A&B	OMM	SC	SC	OMM	OMM	N	
Front Street Holder Station	OMM	OMM	OMM	OMM	OMM	N	
Fulton Municipal	RA	RA	RA	RA	RA	N	
Gowanus Canal RTA1	RA	RA	RA	RA	RA	N	
RTA2	RP	RP	RP	RP	RP	N	
RTA3	RP	RP	RP	RP	RP	N	
Greenpoint Energy Center	RI/RP	RI/RP	RI/RP	RI/RP	RI/RP	N	
Jamaica Gas Light	TBD	TBD	TBD	TBD	TBD	N	
Kings County MGP	OMM	OMM	OMM	OMM	OMM	N	
Metropolitan Works MGP	RD	RI/FS	RD	RD	RD	N	
Nassau Works (Brooklyn Navy Yard) MGP	RD	RD	RD	RD	RD	N	
Newtown Creek MGP		RI/FS	FS	FS	FS	N	
People's Works	TBD	TBD	TBD	TBD	TBD	N	
Plymouth Street Holder Station	SC	SC	SC	SC	SC	N	
Scholes St Station	OMM	OMM	OMM	OMM	OMM	N	
Skillman St Station	OMM	OMM	OMM	OMM	OMM	N	
Union Station	OMM	OMM	OMM	OMM	OMM	N	
Williamsburg Works MGP	RI	RI	RI	RI	RI	N	
Whythe Ave Station OU1	OMM	OMM	OMM	OMM	OMM	N	

Site Characterization	SC
Remedial Investigation	RI
Remedial Planning	RP
Remedial Action	RA
Operation, Maintenance and Monitoring	OMM
No Further Action Required	NFA
To Be Determined	TBD
Withdrew from Order	WDW

Sites no longer reported:
- Jamaica Holder
- Keap Street Holder
- Newtown Holder Station
- Rutledge MGP

Exhibit ___ (CFW-8)

KEDNY's past SIR program spend on an annual basis for 2019-2021

SIR KEDNY Spend 2019-2021			
SITE	**2019**	**2020**	**2021**
Bay Ridge Holder Station A,B	$63,134.59	$32,713.01	$19,422.48
Belmont Holder Station	$84.82		
Brooklyn Gas Light Works	$84.82	$473.96	($697.02)
Citizens Gas Works MGP	$7,676,004.28	$36,145,937.55	$16,547,483.61
Clifton Works	$558,748.41	$2,412,650.09	$750,602.26
Coney Island MGP	$357,572.36	$275,097.77	$232,414.13
Dangman Park MGP	$61,040.19	$95,067.92	$93,234.49
Equity Works MGP	$681,876.71	$415,257.15	$795,265.43
Flatbush Holder Station A, B	$18,408.00	$1,070,125.56	$580,092.07
Front Street Holder Station	$7,527.77	$24,118.16	$267,972.14
Fulton Municipal Works MGP	$6,606,991.35	$24,138,413.24	$4,017,176.37
Gowanus Canal	$11,993,611.77	$73,019,640.52	$41,683,236.46
Greenpoint Energy Center	$725,720.70	$470,392.85	$652,858.22
Jamaica Gas Light Company MGP	$4,327.50	$5,127.68	$6,382.20
Kings County Works MGP	$84,122.72	$15,918.33	$9,556.08
Metropolitan MGP	$164,110.82	$129,101.21	$386,545.00
Nassau MGP	$79,686.57	$158,513.76	$62,058.57
Newtown Creek	$8,566,548.95	$5,187,084.06	$6,592,482.26
Peoples Works MGP	$23,134.54	$45,552.23	$3,459.27
Plymouth Street Holder Station	$84.82		$0.00
Scholes St Holder Station	$759.32	$16,975.62	$16,459.41
Skillman St Holder Station	$53,885.50	$29,782.49	$29,871.52
Union Station (Citizens Branch) Holder Station	$23,137.64	$24,050.44	$30,383.61
Williamsburg Works MGP	$924,312.10	$409,337.06	$393,483.85
Wythe Ave Holder Station	$46,310.66	$45,773.08	$34,525.27
General SIR	$3,452,198.38	$8,789,733.64	$4,484,446.75
Total	**$42,173,425.29**	**$152,956,837.38**	**$77,688,714.43**

Exhibit ___ (CFW-9)

KEDNY's SIR program spend in the Historic Test Year

KEDNY Historic Test Year Spend

Site Name	2022
Bay Ridge Holder Station A,B	$14,120.71
Belmont Holder Station	$1,101.75
Brooklyn Gas Light Works	$1,341.89
Citizens Gas Works MGP	$8,987,180.52
Clifton Works	$195,365.86
Coney Island MGP	$189,060.44
Dangman Park MGP	$140,159.29
Equity Works MGP	$810,869.97
Flatbush Holder Station A, B	$77,116.52
Front Street Holder Station	$4,945,368.73
Fulton Municipal Works MGP	$6,471,416.21
Gowanus Canal	$37,829,328.83
Greenpoint Energy Center	$514,071.15
Jamaica Gas Light Company MGP	$16,376.88
Kings County Works MGP	$5,547.57
Metropolitan MGP	$479,424.22
Nassau MGP	$358,962.37
Newtown Creek	$6,573,421.78
Peoples Works MGP	$4,989.08
Plymouth Street Holder Station	$1,326.24
Scholes St Holder Station	$17,746.20
Skillman St Holder Station	$20,095.15
Union Station (Citizens Branch) Holder Station	$4,615.25
Williamsburg Works MGP	$1,940,522.58
Wythe Ave Holder Station	$35,594.87
General SIR Program	$6,445,363.02
Overall SIR Program	**$76,080,487.08**

Exhibit ___ (CFW-10)

KEDNY's forecast SIR program spend in the Rate Year and Data Years

SITE	Jan-Mar 2023 Forecast	FY24 Forecast	FY25 Forecast	FY26 Forecast	FY27 Forecast	FY28 Forecast
Bay Ridge Holder Station A,B	$18,000	$20,000	$17,000	$22,000	$22,000	$23,000
Belmont Holder Station	$1,000	$1,000	$7,000	$8,000	$9,000	$9,000
Brooklyn Gas Light Works	$1,000	$214,000	$55,000	$19,000	$17,000	$18,000
Citizens Gas Works MGP	$1,844,000	$1,840,000	$10,299,000	$8,943,000	$470,000	$484,000
Clifton Works	$54,000	$330,000	$186,000	$188,000	$194,000	$199,000
Coney Island MGP	$440,000	$479,000	$121,000	$106,000	$109,000	$90,000
Dangman Park MGP	$1,000	$483,000	$50,000	$213,000	$114,000	$91,000
Equity Works MGP	$617,000	$15,362,000	$8,501,000	$424,000	$205,000	$203,000
Flatbush Holder Station A, B	$77,000	$235,000	$24,000	$25,000	$26,000	$27,000
Front Street Holder Station	$191,000	$100,000	$13,000	$14,000	$14,000	$14,000
Fulton Municipal Works MGP	$6,850,000	$17,152,000	$30,628,000	$35,872,000	$40,617,000	$43,192,000
Greenpoint Energy Center	$216,000	$608,000	$2,421,000	$3,554,000	$3,825,000	$4,052,000
Jamaica Gas Light Company MGP	$4,000	$17,000	$29,000	$217,000	$409,000	$7,272,000
Kings County Works MGP	$4,000	$53,000	$358,000	$303,000	$273,000	$225,000
Metropolitan MGP	$390,000	$1,140,000	$14,865,000	$8,310,000	$185,000	$190,000
Nassau MGP	$54,000	$176,000	$1,720,000	$451,000	$546,000	$840,000
Peoples Works MGP	$2,000	$24,000	$0	$0	$0	$0
Plymouth Street Holder Station	$1,000	$1,000	$162,000	$162,000	$166,000	$135,000
Scholes St Holder Station	$1,000	$17,000	$18,000	$17,000	$17,000	$18,000
Skillman St Holder Station	$104,000	$205,000	$41,000	$42,000	$43,000	$44,000
Union Station (Citizens Branch) Holder Station	$8,000	$58,000	$34,000	$30,000	$31,000	$43,000
Williamsburg Works MGP	$571,000	$944,000	$3,020,000	$3,045,000	$29,972,000	$30,841,000
Wythe Ave Holder Station	$103,000	$173,000	$56,000	$58,000	$59,000	$41,000
General SIR Program	$1,718,000	$6,300,000	$6,489,000	$6,684,000	$6,884,000	$7,091,000
Adjustment General SIR Program	$0	$0	($6,489,000)	($6,684,000)	($6,884,000)	($7,091,000)
Subtotal MGP Sites	$13,270,000	$45,932,000	$72,625,000	$62,023,000	$77,323,000	$88,051,000
Adjustment - 20%		($9,186,000)	($14,525,000)	($12,405,000)	($15,465,000)	($17,610,000)
Adjusted MGP sites	**$13,270,000**	**$36,746,000**	**$58,100,000**	**$49,618,000**	**$61,858,000**	**$70,441,000**
Superfund Sites HTY	$6,828,000	$32,686,400	$64,300,000	$66,200,000	$68,200,000	$70,200,000
KEDNY Total Site Spends	**$20,098,000**	**$69,432,400**	**$122,400,000**	**$115,818,000**	**$130,058,000**	**$140,641,000**

Notes:

1. An adjustment of -20% is applied to forecasts based on average 3 calendar year underspend of forecast. The adjustment is applied to the superfund sites in FY24, but not FY25-FY28 as these are based on average annual actual costs from the past 3 calendar years.